Proofs From the Ground Up

A Deep Conversational Reader Before Velleman

Engineering Mathematics to Pure Mathematics

May 2026

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1 How This Book Should Be Read: The Change in Game

This book begins in a place that can feel almost too simple: sentences, variables, words like "every" and "some." That is deliberate. The early foundations of proof are not advanced because they are complicated. They are advanced because they are easy to underestimate.

Many students arrive from calculation-heavy mathematics with a strong set of useful reflexes. They can solve systems, differentiate, integrate, diagonalize a matrix, separate a differential equation, or use a numerical method to approximate an answer. Those skills matter. They are not being thrown away. But proof-based mathematics asks for a different kind of control.

The old game often asks:

What is the answer?

The new game asks:

What exactly is being claimed, and why must it be true?

That change is small in wording and large in consequences.

Focus

Do not measure progress through this part by pages completed. Measure it by whether you can close the file and explain the idea without leaning on the notation as a crutch.

The purpose of this opening part is to make the new game visible. We will treat logic not as a cold symbolic machine, but as the grammar of mathematical responsibility. If a theorem says "for every," you must know who chooses the object. If it says "there exists," you must know what counts as producing one. If a variable appears, you must know whether it is free, fixed, arbitrary, bound, or chosen as a witness. If a sentence has no truth value yet, you must not pretend it is a theorem.

This is the level where beginners often feel silly asking questions. Do not feel silly. These are the load-bearing pieces.

1.1 The Reader's Contract

Here is the contract for reading these foundations.

1. Read a paragraph once for the idea.
2. Read it again for the exact mathematical claim.
3. Produce your own example.
4. Produce a non-example.

5. Ask what would break if a word were changed.
6. Try to say the same thing in ordinary language, symbols, and proof language.

The last item is important. A concept is not yours until you can move between languages. For example, consider:

$$\forall n \in \mathbb{Z}, \quad n^2 \geq 0.$$

Ordinary language:

> Every integer has a nonnegative square.

Proof-language opening:

> Let $n \in \mathbb{Z}$ be arbitrary. We must show $n^2 \geq 0$.

Symbolic form:

$$\forall n \in \mathbb{Z},\ P(n), \qquad \text{where } P(n) \text{ is } n^2 \geq 0.$$

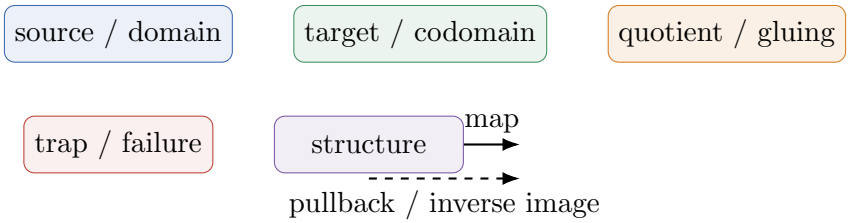
Those are not three unrelated things. They are the same claim viewed through three lenses. Much of proof maturity is the ability to change lenses without changing the claim.

Intuition

Mathematics is not made of symbols. It is made of ideas disciplined by symbols. The discipline matters. So does the idea.

1.2 The Visual Language of This Book

As the book becomes more visual, the diagrams use a consistent language. You do not need to memorize the colors, but the repetition should help your eye.



This matters because diagrams should not become decoration. They are a second notation. The point is to help you see maps, domains, targets, identifications, and failure points before the symbols tighten the idea.

Reading note

When a diagram appears, ask what each region, arrow, and color means formally. If you cannot translate the picture back into definitions, the picture is only a sketch.

1.3 Recurring Motifs

The book also uses small recurring marks as cognitive landmarks.

Mark	Name	Meaning
	compression arrow	examples or repeated moves are being compressed into a definition, theorem, or structure
	local/global marker	a local fact is being assembled into a global conclusion, or a global condition controls local behavior
	hidden quantifier	a phrase is hiding "for every," "there exists," or a dependency of choices
	strategy fork	the proof could go several ways; pause and choose deliberately

These marks are not new mathematics. They are memory handles. When the material gets abstract, familiar landmarks keep you oriented.

1.4 Abstraction as Compression

One of the central themes of the whole reader is this:

> Abstraction is compression with memory.

At first you learn examples. Then a definition compresses many examples into a single object. A theorem compresses many repeated arguments into one reusable tool. A structure, such as group or topology, compresses many different worlds that obey the same rules. A structure-preserving map compresses the idea of "same kind of behavior" across different objects.

This is why mathematics becomes more abstract as it matures. The goal is not to make things vague. The goal is to store more meaning in fewer, more powerful concepts.

Why mathematicians care

Good abstraction feels beautiful because it suddenly makes distant examples look like relatives. Modular arithmetic, symmetries of a square, and invertible matrices are different on the surface, but group theory lets one proof speak to all of them.

Pause

Try this with no symbols. Name three things in your life that are different on the surface but follow the same pattern. A playlist, a task queue, and a proof outline all have order. A map, an API, and a function all send inputs somewhere. This is the emotional beginning of abstraction: noticing repeated shape.

1.5 The Emotional Weather of Learning Proofs

There are several normal states in learning rigorous mathematics:

- **Notation fatigue:** the symbols are individually clear, but the page feels noisy.
- **Delayed understanding:** a definition makes sense only after three examples and one failed proof.
- **Abstraction vertigo:** the text has moved from familiar objects to a structure that contains them.
- **Proof amnesia:** you understood a proof yesterday but cannot reconstruct it today.
- **Semantic blur:** you can read every word but cannot yet say what the theorem is really doing.

None of these are signs that you are not mathematical. They are signs that your mind is building new compression machinery.

Focus

The cure for abstraction vertigo is not speed. It is returning to examples, non-examples, and the exact definition until the abstract word has weight again.

1.6 How To Slow Down Productively

There is an unhelpful way to slow down: stare at a page until the symbols become familiar-looking. There is also a productive way: interrogate the sentence.

When you meet a claim, ask:

- What objects are being discussed?
- What universe do those objects live in?
- Is the sentence true or false once the objects are fixed?
- Which variables are already known, and which are to be chosen?
- Is the claim universal, existential, conditional, or some mixture?

- What would count as a proof?
- What would count as a counterexample?

These questions are not extra decorations. They are how proofs start.

For instance:

$$\forall x \in \mathbb{R}, \exists y \in \mathbb{R}, \quad y > x.$$

This statement is not asking for a single magic number y. It gives you x first, then allows you to answer with a y depending on x. A proof can choose $y = x + 1$. If you miss the order of the quantifiers, the statement feels mysterious. If you see the order, it is almost mechanical.

Walking question

When you walk away from a definition, ask: what are the inputs, what are the outputs, and what promises are being made?

1.7 The Change From Examples To Reasons

Examples are good. Examples are necessary. Examples are not proofs.

Suppose you notice:

$$1 = 1^2, \qquad 1 + 3 = 2^2, \qquad 1 + 3 + 5 = 3^2, \qquad 1 + 3 + 5 + 7 = 4^2.$$

It is natural to guess:

$$1 + 3 + 5 + \cdots + (2n - 1) = n^2.$$

Checking the first hundred cases gives evidence. Checking the first million cases gives more evidence. A proof gives something else: a reason the pattern cannot fail.

One proof is visual. Start with a 1×1 square. To build a 2×2 square, add an L-shaped border of 3 dots. To build a 3×3 square, add an L-shaped border of 5 dots. To build an $n \times n$ square, the last border has $2n - 1$ dots. So the first n odd numbers build an $n \times n$ square.

Another proof uses induction. Another uses the arithmetic series formula. Another counts the number of pairs (i, j) with $1 \leq i, j \leq n$ by grouping them according to the larger of i and j. The theorem is the same, but the proofs reveal different structure.

Intuition

A computation samples the landscape. A proof explains why the landscape has that shape.

This is a central shift. In proof-based mathematics, an answer without a reason is unfinished. A reason without an answer may still be valuable, because it may explain why a whole family of answers behaves as it does.

1.8 What Counts As Understanding

For the early foundations, understanding is not a private feeling of familiarity. It has observable signs.

You understand a statement when you can:

- classify it as a statement, predicate, expression, command, or question;
- identify its variables and their roles;
- state its domain;
- negate it accurately;
- say what a proof would need to do;
- say what a disproof would need to exhibit;
- produce small examples and edge cases.

That is a lot. It is supposed to be. Mathematical reading is active.

Take the statement:

$$\forall n \in \mathbb{Z}, \quad n^2 + n \text{ is even.}$$

Understanding it includes knowing that n is arbitrary, the domain is $\mathbb{Z}$, the claim is universal, and a counterexample would be one integer n for which $n^2 + n$ is odd. It also includes seeing at least two proof routes:

$$n^2 + n = n(n + 1),$$

the product of consecutive integers, one of which is even; or cases, where $n = 2k$ or $n = 2k+1$.

You do not need both proofs every time. But seeing multiple routes makes the terrain more real.

1.9 Formal Does Not Mean Inhuman

Some people talk as if rigor is a punishment: first you enjoy mathematics, then rigor arrives and takes the toys away. That is not the right picture.

Rigor is what lets us keep promises. It lets us distinguish a theorem from a pattern, a definition from a vibe, a proof from a convincing diagram. The goal is not to make mathematics sterile. The goal is to make it honest enough that other people can trust it, build on it, and find exactly where it fails.

History and connection

Greek mathematics, especially Euclid's *Elements*, made proof a public object: not just a private conviction, but a chain of reasons others could inspect. Centuries later, calculus forced mathematicians to sharpen the language again. Infinitesimals, limits, continuity, and convergence all worked brilliantly in practice before their foundations were made precise. The history of rigor is not the history of people being fussy. It is the history of powerful ideas becoming powerful enough to need better grammar.

This history matters because you will sometimes feel that a definition is too careful. "Why do we have to say domain and codomain?" "Why does the order of quantifiers matter?" "Why distinguish a statement from a predicate?" The answer is usually: because later, if we do not, something breaks.

1.10 The Four Views

For each concept in this part, we will keep returning to four views.

The informal view. What is the plain-language idea?

The formal view. What is the exact statement?

The proof view. What kind of action does the statement demand?

The debugging view. How do beginners commonly misuse it?

For example, the universal quantifier $\forall$ has these views:

- Informal: every object in the domain has the property.
- Formal: $\forall x \in D,\ P(x)$.
- Proof: let $x \in D$ be arbitrary and prove $P(x)$.
- Debugging: do not prove it by checking a few favorite values of x.

The existential quantifier $\exists$ has parallel views:

- Informal: at least one object in the domain has the property.
- Formal: $\exists x \in D$ such that $P(x)$.
- Proof: produce or justify a witness.
- Debugging: do not act as if the witness works uniformly unless the statement requires that.

The same idea viewed four ways becomes much harder to confuse.

1.11 A Software Analogy: Types, Scope, and Contracts

If you have programmed, logic will feel less alien when you compare it to types and scope.

A variable with no stated domain is like a parameter with no type. The expression might still be meaningful in someone's head, but the machine cannot safely use it.

The predicate

$$P(x) : x^2 < 10$$

depends on the type of x. If $x \in \mathbb{N}$, the satisfying values are limited. If $x \in \mathbb{Z}$, there are negative values too. If $x \in \mathbb{R}$, there is an interval. If x is a matrix, the expression $x^2 < 10$ may not even mean anything until we define an order.

A bound variable is like a local variable. In

$$\sum_{i=1}^{n} i^2,$$

the i is local to the summation. If the next line says "let i be an integer," that is a new use of the letter unless the context says otherwise. The letter is not sacred; the role is.

A theorem is like a contract. It says: if you give me objects satisfying these hypotheses, I promise this conclusion. A proof is the implementation of the contract.

Common trap

Do not confuse familiar notation with well-defined notation. A symbol you have seen before can become meaningless if the domain changes.

1.12 A Physical Analogy: Instruments and Specifications

Imagine a lab instrument with a specification:

> For every target temperature between 10°C and 40°C, there exists a dial setting that reaches that temperature within one minute.

That has quantifier structure:

$$\forall T \in [10, 40], \exists s \in S, \quad R(s, T).$$

The setting may depend on the target. If the target is 12°C, use one setting. If the target is 35°C, use another.

Now compare:

$$\exists s \in S, \forall T \in [10, 40], \quad R(s, T).$$

This says one fixed setting reaches every target temperature. That is a much stronger and probably false claim.

Physical systems make the order of choices obvious. Mathematics often hides the same issue behind compact notation.

1.13 Beginner-To-Rigor Reading Routine

When you face a theorem, use this routine.

1. Strip the theorem down to its logical skeleton.
2. Name the domain of every variable.
3. Mark which variables are arbitrary and which are chosen.
4. Translate the conclusion into the action needed to prove it.
5. Try one example.
6. Try one non-example or boundary case.
7. Write the first line of the proof before thinking about the clever part.

The first line is often forced. For

$$\forall x \in A, \exists y \in B, \quad P(x, y),$$

the proof probably begins:

> Let $x \in A$ be arbitrary. We must find $y \in B$ such that $P(x, y)$.

That line does not solve the problem, but it puts the problem in the correct shape.

Formal

A proof is not a transcript of how you discovered the idea. It is a structured explanation showing why the conclusion follows from the assumptions.

1.14 Proof Lab: One Claim, Three Readings

Proof lab

Study the claim:

$$\forall n \in \mathbb{Z}, \quad n^2 - n \text{ is even.}$$

Reading 1: examples.

$$3^2 - 3 = 6, \qquad (-4)^2 - (-4) = 20, \qquad 0^2 - 0 = 0.$$

The examples support the claim, but they do not prove it.

Reading 2: structure.

$$n^2 - n = n(n-1).$$

The numbers n and $n-1$ are consecutive integers, so one is even. Therefore their product is even.

Reading 3: proof skeleton. Let $n \in \mathbb{Z}$ be arbitrary. Since n and $n-1$ are consecutive integers, one of them is even. Hence $n(n-1)$ is even. Therefore $n^2 - n$ is even. Since n was arbitrary, the claim holds for every integer n.

Notice the last sentence. It is not decoration. It closes the universal proof by reminding the reader that no special property of n was used.

1.15 Non-Examples Of Proof

The following are not proofs of

$$\forall n \in \mathbb{Z}, \quad n^2 - n \text{ is even.}$$

Several examples.

$$1^2 - 1 = 0, \quad 2^2 - 2 = 2, \quad 3^2 - 3 = 6.$$

Good evidence, not proof.

An unsupported pattern claim.

> It keeps alternating, so it must be even.

What alternates? Why must it continue? What theorem justifies the pattern?

A circular explanation.

> $n^2 - n$ is even because it is always divisible by 2.

That only restates the definition of evenness. It does not show divisibility by 2.

A false general principle.

> Since n^2 and n have the same parity, their difference is even.

This one is actually a valid route if you prove the parity statement first. Without proof, it is a hidden lemma.

Common trap

When a proof says "clearly," ask what theorem, definition, or earlier result makes it clear. Sometimes "clearly" is fine. Sometimes it is where the missing proof is hiding.

1.16 Exercises

1. For the claim $\forall n \in \mathbb{Z},\ n^2 + n$ is even, write one example, one proof idea, and one full proof.
2. Give a computational check for $1 + 3 + \cdots + (2n - 1) = n^2$ when $n = 6$. Then explain why that check is not a proof.
3. Find a formula from calculus or physics that you know. State one assumption under which it is valid and one situation where it may fail.
4. Write the first line of a proof of: $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}$ such that $x + y = 0$.
5. Write the first line of a disproof of: $\exists y \in \mathbb{R}, \forall x \in \mathbb{R}$ such that $x + y = 0$.
6. Explain in ordinary language why the order of choices matters in a thermostat or control-system specification.

Walking question

Think of a theorem as a promise. Who is allowed to choose the inputs? Who must produce the outputs? What would count as breaking the promise?

2 Mathematical Sentences

Before proof, there is grammar. Not grammar in the sense of commas and adjectives, but mathematical grammar: what kind of object is this line of writing?

Some lines assert something. Some merely name an object. Some ask a question. Some issue an instruction. Some contain variables and become true or false only after the variables receive

values. If you do not know which kind of line you are reading, proof becomes blurry before it even starts.

Formal

A *statement* is a sentence that is either true or false.

Examples:

- "7 is prime" is a true statement.
- "9 is prime" is a false statement.
- "There are infinitely many prime numbers" is a true statement.
- "Every continuous function is differentiable" is a false statement.
- "There are infinitely many twin primes" is a statement whose truth is not currently known.

The last example is important. A sentence can be a statement even if humans do not know whether it is true. Truth value and known truth value are different ideas.

2.1 Statements Are Claims, Not Decorations

The expression

$$x^2 + 1$$

is not a statement. It does not claim anything. It is an expression, a mathematical object waiting to be evaluated or used.

The line

$$x^2 + 1 = 10$$

looks more sentence-like, but by itself it is not yet a statement unless x already has a fixed meaning. If $x = 3$, it is true. If $x = 0$, it is false. If x is unspecified, the line is a predicate: a condition that may become true or false after x is supplied.

The command

> Solve $x^2 + 1 = 10$.

is not a statement. It is an instruction.

The question

> Is $x^2 + 1 = 10$?

is not a statement. It asks for one.

Focus

The first proof-reading filter is simple: does this sentence have a truth value yet?

2.2 Predicates

Formal

A *predicate* is a sentence with one or more variables that becomes a statement when the variables are assigned values from a specified domain.

For example,

$$P(n): \quad n \text{ is prime}$$

is a predicate. If $n = 7$, $P(n)$ is true. If $n = 9$, $P(n)$ is false. If $n = 1$, $P(n)$ is false under the standard definition of prime.

Another predicate:

$$Q(x): \quad x^2 < 2.$$

This predicate changes personality depending on the domain.

Over $\mathbb{N} = \{1, 2, 3, \ldots\}$, it is true only for $x = 1$.

Over $\mathbb{Z}$, it is true for $x = -1, 0, 1$ if 0 is included in the integers, as usual.

Over $\mathbb{R}$, it is true exactly when

$$-\sqrt{2} < x < \sqrt{2}.$$

Same symbols. Different domains. Different mathematical situation.

Common trap

Never let a variable float without a domain. A variable without a domain is a character without a setting.

2.3 Expressions, Equations, Inequalities, And Sentences

It helps to separate four categories.

Expression. A mathematical object or recipe for one:

$$x^2 + 1, \qquad \sin t, \qquad A\vec{v}, \qquad \sum_{k=1}^{n} k.$$

Expressions can be simplified, evaluated, transformed, or compared. They are not true or false by themselves.

Equation or inequality with free variables. A condition:

$$x^2 + 1 = 10, \qquad x < y, \qquad \|v\| = 1.$$

These may become statements after variables are assigned or quantified.

Statement. A complete claim:

$$\forall x \in \mathbb{R}, \quad x^2 + 1 > 0.$$

This has a truth value.

Instruction. A command to do something:

$$\text{Find all } x \in \mathbb{R} \text{ such that } x^2 + 1 = 10.$$

The instruction itself is not true or false, although a proposed answer can be correct or incorrect.

In textbooks, these categories are often mixed in quick succession. A problem might say:

> Let $f(x) = x^2 + 1$. Solve $f(x) = 10$. Prove that $f(x) > 0$ for all $x \in \mathbb{R}$.

The first sentence defines notation. The second gives an instruction. The third states a proof task. You need to hear the shift.

2.4 Open Sentences And Closed Sentences

An *open sentence* has a free variable. A *closed sentence* has no free variables. Closed mathematical sentences are candidates for being statements.

Open:

$$x^2 = 4.$$

Closed by assignment:

$$3^2 = 4.$$

This is false.

Closed by quantification:

$$\exists x \in \mathbb{R}, \quad x^2 = 4.$$

This is true.

Closed another way:

$$\forall x \in \mathbb{R}, \quad x^2 = 4.$$

This is false.

The same open sentence can be closed in different ways, producing different statements.

Intuition

An open sentence is like a form with blanks. A quantifier tells you how the blanks are to be filled: by every possible value, by at least one value, or by values chosen in a particular order.

2.5 Truth Is Relative To Interpretation

The sentence

$$2 + 2 = 4$$

feels absolutely true, and in ordinary arithmetic it is. But even here, mathematics is more careful than habit.

In modular arithmetic modulo 3,

$$2 + 2 \equiv 1 \pmod 3.$$

That does not make ordinary arithmetic false. It means the symbols are being interpreted in a different structure.

Similarly,

$$x^2 \geq 0$$

is true for all real numbers x. But if x is complex, the symbol $\geq$ is not naturally defined in the same order-compatible way. The sentence is not automatically false; it may be ill-formed unless an order has been specified.

In linear algebra, the expression $AB = BA$ may be true for some pairs of matrices and false for others. In a commutative ring it is true for all elements. In a noncommutative ring it may fail. The same visual pattern of symbols does not settle the meaning by itself.

History and connection

The nineteenth and twentieth centuries forced mathematicians to become more explicit about interpretation. Non-Euclidean geometry showed that familiar words like “line” and “parallel” could satisfy different axiom systems. Abstract algebra showed that operations resembling addition or multiplication could live in many structures. Logic responded by separating syntax, the written form, from semantics, the interpretation that gives it meaning.

2.6 A Small Truth-Value Table For Sentences

Classify the following lines.

$$5 \in \mathbb{Z}$$

This is a true statement.

$$\pi \in \mathbb{Q}$$

This is a false statement.

$$x \in \mathbb{Q}$$

This is a predicate if x is not already fixed.

$$\text{Let } x \in \mathbb{Q}.$$

This is not a statement. It introduces a variable under a condition.

$$\text{Show that } \sqrt{2} \notin \mathbb{Q}.$$

This is an instruction.

$$\sqrt{2} \notin \mathbb{Q}.$$

This is a true statement.

These distinctions are easy when displayed slowly. In real proofs, they are compressed. A line like "Let $x \in A$" may be followed immediately by "Then $x \in B$," and you need to know that the first line sets context while the second asserts a claim under that context.

2.7 Mathematical Sentences In Proofs

Proofs are written in ordinary language mixed with symbols. That does not mean every sentence in a proof is a theorem. Proof text contains:

- assumptions,
- variable introductions,
- definitions being unfolded,
- intermediate claims,
- justifications,
- conclusions.

Consider this short proof:

> Let $n \in \mathbb{Z}$ be even. Then $n = 2k$ for some $k \in \mathbb{Z}$. Hence $n^2 = 4k^2 = 2(2k^2)$, and since $2k^2 \in \mathbb{Z}$, n^2 is even.

The sentence "Let $n \in \mathbb{Z}$ be even" introduces a fixed but arbitrary integer satisfying a hypothesis. The sentence "$n = 2k$ for some $k \in \mathbb{Z}$" is a claim justified by the definition of even. The equation $n^2 = 4k^2 = 2(2k^2)$ is a chain of equalities. The final phrase closes the definition.

Reading a proof well means hearing which role each line plays.

2.8 Claims And Their Social Names

Mathematics uses several names for proved statements.

- A *theorem* is usually a major result.
- A *proposition* is a result of moderate importance.
- A *lemma* is a helper result used to prove something else.
- A *corollary* follows quickly from an earlier result.
- A *claim* is often a local statement inside a proof.

These names are cultural labels, not rigid biological species. One author's lemma may be another author's theorem. The logical status is the same: each is a statement intended to be proved from accepted assumptions, definitions, and previous results.

2.9 Non-Statements That Look Mathematical

Here are common non-statements.

A bare expression:

$$\frac{x^2 - 1}{x - 1}.$$

It may simplify to $x + 1$ when $x \neq 1$, but by itself it is not true or false.

A definition command:

$$\text{Define } f : \mathbb{R} \to \mathbb{R} \text{ by } f(x) = x^2.$$

This sets notation.

A problem instruction:

$$\text{Determine whether } A \subseteq B.$$

This asks you to produce a statement, either $A \subseteq B$ or $A \not\subseteq B$, with justification.

A dangling implication:

$$\text{If } x > 0.$$

This is incomplete. An implication needs a conclusion.

A type declaration:

$$x \in \mathbb{R}.$$

This can be a statement if x has already been named. It can also be a variable declaration if it appears as "Let $x \in \mathbb{R}$." Context matters.

Common trap

The line $x \in \mathbb{R}$ is not automatically one thing in all contexts. In a proof, "Let $x \in \mathbb{R}$" introduces a variable. In a list of assumptions, $x \in \mathbb{R}$ is a condition. In a completed argument about a previously defined object x, it may be a claim.

2.10 Worked Example: Sorting The Sentence Types

Classify each line.

$$4 + 6 = 10.$$

Statement, true.

$$n + 6 = 10.$$

Predicate, unless n has already been assigned.

$$n + 6.$$

Expression.

$$\text{Find } n \in \mathbb{Z} \text{ such that } n + 6 = 10.$$

Instruction.

$$\forall n \in \mathbb{Z}, \quad n + 6 = 10.$$

Statement, false.

$$\exists n \in \mathbb{Z}, \quad n + 6 = 10.$$

Statement, true.

$$\text{Is there an } n \in \mathbb{Z} \text{ such that } n + 6 = 10?$$

Question.

The open condition $n + 6 = 10$ can be converted into different statements by different quantifiers. This is why classifying the sentence comes before proving it.

2.11 Worked Example: Domain Changes Meaning

Consider:

$$P(x): \quad x^2 = 2.$$

Over $\mathbb{N}$, no x satisfies $P(x)$.

Over $\mathbb{Z}$, no x satisfies $P(x)$.

Over $\mathbb{Q}$, no x satisfies $P(x)$, though proving that requires the irrationality of $\sqrt{2}$.

Over $\mathbb{R}$, two values satisfy it: $x = \sqrt{2}$ and $x = -\sqrt{2}$.

Over $\mathbb{C}$, the same two real values satisfy it, but now they are viewed as complex numbers with imaginary part 0.

So the statement

$$\exists x, \quad x^2 = 2$$

is incomplete until the domain of x is known. Over $\mathbb{Q}$ it is false. Over $\mathbb{R}$ it is true.

Focus

Whenever you see "there exists x," train yourself to ask: exists where?

2.12 Software Analogy: Boolean Functions And Type Errors

A predicate resembles a Boolean-valued function:

$$\texttt{isPrime}(n)$$

returns true or false when n is a suitable input. But if you call

$$\texttt{isPrime("hello")}$$

you have not discovered a false prime. You have made a type error.

Mathematics has similar issues. The predicate

$$x < 5$$

is natural for real numbers. It is not automatically meaningful for vectors, matrices, functions, or sets. You might define an order on some of these objects, but that definition is extra structure. Without it, the sentence is not ready.

This is one reason pure mathematics is careful about domains. The domain is not bureaucratic paperwork. It is part of the meaning.

2.13 Physical Analogy: Measuring Devices

A measuring device produces meaningful readings only under a protocol. A thermometer reading "37" means different things depending on Celsius, Fahrenheit, calibration, and where the sensor is placed. The number alone is not the full statement.

Similarly,

$$x = 37$$

means little until x is known. Is x a temperature, an index, a voltage, a dimension, a solution to an equation, an element of a finite field? The symbol can be reused, but meaning comes from context.

Mathematical notation is compact because context carries weight. Beginners often blame themselves for not understanding a symbol, when the real problem is that they have not identified the surrounding sentence.

2.14 Proof Lab: From Predicate To Theorem

Proof lab

Start with the predicate:

$$P(n): \quad n^2 + n \text{ is even.}$$

Version 1: universal theorem.

$$\forall n \in \mathbb{Z}, \quad P(n).$$

This says every integer satisfies the predicate. A proof must handle an arbitrary integer.
Version 2: existential theorem.

$$\exists n \in \mathbb{Z} \quad P(n).$$

This says at least one integer satisfies the predicate. A proof may simply choose $n = 1$, since $1^2 + 1 = 2$ is even.
Version 3: false universal theorem over a changed predicate.

$$\forall n \in \mathbb{Z}, \quad n^2 + n + 1 \text{ is even.}$$

This is false. A counterexample is $n = 0$, giving 1.
Lesson. The predicate is the raw condition. The quantifier turns it into a theorem-shaped claim.

2.15 Common Traps

Common trap

Trap 1: Treating an equation with variables as automatically true. The line $x^2 = 9$ is not true for all x. It is a condition satisfied by some values and not others.
Trap 2: Forgetting the domain. The statement "$x^2 = 2$ has a solution" is true over $\mathbb{R}$ and false over $\mathbb{Q}$.
Trap 3: Confusing a command with a claim. "Find x" is not something you prove. A proposed answer is what gets checked.
Trap 4: Thinking unknown means neither true nor false. The twin prime conjecture is a statement even though its truth is not known.

2.16 Exercises

Classify each as statement, predicate, expression, command, question, or context-setting phrase. If it is a predicate, give two possible domains and explain how the truth set changes.

1. $3 + 4 = 7$.
2. $x + 4 = 7$.
3. $x + 4$.
4. Find all $x \in \mathbb{R}$ such that $x + 4 = 7$.
5. Is every integer a rational number?
6. Every integer is a rational number.
7. $A \subseteq B$.
8. Let A and B be sets.
9. $\sin^2 t + \cos^2 t = 1$.
10. $\sin^2 t + \cos^2 t$.
11. $\forall t \in \mathbb{R},\ \sin^2 t + \cos^2 t = 1$.
12. There exists an integer n such that $n^2 = 12$.
13. There exists a real number x such that $x^2 = 12$.
14. Define $f : \mathbb{R} \to \mathbb{R}$ by $f(x) = x^2$.
15. f is injective.

For items involving A, B, or f, state what extra context is needed before the line has a definite truth value.

Walking question

During ordinary reading, pause at a displayed formula and ask: is this an object, a condition, a claim, or an instruction?

3 Variables Have Roles

Beginners often treat variables as generic letters. In proofs, variables have jobs. Confusing those jobs is one of the fastest ways to get lost.

The same letter can play different roles in different sentences. The role is not determined by the shape of the letter. It is determined by the logic around it.

Focus

When a variable appears, ask: who introduced it, what domain does it live in, and am I allowed to choose it?

3.1 Fixed But Unknown

Suppose a proof begins:

> Let $n \in \mathbb{Z}$.

Now n is fixed for the duration of the local argument. We may not know whether n is 5, -12, or 0, but it is one particular integer. It should not quietly change values halfway through the proof.

This is similar to a function parameter in software. If a function is called with an input, that input may be unknown to the programmer while writing the function, but during one execution it has one value.

```
def f(n):
```

Inside the function body, n is not all integers at once. It is the input value for that call. A proof often treats a variable similarly: fixed but unspecified.

For example:

> Let $n \in \mathbb{Z}$. Suppose n is even.

Now n is a fixed integer with an additional property. From evenness we may write:

$$n = 2k \quad \text{for some } k \in \mathbb{Z}.$$

The k is not arbitrary. It is chosen as a witness to the evenness of this fixed n.

3.2 Arbitrary

When a proof says:

> Let $n \in \mathbb{Z}$ be arbitrary.

it means we are not allowed to use any special feature of n beyond membership in $\mathbb{Z}$ and whatever hypotheses are stated. If the argument works for such an arbitrary n, it works for all integers.

This is the standard opening move for proving a universal statement:

$$\forall n \in \mathbb{Z}, \quad P(n).$$

Proof skeleton:

> Let $n \in \mathbb{Z}$ be arbitrary. We will show $P(n)$.

The word "arbitrary" is not magic, but it is a warning label. It says: do not secretly choose $n = 2$ because that is convenient. Do not assume n is positive unless positivity is part of the hypothesis. Do not assume n is nonzero unless nonzero is part of the hypothesis.

Common trap

To prove a statement about every integer, you may not prove it only for your favorite integer. "Let $n = 4$" proves something about 4, not about all $n \in \mathbb{Z}$.

3.3 Chosen As A Witness

Existential statements ask for witnesses.

$$\exists n \in \mathbb{Z} \quad n^2 = 49.$$

A witness is an object that makes the existential claim true. Here $n = 7$ is a witness, and so is $n = -7$.

To prove:

$$\exists x \in \mathbb{R} \quad x^2 - 5x + 6 = 0,$$

you can choose $x = 2$ and verify:

$$2^2 - 5 \cdot 2 + 6 = 4 - 10 + 6 = 0.$$

That is enough. You do not need to find every solution unless the problem asks for all solutions. Existential proof requires at least one witness, not a census.

Sometimes the witness is built from earlier variables. To prove:

$$\forall x \in \mathbb{R}, \exists y \in \mathbb{R} \quad x + y = 0,$$

you cannot choose one y before seeing x. After x is given, choose:

$$y = -x.$$

Then $x + y = x + (-x) = 0$.

Intuition

A witness is evidence with a name. For an existential claim, one correctly verified witness is enough.

3.4 Bound Variables

A variable is *bound* when its meaning is controlled by a quantifier or a local operator.

In

$$\forall x \in \mathbb{R}, \quad x^2 \geq 0,$$

the variable x is bound by $\forall x \in \mathbb{R}$.

In

$$\exists k \in \mathbb{Z}, \quad n = 2k,$$

the variable k is bound by $\exists k \in \mathbb{Z}$.

In

$$\sum_{i=1}^{n} i^2,$$

the variable i is bound by the summation. It is local to the sum.

In

$$\int_a^b f(t)\, dt,$$

the variable t is bound by the integral. It is a dummy variable. The following expressions are the same:

$$\int_a^b f(t)\,dt \qquad \text{and} \qquad \int_a^b f(u)\,du.$$

The letter changed; the role did not.

Formal

Bound variables can usually be renamed, as long as the renaming does not collide with another variable and change the meaning.

3.5 Free Variables

A variable is *free* when it is not bound by a quantifier or local operator and has not been fixed by surrounding context.

In

$$x^2 \geq 0,$$

x is free unless earlier text has introduced it.

In

$$\forall x \in \mathbb{R}, \quad x < y,$$

x is bound, but y is free unless context defines it. This is not a complete statement by itself. It becomes meaningful if, for example, y has already been fixed as a real number. Then it says every real number is less than that fixed y, which is false.

Free variables are not bad. They are how we write predicates, definitions, and local claims. But they require context.

3.6 Dummy Variables

Some bound variables are called dummy variables because their names carry no lasting meaning.

$$\sum_{i=1}^{n} i = \sum_{j=1}^{n} j.$$

The i and j are dummy variables. They are placeholders used by the summation notation.

Likewise:

$$\{x \in \mathbb{R} : x^2 < 1\} = \{t \in \mathbb{R} : t^2 < 1\}.$$

Both describe the open interval $(-1, 1)$.

But not every renaming is safe. Compare:

$$\forall x \in \mathbb{R}, \exists y \in \mathbb{R}, \quad y > x.$$

If we foolishly rename y to x inside the same scope, we get:

$$\forall x \in \mathbb{R}, \exists x \in \mathbb{R}, \quad x > x,$$

which is broken and misleading. The inner x shadows the outer x, and the condition $x > x$ is never true. The renaming caused a collision.

Common trap

Renaming bound variables is safe only when it preserves scope. Avoid reusing a letter if the old and new variables need to be compared.

3.7 Parameters

A parameter is a fixed object that controls a family of statements or objects.

For example:

$$f_a(x) = ax$$

defines a family of functions indexed by a. Here x is the input variable, while a is a parameter. If $a = 2$, the function is $f_2(x) = 2x$. If $a = -1$, the function is $f_{-1}(x) = -x$.

In the statement:

$$\forall a \in \mathbb{R}, \forall x \in \mathbb{R}, \quad f_a(x) = ax,$$

both a and x are bound by universal quantifiers. But conceptually, a chooses a member of the family and x chooses an input to that member.

Parameters are common in analysis:

$$\lim_{x \to a} f(x) = L.$$

Here a and L are often fixed while x varies toward a. If you blur these roles, epsilon-delta proofs become painful.

3.8 Constants And Named Objects

A constant is a fixed named object. Some constants are universal, like π or e. Others are local:

Let $c = \sqrt{2}$.

Now c has a fixed meaning in the current context.

But a letter is not permanently a constant just because it was once used that way. In one proof c may be a fixed real number. In another proof c may be a witness chosen after an epsilon is given. The local text decides.

Mathematical letters are reusable tools. Do not attach too much personality to them.

3.9 Variables In Definitions

Definitions often use variables in a special way. Consider:

$$n \text{ is even} \quad \Longleftrightarrow \quad \exists k \in \mathbb{Z} \text{ such that } n = 2k.$$

Here n is the object being defined. The variable k is a witness variable inside the definition. To prove n is even, you must produce such a k. To use the fact that n is even, you are allowed to introduce such a k.

Example:

> Suppose n is even. Then there exists $k \in \mathbb{Z}$ such that $n = 2k$.

At this point, you often write:

> Choose $k \in \mathbb{Z}$ such that $n = 2k$.

This k depends on n. If $n = 10$, one possible k is 5. If $n = -6$, one possible k is -3.

Focus

Definitions are machines with variables inside. To use a definition well, know which variables are inputs and which variables must be supplied as witnesses.

3.10 Scope

Scope is the region of text where a variable has a particular meaning.

In a proof:

> Let $x \in A$. Since $A \subseteq B$, we have $x \in B$.

The variable x is in scope for those sentences. If the proof later starts a new paragraph:

Now let $x \in C$.

the writer may be reusing the same letter for a new object. Good writers avoid confusing reuse, but textbooks sometimes rely on context.

In nested quantifiers, scope is even more important:

$$\forall x \in A, \exists y \in B, \forall z \in C, \quad R(x, y, z).$$

The y may depend on x, because y is chosen after x. The z is chosen after both x and y, but because it is universal, the claim must work for all z.

3.11 Dependency

Dependency is one of the most important ideas hidden in variable roles.

Compare:

$$\forall x \in \mathbb{R}, \exists y \in \mathbb{R}, \quad y = x^2$$

and

$$\exists y \in \mathbb{R}, \forall x \in \mathbb{R}, \quad y = x^2.$$

The first is true. Given x, choose $y = x^2$. The witness y depends on x.

The second is false. It asks for one fixed real number y equal to every square x^2. No single number can be both 0^2 and 2^2.

Dependency is not always written explicitly. Sometimes authors write:

$$\forall x \in X, \exists y_x \in Y, \quad P(x, y_x).$$

The subscript y_x is a reminder that the chosen y may depend on x. But the quantifier order already said that. The notation only makes the dependency visible.

Intuition

Quantifier order is a dependency diagram in disguise.

3.12 Worked Parsing: A Simple Nested Claim

Consider:

$$\forall x \in \mathbb{R}, \exists y \in \mathbb{R} \text{ such that } y > x.$$

Read it slowly.

1. The challenger gives an arbitrary real number x.

2. After seeing x, we must produce a real number y.
3. The produced y must satisfy $y > x$.

A proof:

> Let $x \in \mathbb{R}$ be arbitrary. Choose $y = x + 1$. Then $y \in \mathbb{R}$, and since $x + 1 > x$, we have $y > x$. Therefore for every real x there exists a real y such that $y > x$.

Variable roles:

- x is arbitrary.
- y is a witness.
- y depends on x.

Now reverse the quantifiers:

$$\exists y \in \mathbb{R}, \forall x \in \mathbb{R} \text{ such that } y > x.$$

Now y must be chosen first. It cannot depend on x. The claim asks for a real number strictly greater than every real number. But taking $x = y$ would require $y > y$, which is impossible; hence no such y exists and the statement is false.

3.13 Worked Parsing: Epsilon Language

Analysis uses variable roles intensely. The statement

$$\lim_{n \to \infty} a_n = L$$

means:

$$\forall \varepsilon > 0, \exists N \in \mathbb{N}, \forall n \in \mathbb{N}, \quad n \geq N \Rightarrow |a_n - L| < \varepsilon.$$

Roles:

- ε is chosen by the challenger. It is an arbitrary positive tolerance.
- N is chosen in response to ε.
- n is arbitrary after N is chosen, subject to $n \geq N$.
- L is usually a fixed proposed limit.

The dependency is:

$$N = N(\varepsilon).$$

The N may depend on ε. It may not depend on n, because N is chosen before the final universal n.

Common trap

In an epsilon proof, choosing N in terms of n is a scope error. The proof must find one threshold N that works for all later $n \geq N$.

3.14 Worked Parsing: Set Inclusion

The statement

$$A \subseteq B$$

means:

$$\forall x, \quad x \in A \Rightarrow x \in B.$$

In a proof of $A \subseteq B$, the variable x is arbitrary:

> Let $x \in A$ be arbitrary. We must show $x \in B$.

This is why set-inclusion proofs often feel repetitive. They are rehearsing the logic of the subset symbol.

To disprove $A \subseteq B$, you need a witness:

$$\exists x \quad x \in A \text{ and } x \notin B.$$

The same variable x is arbitrary in the proof of inclusion and a witness in the disproof. The role changes with the task.

3.15 Physical Analogy: Knobs And Readings

In a lab setup, some quantities are controlled, some are measured, and some are fixed by the apparatus.

Suppose a statement says:

> For every input voltage V, there exists a resistor setting R such that the current is below 1 amp.

Here V is arbitrary; R is a response. The setting R may depend on V.

If the statement says:

> There exists a resistor setting R such that for every input voltage V, the current is below 1 amp.

then R is fixed before V varies. That is a different engineering promise.

Mathematical variables behave the same way. Some are knobs. Some are readings. Some are universal disturbances. Some are chosen controls.

3.16 Software Analogy: Scope Bugs

Consider pseudocode:

```
for x in X:
choose y depending on x
```

This corresponds to:

$$\forall x \in X, \exists y \in Y, \quad P(x, y).$$

Now compare:

```
choose y
for x in X:
check P(x,y)
```

This corresponds to:

$$\exists y \in Y, \forall x \in X, \quad P(x, y).$$

The code is different. The mathematics is different. The order of lines controls what data is available.

3.17 Common Variable Traps

Common trap

Trap 1: Changing an arbitrary variable into a special one. If n is arbitrary, do not later say "take $n = 2$" unless you are switching to an example.
Trap 2: Choosing a witness too early. In $\forall x \exists y$, y can depend on x. In $\exists y \forall x$, it cannot.
Trap 3: Reusing a bound variable outside its scope. The i in $\sum_{i=1}^{n} i^2$ has no life after the summation ends.
Trap 4: Treating a parameter as a variable of motion. In $\lim_{x \to a} f(x) = L$, the variable x moves; a and L are fixed in the statement.

3.18 Proof Lab: Repairing Variable Misuse

Proof lab

Bad proof attempt. Claim: $\forall n \in \mathbb{Z},\ n^2 + n$ is even.
"Let $n = 3$. Then $n^2 + n = 12$, which is even. Therefore $n^2 + n$ is even for all integers."
Diagnosis. The proof chooses a special n. It proves the statement for $n = 3$, not for arbitrary n.
Repaired proof. Let $n \in \mathbb{Z}$ be arbitrary. Then

$$n^2 + n = n(n + 1).$$

The integers n and $n+1$ are consecutive, so one of them is even. Therefore their product is even. Hence $n^2 + n$ is even. Since n was arbitrary, the result holds for all $n \in \mathbb{Z}$.

3.19 Proof Lab: Witnesses Depending On Earlier Choices

Proof lab

Claim:

$$\forall a \in \mathbb{R}, \exists b \in \mathbb{R}, \quad a + b = 5.$$

Proof. Let $a \in \mathbb{R}$ be arbitrary. Choose $b = 5 - a$. Since $a \in \mathbb{R}$, we have $5 - a \in \mathbb{R}$. Also

$$a + b = a + (5 - a) = 5.$$

Thus for every real number a, there exists a real number b such that $a + b = 5$.
Role check. The variable a is arbitrary. The variable b is a witness. The witness depends on a.

3.20 Exercises

For each statement, identify the role of every variable: arbitrary, fixed parameter, witness, bound dummy variable, or free variable needing context.

1. $\forall n \in \mathbb{Z}, \exists k \in \mathbb{Z}$ such that $n^2 + n = 2k$.
2. $\exists k \in \mathbb{Z}, \forall n \in \mathbb{Z}$ such that $n^2 + n = 2k$.
3. $\sum_{i=1}^{n} i^2$.
4. $\int_0^1 x^n \, dx$.
5. $\forall \varepsilon > 0, \exists \delta > 0, \forall x \in \mathbb{R},\ |x - a| < \delta \Rightarrow |f(x) - f(a)| < \varepsilon$.
6. $A \subseteq B$.
7. $\forall x \in A, \exists y \in B, \forall z \in C,\ R(x, y, z)$.
8. $\exists y \in B, \forall x \in A, \forall z \in C,\ R(x, y, z)$.

Then answer:

1. In which statements may the existential variable depend on an earlier universal variable?
2. In which statements must one object work for all later choices?
3. Which variables can be safely renamed?
4. Which variables would cause confusion if renamed carelessly?

Walking question

When reading a proof, track variables like characters entering and leaving a scene. Who is on stage? Who has already been chosen? Who is allowed to respond to whom?

4 Quantifiers

The symbols $\forall$ and $\exists$ are small, but they control the shape of entire proofs.

$$\forall = \text{for every}, \qquad \exists = \text{there exists}.$$

They are not merely shorthand. They are instructions for a mathematical game of choices.

Focus

To understand a quantified statement, read it as an order of play: who chooses first, who responds, and what must be true after the choices are made?

4.1 Universal Statements

A universal statement has the form:

$$\forall x \in D, \quad P(x).$$

It says every object in the domain D satisfies property P.

To prove it, use the universal proof pattern:

> Let $x \in D$ be arbitrary. We must show $P(x)$.

To disprove it, find one counterexample:

> Find $x \in D$ such that $P(x)$ is false.

Example:

$$\forall n \in \mathbb{Z}, \quad n^2 \geq 0.$$

True, because the square of any integer is nonnegative.

False universal statement:

$$\forall n \in \mathbb{Z}, \quad n > 0.$$

A counterexample is $n = 0$, or $n = -1$.

The counterexample does not need to be dramatic. One valid failure is enough.

4.2 Existential Statements

An existential statement has the form:

$$\exists x \in D \quad P(x).$$

It says at least one object in D satisfies P.

To prove it, produce a witness:

> Choose $x =$ something in D. Verify $P(x)$.

To disprove it, prove a universal negative:

> Let $x \in D$ be arbitrary. Show $P(x)$ is false.

Example:

$$\exists n \in \mathbb{Z}, \quad n^2 = 16.$$

True. A witness is $n = 4$.

False existential statement:

$$\exists n \in \mathbb{Z}, \quad n^2 = 2.$$

False, because no integer has square 2.

Notice the asymmetry:

- A universal statement needs all cases for proof, one case for disproof.
- An existential statement needs one case for proof, all cases for disproof.

4.3 The Quantifier Game

Think of $\forall$ as a challenger and $\exists$ as a responder.

For

$$\forall x \in \mathbb{R}, \exists y \in \mathbb{R}, \quad x + y = 0,$$

the game goes:

1. Challenger chooses $x \in \mathbb{R}$.
2. You choose $y \in \mathbb{R}$.
3. You must make $x + y = 0$ true.

Winning strategy:

$$y = -x.$$

For

$$\exists y \in \mathbb{R}, \forall x \in \mathbb{R}, \quad x + y = 0,$$

the game goes:

1. You choose one $y \in \mathbb{R}$.
2. Challenger chooses $x \in \mathbb{R}$.
3. The equation $x + y = 0$ must be true.

There is no winning strategy. After you choose y, the challenger can choose $x = 1 - y$, giving

$$x + y = 1.$$

The two statements use the same equation but have different quantifier order. One is true; the other is false.

Intuition

Quantifiers are not seasoning sprinkled on a formula. They are part of the claim.

4.4 Order Matters

Compare:

$$\forall x \in \mathbb{R}, \exists y \in \mathbb{R}, \quad y > x$$

and

$$\exists y \in \mathbb{R}, \forall x \in \mathbb{R}, \quad y > x.$$

First statement: true. Given x, choose $y = x + 1$.

Second statement: false. It asks for a real number larger than every real number.

Compare:

$$\forall x \in \mathbb{R}, \exists y \in \mathbb{R}, \quad y^2 = x$$

and

$$\exists y \in \mathbb{R}, \forall x \in \mathbb{R}, \quad y^2 = x.$$

The first is false over $\mathbb{R}$, because negative x values have no real square roots. The second is also false, because one fixed square cannot equal every real number.

Now change the domain:

$$\forall x \in [0, \infty), \exists y \in \mathbb{R}, \quad y^2 = x.$$

This is true. Given $x \geq 0$, choose $y = \sqrt{x}$.

Focus

Quantifier order and domain are two of the main controls on mathematical meaning.

4.5 Restricted Quantifiers

The notation

$$\forall x \in D, \quad P(x)$$

is shorthand for:

$$\forall x, \quad x \in D \Rightarrow P(x).$$

The notation

$$\exists x \in D \quad P(x)$$

is shorthand for:

$$\exists x, \quad x \in D \text{ and } P(x).$$

This difference matters when negating statements later. Universal restrictions pair naturally with implication; existential restrictions pair naturally with "and."

Example:

$$\forall x \in \mathbb{R}, \quad x^2 \geq 0$$

means:

$$\forall x, \quad x \in \mathbb{R} \Rightarrow x^2 \geq 0.$$

Example:

$$\exists x \in \mathbb{R} \quad x^2 = 2$$

means:

$$\exists x, \quad x \in \mathbb{R} \text{ and } x^2 = 2.$$

The restricted notation is easier to read, but the expanded form reveals the logic.

4.6 Vacuous Truth

Universal statements over empty domains are true.

$$\forall x \in \emptyset, \quad P(x)$$

is true, no matter what P is.

This feels strange at first. The reason is counterexamples. To make a universal statement false, you need an object in the domain that fails the property. If the domain is empty, there is no such object.

Example:

$$\forall x \in \emptyset, \quad x^2 = 17$$

is true.

But:

$$\exists x \in \emptyset, \quad x^2 = 17$$

is false.

A physical analogy: "Every device in the empty box is broken" and "Every device in the empty box works" are both vacuously true if there are no devices in the box. The existential claim "There is a working device in the empty box" is false.

Common trap

Vacuous truth is not a loophole. It is what keeps universal statements logically consistent when domains are empty.

4.7 Uniqueness

Sometimes existence is not enough. We want exactly one object.

The notation

$$\exists! x \in D \quad P(x)$$

means:

> There exists exactly one $x \in D$ such that $P(x)$.

This contains two claims:

1. Existence: at least one such x exists.
2. Uniqueness: any two such objects must be equal.

Formal expansion:

$$\exists x \in D \, (P(x) \text{ and } \forall y \in D, \; P(y) \Rightarrow y = x) \, .$$

Example:

$$\exists! x \in \mathbb{R} \quad x + 3 = 10.$$

True, because $x = 7$ works, and if $y + 3 = 10$, then $y = 7$.

Example:

$$\exists! x \in \mathbb{R} \quad x^2 = 4.$$

False, because both $x = 2$ and $x = -2$ work.

Proof lab

To prove uniqueness, do not merely display one solution. After existence, suppose u and v are both solutions, then prove $u = v$.

4.8 At Least, At Most, Exactly

Quantifiers also express counting claims.

At least one:

$$\exists x \in D \quad P(x).$$

At most one:

$$\forall x, y \in D, \quad (P(x) \text{ and } P(y)) \Rightarrow x = y.$$

Exactly one:

$$\exists x \in D \quad P(x)$$

and

$$\forall x, y \in D, \quad (P(x) \text{ and } P(y)) \Rightarrow x = y.$$

Example:

The equation $x^2 = 0$ has exactly one real solution. Existence: $x = 0$ works. At most one: if $u^2 = 0$ and $v^2 = 0$, then $u = 0$ and $v = 0$, so $u = v$.

The equation $x^2 = 1$ has at least one real solution, but not at most one, because 1 and -1 both work.

4.9 Quantifiers And Implication

Many mathematical statements have the form:

$$\forall x \in D, \quad P(x) \Rightarrow Q(x).$$

This says every object in D that satisfies P also satisfies Q.

Example:

$$\forall n \in \mathbb{Z}, \quad n \text{ is divisible by } 4 \Rightarrow n \text{ is even.}$$

To prove it:

> Let $n \in \mathbb{Z}$ be arbitrary. Suppose n is divisible by 4. Show n is even.

You do not need to prove every integer is divisible by 4. You assume the hypothesis for an arbitrary integer and prove the conclusion under that assumption.

To disprove it, you need:

$$\exists n \in \mathbb{Z} \quad n \text{ is divisible by } 4 \text{ and } n \text{ is not even.}$$

No such n exists, so the original statement is true.

4.10 Common Speech Versus Mathematical Speech

Ordinary language is flexible. Mathematical language is less forgiving.

"Everyone has someone they admire" usually means:

$$\forall p \in P, \exists q \in P, \quad A(p, q).$$

Each person may admire a different person.

"There is someone everyone admires" means:

$$\exists q \in P, \forall p \in P, \quad A(p, q).$$

One person is admired by all.

These are very different social claims.

Likewise:

> Every problem has a method that solves it.

means:

$$\forall p, \exists m, \quad m \text{ solves } p.$$

But:

> There is a method that solves every problem.

means:

$$\exists m, \forall p, \quad m \text{ solves } p.$$

The second is much stronger.

4.11 History And Connections

History and connection

Quantifiers became explicit relatively late in the history of mathematics. Mathematicians had always used words like "all" and "some," but modern symbolic logic, especially in the work of Frege, Peano, Russell, Hilbert, and others, made quantifier structure an object of study. This mattered because mathematics was becoming more abstract. Once you study arbitrary sets, functions, spaces, and structures, you need a language that says exactly which objects are being varied and which are fixed.

Quantifiers later appear everywhere.

In analysis:

$$\forall \varepsilon > 0, \exists \delta > 0, \forall x, \quad |x - a| < \delta \Rightarrow |f(x) - f(a)| < \varepsilon.$$

In algebra:

$$\forall a \in G, \exists b \in G, \quad ab = ba = e.$$

In topology:

$$\forall x \in U, \exists r > 0, \quad B(x, r) \subseteq U.$$

In computer science:

$$\forall \text{ inputs } i, \exists \text{ output } o, \quad \text{Spec}(i, o).$$

Quantifiers are not a chapter you leave behind. They are the operating system.

4.12 Worked Example: Proving A Universal-Existential Statement

Claim:

$$\forall n \in \mathbb{Z}, \exists m \in \mathbb{Z}, \quad n + m = 0.$$

Proof:

> Let $n \in \mathbb{Z}$ be arbitrary. Choose $m = -n$. Since $n \in \mathbb{Z}$, we have $-n \in \mathbb{Z}$, so $m \in \mathbb{Z}$. Also
>
> $$n + m = n + (-n) = 0.$$
>
> Therefore for every integer n, there exists an integer m such that $n + m = 0$.

What to notice:

- The proof starts with an arbitrary n because the first quantifier is universal.
- The witness m is chosen after n.
- The proof verifies both membership $m \in \mathbb{Z}$ and the required equation.

4.13 Worked Example: Disproving A Reversed Statement

Claim:

$$\exists m \in \mathbb{Z}, \forall n \in \mathbb{Z}, \quad n + m = 0.$$

This is false.

Disproof:

> Suppose, for contradiction, that there exists $m \in \mathbb{Z}$ such that for all $n \in \mathbb{Z}$, $n + m = 0$. Since the statement holds for all integers n, it holds for $n = 0$ and $n = 1$. Taking $n = 0$ gives $m = 0$. Taking $n = 1$ gives $1 + m = 0$, so $m = -1$. Thus $m = 0$ and $m = -1$, a contradiction. Therefore no such integer m exists.

You can also give a direct disproof:

> Let $m \in \mathbb{Z}$ be arbitrary. Choose $n = 1 - m$. Then $n \in \mathbb{Z}$, but $n + m = 1 \neq 0$. So this m does not work. Since m was arbitrary, no integer m works for all n.

The direct disproof is especially instructive: to defeat a proposed universal property of m, choose a counterexample n depending on m.

4.14 Worked Example: A Three-Quantifier Statement

Consider:

$$\forall a \in \mathbb{R}, \forall b \in \mathbb{R}, \exists x \in \mathbb{R}, \quad x > a \text{ and } x > b.$$

Plain language:

> Given any two real numbers, there is a real number larger than both.

Proof:

> Let $a, b \in \mathbb{R}$ be arbitrary. Choose
>
> $$x = \max\{a, b\} + 1.$$
>
> Then $x \in \mathbb{R}$, and $x > a$ and $x > b$. Therefore for any real numbers a and b, there exists a real number larger than both.

If you do not want to use max, use:

$$x = a^2 + b^2 + |a| + |b| + 1.$$

That is overkill, but it works. Proofs do not always need the prettiest witness; they need a correct witness.

4.15 Worked Example: A False Three-Quantifier Statement

Consider:

$$\exists x \in \mathbb{R}, \forall a \in \mathbb{R}, \forall b \in \mathbb{R}, \quad x > a \text{ and } x > b.$$

This says there is one real number larger than all pairs of real numbers. In particular, it would be larger than every real number. False.

Disproof:

> Let $x \in \mathbb{R}$ be arbitrary. Choose $a = x + 1$ and $b = 0$. Then $a \in \mathbb{R}$ and $b \in \mathbb{R}$, but $x > a$ is false. Therefore this x does not satisfy the required property. Since x was arbitrary, no such real x exists.

Again, the counterexample may depend on the proposed witness. That is legal because in disproving $\exists x \forall a \forall b$, we start by considering an arbitrary candidate x and then show it fails.

4.16 Quantifiers In Definitions

Definitions often hide quantifiers.

Even integer.

$$n \text{ is even} \iff \exists k \in \mathbb{Z},\ n = 2k.$$

Subset.

$$A \subseteq B \iff \forall x,\ x \in A \Rightarrow x \in B.$$

Function equality. If $f, g : A \to B$, then

$$f = g \iff \forall x \in A,\ f(x) = g(x).$$

Injective function.

$$f : A \to B \text{ is injective} \iff \forall x_1, x_2 \in A,\ f(x_1) = f(x_2) \Rightarrow x_1 = x_2.$$

Surjective function.

$$f : A \to B \text{ is surjective} \iff \forall y \in B, \exists x \in A,\ f(x) = y.$$

Surjectivity is a quantifier statement. The output target y is arbitrary; the input x is a witness that may depend on y.

Common trap

Many proof mistakes are definition mistakes: the writer remembers the slogan but not the quantifiers inside the definition.

4.17 Negating Quantifiers Briefly

Although negation gets its own deeper treatment later, quantifiers already demand the basic rules:

$$\neg(\forall x \in D,\ P(x)) \iff \exists x \in D \text{ such that } \neg P(x).$$

$$\neg(\exists x \in D,\ P(x)) \iff \forall x \in D,\ \neg P(x).$$

Not every integer is positive means there exists an integer that is not positive.

There is no integer whose square is 2 means every integer has square not equal to 2.

For nested quantifiers, flip each quantifier as the negation passes through:

$$\neg\left(\forall x \in A, \exists y \in B,\ P(x,y)\right)$$

becomes

$$\exists x \in A, \forall y \in B,\ \neg P(x,y).$$

Plain language:

> It is not true that every x has some good y.

means:

> There is a particular x for which every y fails.

4.18 Proof Lab: Translating A Definition

Proof lab

Definition:

$$f : A \to B \text{ is surjective} \quad \Longleftrightarrow \quad \forall b \in B, \exists a \in A,\ f(a) = b.$$

Proof skeleton for surjectivity. Let $b \in B$ be arbitrary. We must find $a \in A$ such that $f(a) = b$.
Disproof skeleton. Find $b \in B$ such that for every $a \in A$, $f(a) \neq b$.
Example. Let $f : \mathbb{R} \to \mathbb{R}$ be $f(x) = x^3$. To prove f is surjective, let $b \in \mathbb{R}$ be arbitrary and choose $a = \sqrt[3]{b}$. Then $f(a) = a^3 = b$.
Non-example. Let $g : \mathbb{R} \to \mathbb{R}$ be $g(x) = x^2$. It is not surjective, because $b = -1 \in \mathbb{R}$, and for every $a \in \mathbb{R}$, $a^2 \neq -1$.

4.19 Proof Lab: Quantifier Order In Limits

Proof lab

The sequence statement

$$\lim_{n\to\infty} a_n = L$$

means:

$$\forall \varepsilon > 0, \exists N \in \mathbb{N}, \forall n \in \mathbb{N}, \quad n \geq N \Rightarrow |a_n - L| < \varepsilon.$$

Correct dependency. N may depend on ε.
Incorrect dependency. N may not depend on n, because the statement requires the same N to work for all $n \geq N$.
Example. For $a_n = 1/n$ and $L = 0$, given $\varepsilon > 0$, choose N so large that $N > 1/\varepsilon$. Then for every $n \geq N$,

$$\left|\frac{1}{n} - 0\right| = \frac{1}{n} \leq \frac{1}{N} < \varepsilon.$$

4.20 Common Quantifier Traps

Common trap

Trap 1: Reversing quantifiers casually. $\forall x \exists y$ and $\exists y \forall x$ usually say different things.
Trap 2: Proving an existential when the theorem is universal. One example cannot prove an all-claim.
Trap 3: Proving a universal when only existence was needed. This is not logically wrong, but it may waste effort. If one witness is enough, one witness is enough.
Trap 4: Forgetting membership. When you choose a witness, verify it lies in the required domain.
Trap 5: Letting later variables influence earlier choices. In $\forall \varepsilon \exists N \forall n$, the N cannot depend on n.

4.21 A Quantifier Checklist

Before proving a quantified statement, write:

1. The quantifiers in order.
2. The role of each variable.
3. What you are allowed to choose.
4. What must be arbitrary.
5. What the final condition is.

6. What a counterexample would look like.

For:

$$\forall x \in X, \exists y \in Y, \forall z \in Z, \quad P(x, y, z),$$

the checklist says:

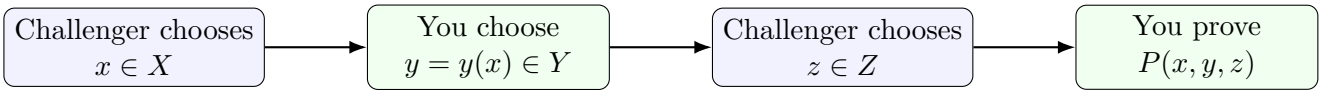
- x is arbitrary.
- y is chosen after x and may depend on x.
- z is arbitrary after y is chosen.
- The proof must show $P(x, y, z)$ for every $z \in Z$.
- A disproof needs one $x \in X$ such that every $y \in Y$ fails for at least one $z \in Z$.

The negation makes the last item precise:

$$\exists x \in X, \forall y \in Y, \exists z \in Z, \quad \neg P(x, y, z).$$

4.22 Visualizing Quantifier Flow

Quantifiers are easier to read when you stop seeing them as a string of symbols and start seeing them as a flow of responsibility. The order tells you who moves first, who responds, and what information is available at each step.



For $\forall x \in X, \exists y \in Y, \forall z \in Z,\ P(x, y, z)$,
the witness y may depend on x, but
it must be chosen before z is revealed.

That one sentence, "chosen before z is revealed," is often the difference between a valid proof and a proof that accidentally cheats.

Common trap

False intuition: "If I can choose a good object eventually, the existential quantifier is satisfied." Not always. You must choose it at the point where the quantifier appears. In $\exists y\, \forall x$, the y is chosen before seeing x. In $\forall x\, \exists y$, the y may respond to x.

4.23 Symbolic Execution for Quantifiers

Software engineers are used to tracing a program with inputs, local variables, and assertions. Quantified statements can be traced the same way.

For the statement

$$\forall x \in X, \exists y \in Y, \forall z \in Z, \quad P(x, y, z),$$

the execution trace is:

1. Input arrives: an arbitrary $x \in X$.
2. Your code must compute or specify a $y \in Y$. This y may use x.
3. Another input arrives: an arbitrary $z \in Z$.
4. Your assertion must pass: $P(x, y, z)$.

The negation is the failing test case:

$$\exists x \in X, \forall y \in Y, \exists z \in Z, \quad \neg P(x, y, z).$$

It says there is a particular x that breaks every possible response y by producing some bad z.

Proof lab

Proof anatomy drill. For each quantified statement, write the execution trace and the failing test case.

1. $\forall n \in \mathbb{Z}, \exists m \in \mathbb{Z},\ n < m.$
2. $\forall \varepsilon > 0, \exists N \in \mathbb{N}, \forall n \geq N,\ |a_n - L| < \varepsilon.$
3. $\exists \delta > 0, \forall x, y \in \mathbb{R},\ |x - y| < \delta \Rightarrow |f(x) - f(y)| < 1.$

4.24 Exercises

Decide whether each statement is true or false. Prove it or give a counterexample.

1. $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}, \quad x < y.$
2. $\exists y \in \mathbb{R}, \forall x \in \mathbb{R}, \quad x < y.$
3. $\forall n \in \mathbb{Z}, \exists m \in \mathbb{Z}, \quad n < m.$
4. $\exists m \in \mathbb{Z}, \forall n \in \mathbb{Z}, \quad n < m.$
5. $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}, \quad x + y = 10.$
6. $\exists y \in \mathbb{R}, \forall x \in \mathbb{R}, \quad x + y = 10.$
7. $\forall x \in [0, \infty), \exists y \in \mathbb{R}, \quad y^2 = x.$

8. $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}, \quad y^2 = x.$
9. $\exists x \in \mathbb{R}, \forall y \in \mathbb{R}, \quad x + y = y.$
10. $\forall y \in \mathbb{R}, \exists x \in \mathbb{R}, \quad x + y = y.$

Translate each into symbols.

1. Every real number has an additive inverse.
2. Some real number is larger than every integer.
3. For every integer, there is a larger integer.
4. There is a smallest natural number.
5. Every nonempty finite set has an element.
6. There is a function from $\mathbb{R}$ to $\mathbb{R}$ that sends every real number to 0.

Negate each statement.

1. $\forall x \in A,\ P(x).$
2. $\exists x \in A,\ P(x).$
3. $\forall x \in A, \exists y \in B,\ P(x, y).$
4. $\exists y \in B, \forall x \in A,\ P(x, y).$
5. $\forall \varepsilon > 0, \exists N \in \mathbb{N}, \forall n \geq N,\ |a_n - L| < \varepsilon.$

Walking question

Pick a sentence from ordinary life containing "everyone," "someone," or "always." Translate it into quantifiers. Then reverse the quantifiers and notice how the meaning changes.

5 Definitions as Machines

Definitions are the machinery of proof. A definition is not decoration around a theorem. It is not a vocabulary word you memorize so that you can understand the sentence. In proof-based mathematics, a definition is an exact little machine with two doors.

One door is the *use* door. If you are told that an object satisfies a definition, you may open the definition and take out the data it promises.

The other door is the *prove* door. If you want to show that an object satisfies a definition, the definition tells you exactly what data you must build.

For example, the definition of even integer is:

$$n \text{ is even} \quad \Longleftrightarrow \quad \exists k \in \mathbb{Z} \text{ such that } n = 2k.$$

If someone gives you "n is even," you may write $n = 2k$ for some integer k. If someone asks you to prove "n is even," you must produce an integer k such that $n = 2k$.

Focus

A definition is a reversible contract: it says what you get to use and what you must prove.

This is a deep shift from calculation-based mathematics. In a calculation course, a phrase like "continuous function" may often mean "a function whose graph has no breaks." That picture is useful. But in proof, the picture is not the machine. The machine is the definition. Depending on the course, continuity may mean an epsilon-delta condition, preservation of limits, inverse images of open sets, or something else equivalent in a particular setting. The proof has to use the machine that has actually been installed.

5.1 Why definitions feel weird at first

Beginners often experience definitions as strangely artificial. Why say "there exists an integer k such that $n = 2k$" when we already know what even means? Why say "a divides b if $b = ak$ for some integer k" when everyone knows division?

The answer is that ordinary understanding is not stable enough for proof. It is too full of pictures, memories, and unstated assumptions. Proof needs handles.

The definition

$$n = 2k \quad \text{for some } k \in \mathbb{Z}$$

gives us a handle. It turns the word "even" into an equation. Once we have an equation, we can substitute, add, multiply, factor, and compare. A vague feeling becomes an object we can use.

Intuition

Definitions are adapters. They convert ordinary mathematical words into proof-compatible data.

The same thing happens in software. A type signature does not tell you the full personality of a function, but it tells you how the function can be connected to other code. A function declared as

```
isEven :  Integer -> Boolean
```

is not the same kind of thing as a function declared as

```
halve :  EvenInteger -> Integer.
```

The contract matters. A compiler is strict because ambiguity makes large systems fragile. Mathematics is strict for the same reason.

5.2 Opening a definition

To open a definition is to replace a word by the data the definition guarantees.

Suppose a proof begins:

> Let n be an even integer.

The next legal move is:

$$n = 2k \text{ for some } k \in \mathbb{Z}.$$

Not because we guessed it. Not because it is usually true. Because that is exactly what "even" means.

Here is the same idea with divisibility.

$$a \mid b$$

is read "a divides b." The definition is:

$$a \mid b \quad \Longleftrightarrow \quad \exists k \in \mathbb{Z} \text{ such that } b = ak.$$

So if a proof says $a \mid b$, you may introduce an integer k with $b = ak$.

Formal

Opening patterns:

$$\begin{aligned} n \text{ is even} &\Rightarrow n = 2k \text{ for some } k \in \mathbb{Z}, \\ n \text{ is odd} &\Rightarrow n = 2k+1 \text{ for some } k \in \mathbb{Z}, \\ a \mid b &\Rightarrow b = ak \text{ for some } k \in \mathbb{Z}, \\ x \in A \cap B &\Rightarrow x \in A \text{ and } x \in B, \\ x \in A \cup B &\Rightarrow x \in A \text{ or } x \in B. \end{aligned}$$

Opening a definition is often the first real move in a proof. A theorem gives you hypotheses. The hypotheses are wrapped packages. Definitions unwrap them.

5.3 Closing a definition

Closing a definition goes in the other direction. You have built the data, and now you are allowed to name the property.

Suppose you have reached

$$m + n = 2(r + s)$$

where $r + s \in \mathbb{Z}$. Then you may conclude $m + n$ is even. You are closing the definition of even.

Suppose you have reached

$$c = a(mn)$$

where $mn \in \mathbb{Z}$. Then you may conclude $a \mid c$. You are closing the definition of divisibility.

Focus

Opening turns a word into usable data. Closing turns constructed data into the word you were trying to prove.

This open-close rhythm is everywhere:

- To use injectivity, open it: from $f(x) = f(y)$ conclude $x = y$.
- To prove injectivity, close it: start with arbitrary x, y and show $f(x) = f(y) \Rightarrow x = y$.
- To use subset inclusion, open it: from $A \subseteq B$ and $x \in A$ conclude $x \in B$.
- To prove subset inclusion, close it: let $x \in A$ be arbitrary and show $x \in B$.

The words change. The machine rhythm stays the same.

5.4 Worked proof: even plus even

Theorem 1. *The sum of two even integers is even.*

Proof. Let $a, b \in \mathbb{Z}$ be even. Since a is even, there exists $m \in \mathbb{Z}$ such that $a = 2m$. Since b is even, there exists $n \in \mathbb{Z}$ such that $b = 2n$. Then

$$a + b = 2m + 2n = 2(m + n).$$

Because $m + n \in \mathbb{Z}$, this shows that $a + b$ is twice an integer. Therefore $a + b$ is even. □

This proof is small, but it contains the whole pattern.

We opened the definition twice:

$$a = 2m, \qquad b = 2n.$$

We used algebra:

$$a + b = 2(m + n).$$

Then we closed the definition:

$$a + b \text{ is even.}$$

The proof is not proving that $2m + 2n = 2(m + n)$ in isolation. That algebra line is just the hinge. The real proof structure is definition, algebra, definition.

Walking question

When you read a proof, ask: which definition was opened here, and which definition was closed at the end?

5.5 The integer witness matters

The phrase "for some integer k" is not filler. It is part of the machine.

To prove n is even, it is not enough to write

$$n = 2 \cdot \frac{n}{2}.$$

That is true for every real number n, but it does not prove that every real number is even. The witness must be an integer.

For instance, take $n = 3$. Then

$$3 = 2 \cdot \frac{3}{2}.$$

But $\frac{3}{2} \notin \mathbb{Z}$, so this does not show that 3 is even.

Common trap

Whenever a definition says "there exists $k \in \mathbb{Z}$," the membership condition $k \in \mathbb{Z}$ is not optional. Many false proofs accidentally use a witness from the wrong universe.

Here is a correct closing step:

$$6n + 10 = 2(3n + 5).$$

If $n \in \mathbb{Z}$, then $3n + 5 \in \mathbb{Z}$. Therefore $6n + 10$ is even.

Here is a suspicious closing step:

$$3n + 1 = 2\left(\frac{3n + 1}{2}\right).$$

This does not prove $3n + 1$ is even unless you already know $\frac{3n+1}{2}$ is an integer. For $n = 1$, it is 2; for $n = 2$, it is $\frac{7}{2}$. The witness test catches the mistake.

5.6 Worked proof: odd plus odd

Theorem 2. *The sum of two odd integers is even.*

Proof. Let $a, b \in \mathbb{Z}$ be odd. Since a is odd, there exists $m \in \mathbb{Z}$ such that $a = 2m + 1$. Since b is odd, there exists $n \in \mathbb{Z}$ such that $b = 2n + 1$. Then

$$a + b = (2m + 1) + (2n + 1) = 2m + 2n + 2 = 2(m + n + 1).$$

Since $m + n + 1 \in \mathbb{Z}$, the sum $a + b$ is twice an integer. Therefore $a + b$ is even. □

Notice that the proof did not say "odd plus odd is obviously even." It translated oddness into equations and translated the final equation into evenness. The obvious idea was real, but the definition made it proof.

5.7 Worked proof: divisibility is transitive

Theorem 3. *Let $a, b, c \in \mathbb{Z}$. If $a \mid b$ and $b \mid c$, then $a \mid c$.*

Proof. Suppose $a \mid b$ and $b \mid c$. Since $a \mid b$, there exists $m \in \mathbb{Z}$ such that $b = am$. Since $b \mid c$, there exists $n \in \mathbb{Z}$ such that $c = bn$. Substituting $b = am$ into $c = bn$, we get

$$c = (am)n = a(mn).$$

Because $mn \in \mathbb{Z}$, this shows that c is an integer multiple of a. Hence $a \mid c$. □

The proof is just the definition of divisibility doing its job. The input data gives:

$$b = am, \qquad c = bn.$$

The desired conclusion is $a \mid c$, so we want:

$$c = a(\text{integer}).$$

Substitution gives exactly that.

Common trap

The statement $a \mid b$ means $b = ak$, not $a = bk$. The number being divided is the one that becomes a multiple.

5.8 Bad proof debugged: divisibility

Consider the claim:

Theorem 4. *If $a \mid b$ and $a \mid c$, then $a \mid (b + c)$.*

Here is a bad proof:

> Since $a \mid b$ and $a \mid c$, divide both by a. Then b/a and c/a are integers. So $(b+c)/a$ is an integer. Therefore $a \mid (b + c)$.

The idea is not terrible, but the proof is slippery. It uses fractions where integer equations would be cleaner. It also hides the integer witnesses.

Here is the definition-machine version.

Proof. Suppose $a \mid b$ and $a \mid c$. Since $a \mid b$, there exists $m \in \mathbb{Z}$ such that $b = am$. Since $a \mid c$, there exists $n \in \mathbb{Z}$ such that $c = an$. Then

$$b + c = am + an = a(m + n).$$

Because $m + n \in \mathbb{Z}$, we have $a \mid (b + c)$. □

The corrected proof exposes the structure:

- $a \mid b$ gives one integer witness m.
- $a \mid c$ gives another integer witness n.
- The new witness for $a \mid (b + c)$ is $m + n$.

Proof lab

Debug this proof:

> If $a \mid b$, then $a \mid bc$. Since $a \mid b$, we know $a = bk$. Multiplying by c gives $ac = bkc$, so $a \mid bc$.

Find the exact line where the definition was opened incorrectly. Then rewrite the proof using the correct witness.

5.9 Definitions with hidden quantifiers

Many definitions hide quantifiers inside ordinary-sounding words.

$$A \subseteq B$$

means:

$$\forall x,\ x \in A \Rightarrow x \in B.$$

To use $A \subseteq B$, you need an element $x \in A$. Then you can conclude $x \in B$.

To prove $A \subseteq B$, you must show that every element of A belongs to B. The usual proof pattern is:

> Let $x \in A$ be arbitrary. We will show $x \in B$.

That one sentence is not decorative. It is the machine for proving a universal statement.

Theorem 5. *For all sets A and B, $A \cap B \subseteq A$.*

Proof. Let $x \in A \cap B$ be arbitrary. By definition of intersection, $x \in A$ and $x \in B$. In particular, $x \in A$. Therefore $A \cap B \subseteq A$. □

Again the proof is open-close:

- To prove a subset claim, open the subset target: let x be an arbitrary element of the left side.
- Open the definition of intersection.
- Close the subset claim by showing x belongs to the right side.

5.10 Definitions in analysis: bounded sequences

Let us look ahead to real analysis.

Definition 1. A sequence (a_n) of real numbers is bounded above if there exists $M \in \mathbb{R}$ such that $a_n \leq M$ for every $n \in \mathbb{N}$.

This definition has two important pieces:

- there exists one real number M,
- that same M works for every index n.

To use bounded-above-ness, you may introduce a real number M and say $a_n \leq M$ for all n.

To prove bounded-above-ness, you must produce one number M that works for all terms.

Theorem 6. *The sequence $a_n = 1 - \frac{1}{n}$ is bounded above.*

Proof. For every $n \in \mathbb{N}$, we have $\frac{1}{n} > 0$, so $1 - \frac{1}{n} < 1$. Thus $a_n \leq 1$ for every $n \in \mathbb{N}$. Therefore the sequence is bounded above, with upper bound $M = 1$. □

The proof closes the definition by presenting a witness: $M = 1$.

Could we choose $M = 100$? Yes. A bound does not have to be the best bound. To prove bounded above, one working ceiling is enough.

Could we choose $M = 1 - \frac{1}{n}$? No. That expression depends on n. The definition asks for one M that works for every n.

Common trap

When a definition says $\exists M\ \forall n$, the M must be chosen before n. It cannot secretly change from term to term.

5.11 Equivalent definitions and why they are allowed

Sometimes a concept has multiple definitions that turn out to be equivalent. This can be confusing: if definitions are supposed to be exact machines, how can there be more than one?

The answer is that a mathematical theory may prove two machines always accept the same inputs. Once that theorem is established, either machine can be used.

For example, for an integer n, these are equivalent:

$$n \text{ is even}$$

and

$$n^2 \text{ is even.}$$

But we should not silently use the second as the definition unless it has been stated or proved. The usual definition of even is $n = 2k$ for some integer k. The statement "n^2 is even iff n is even" is a theorem.

Another example comes from continuity. In a first analysis course, continuity at a point often means:

$$\forall \varepsilon > 0, \exists \delta > 0, \forall x,\ |x - a| < \delta \Rightarrow |f(x) - f(a)| < \varepsilon.$$

In topology, continuity is often defined by:

$$U \text{ open in the codomain} \Rightarrow f^{-1}(U) \text{ open in the domain.}$$

For functions $\mathbb{R} \to \mathbb{R}$ with the usual topology, these definitions agree. But proving that agreement is mathematics, not word magic.

Formal

Use the definition your context gives you. Use an equivalent characterization only after it has been proved, cited, or explicitly allowed.

5.12 Bad proof debugged: rational numbers

Definition 2. A real number r is rational if there exist integers p, q with $q \neq 0$ such that

$$r = \frac{p}{q}.$$

Claim:

Theorem 7. *The sum of two rational numbers is rational.*

Bad proof:

> Let a and b be rational. Then $a = p/q$ and $b = p/q$. So $a + b = 2p/q$, which is rational.

The proof accidentally uses the same numerator and denominator for both rational numbers. The definition gives each rational number its own witnesses. There is no reason the witnesses must match.

Correct proof:

Proof. Let $a, b \in \mathbb{R}$ be rational. Since a is rational, there exist integers p, q with $q \neq 0$ such that $a = p/q$. Since b is rational, there exist integers r, s with $s \neq 0$ such that $b = r/s$. Then

$$a + b = \frac{p}{q} + \frac{r}{s} = \frac{ps + rq}{qs}.$$

Now $ps + rq \in \mathbb{Z}$, $qs \in \mathbb{Z}$, and $qs \neq 0$. Therefore $a + b$ is rational. □

The repaired proof respects the witnesses:

$$a = \frac{p}{q}, \qquad b = \frac{r}{s}.$$

Different objects usually need different witness names. Reusing letters is not illegal when done carefully, but it is a common way to lie to yourself.

Common trap

If two hypotheses both say "there exists," do not assume they give the same object. Give the witnesses different names unless there is a reason they must coincide.

5.13 Definitions as tests

Definitions also tell you how to find examples and non-examples.

To test whether 18 is divisible by 6, ask whether

$$18 = 6k$$

for some integer k. Yes: $k = 3$.

To test whether 20 is divisible by 6, ask whether

$$20 = 6k$$

for some integer k. No integer k works.

To test whether the set

$$A = \{2, 4, 6\}$$

is a subset of

$$B = \{1, 2, 3, 4, 5, 6\},$$

check every element of A. Yes, each belongs to B.

To test whether $B \subseteq A$, find one element of B not in A. The number 1 works, so $B \not\subseteq A$.

The definition tells you what evidence counts.

5.14 A larger worked proof: product of odds

Theorem 8. *The product of two odd integers is odd.*

Proof. Let $a, b \in \mathbb{Z}$ be odd. Then there exist $m, n \in \mathbb{Z}$ such that

$$a = 2m + 1, \qquad b = 2n + 1.$$

Thus

$$ab = (2m + 1)(2n + 1) = 4mn + 2m + 2n + 1.$$

Factor out 2 from all terms except the final 1:

$$ab = 2(2mn + m + n) + 1.$$

Since $2mn + m + n \in \mathbb{Z}$, the product ab has the form $2k + 1$ for an integer k. Therefore ab is odd. □

The witness for the final oddness is

$$k = 2mn + m + n.$$

The proof is easier to trust when you can point to the witness.

5.15 Walking questions

Walking question

Pick a definition you know: prime, continuous, invertible, independent, open, closed, differentiable. Ask:

1. What data do I get if I am told the definition holds?
2. What data must I build if I want to prove it holds?
3. Which words in the definition are easy to ignore but actually essential?

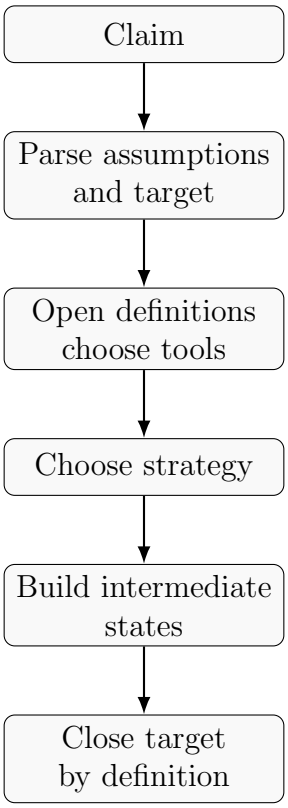
6 Proof Anatomy and Proof Compression

Real mathematical writing is compressed. A textbook proof may use three lines where a beginner needs a page. A research paper may use one phrase where a textbook uses a paragraph. This is not because mathematicians enjoy hiding the ball. It is because communities develop shared proof idioms. To read serious mathematics, you need to learn how to expand compressed proofs and how to compress your own proofs responsibly.

6.1 Proof anatomy

Every proof has anatomy:

1. **Claim:** What exactly is being proved?
2. **Assumptions:** What are you allowed to use?
3. **Target:** What must be shown by definition?
4. **Available tools:** Which definitions, previous theorems, and algebraic facts are relevant?
5. **Strategy:** Direct, contrapositive, contradiction, cases, induction, construction, or two inclusions?
6. **Intermediate states:** What smaller claims bridge assumptions to target?
7. **Closure:** Where exactly does the proof reach the target?



6.2 Anatomy example: composition of injective functions

Claim:

$$\text{If } f : A \to B \text{ and } g : B \to C \text{ are injective, then } g \circ f \text{ is injective.}$$

Assumptions.

- $f : A \to B$.
- $g : B \to C$.
- f is injective.
- g is injective.

Target.

To prove $g \circ f$ is injective, we must show:

$$(g \circ f)(a_1) = (g \circ f)(a_2) \Rightarrow a_1 = a_2$$

for arbitrary $a_1, a_2 \in A$.

Available tools.

Injectivity of g says:

$$g(b_1) = g(b_2) \Rightarrow b_1 = b_2.$$

Injectivity of f says:

$$f(a_1) = f(a_2) \Rightarrow a_1 = a_2.$$

Intermediate states.

Starting from

$$g(f(a_1)) = g(f(a_2)),$$

injectivity of g gives

$$f(a_1) = f(a_2).$$

Then injectivity of f gives

$$a_1 = a_2.$$

Compressed proof.

> Suppose $(g \circ f)(a_1) = (g \circ f)(a_2)$. Then $g(f(a_1)) = g(f(a_2))$, so $f(a_1) = f(a_2)$ by injectivity of g. By injectivity of f, $a_1 = a_2$. Hence $g \circ f$ is injective.

6.3 What "obvious" often hides

When a proof says "clearly," it may be compressing one of several standard moves:

- applying a definition,
- using a previously proved lemma,
- doing algebra,
- checking a domain condition,
- handling a trivial case,
- invoking a standard theorem.

Your job as a reader is to expand the word "clearly" into the actual move. Your job as a writer is to avoid compressing a step before you know exactly what it contains.

Common trap

False intuition: short proofs are automatically better. A short proof is better only when the reader can reconstruct the omitted steps. Premature compression produces mystery, not elegance.

6.4 Proof sketch culture

Advanced mathematics often uses proof sketches. A proof sketch is not a fake proof; it is a map of the main route, leaving routine checks to the reader. But what counts as routine depends on audience.

For a beginner, "routine" may include too much. For an expert, writing every algebraic rearrangement can hide the main idea. Mathematical writing is calibrated to a community.

Proof lab

Compression drill. Take a one-page proof you have written. Produce three versions:

1. expanded beginner version, with every definition opened;
2. textbook version, with standard moves named;
3. proof sketch, with only the strategic spine.

Then mark which audience each version serves.

6.5 Exercises

1. Prove that the product of two even integers is even.
2. Prove that the sum of an even integer and an odd integer is odd.
3. Prove that if $a \mid b$ and $a \mid c$, then $a \mid (b - c)$.
4. Prove that if $a \mid b$, then $a \mid bc$ for every integer c.
5. Prove that if $a, b, c \in \mathbb{Z}$, $a \mid b$, and $b \mid c$, then $a \mid c$.
6. Prove that the product of two rational numbers is rational.
7. Let A, B, C be sets. Prove that if $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$.
8. Let A and B be sets. Prove that $A \cap B \subseteq A \cup B$.
9. Give a definition-machine proof that if a sequence is bounded above by M and bounded below by m, then every term lies in the interval $[m, M]$.
10. Write a bad proof of one of the previous results, then annotate exactly where the definition machine was misused.

7 If-Then Statements and Proof Strategy

Most theorems are promises of the form

$$P \Rightarrow Q.$$

This is read "if P, then Q." The statement P is the hypothesis. The statement Q is the conclusion.

The proof task is not to prove P. It is not to prove Q in empty space. The task is to prove that Q must hold in every situation where P holds.

Focus

An implication is a bridge. You do not prove that everyone is standing at the entrance. You prove that anyone who starts at the entrance can get to the exit.

This distinction is subtle at first, but it explains many proof strategies. A direct proof starts at P and walks to Q. A contrapositive proof starts at $\neg Q$ and walks to $\neg P$. A contradiction proof assumes the theorem fails and shows the assumption breaks the system.

7.1 The anatomy of an implication

Consider:

$$\text{If } n \text{ is divisible by 12, then } n \text{ is divisible by 3.}$$

The hypothesis is:

$$n \text{ is divisible by 12.}$$

The conclusion is:

$$n \text{ is divisible by 3.}$$

A proof should begin by assuming the hypothesis:

> Suppose n is divisible by 12.

Then it should aim at the conclusion:

> We will show n is divisible by 3.

Now the definitions tell us what to do. "Divisible by 12" means

$$n = 12k$$

for some integer k. "Divisible by 3" means we want

$$n = 3m$$

for some integer m.

Since $12k = 3(4k)$, the witness is $m = 4k$.

Proof. Suppose n is divisible by 12. Then $n = 12k$ for some $k \in \mathbb{Z}$. Since

$$n = 12k = 3(4k)$$

and $4k \in \mathbb{Z}$, it follows that n is divisible by 3. □

That is direct proof in its cleanest form: assume the input condition, unfold it, and build the output condition.

7.2 Truth of an implication

An implication

$$P \Rightarrow Q$$

is false only in one case: P is true and Q is false.

This matches ordinary promises. Suppose a repair shop says:

> If you bring the car before noon, we will inspect it today.

The promise is broken only if you bring the car before noon and they do not inspect it today. If you do not bring the car before noon, the promise was not tested.

Mathematical implications work the same way. The statement

$$\text{If } n \text{ is divisible by 12, then } n \text{ is divisible by 3}$$

does not claim that every integer is divisible by 12. It says that every integer inside the hypothesis zone is also inside the conclusion zone.

Formal

The implication $P \Rightarrow Q$ is equivalent to saying:

$$\text{there is no case where } P \text{ is true and } Q \text{ is false.}$$

This is why counterexamples to implications have a specific shape. To disprove

$$P \Rightarrow Q,$$

you must find an example where P holds and Q fails.

7.3 Vacuous truth without the fog

The statement

$$\text{If } x \in \emptyset, \text{ then } x^2 = 17$$

is true for every real number x if we read it as a universal implication:

$$\forall x \in \mathbb{R},\ x \in \emptyset \Rightarrow x^2 = 17.$$

Why? Because there is no $x \in \emptyset$. There is no case with a true hypothesis and false conclusion.

This can feel like a trick, but it is not a trick. It is bookkeeping. An implication says that all cases satisfying the hypothesis also satisfy the conclusion. If there are no cases satisfying the hypothesis, the containment is empty and therefore unbroken.

A physical analogy: suppose a sign says,

> All passengers in row 99 must wear blue wristbands.

If the theater has no row 99, the rule has not been violated. There is nobody in row 99 without a wristband.

Common trap

Vacuous truth does not mean the conclusion is magically true. It means the conditional promise has no counterexample because the hypothesis never occurs.

7.4 Direct proof as forward engineering

Direct proof is the first strategy to try when the hypothesis contains useful structure and the conclusion asks for structure you can build from it.

The pattern:

> Assume P. Use definitions, known facts, and algebra to prove Q.

Here is a classic example.

Theorem 9. *If n is an odd integer, then n^2 is odd.*

Proof. Suppose n is odd. Then there exists $k \in \mathbb{Z}$ such that $n = 2k + 1$. Squaring,

$$n^2 = (2k+1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1.$$

Since $2k^2 + 2k \in \mathbb{Z}$, the number n^2 is odd. □

The hypothesis "n is odd" opens into a useful expression. The conclusion "n^2 is odd" asks for an expression of the form 2(integer) + 1. Direct proof is natural.

Intuition

Direct proof feels like starting with raw material and machining it into the required shape.

7.5 When direct proof gets stuck

Consider:

Theorem 10. *If n^2 is even, then n is even.*

A direct proof would start:

> Suppose n^2 is even. Then $n^2 = 2k$ for some integer k.

We want to prove $n = 2m$ for some integer m.

Can we take square roots and write

$$n = \sqrt{2k}?$$

That does not prove n is twice an integer. The expression $\sqrt{2k}$ is usually not helpful.

The hypothesis gives information about n^2, but the conclusion is about n. Direct proof is possible with a stronger theorem about prime divisibility, but at this stage there is a cleaner strategy: prove the contrapositive.

7.6 Contrapositive

The contrapositive of

$$P \Rightarrow Q$$

is

$$\neg Q \Rightarrow \neg P.$$

These two statements are logically equivalent. They always have the same truth value.

For the theorem

$$\text{If } n^2 \text{ is even, then } n \text{ is even,}$$

the contrapositive is:

$$\text{If } n \text{ is not even, then } n^2 \text{ is not even.}$$

For integers, "not even" means odd. So we prove:

$$\text{If } n \text{ is odd, then } n^2 \text{ is odd.}$$

We already know how to prove that directly.

Proof. We prove the contrapositive. Suppose n is not even. Since n is an integer, n is odd. Then $n = 2k + 1$ for some $k \in \mathbb{Z}$. Hence

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1.$$

Thus n^2 is odd, so n^2 is not even. Therefore, by contrapositive, if n^2 is even, then n is even. □

The contrapositive turns the awkward direction into the natural one.

Formal

$$P \Rightarrow Q$$

and

$$\neg Q \Rightarrow \neg P$$

are equivalent. The converse

$$Q \Rightarrow P$$

is a different statement. The inverse

$$\neg P \Rightarrow \neg Q$$

is also a different statement.

7.7 Converse confusion

Beginners often prove the converse by accident.

Original theorem:

$$\text{If } n^2 \text{ is even, then } n \text{ is even.}$$

Converse:

$$\text{If } n \text{ is even, then } n^2 \text{ is even.}$$

The converse happens to be true here, but proving it does not prove the original. In many cases the converse is false.

Original:

$$\text{If } n \text{ is divisible by 12, then } n \text{ is divisible by 3.}$$

Converse:

$$\text{If } n \text{ is divisible by 3, then } n \text{ is divisible by 12.}$$

False. Take $n = 6$. Then 6 is divisible by 3, but not by 12.

Common trap

When proving an implication, keep the direction visible. Write the hypothesis and conclusion separately before you begin.

7.8 Software analogy: preconditions and postconditions

An implication behaves like a function contract.

Precondition: the input array is sorted.

Postcondition: binary search returns the index of the target if it exists.

The contract does not say every array is sorted. It says: under this precondition, the algorithm guarantees this postcondition.

If you test binary search on an unsorted array and get nonsense, that does not refute the contract. You violated the hypothesis. If you test it on a sorted array and it fails to find an existing target, then you have a real counterexample.

The same discipline applies to theorems. When a theorem says

$$P \Rightarrow Q,$$

the hypothesis P is the precondition. The conclusion Q is the postcondition.

7.9 Strategy selection

Before writing a proof, pause and classify the statement.

- Is it a universal statement? You probably start with an arbitrary object.
- Is it an existential statement? You probably need to build or identify a witness.
- Is it an implication? You probably assume the hypothesis.
- Is the conclusion a definition? You probably need to close that definition.
- Is the conclusion negative? Contradiction or contrapositive may help.
- Does the hypothesis look weak but the negated conclusion looks concrete? Try contrapositive.

Proof strategy is not a bag of tricks. It is reading the grammar of the statement and letting the grammar tell you what the first move should be.

Proof lab

For each theorem below, write only the first proof sentence and the final target shape.

1. If n is even, then $n^2 + 4n + 6$ is even.
2. If $a \mid b$ and $b \mid c$, then $a \mid c$.
3. If $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$.
4. If x is rational and y is rational, then $x - y$ is rational.

7.10 Worked proof: rational minus rational

Theorem 11. *If x and y are rational numbers, then $x - y$ is rational.*

Proof. Suppose x and y are rational. Then there exist integers a, b, c, d with $b \neq 0$ and $d \neq 0$ such that

$$x = \frac{a}{b}, \qquad y = \frac{c}{d}.$$

Then

$$x - y = \frac{a}{b} - \frac{c}{d} = \frac{ad - bc}{bd}.$$

Since $ad - bc \in \mathbb{Z}$, $bd \in \mathbb{Z}$, and $bd \neq 0$, the number $x - y$ is rational. □

The hypothesis gives two rational witnesses. The conclusion asks for one rational witness for $x - y$. The proof builds it:

$$\frac{ad - bc}{bd}.$$

7.11 Worked proof: subset transitivity

Theorem 12. *If $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$.*

Proof. Suppose $A \subseteq B$ and $B \subseteq C$. Let $x \in A$ be arbitrary. Since $A \subseteq B$ and $x \in A$, we have $x \in B$. Since $B \subseteq C$ and $x \in B$, we have $x \in C$. Therefore every element of A is an element of C, so $A \subseteq C$. □

The proof is short because subset inclusion is a transport machine:

$$x \in A \quad \Rightarrow \quad x \in B \quad \Rightarrow \quad x \in C.$$

7.12 Proof by cases

Sometimes an implication has a hypothesis that naturally splits into alternatives. Then proof by cases is appropriate.

Theorem 13. *For every integer n, the product $n(n+1)$ is even.*

Proof. Let $n \in \mathbb{Z}$ be arbitrary. Every integer is either even or odd.

Case 1: n is even. Then $n = 2k$ for some $k \in \mathbb{Z}$, so

$$n(n+1) = 2k(n+1) = 2(k(n+1)).$$

Since $k(n+1) \in \mathbb{Z}$, the product is even.

Case 2: n is odd. Then $n = 2k+1$ for some $k \in \mathbb{Z}$. Therefore $n+1 = 2k+2 = 2(k+1)$, so

$$n(n+1) = n \cdot 2(k+1) = 2(n(k+1)).$$

Since $n(k+1) \in \mathbb{Z}$, the product is even.

In both cases, $n(n+1)$ is even. □

The cases are exhaustive because every integer is even or odd. They are useful because each case provides a different form that makes the conclusion easy.

Common trap

In proof by cases, the cases must cover every possibility allowed by the hypothesis. A proof that checks only convenient cases is just a partial proof.

7.13 Biconditionals

A biconditional

$$P \iff Q$$

means both:

$$P \Rightarrow Q$$

and

$$Q \Rightarrow P.$$

The usual proof structure is therefore two implications.

Theorem 14. *For an integer n, n is even if and only if n^2 is even.*

Proof. First suppose n is even. Then $n = 2k$ for some $k \in \mathbb{Z}$. Hence

$$n^2 = (2k)^2 = 4k^2 = 2(2k^2).$$

Since $2k^2 \in \mathbb{Z}$, n^2 is even.

Conversely, suppose n^2 is even. We prove this direction by contrapositive. If n is not even, then n is odd. So $n = 2k + 1$ for some $k \in \mathbb{Z}$. Then

$$n^2 = (2k + 1)^2 = 2(2k^2 + 2k) + 1,$$

which is odd. Thus n^2 is not even. Therefore if n^2 is even, then n is even.

Combining the two directions, n is even if and only if n^2 is even. □

The phrase "if and only if" is strong. It says the two conditions are perfectly synchronized.

Intuition

A biconditional says two tests recognize exactly the same objects.

7.14 Bad proof debugged: proving the wrong direction

Claim:

Theorem 15. *If n^2 is odd, then n is odd.*

Bad proof:

> Suppose n is odd. Then $n = 2k + 1$. Squaring gives $n^2 = 2(2k^2 + 2k) + 1$, so n^2 is odd. Therefore if n^2 is odd, then n is odd.

The proof establishes:

$$n \text{ odd} \Rightarrow n^2 \text{ odd}.$$

That is the converse of the desired statement.

A correct proof can use the contrapositive:

Proof. We prove the contrapositive. Suppose n is not odd. Since n is an integer, n is even. Then $n = 2k$ for some $k \in \mathbb{Z}$. Hence

$$n^2 = 4k^2 = 2(2k^2).$$

So n^2 is even, and therefore n^2 is not odd. By contrapositive, if n^2 is odd, then n is odd. □

The repair is not longer because mathematics enjoys ceremony. It is longer because direction matters.

7.15 History and connection: implication as structure

History and connection

The formal study of implication goes back to ancient logic, especially Aristotle's work on syllogisms, but the modern symbolic treatment grew through Boole, Frege, Peano, Russell, Hilbert, and others. In the twentieth century, implication became central not only in mathematics but also in computing: proof systems, type systems, program verification, and automated theorem provers all depend on precise conditional structure.

A theorem is a reusable implication. Once proved, it becomes a component. If a later proof can establish the theorem's hypothesis, it can import the theorem's conclusion.

This is exactly how large mathematical theories scale. You do not reprove everything. You prove reliable implications and compose them.

7.16 A proof-strategy checklist

When facing a new theorem, write:

$$\begin{aligned} \text{Hypotheses:} &\quad \cdots \\ \text{Conclusion:} &\quad \cdots \end{aligned}$$

Then ask:

1. What definitions can I open from the hypotheses?
2. What definition must I close for the conclusion?
3. Is the statement universal? If yes, what arbitrary object should I choose?
4. Is the statement existential? If yes, what witness should I build?
5. Is direct proof natural?
6. Would the contrapositive be clearer?
7. Would contradiction turn the target into a useful false assumption?
8. Are there cases that cover all possibilities?

This checklist is not training wheels. Experienced mathematicians do a compressed version of it mentally.

7.17 Exercises

1. Prove directly: if n is even, then $5n + 8$ is even.
2. Prove directly: if n is odd, then $3n + 2$ is odd.
3. Prove by contrapositive: if n^2 is odd, then n is odd.
4. Prove by cases: for every integer n, $n^2 + n$ is even.
5. Prove: if $a \mid b$ and $c \mid d$, then $ac \mid bd$.
6. Prove: if x and y are rational, then $3x - 5y$ is rational.
7. Prove: if $A \subseteq B$, then $A \cap C \subseteq B \cap C$.
8. Find a counterexample to the converse: if n is divisible by 4, then n is even.
9. For each statement in this section, identify whether the proof used direct proof, contrapositive, cases, or a combination.
10. Write a one-paragraph strategy for proving: if $n^2 - 1$ is not divisible by 8, then n is even.

8 Proof by Contradiction

Proof by contradiction is one of the most dramatic methods in mathematics. It begins by denying what we want to prove. Then it follows that denial carefully until something impossible appears.

The basic form is:

$$\text{To prove } P, \text{ assume } \neg P \text{ and derive a contradiction.}$$

Once the contradiction appears, the assumption $\neg P$ cannot be maintained. Therefore P must be true.

Focus

Contradiction is not a license to be vague. The contradiction must be a precise collision: a statement and its negation, or an impossible fact such as $0 = 1$, an integer being both even and odd, or a positive number being less than itself.

8.1 A tiny contradiction proof

Theorem 16. *There is no smallest positive real number.*

Proof. Suppose, for contradiction, that there is a smallest positive real number r. Since $r > 0$, the number $r/2$ is also positive. But

$$0 < \frac{r}{2} < r.$$

This contradicts the assumption that r is the smallest positive real number. Therefore there is no smallest positive real number. □

The impossible collision is exact:

- Assumption: r is the smallest positive real number.
- Derived fact: $r/2$ is a positive real number smaller than r.

Those cannot both be true.

8.2 Contradiction versus contrapositive

Contradiction and contrapositive are related, but they feel different in practice.

To prove an implication

$$P \Rightarrow Q$$

by contrapositive, prove:

$$\neg Q \Rightarrow \neg P.$$

To prove it by contradiction, assume:

$$P \text{ and } \neg Q$$

and derive an impossibility.

For example, to prove

$$\text{if } n^2 \text{ is even, then } n \text{ is even,}$$

contrapositive assumes n is odd and proves n^2 is odd.

Contradiction assumes n^2 is even and n is odd, then derives that n^2 is odd, contradicting that n^2 is even.

Both are valid. The contrapositive proof often feels cleaner because it produces a positive conclusion rather than carrying a contradiction flag.

Intuition

Contrapositive says, "If the destination is absent, the starting condition was absent." Contradiction says, "Suppose the starting condition holds but the destination fails; now watch the system break."

8.3 The contradiction contract

A contradiction proof has a contract:

1. State what you are assuming for contradiction.
2. Use that assumption together with known facts.
3. Derive a clear impossibility.
4. Identify the impossibility.
5. Conclude that the contradictory assumption was false.

Skipping step 4 is a common beginner mistake. A proof that ends with "which is a contradiction" but does not say what contradicted what leaves the reader to do the proof's final job.

Common trap

Do not use contradiction as a fog machine. If you cannot name the contradictory pair, the proof is not finished.

8.4 Classic proof: irrationality of square root of 2

Theorem 17. *$\sqrt{2}$ is irrational.*

Proof. Suppose, for contradiction, that $\sqrt{2}$ is rational. Then there exist integers a, b with $b \neq 0$ such that

$$\sqrt{2} = \frac{a}{b}.$$

Choose such a fraction in lowest terms, so a and b have no common factor greater than 1.

Squaring both sides gives

$$2 = \frac{a^2}{b^2},$$

so

$$a^2 = 2b^2.$$

Thus a^2 is even. Since a^2 is even, a is even. Therefore $a = 2k$ for some $k \in \mathbb{Z}$.

Substitute $a = 2k$ into $a^2 = 2b^2$:

$$(2k)^2 = 2b^2.$$

So

$$4k^2 = 2b^2,$$

and hence

$$b^2 = 2k^2.$$

Thus b^2 is even, so b is even.

We have shown that both a and b are even. Therefore they share a factor of 2, contradicting the choice of a/b in lowest terms. Hence $\sqrt{2}$ is irrational. □

This proof is famous because it turns a geometric fact into arithmetic pressure. The diagonal of a unit square has length $\sqrt{2}$. If that length were a ratio of whole numbers, the ratio could be chosen in lowest terms. But the equation forces both numerator and denominator to be even, so the ratio was not in lowest terms after all.

History and connection

The discovery of irrational magnitudes is traditionally linked to the Pythagorean school. The historical details are hard to separate from legend, but the mathematical shock is real: geometry produced lengths that ratios of whole numbers could not express. This forced a major expansion in the concept of number.

8.5 Why lowest terms matters

In the proof of irrationality of $\sqrt{2}$, the phrase "choose the fraction in lowest terms" is essential.

If we only assume

$$\sqrt{2} = \frac{a}{b}$$

for some integers a, b, then showing a and b are both even is not immediately impossible. Fractions are often not in lowest terms:

$$\frac{4}{6} = \frac{2}{3}.$$

The contradiction requires the extra condition that the representation has no common factor.

Why are we allowed to choose a lowest-terms representation? Because every rational number can be written as a fraction of integers with no common factor greater than 1. That fact comes from the well-ordering principle or the Euclidean algorithm.

The contradiction is:

$$a \text{ and } b \text{ have no common factor greater than } 1$$

and

$$a \text{ and } b \text{ are both divisible by } 2.$$

Formal

A contradiction proof often needs a carefully strengthened assumption. "Let a/b be in lowest terms" is not decoration; it creates the exact condition that will later be contradicted.

8.6 Bad proof debugged: square root of 2

Bad proof:

> Suppose $\sqrt{2} = a/b$. Then $a^2 = 2b^2$. So a^2 is even, hence a is even. Similarly b is even. This is impossible. Therefore $\sqrt{2}$ is irrational.

There are two problems.

First, "similarly b is even" is not immediate. You need to substitute $a = 2k$ and derive $b^2 = 2k^2$.

Second, "This is impossible" is false unless the fraction was chosen in lowest terms. It is not impossible for two integers to both be even. It is impossible for numerator and denominator of a lowest-terms fraction to both be even.

Corrected proof skeleton:

1. Assume $\sqrt{2} = a/b$ in lowest terms.
2. Derive $a^2 = 2b^2$.
3. Conclude a is even, so $a = 2k$.
4. Substitute and conclude b is even.
5. Contradict lowest terms.

8.7 Contradiction with divisibility

Theorem 18. *There is no integer n such that n is both even and odd.*

Proof. Suppose, for contradiction, that there exists an integer n that is both even and odd. Since n is even, $n = 2a$ for some $a \in \mathbb{Z}$. Since n is odd, $n = 2b + 1$ for some $b \in \mathbb{Z}$. Thus

$$2a = 2b + 1.$$

Rearranging gives

$$2(a - b) = 1.$$

This says that 1 is even, since $a - b \in \mathbb{Z}$. But 1 is not even. Contradiction. Therefore no integer is both even and odd. □

The contradiction here is not emotional. It is exact:

$$1 = 2(a - b)$$

would make 1 divisible by 2, which it is not.

8.8 Existence proofs by contradiction

Contradiction can prove existence. To prove that some object exists, assume no such object exists and show impossibility.

Theorem 19. *There exist irrational numbers x and y such that $x + y$ is rational.*

This theorem actually has a direct proof:

$$x = \sqrt{2}, \qquad y = -\sqrt{2}.$$

Both are irrational, but

$$x + y = 0,$$

which is rational.

Contradiction is not needed.

This matters because beginners sometimes use contradiction whenever a theorem feels hard. But contradiction is not always the cleanest method. For existence claims, a witness is usually better than a contradiction if you can find one.

Focus

If you can prove existence by showing the object, show the object. A witness beats a thunderstorm.

8.9 Nonconstructive contradiction

Some existence proofs by contradiction do not produce a concrete object. They show that nonexistence is impossible.

Here is a simple finite example.

Theorem 20. *Among any three integers, two have the same parity.*

Proof. Suppose, for contradiction, that among three integers no two have the same parity. Then at most one of the integers is even and at most one is odd. That accounts for at most two integers total. But we have three integers. Contradiction. Therefore at least two of the three integers have the same parity. □

This is a pigeonhole argument in miniature. There are two parity boxes, even and odd. Three integers cannot occupy two boxes without some box receiving at least two integers.

History and connection

The pigeonhole principle is often associated with Dirichlet, who used counting arguments to prove deep results in number theory. Its simplest form says that if more objects than boxes are distributed among the boxes, at least one box contains more than one object.

8.10 Infinite primes

Euclid's proof that there are infinitely many primes is one of the great contradiction proofs.

Theorem 21. *There are infinitely many prime numbers.*

Proof. Suppose, for contradiction, that there are only finitely many primes. List them as

$$p_1, p_2, \dots, p_n.$$

Now form the number

$$N = p_1 p_2 \cdots p_n + 1.$$

The integer N is greater than 1, so it has a prime divisor q.

Since $p_1, \ldots, p_n$ are assumed to be all the primes, q must equal one of them, say $q = p_i$. Because q divides the product $p_1 p_2 \cdots p_n$, and q also divides N, it follows that q divides the difference

$$N - p_1 p_2 \cdots p_n = 1.$$

But no prime divides 1. This is a contradiction. Therefore there are infinitely many primes. □

The engine is clever:

$$N = p_1 p_2 \cdots p_n + 1.$$

This number is designed to leave remainder 1 when divided by every prime on the supposed complete list. So it either is prime itself or has a prime factor not on the list. Either way, the list was incomplete.

Intuition

Euclid's construction is like checking a supposedly complete registry, then building a new object that evades every entry in the registry.

8.11 The exact contradiction in Euclid's proof

The contradiction is sometimes misstated as:

> N is not divisible by any prime, which is impossible.

That is close, but the clean version is:

- Every integer greater than 1 has a prime divisor.
- If the list contains all primes, that divisor must be one of $p_1, \ldots, p_n$.
- But no p_i divides N, because N leaves remainder 1 when divided by p_i.

The contradiction is between "q is on the complete list" and "q cannot be any prime on the list."

Common trap

Euclid's proof does not say $p_1 p_2 \cdots p_n + 1$ is always prime. For $2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13 + 1 = 30031$, the result is composite. The point is that its prime factors are not on the original list.

8.12 Contradiction and minimal counterexamples

A powerful method combines contradiction with choosing a smallest counterexample.

The rough pattern:

1. Suppose the theorem is false.
2. Then there is at least one counterexample.
3. Choose the smallest counterexample.
4. Use it to build an even smaller counterexample.
5. Contradiction.

Here is a gentle example.

Theorem 22. *Every integer $n \geq 2$ has a prime divisor.*

Proof. Suppose, for contradiction, that some integer $n \geq 2$ has no prime divisor. By the well-ordering principle, choose the smallest such integer and call it m.

If m is prime, then m has a prime divisor, namely itself. That contradicts the choice of m. Therefore m is not prime.

Since $m \geq 2$ and m is not prime, it is composite. So $m = ab$ for integers a, b with

$$1 < a < m, \qquad 1 < b < m.$$

Because $a < m$ and m was the smallest integer at least 2 with no prime divisor, the integer a must have a prime divisor p. Since $p \mid a$ and $a \mid m$, we get $p \mid m$. This contradicts the choice of m as having no prime divisor.

Therefore every integer $n \geq 2$ has a prime divisor. □

Minimal counterexample arguments are common in number theory, combinatorics, graph theory, and induction. They are a formal way of saying: if failure happens, let us inspect the first failure. Often the first failure cannot exist.

8.13 Danger: proving too much

A contradiction proof can go wrong by using an assumption that is stronger than the negation of the theorem.

Suppose you want to prove:

$$\text{There exists an integer } n \text{ such that } n^2 > 10.$$

The negation is:

$$\text{For every integer } n,\ n^2 \leq 10.$$

It is not:

$$\text{For every integer } n,\ n^2 < 0.$$

If you assume something stronger than the actual negation and derive a contradiction, you have not proved the theorem. You only proved that your stronger assumption was false.

Formal

To prove P by contradiction, assume exactly $\neg P$, or at least an assumption that follows from $\neg P$. Do not invent a stronger denial because it is easier to break.

8.14 Negating statements before contradiction

Contradiction requires accurate negation.

Original:

$$\forall x \in A,\ P(x).$$

Negation:

$$\exists x \in A \text{ such that } \neg P(x).$$

Original:

$$\exists x \in A \text{ such that } P(x).$$

Negation:

$$\forall x \in A,\ \neg P(x).$$

Original:

$$P \Rightarrow Q.$$

Negation:

$$P \text{ and } \neg Q.$$

If the negation is wrong, the contradiction proof is proving the wrong thing.

Proof lab

Write the negation you would assume for contradiction:

1. Every bounded sequence converges.
2. There exists a largest integer.
3. If n^2 is divisible by 3, then n is divisible by 3.
4. For every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that $n \geq N$ implies $|a_n - L| < \varepsilon$.

8.15 Bad proof debugged: no largest integer

Claim:

Theorem 23. *There is no largest integer.*

Bad proof:

> Suppose there is a largest integer N. Then $N + 1$ is larger. Contradiction.

This is almost fine, but it skips one small piece: $N+1$ must be an integer. The largest-integer assumption concerns integers, not all real numbers. A polished proof says:

Proof. Suppose, for contradiction, that there is a largest integer N. Since $N \in \mathbb{Z}$, we have $N + 1 \in \mathbb{Z}$. But $N + 1 > N$, contradicting the assumption that N is the largest integer. Therefore there is no largest integer. □

This may feel picky. It is picky in the same way bridge inspection is picky. The missing sentence is exactly what connects the constructed object back to the domain of the claim.

8.16 Contradiction in later mathematics

Contradiction appears everywhere.

In real analysis, it proves uniqueness of limits:

> Assume a sequence has two different limits L and M. Choose disjoint neighborhoods around L and M. Eventually the sequence must be inside both, impossible.

In linear algebra, it proves uniqueness of inverses:

> Assume a matrix has two inverses B and C. Then $B = BI = B(AC) = (BA)C = IC = C$.

In topology, it proves separation results. In algebra, it proves nonexistence results. In computability, diagonal arguments often have contradiction at their core.

History and connection

Proof by contradiction is classically accepted in standard mathematics, but it has also been philosophically debated. Constructivist mathematicians distinguish between proving that nonexistence is impossible and actually constructing an object. This debate is not just philosophy; it affects proof assistants, type theory, and the meaning of existence in computational mathematics.

8.17 Walking questions

Walking question

When you see a contradiction proof, ask:

1. What exact statement is being denied?
2. Is the denial correct?
3. Which assumptions are used after the denial?
4. What is the contradictory pair?
5. Could the proof be rewritten as a contrapositive or direct proof?

8.18 Exercises

1. Prove by contradiction that there is no largest real number.
2. Prove by contradiction that there is no smallest positive integer greater than 1 that is divisible by every positive integer.
3. Prove that $\sqrt{3}$ is irrational. You may use the fact that if $3 \mid n^2$, then $3 \mid n$.
4. Prove that $\sqrt{5}$ is irrational. You may use the analogous divisibility fact.
5. Prove that there do not exist $a, b \in \mathbb{Z}$ such that $a^2 - 4b = 2$.
6. Prove there are infinitely many integers divisible by 7.
7. Prove that if $x + y = 0$ and $x > 0$, then $y < 0$ by contradiction.
8. Write a contradiction proof of: if $A \subseteq B$ and $x \notin B$, then $x \notin A$.
9. Find the flaw in this proof: "Suppose n is the smallest integer. Then $n - 1 < n$, contradiction. Therefore no integer is smallest." Repair it.
10. For each contradiction proof in this section, identify the exact contradiction in one sentence.

9 Examples, Non-examples, and Counterexamples

Examples are not the snack after the theorem. They are part of the meal.

A definition without examples is a machine you have never seen run. A theorem without examples is a road sign for a road you have never walked. Counterexamples are stress tests. Non-examples are boundary markers. Together they build mathematical taste.

Focus

Examples teach you what a definition includes. Non-examples teach you what it excludes. Counterexamples teach you how statements fail.

This section is about a habit: whenever you meet a new concept, build a small zoo around it. Do not collect only pretty examples. Collect edge cases, boring cases, tiny cases, broken cases, and almost-cases.

9.1 Examples versus proofs

An example can prove an existential statement.

Statement:

$$\exists n \in \mathbb{Z} \text{ such that } n^2 = 25.$$

Proof:

$$n = 5.$$

That is enough. An existential statement asks for at least one witness. A single valid witness proves it.

But an example cannot prove a universal statement.

Statement:

$$\forall n \in \mathbb{Z},\ n^2 \geq 0.$$

Checking $n = 5$ is not a proof. Checking $n = -100$ is not a proof. Checking a million integers by computer is evidence, but not proof. The statement concerns every integer, and there are infinitely many.

To prove it, use a general argument:

Proof. Let $n \in \mathbb{Z}$ be arbitrary. If $n \geq 0$, then $n^2 \geq 0$. If $n < 0$, then $-n > 0$, and $n^2 = (-n)^2 > 0$. Therefore $n^2 \geq 0$ for every integer n. □

The proof handles all integers at once because it reasons from arbitrary structure, not from sampled values.

Common trap

An example proves "there exists." A counterexample disproves "for all." Neither one proves a universal statement.

9.2 Non-examples

A non-example is an object that fails a definition.

Definition:

$$n \text{ is even} \quad \Longleftrightarrow \quad n = 2k \text{ for some } k \in \mathbb{Z}.$$

Example: 10 is even because $10 = 2 \cdot 5$.

Non-example: 11 is not even because there is no integer k such that $11 = 2k$.

Non-examples are useful because they force you to use the exact definition. Saying "11 feels odd" is not proof. Saying "if $11 = 2k$, then $k = 11/2$, which is not an integer" is proof.

Intuition

A non-example is not merely absent from the club. It fails the entry test, and you should know which part of the test it fails.

9.3 Counterexamples

A counterexample is an example that disproves a claim.

Claim:

$$\forall x, y \in \mathbb{R},\ x^2 = y^2 \Rightarrow x = y.$$

Counterexample:

$$x = 1, \qquad y = -1.$$

Then

$$x^2 = 1 = y^2,$$

but

$$x \neq y.$$

The counterexample has the exact shape needed to refute the implication:

hypothesis true, conclusion false.

For a universal implication

$$\forall x,\ P(x) \Rightarrow Q(x),$$

a counterexample must provide an x such that $P(x)$ is true and $Q(x)$ is false.

Formal

To disprove

$$\forall x \in D,\ P(x),$$

find $x \in D$ such that $\neg P(x)$.
To disprove

$$\forall x \in D,\ P(x) \Rightarrow Q(x),$$

find $x \in D$ such that $P(x)$ is true and $Q(x)$ is false.

9.4 The domain is part of the counterexample

Claim:

$$\forall x,\ x^2 \geq x.$$

Is this true? It depends on the domain.

If $x \in \mathbb{Z}$, the statement is true. For integers, either $x \leq 0$, in which case $x^2 \geq 0 \geq x$, or $x \geq 1$, in which case $x^2 \geq x$.

If $x \in \mathbb{R}$, the statement is false. Take

$$x = \frac{1}{2}.$$

Then

$$x^2 = \frac{1}{4} < \frac{1}{2} = x.$$

So the counterexample is not just "1/2." It is "1/2 in the real-number domain." If the theorem was about integers, 1/2 is not even eligible.

Common trap

A counterexample must live inside the stated domain. An object outside the domain refutes nothing.

9.5 Near misses

A near miss almost satisfies a definition but fails one condition. Near misses are valuable because they reveal why each condition exists.

Definition:

A function $f : A \to B$ is injective if for all $x, y \in A$, $f(x) = f(y)$ implies $x = y$.

Example:

$$f : \mathbb{R} \to \mathbb{R}, \qquad f(x) = 2x + 1$$

is injective.

Non-example:

$$g : \mathbb{R} \to \mathbb{R}, \qquad g(x) = x^2$$

is not injective because $g(1) = g(-1)$ but $1 \neq -1$.

Near miss:

$$h : [0, \infty) \to \mathbb{R}, \qquad h(x) = x^2$$

is injective. Same formula, different domain.

This near miss teaches a serious lesson: the function is not just the formula. The domain and codomain are part of the object.

9.6 Injective and surjective examples

Let

$$f : \mathbb{N} \to \mathbb{N}, \qquad f(n) = n + 1,$$

where $\mathbb{N} = \{1, 2, 3, \ldots\}$.

This function is injective. If

$$f(a) = f(b),$$

then

$$a + 1 = b + 1,$$

so $a = b$.

It is not surjective. There is no $n \in \mathbb{N}$ such that

$$n + 1 = 1.$$

The codomain element 1 is missed.

Now let

$$g : \{1, 2\} \to \{a\}$$

be defined by

$$g(1) = a, \qquad g(2) = a.$$

This function is surjective because the only codomain element, a, is hit. It is not injective because two different inputs have the same output.

These examples separate two ideas that beginners often blur:

- Injective means no merging.
- Surjective means no missing codomain elements.

9.7 Counterexamples to converses

If a theorem says

$$P \Rightarrow Q,$$

the converse is

$$Q \Rightarrow P.$$

The converse may be true or false. A counterexample to the converse must have Q true and P false.

Theorem:

$$\text{If } n \text{ is divisible by } 12, \text{ then } n \text{ is divisible by } 3.$$

Converse:

$$\text{If } n \text{ is divisible by } 3, \text{ then } n \text{ is divisible by } 12.$$

Counterexample:

$$n = 6.$$

Then 6 is divisible by 3, but 6 is not divisible by 12.

This is a clean counterexample because it satisfies the converse hypothesis and fails the converse conclusion.

Walking question

For any implication you meet today, say its converse out loud. Then ask whether the converse is true, false, or uncertain. If false, try to build a counterexample.

9.8 Counterexamples to overgeneralization

A common source of false statements is taking a true theorem and removing a hypothesis.

True theorem:

> If f is continuous on a closed bounded interval $[a, b]$, then f attains a maximum on $[a, b]$.

Remove "closed":

$$f(x) = x \quad \text{on } (0, 1).$$

The function is continuous and bounded, but it does not attain a maximum. Its supremum is 1, but 1 is not in the domain.

Remove "bounded interval":

$$f(x) = x \quad \text{on } [0, \infty).$$

The function is continuous on a closed set, but it has no maximum.

Remove "continuous":

$$f(x) = \begin{cases} x, & 0 \le x < 1, \\ 0, & x = 1 \end{cases} \quad \text{on } [0,1].$$

This function is defined on a closed bounded interval, but it does not attain its supremum 1.

The theorem's assumptions are not fancy wrapping. They are load-bearing.

History and connection

Nineteenth-century analysis was shaped by examples that broke plausible statements about functions. Mathematicians learned that continuity, compactness, differentiability, and convergence needed precise hypotheses. Counterexamples did not weaken analysis; they made it stronger.

9.9 Bad counterexamples

Claim:

$$\text{Every prime number is odd.}$$

A student says:

> Counterexample: 9, because 9 is odd but not prime.

This is not a counterexample. The claim has the form:

$$\text{prime} \Rightarrow \text{odd}.$$

To refute it, we need:

$$\text{prime and not odd}.$$

The correct counterexample is 2, because 2 is prime and even.

Another bad counterexample:

Claim:

$$\forall x \in \mathbb{R},\ x^2 + 1 > 0.$$

Counterexample offered:

$$x = i.$$

But $i \notin \mathbb{R}$. The proposed object is outside the domain. It does not refute the real-number claim.

Common trap

For an implication, a counterexample must satisfy the hypothesis. For a quantified statement, a counterexample must belong to the quantified domain.

9.10 Examples that prove existence

Statement:

$$\exists a, b \in \mathbb{Z} \text{ such that } a^2 - b^2 = 15.$$

Try small numbers:

$$4^2 - 1^2 = 16 - 1 = 15.$$

So $a = 4$, $b = 1$ proves the statement.

Statement:

$$\exists x \in \mathbb{R} \text{ such that } x^2 = 2.$$

Take $x = \sqrt{2}$. This proves existence in $\mathbb{R}$ because $\sqrt{2}$ is a real number.

Statement:

$$\exists x \in \mathbb{Q} \text{ such that } x^2 = 2.$$

The same candidate does not work because $\sqrt{2} \notin \mathbb{Q}$. In fact the statement is false.

Again, the domain matters.

9.11 Families of examples

One example can prove existence, but a family of examples builds understanding.

Even integers:

$$\ldots, -4, -2, 0, 2, 4, \ldots$$

Family form:

$$2k, \qquad k \in \mathbb{Z}.$$

Odd integers:

$$\ldots, -3, -1, 1, 3, 5, \ldots$$

Family form:

$$2k + 1, \qquad k \in \mathbb{Z}.$$

Multiples of 7:

$$7k, \qquad k \in \mathbb{Z}.$$

Rational numbers:

$$\frac{p}{q}, \qquad p, q \in \mathbb{Z},\ q \neq 0.$$

Upper bounds for $(1 - 1/n)$:

$$1,\ 2,\ 100,\ \pi$$

are all upper bounds. The least upper bound is 1, but being an upper bound is a weaker condition.

Families reveal the degrees of freedom in a definition. They show what can vary while the structure remains.

9.12 Boundary examples

Boundary examples live at the edge of a definition.

Is 0 even? Yes:

$$0 = 2 \cdot 0.$$

Is 1 prime? No. A prime number has exactly two positive divisors, 1 and itself. The number 1 has only one positive divisor.

Is the empty set a subset of every set? Yes. To prove $\emptyset \subseteq A$, we must show every element of $\emptyset$ belongs to A. There are no elements of $\emptyset$, so there is no counterexample.

Is the empty function from $\emptyset$ to B a function? Yes. It assigns to every element of $\emptyset$ exactly one element of B, because there are no elements requiring assignment. This is another appearance of vacuous truth.

Boundary examples often feel strange because they test whether you are following the definition or relying on a picture.

Intuition

If an example feels silly but satisfies the definition, keep it. Silly examples are excellent stress tests.

9.13 The role of small finite models

When learning set theory, relations, functions, or algebra, small finite examples are priceless.

Let

$$A = \{1, 2, 3\}, \qquad B = \{2, 3, 4\}.$$

Then

$$A \cap B = \{2, 3\},$$

$$A \cup B = \{1, 2, 3, 4\},$$
$$A \setminus B = \{1\}.$$

These examples are not deep, but they calibrate the definitions.

For a relation example, define R on $\{1, 2, 3\}$ by

$$xRy \quad \Longleftrightarrow \quad x \leq y.$$

This relation is reflexive and transitive, but not symmetric.

Define S by

$$xSy \quad \Longleftrightarrow \quad x - y \text{ is even.}$$

This relation is reflexive, symmetric, and transitive. It groups the set into parity classes.

Tiny models let you see properties without drowning in notation.

9.14 Counterexample search strategies

When a statement seems false, search systematically.

First, identify the shape of failure.

For

$$\forall x,\ P(x) \Rightarrow Q(x),$$

you need $P(x)$ true and $Q(x)$ false.

Second, try boundary cases:

$$0, \quad 1, \quad -1, \quad \emptyset, \quad \{0\}, \quad \text{constant functions.}$$

Third, try changing the domain or codomain. Many false statements about functions fail because of codomain choices.

Fourth, try degenerate examples:

- empty sets,
- singleton sets,
- zero maps,
- identity maps,
- constant sequences,
- sequences with alternating signs.

Fifth, remove one assumption from a true theorem and see what breaks.

Proof lab

Find counterexamples to these claims. For each one, write the hypothesis true/false and conclusion true/false.

1. If $ab = 0$, then $a = 0$ and $b = 0$.
2. If $x^2 > 4$, then $x > 2$.
3. If a function is not injective, then it is not surjective.
4. If $A \cup B = A \cup C$, then $B = C$.
5. Every bounded sequence is convergent.

9.15 Worked counterexamples

Claim:

$$ab = 0 \Rightarrow a = 0 \text{ and } b = 0.$$

Counterexample:

$$a = 0, \qquad b = 5.$$

Then $ab = 0$, but $b \neq 0$, so the conclusion "$a = 0$ and $b = 0$" is false.

The true theorem is:

$$ab = 0 \Rightarrow a = 0 \text{ or } b = 0.$$

Claim:

$$x^2 > 4 \Rightarrow x > 2.$$

Counterexample:

$$x = -3.$$

Then $x^2 = 9 > 4$, but $x > 2$ is false.

The true conclusion would be:

$$x > 2 \text{ or } x < -2.$$

Claim:

$$A \cup B = A \cup C \Rightarrow B = C.$$

Counterexample:

$$A = \{1, 2\}, \qquad B = \{1\}, \qquad C = \{2\}.$$

Then

$$A \cup B = \{1, 2\} = A \cup C,$$

but

$$B \neq C.$$

The set A hides the difference between B and C because it already contains both elements.

9.16 Examples in proof discovery

Examples are not only for checking final answers. They help you discover proofs.

Suppose you want to prove:

The sum of the first n odd positive integers is n^2.

Compute:

$$1 = 1,$$
$$1 + 3 = 4,$$
$$1 + 3 + 5 = 9,$$
$$1 + 3 + 5 + 7 = 16.$$

The examples suggest squares. More importantly, they suggest a visual proof: start with a 1×1 square, then add an L-shaped border of 3 dots, then one of 5, then one of 7. Each odd number grows the square by one layer.

Examples can also suggest induction:

$$1 + 3 + \cdots + (2n - 1) = n^2.$$

Then the next term is

$$2(n + 1) - 1 = 2n + 1,$$

and

$$n^2 + (2n + 1) = (n + 1)^2.$$

The examples did not prove the theorem, but they revealed the mechanism.

9.17 Examples in debugging definitions

Suppose someone defines a prime number as:

> A number divisible only by 1 and itself.

This is close, but incomplete. What kind of number? Positive integer? Integer greater than 1?

If negative integers are allowed, is -5 prime? Its divisors include $-1, 1, -5, 5$. If 1 is allowed, the definition accidentally makes 1 look prime unless we require exactly two positive divisors or require the number to be greater than 1.

A cleaner definition:

Definition 3. A positive integer p is prime if $p > 1$ and its only positive divisors are 1 and p.

Examples force definitions to become honest. They expose missing domain restrictions and ambiguous words.

History and connection

Mathematics often advances when examples pressure definitions. The modern definition of function had to grow broad enough to include strange functions. The definition of continuity had to survive pathological examples. The definition of manifold had to separate local Euclidean behavior from global shape. Good definitions are not guessed fully formed; they are tested.

9.18 Physical analogy: crash testing

Engineers do not test a design only under pleasant conditions. They test load, heat, vibration, fatigue, edge cases, and failure modes.

Counterexamples are mathematical crash tests. A theorem that survives only nice examples is not a theorem yet. A definition that behaves well only for familiar examples is not mature yet.

This is why the empty set, zero, identity functions, constant functions, and weird sequences appear so often. They are not annoyances. They are test fixtures.

9.19 Software analogy: unit tests and property tests

An example is like a unit test:

$$f(2) = 4.$$

It checks one case.

A theorem is like a property test:

$$\forall x,\ f(x) \geq 0.$$

It describes a behavior over a whole domain.

A counterexample is a failing test case with a stack trace. It tells you not merely that something failed, but where:

- domain issue,
- missing hypothesis,
- wrong direction,
- too-strong conclusion,
- hidden edge case.

Good mathematicians keep a mental test suite. When they see a proposed theorem about functions, they test constant functions, identity functions, squaring, absolute value, step functions, and empty-domain functions. When they see a theorem about sets, they test empty sets, equal sets, disjoint sets, and nested sets.

9.20 Non-examples in analysis

Definition:

> A sequence (a_n) converges to L if for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $n \geq N$, $|a_n - L| < \varepsilon$.

Example:

$$a_n = \frac{1}{n}$$

converges to 0.

Non-example:

$$a_n = (-1)^n$$

does not converge. It keeps jumping between 1 and -1.

Near miss:

$$a_n = \frac{(-1)^n}{n}$$

does converge to 0. It alternates, but the jumps shrink.

The near miss teaches that oscillation alone does not prevent convergence. Persistent oscillation does.

9.21 How to present a counterexample

A counterexample should be written as a tiny proof.

Bad:

> Counterexample: -3.

Better:

> The claim "if $x^2 > 4$, then $x > 2$" is false. Take $x = -3$. Then $x^2 = 9 > 4$, so the hypothesis is true, but $x > 2$ is false. Therefore the implication is false.

This format removes doubt. It verifies membership in the domain, checks the hypothesis, and checks the failure of the conclusion.

Formal

Counterexample template:

> Take [object]. It belongs to [domain]. It satisfies [hypothesis] because [reason]. It fails [conclusion] because [reason]. Therefore the statement is false.

9.22 Examples are not random

At first, finding examples may feel like guessing. Over time, it becomes structured.

To find an injective but not surjective function, you want no merging but some missing codomain element. So shift the natural numbers:

$$n \mapsto n + 1.$$

To find a surjective but not injective function, you want full coverage but merging. So collapse a larger domain onto a smaller codomain:

$$\{1, 2\} \to \{a\}.$$

To find a continuous function that is not differentiable, you want no jump but a sharp corner:

$$f(x) = |x|.$$

To find a bounded sequence that does not converge, you want it trapped but not settling:

$$a_n = (-1)^n.$$

To find two different sets B, C with $A \cup B = A \cup C$, make A large enough to hide the difference.

This is design, not guessing.

10 How Mathematicians Choose a Proof Strategy

Beginners often ask, "How was I supposed to know to try that?" That is one of the right questions. Proof discovery can look magical when books only show the finished proof. In practice, mathematicians try strategies because the logical shape of the claim suggests them.

Reading note

⇝ Strategy fork. When you see this mark, stop reading for a moment and ask which proof path the target suggests.

10.1 Read the target first

The conclusion usually tells you the outer proof shape.

Target shape	Natural strategy	First move
$\forall x \in A,\ P(x)$	arbitrary object	Let $x \in A$ be arbitrary
$\exists x \in A,\ P(x)$	construction	Choose or construct a witness
$P \Rightarrow Q$	direct or contrapositive	Suppose P; or suppose $\neg Q$
$P \Leftrightarrow Q$	two directions	Prove $P \Rightarrow Q$ and $Q \Rightarrow P$
$A = B$ for sets	two inclusions	Prove $A \subseteq B$ and $B \subseteq A$
"unique"	two-candidate argument	Suppose two objects satisfy the property
Statement over $\mathbb{N}$	induction	Prove base case and step
Nonexistence	contradiction or universal negation	Assume existence and derive conflict

10.2 When contradiction is useful

Contradiction is useful when denying the conclusion gives concrete data. For example, to prove there is no smallest positive real number, assuming there is one gives an object $r > 0$ with a strong property. You can then construct $r/2$, which violates that property.

Contradiction is also useful when the conclusion is negative:

$$\text{There is no object with property } P.$$

Assume such an object exists, then force an impossibility.

Common trap

False intuition: contradiction is the strongest proof method, so use it whenever stuck. Better intuition: contradiction is useful when the negation of the target gives usable structure. If direct proof gives a clean path, use it.

10.3 How experts discover proofs

Experts often work backward in scratch:

1. What do I need at the end?
2. Which definition would close the target?
3. What data would be enough to close that definition?
4. Do the hypotheses provide that data?
5. If not, can I prove a bridge lemma?

Then they write the final proof forward.

Proof lab

Discovery drill. For each claim, write backward scratch work before writing the proof.

1. If $a \mid b$ and $a \mid c$, then $a \mid (3b - 5c)$.
2. If f and g are injective, then $g \circ f$ is injective.
3. If $U, V \subseteq \mathbb{R}$ are open, then $U \cap V$ is open.
4. If $a_n \to 0$, then $|a_n| \to 0$.

10.4 Intermediate states

Many proofs become easier when you name the missing middle. For composition of injective functions, the final target is $a_1 = a_2$. The assumption gives $g(f(a_1)) = g(f(a_2))$. The missing middle is:

$$f(a_1) = f(a_2).$$

Once you see that intermediate state, the proof becomes a two-step chain.

Focus

When stuck, ask: what statement would put me halfway between the assumptions and the target?

10.5 A Recovery Page: Tiny Proofs, No Philosophy

After a dense strategy discussion, return to small proofs. They are the scales and arpeggios of proof writing.

Exercise 1. Prove that if n is even, then $n + 2$ is even.

Proof. Suppose n is even. Then $n = 2k$ for some $k \in \mathbb{Z}$. Thus

$$n + 2 = 2k + 2 = 2(k + 1).$$

Since $k + 1 \in \mathbb{Z}$, $n + 2$ is even. □

Exercise 2. Prove that if $A \subseteq B$ and $x \in A$, then $x \in B$.

Proof. This is exactly the definition of subset. Since $A \subseteq B$, every element of A is an element of B. Since $x \in A$, it follows that $x \in B$. □

Pause

When the book gets conceptually heavy, do two tiny proofs. Small correct moves restore trust in the machinery.

10.6 Walking questions

Walking question

For every new definition, build the following list:

1. one tiny example,
2. one large or infinite example,
3. one non-example,
4. one near miss,
5. one boundary case,
6. one theorem where the definition is useful,
7. one false statement the definition helps refute.

10.7 Exercises

1. Give three examples and three non-examples of even integers. For each non-example, explain why the definition fails.

2. Give an example of two irrational numbers whose sum is rational.
3. Give an example of a rational number and an irrational number whose sum is irrational.
4. Find a counterexample: if $x^2 = y^2$, then $x = y$.
5. Find a counterexample: if $x < y$, then $x^2 < y^2$.
6. Find a counterexample: every function $\mathbb{R} \to \mathbb{R}$ is injective.
7. Find a counterexample: every injective function $\mathbb{N} \to \mathbb{N}$ is surjective.
8. Find a counterexample: every surjective function is injective.
9. Find a counterexample: every bounded sequence converges.
10. Find a counterexample: every continuous function is differentiable.
11. Give a near miss for the definition of prime number and explain exactly what fails.
12. Give a near miss for linear independence in $\mathbb{R}^2$ or $\mathbb{R}^3$.
13. For the theorem "a continuous function on $[a, b]$ attains a maximum," give one counterexample for each missing hypothesis: continuity, closed interval, bounded interval.
14. For each false statement above, rewrite it into a nearby true statement.
15. Choose a definition from a later course you care about and build a mini zoo: examples, non-examples, counterexamples, and boundary cases.

11 Sets

11.1 Why sets are the grammar of modern mathematics

A set is one of the first ideas in rigorous mathematics that feels both too simple and too slippery. Too simple, because a set is just a collection of objects. Too slippery, because that sentence hides nearly everything that later mathematics needs: membership, equality, containment, construction, size, structure, and the discipline of saying exactly what kind of object we are talking about.

When we say

$$A = \{1, 2, 3\},$$

we are not naming three numbers separately. We are naming one object whose members are the numbers 1, 2, and 3. This shift is small but powerful. A set lets us treat many things as one thing.

That is already familiar from software. A list, array, database table, hash set, type, interface, and collection all let us talk about many possible values as a single object. But mathematical sets are not arrays. A set has no order and no repeated entries. The set $\{1, 2, 3\}$ is the same set as $\{3, 2, 1\}$ and as $\{1, 1, 2, 3, 3\}$.

Focus

A set is determined by its members, not by how those members are written down.

The membership symbol

$$x \in A$$

means that x is an element of A. Its negation

$$x \notin A$$

means that x is not an element of A.

The statement $2 \in \{1, 2, 3\}$ is true. The statement $4 \in \{1, 2, 3\}$ is false. The statement $\{2\} \in \{1, 2, 3\}$ is false, because $\{2\}$ is a set, not the number 2.

Common trap

Do not confuse an object with a set containing that object. Usually $2 \neq \{2\}$. The first is a number. The second is a set whose only member is that number.

Sets become essential because most later definitions begin by saying what kind of objects are allowed. A function has a domain and codomain, both sets. A sequence is a function whose domain is often $\mathbb{N}$. A metric space is a set equipped with a distance function. A group is a set equipped with an operation. A topological space is a set equipped with a collection of open subsets. A vector space is a set equipped with addition and scalar multiplication.

That pattern is worth noticing:

$$\text{mathematical structure} = \text{set} + \text{extra data} + \text{rules}.$$

Before we can understand the extra data, we need to understand the set underneath it.

11.2 Membership, equality, and extensionality

The most basic relation between an object and a set is membership. The most basic relation between two sets is equality.

Formal

Two sets A and B are equal if and only if they have exactly the same elements:

$$A = B \quad \iff \quad \forall x,\ (x \in A \iff x \in B).$$

This principle is called extensionality. It says that a set is determined by its extension, meaning the objects that belong to it.

For example,

$$\{n \in \mathbb{Z} : n^2 = 1\} = \{-1, 1\}.$$

The left side describes the set by a property. The right side lists its elements. They are the same set because they contain the same objects.

This is also why order and repetition do not matter:

$$\{a, b, c\} = \{c, a, b\} = \{a, a, b, c, c\}.$$

All three have the same members.

Intuition

A set is like a yes-or-no test for membership. For each object x under discussion, the set A answers: yes, $x \in A$, or no, $x \notin A$.

That membership-test perspective will later connect directly to logic and software. A subset of a universe U can be represented by a Boolean-valued predicate on U. If $A \subseteq U$, then A corresponds to the predicate

$$P_A(x) : x \in A.$$

In a program, this might be a function returning true or false. In mathematics, it is a property that determines membership.

The phrase "under discussion" matters. In ordinary set theory, membership is not restricted to one universal type. But in actual proofs, we almost always work inside a domain: integers, real numbers, functions, matrices, points in a space, ideals in a ring. Your proof becomes clearer when you keep that ambient universe visible.

Walking question

Pick a set you know, such as $\mathbb{Z}$, $\mathbb{Q}$, or the set of continuous functions $\mathbb{R} \to \mathbb{R}$. Ask: what counts as an element of this set, and what would be a thing that looks related but is not an element?

11.3 Roster notation and set-builder notation

There are two common ways to specify sets. Roster notation lists members:

$$A = \{2, 4, 6, 8\}.$$

Set-builder notation describes members by a condition:

$$B = \{n \in \mathbb{Z} : n \text{ is even and } 2 \leq n \leq 8\}.$$

Here $A = B$.

Set-builder notation has the general shape

$$\{x \in D : P(x)\},$$

read as "the set of all x in D such that $P(x)$."

The domain D is not decoration. Compare:

$$\{x \in \mathbb{R} : x^2 < 2\}$$

with

$$\{x \in \mathbb{Z} : x^2 < 2\}.$$

The first is the interval $(-\sqrt{2}, \sqrt{2})$. The second is $\{-1, 0, 1\}$. The formula $x^2 < 2$ did not change, but the set changed because the allowed inputs changed.

Common trap

The expression $\{x : x^2 < 2\}$ is often understandable from context, but it is dangerous when no universe is clear. In early proof writing, include the domain.

Set-builder notation also trains you to read quantifiers. The statement

$$y \in \{x \in D : P(x)\}$$

means exactly:

$$y \in D \text{ and } P(y).$$

Nothing mystical happens inside the braces. The bound variable x is local. We could write

$$\{t \in D : P(t)\}$$

and get the same set.

Examples:

$$\{n \in \mathbb{N} : n \leq 5\} = \{1, 2, 3, 4, 5\}$$

if $\mathbb{N} = \{1, 2, 3, \ldots\}$.

$$\{x \in \mathbb{R} : x^2 + 1 = 0\} = \varnothing.$$

$$\{z \in \mathbb{C} : z^2 + 1 = 0\} = \{i, -i\}.$$

The last two examples are a useful warning. A set can be empty in one universe and nonempty in another. Algebra, analysis, and geometry often differ precisely because they change the universe.

11.4 The empty set and singleton sets

The empty set is the set with no elements. It is written

$$\varnothing.$$

It is not nothing. It is a set. It is the unique set satisfying

$$\forall x,\ x \notin \varnothing.$$

This can feel pedantic, but the empty set behaves like zero in arithmetic. It is the identity for union:

$$A \cup \varnothing = A.$$

It is the absorbing element for intersection:

$$A \cap \varnothing = \varnothing.$$

A singleton is a set with exactly one element:

$$\{a\}.$$

Again, a and $\{a\}$ are different kinds of objects. This distinction becomes unavoidable when sets contain sets.

For example, let

$$A = \varnothing, \qquad B = \{\varnothing\}.$$

Then $A \neq B$. The set A has no elements. The set B has one element, namely $\varnothing$.

We can keep going:

$$\varnothing, \quad \{\varnothing\}, \quad \{\{\varnothing\}\}, \quad \{\varnothing, \{\varnothing\}\}.$$

These are all different sets. This kind of construction is not merely playful. In one standard foundation of mathematics, natural numbers can be built from the empty set:

$$0 = \varnothing, \quad 1 = \{\varnothing\}, \quad 2 = \{\varnothing, \{\varnothing\}\}, \quad \dots$$

You do not need that construction for ordinary proof work, but it shows how much can be encoded using only sets.

History and connection

Georg Cantor developed set theory in the late nineteenth century while studying questions about trigonometric series and infinite sets of real numbers. What began as a tool for analysis became a foundation for much of modern mathematics. The empty set, singleton sets, and sets of sets are part of that foundational language.

11.5 Subsets and containment

Membership relates an object to a set. Containment relates a set to a set.

Formal

Let A and B be sets. We say A is a subset of B, written $A \subseteq B$, if every element of A is an element of B:

$$A \subseteq B \quad \iff \quad \forall x,\ (x \in A \Rightarrow x \in B).$$

For example,

$$\{1, 3\} \subseteq \{1, 2, 3, 4\}.$$

Also,

$$\mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}.$$

The symbol $\subseteq$ allows equality. Thus $A \subseteq A$ for every set A.

Some authors use $A \subset B$ to mean "proper subset," while others use it the same way as $\subseteq$. To avoid ambiguity, this reader uses:

$$A \subseteq B$$

for subset, and

$$A \subsetneq B$$

for proper subset.

A proper subset satisfies

$$A \subseteq B \quad \text{and} \quad A \neq B.$$

Equivalently, every element of A lies in B, and at least one element of B does not lie in A.

$$A \subsetneq B \quad \Longleftrightarrow \quad A \subseteq B \text{ and } \exists b \in B \text{ such that } b \notin A.$$

Common trap

The statement $a \in B$ and the statement $\{a\} \subseteq B$ are closely related but not identical in grammar. The first says an object belongs to B. The second says a one-element set is contained in B. They are equivalent when $\{a\}$ is the singleton containing a, but they are not the same sentence.

Subset proofs are the first place many students learn the standard proof pattern:

> To prove $A \subseteq B$, let $x \in A$ be arbitrary. Then show $x \in B$.

That sentence is worth memorizing, not as a trick, but as a direct unpacking of the definition.

11.6 Power sets and sets of sets

The power set of a set A, written $\mathcal{P}(A)$, is the set of all subsets of A:

$$\mathcal{P}(A) = \{S : S \subseteq A\}.$$

If

$$A = \{1, 2, 3\},$$

then

$$\mathcal{P}(A) = \{\varnothing, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}.$$

Notice the levels. The elements of $\mathcal{P}(A)$ are themselves sets. For example:

$$\{1, 2\} \in \mathcal{P}(A),$$

but

$$1 \notin \mathcal{P}(A)$$

because 1 is not a subset of A.

This level discipline matters later. A topology on a set X is a collection of subsets of X. That means a topology is a subset of $\mathcal{P}(X)$:

$$\mathcal{T} \subseteq \mathcal{P}(X).$$

The elements of $\mathcal{T}$ are open sets, and the elements of those open sets are points of X. Losing track of those levels causes confusion in topology.

In algebra, a quotient group is built from a set of equivalence classes. Each equivalence class is a subset of the original group. So again, we handle sets whose elements are sets.

Intuition

If A is a drawer of objects, then $\mathcal{P}(A)$ is the catalog of every possible subdrawer you could make from objects in A.

For finite sets, power sets have a clean size formula. If A has n elements, then $\mathcal{P}(A)$ has 2^n elements. Each element of A gives a yes-or-no choice: include it in a subset or do not include it.

This is exactly the same count as bit strings of length n. A subset of $\{1, \ldots, n\}$ can be encoded by a binary string: 1 means present, 0 means absent. This is why power sets appear naturally in combinatorics and computer science.

11.7 Union, intersection, difference, and complement

The basic operations on sets mirror logical connectives.

Formal

For sets A and B:

$$A \cup B = \{x : x \in A \text{ or } x \in B\},$$

$$A \cap B = \{x : x \in A \text{ and } x \in B\},$$

$$A \setminus B = \{x : x \in A \text{ and } x \notin B\}.$$

If a universe U is fixed, the complement of A in U is

$$A^c = U \setminus A.$$

The word "or" in mathematics is inclusive unless stated otherwise. Thus $x \in A \cup B$ if x is in A, in B, or in both.

Example:

$$A = \{1, 2, 3\}, \qquad B = \{3, 4, 5\}.$$

Then

$$A \cup B = \{1, 2, 3, 4, 5\},$$
$$A \cap B = \{3\},$$
$$A \setminus B = \{1, 2\},$$
$$B \setminus A = \{4, 5\}.$$

Set difference is not symmetric. Usually

$$A \setminus B \neq B \setminus A.$$

Complement depends on the universe. If the universe is $\mathbb{R}$ and $A = [0, 1]$, then

$$A^c = (-\infty, 0) \cup (1, \infty).$$

If the universe is $\mathbb{Z}$ and $A = \{0, 1\}$, then

$$A^c = \mathbb{Z} \setminus \{0, 1\}.$$

Common trap

Never write A^c unless the ambient universe is clear. The complement of a set is not absolute; it is relative to where you are working.

The set operations are logic in disguise:

$$x \in A \cap B \iff (x \in A) \wedge (x \in B),$$

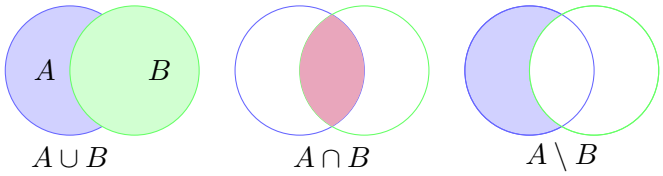
$$x \in A \cup B \iff (x \in A) \vee (x \in B),$$

$$x \in A \setminus B \iff (x \in A) \wedge \neg(x \in B).$$

This is the main reason set proofs and logic proofs feel so similar. They are similar.

11.8 Visual Layer: Set Operations

Pictures do not replace proofs, but they train the eye. A set diagram is a way to see which membership conditions are being combined.



The formal proof still comes from membership:

$$x \in A \cap B \quad \text{means} \quad x \in A \text{ and } x \in B.$$

The diagram helps you guess the identity. The element proof verifies it for arbitrary sets, not just friendly circles on a page.

Common trap

False intuition: Venn diagrams prove all set identities. They are excellent intuition for a few sets, but they can hide details about empty sets, indexed families, ambient universes, and arbitrary collections. Use diagrams to see; use definitions to prove.

11.9 Intervals and common number sets

Many early examples use subsets of $\mathbb{R}$. The standard intervals are:

$$(a, b) = \{x \in \mathbb{R} : a < x < b\},$$

$$[a, b] = \{x \in \mathbb{R} : a \leq x \leq b\},$$

$$(a, b] = \{x \in \mathbb{R} : a < x \leq b\},$$

$$[a, b) = \{x \in \mathbb{R} : a \leq x < b\}.$$

Infinite intervals work similarly:

$$(a, \infty) = \{x \in \mathbb{R} : x > a\},$$

$$(-\infty, b] = \{x \in \mathbb{R} : x \leq b\}.$$

The symbols ∞ and $-\infty$ are not real numbers here. They are notation indicating that the interval has no finite endpoint in that direction.

Some useful number sets:

$$\mathbb{N} = \text{natural numbers},$$

$$\mathbb{Z} = \text{integers},$$

$$\mathbb{Q} = \text{rational numbers},$$

$$\mathbb{R} = \text{real numbers},$$

$$\mathbb{C} = \text{complex numbers}.$$

Be alert: some authors include 0 in $\mathbb{N}$, and some do not. If a proof depends on this, say which convention you are using.

Set language makes familiar facts precise:

$$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}.$$

Also,

$$[0,1] \cap [1,2] = \{1\},$$

while

$$(0,1) \cap (1,2) = \varnothing.$$

History and connection

Intervals are the bridge from elementary set language to analysis and topology. In calculus, intervals are where functions are continuous or differentiable. In topology, intervals generate the usual open sets of the real line. In measure theory, intervals are the first pieces from which length and integration are built.

11.10 Indexed unions and intersections

Union and intersection are not limited to two sets. If $\{A_i : i \in I\}$ is a family of sets indexed by a set I, then

$$\bigcup_{i \in I} A_i = \{x : \exists i \in I \text{ such that } x \in A_i\},$$

and

$$\bigcap_{i \in I} A_i = \{x : \forall i \in I,\ x \in A_i\}.$$

The quantifiers are the point. Union uses an existential quantifier. Intersection uses a universal quantifier.

Example:

$$\bigcup_{n \in \mathbb{N}} \left(0, 1 - \frac{1}{n}\right) = (0,1)$$

if $\mathbb{N} = \{1, 2, 3, \ldots\}$. The interval for $n = 1$ is empty, for $n = 2$ is $(0, 1/2)$, for $n = 3$ is $(0, 2/3)$, and so on. Every real number between 0 and 1 eventually appears in one of these intervals.

By contrast,

$$\bigcap_{n \in \mathbb{N}} \left(0, 1 + \frac{1}{n}\right) = (0,1].$$

Every number in $(0, 1]$ lies in all the intervals. A number larger than 1 eventually gets excluded.

Here is a discrete example. Let

$$A_n = \{k \in \mathbb{N} : k \geq n\}.$$

Then

$$\bigcup_{n\in\mathbb{N}} A_n = \mathbb{N},$$

because every natural number is at least 1. But

$$\bigcap_{n\in\mathbb{N}} A_n = \varnothing,$$

because no natural number is at least every natural number.

Common trap

When reading indexed unions and intersections, translate them into quantifiers. Most mistakes come from treating "some index" and "every index" as if they were the same.

11.11 Cartesian products and ordered pairs

Sets can be combined by forming ordered pairs. The Cartesian product of A and B is

$$A \times B = \{(a, b) : a \in A \text{ and } b \in B\}.$$

If

$$A = \{1, 2\}, \qquad B = \{x, y, z\},$$

then

$$A \times B = \{(1, x), (1, y), (1, z), (2, x), (2, y), (2, z)\}.$$

Order matters:

$$(1, x) \neq (x, 1)$$

unless the two coordinates happen to be equal in a context where that comparison even makes sense. Also,

$$A \times B$$

is generally not equal to

$$B \times A.$$

The plane $\mathbb{R}^2$ is the Cartesian product

$$\mathbb{R} \times \mathbb{R}.$$

The space $\mathbb{R}^3$ is

$$\mathbb{R} \times \mathbb{R} \times \mathbb{R}.$$

Cartesian products are where set theory starts to feel geometric. A subset of $\mathbb{R}^2$ is a set of points in the plane. A graph of a function is a subset of a Cartesian product. A relation from A to B is a subset of $A \times B$.

This viewpoint will matter when we define functions. A function can be understood as a special kind of relation, hence as a special subset of $A \times B$.

Intuition

The product $A \times B$ is the set of all possible input-output pairs if the first coordinate must come from A and the second from B.

11.12 Partitions and equivalence classes

A partition of a set A is a way of breaking A into nonempty, nonoverlapping pieces whose union is all of A.

Formal

A collection $\mathcal{P}$ of subsets of A is a partition of A if:

$$\forall P \in \mathcal{P},\ P \neq \varnothing,$$

$$\forall P, Q \in \mathcal{P},\ P \neq Q \Rightarrow P \cap Q = \varnothing,$$

and

$$\bigcup_{P \in \mathcal{P}} P = A.$$

Example: the even and odd integers form a partition of $\mathbb{Z}$:

$$E = \{n \in \mathbb{Z} : n \text{ is even}\},$$

$$O = \{n \in \mathbb{Z} : n \text{ is odd}\},$$

and

$$\{E, O\}$$

is a partition of $\mathbb{Z}$.

Partitions connect to equivalence relations. If we define $a \sim b$ to mean that a and b have the same remainder when divided by 3, then the equivalence classes are

$$3\mathbb{Z}, \quad 3\mathbb{Z} + 1, \quad 3\mathbb{Z} + 2.$$

These three sets partition $\mathbb{Z}$.

This is the first glimpse of quotient constructions. In algebra, quotient groups and quotient rings collect elements into equivalence classes. In topology, quotient spaces glue points together. In software, partitioning appears in hashing, bucketing, grouping records by a key, and classifying states by an invariant.

History and connection

Cantor's work on sets did not only create a language for collections; it made possible a more flexible way of identifying when two objects should count as the same for a given purpose. Equivalence classes and quotient structures are now central in algebra, topology, geometry, and theoretical computer science.

11.13 A first look at size

For finite sets, size is counted by the number of elements. We write $|A|$ for the cardinality of A. If

$$A = \{a, b, c\},$$

then

$$|A| = 3.$$

For finite sets:

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

The subtraction prevents double-counting the overlap.

If A and B are disjoint, meaning $A \cap B = \varnothing$, then

$$|A \cup B| = |A| + |B|.$$

For Cartesian products:

$$|A \times B| = |A||B|$$

when A and B are finite.

Infinite sets require a deeper idea. Cantor's insight was that two sets have the same size if there is a bijection between them. This means the natural numbers and the even natural numbers have the same infinite size, because

$$f : \mathbb{N} \to 2\mathbb{N}, \qquad f(n) = 2n$$

matches every natural number with exactly one even natural number and misses no even natural number.

At first that seems impossible. The evens are a proper subset of the naturals. But infinite size does not behave like finite size. This is one of the reasons functions matter so much: they let us compare sets by pairing elements.

Focus

The deep bridge from sets to functions is this: to understand a set, we often study the maps into it, out of it, and between it and other sets.

11.14 Exercises for sets

Exercise 3. For each pair, decide whether the objects are equal.

$$\{1,2,3\} \quad \text{and} \quad \{3,2,1,1\},$$

$$\{1\} \quad \text{and} \quad 1,$$

$$\varnothing \quad \text{and} \quad \{\varnothing\},$$

$$\{x \in \mathbb{Z} : x^2 = 4\} \quad \text{and} \quad \{-2,2\}.$$

Exercise 4. Let $A = \{1,2,3,4\}$ and $B = \{3,4,5\}$. Compute $A \cup B$, $A \cap B$, $A \setminus B$, $B \setminus A$, and $\mathcal{P}(\{1,2\})$.

Exercise 5. Write each set in set-builder notation:

$$\{0,2,4,6,8,\ldots\},$$

$$(-3,5],$$

$$\{1,4,9,16,25,\ldots\}.$$

State the domain in each case.

Exercise 6. Let $A_n = (0, 1/n)$ for $n \in \mathbb{N}$. Find

$$\bigcup_{n\in\mathbb{N}} A_n \quad \text{and} \quad \bigcap_{n\in\mathbb{N}} A_n.$$

Explain your answer using quantifiers.

Walking question

As you move through the day, look for examples of sets: possible passwords, open browser tabs, files in a directory, accepted inputs to a function, rows returned by a query. For each one, ask what membership means.

12 Set Proofs

12.1 The basic method: element chasing

Most elementary set proofs are element-chasing proofs. This means we prove statements about sets by taking an arbitrary element and following what membership means.

To prove

$$A \subseteq B,$$

we prove:

$$\forall x,\ x \in A \Rightarrow x \in B.$$

The proof pattern is:

Let $x \in A$ be arbitrary. We will show $x \in B$.

To prove

$$A = B,$$

we usually prove both containments:

$$A \subseteq B \quad \text{and} \quad B \subseteq A.$$

The proof pattern is:

First let $x \in A$ and show $x \in B$. Then let $x \in B$ and show $x \in A$.

Focus

Set equality is usually proved by two subset proofs.

This is not merely a recipe. It comes from the definition:

$$A = B \iff \forall x,\ (x \in A \iff x \in B).$$

A biconditional can be proved by proving both directions.

12.2 Worked proof: intersection is contained in each set

Theorem 24. *For all sets A and B,*

$$A \cap B \subseteq A$$

and

$$A \cap B \subseteq B.$$

Proof. We prove $A \cap B \subseteq A$. Let $x \in A \cap B$ be arbitrary. By the definition of intersection, $x \in A$ and $x \in B$. In particular, $x \in A$. Therefore $A \cap B \subseteq A$.

The proof that $A \cap B \subseteq B$ is similar. Let $x \in A \cap B$ be arbitrary. Then $x \in A$ and $x \in B$, so in particular $x \in B$. Hence $A \cap B \subseteq B$. □

This proof is short because the theorem is mostly a direct unpacking of definitions. Many early proofs are like this. Do not be disappointed by that. You are learning how definitions move.

Intuition

If x is in the overlap of A and B, then x is certainly in A and certainly in B.

The formal proof is the disciplined version of that sentence.

12.3 Worked proof: distributivity

Set operations distribute over each other in ways that mirror logic.

Theorem 25. *For all sets A, B, and C,*

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C).$$

Proof. We prove equality by proving both containments.

First, let $x \in A \cap (B \cup C)$. Then $x \in A$ and $x \in B \cup C$. Since $x \in B \cup C$, we know $x \in B$ or $x \in C$.

If $x \in B$, then $x \in A$ and $x \in B$, so $x \in A \cap B$. Therefore $x \in (A \cap B) \cup (A \cap C)$.

If $x \in C$, then $x \in A$ and $x \in C$, so $x \in A \cap C$. Therefore $x \in (A \cap B) \cup (A \cap C)$.

In either case, $x \in (A \cap B) \cup (A \cap C)$. Hence

$$A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C).$$

Conversely, let $x \in (A \cap B) \cup (A \cap C)$. Then $x \in A \cap B$ or $x \in A \cap C$.

If $x \in A \cap B$, then $x \in A$ and $x \in B$. Since $x \in B$, we have $x \in B \cup C$. Thus $x \in A \cap (B \cup C)$.

If $x \in A \cap C$, then $x \in A$ and $x \in C$. Since $x \in C$, we have $x \in B \cup C$. Thus $x \in A \cap (B \cup C)$.

In either case, $x \in A \cap (B \cup C)$. Hence

$$(A \cap B) \cup (A \cap C) \subseteq A \cap (B \cup C).$$

Therefore

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C).$$

□

Read the proof again and notice the structure. Every step is membership translation:

$$x \in A \cap (B \cup C)$$

becomes

$$x \in A \quad \text{and} \quad (x \in B \text{ or } x \in C).$$

That is logical distributivity:

$$P \wedge (Q \vee R) \iff (P \wedge Q) \vee (P \wedge R).$$

12.4 De Morgan's laws

De Morgan's laws explain how complement interacts with union and intersection. They are among the most useful set identities in analysis, topology, probability, and logic.

Theorem 26. *Let A and B be subsets of a fixed universe U. Then*

$$(A \cup B)^c = A^c \cap B^c$$

and

$$(A \cap B)^c = A^c \cup B^c.$$

Proof. We prove the first identity. Let $x \in U$. Then

$$x \in (A \cup B)^c$$

if and only if

$$x \notin A \cup B.$$

This holds if and only if it is not true that $x \in A$ or $x \in B$. By De Morgan's law for logic, this is equivalent to

$$x \notin A \text{ and } x \notin B.$$

That is equivalent to

$$x \in A^c \text{ and } x \in B^c,$$

which is equivalent to

$$x \in A^c \cap B^c.$$

Therefore $(A \cup B)^c = A^c \cap B^c$.

The proof of the second identity is similar, using the logical equivalence

$$\neg(P \wedge Q) \iff (\neg P) \vee (\neg Q).$$

□

This proof used a chain of biconditionals rather than separate containment proofs. That is also legitimate. If for every x in the universe we show

$$x \in S \iff x \in T,$$

then $S = T$.

Common trap

Do not write $(A \cup B)^c = A^c \cup B^c$. Complement flips union into intersection and intersection into union.

12.5 Proofs involving the empty set

The empty set often feels strange because universal statements about it are automatically true. The statement

$$\forall x \in \varnothing,\ P(x)$$

is true, because there is no $x \in \varnothing$ that could violate P.

This is called vacuous truth. It is not a trick. The implication

$$x \in \varnothing \Rightarrow P(x)$$

is true for every x, because the premise is always false.

Theorem 27. *For every set A,*

$$\varnothing \subseteq A.$$

Proof. To show $\varnothing \subseteq A$, we must show that every element of $\varnothing$ is an element of A.

Let $x \in \varnothing$ be arbitrary. But there is no such x. Therefore there is no counterexample to the statement that every element of $\varnothing$ lies in A. Hence $\varnothing \subseteq A$. □

Some readers dislike the sentence "Let $x \in \varnothing$" because no such x exists. A more formal version avoids that discomfort:

Proof. For every object x, the implication

$$x \in \varnothing \Rightarrow x \in A$$

is true, since $x \in \varnothing$ is false. Therefore $\varnothing \subseteq A$. □

Intuition

To disprove $\varnothing \subseteq A$, you would need to find an element of $\varnothing$ that is not in A. There are no elements of $\varnothing$ to find.

12.6 How to disprove a set statement

To disprove a universal set identity, find a counterexample. This can mean finding specific sets for which the proposed identity fails, or finding an element that belongs to one side but not the other.

False claim:

$$A \cup B = A \cap B$$

for all sets A and B.

Counterexample: let

$$A = \{1\}, \qquad B = \{2\}.$$

Then

$$A \cup B = \{1, 2\}$$

while

$$A \cap B = \varnothing.$$

So the equality fails.

False claim:

$$A \setminus (B \cup C) = (A \setminus B) \cup (A \setminus C)$$

for all sets A, B, and C.

Take

$$A = \{1\}, \qquad B = \{1\}, \qquad C = \varnothing.$$

Then

$$A \setminus (B \cup C) = \varnothing,$$

but

$$(A \setminus B) \cup (A \setminus C) = \varnothing \cup \{1\} = \{1\}.$$

The statement is false.

The correct identity is

$$A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C).$$

This is De Morgan's law in another form.

Common trap

When a statement says "for all sets," one small concrete counterexample is enough to disprove it. You do not need to explain all possible failures.

12.7 Common algebra of sets

Many set identities resemble algebraic identities:

$$A \cup B = B \cup A,$$

$$A \cap B = B \cap A,$$

$$A \cup (B \cup C) = (A \cup B) \cup C,$$

$$A \cap (B \cap C) = (A \cap B) \cap C,$$

$$A \cup A = A,$$

$$A \cap A = A,$$

$$A \cup \varnothing = A,$$

$$A \cap \varnothing = \varnothing.$$

These can all be proved by element chasing. For example:

Theorem 28. *For every set A,*

$$A \cup A = A.$$

Proof. Let x be arbitrary. Then

$$x \in A \cup A$$

if and only if

$$x \in A \text{ or } x \in A,$$

which is equivalent to $x \in A$. Therefore $A \cup A = A$. □

The corresponding logical law is

$$P \vee P \iff P.$$

There is a useful analogy with Boolean algebra. If we think of a set A as a truth-valued function on a universe U, then union corresponds to OR, intersection to AND, and complement to NOT. This is not just analogy. It is a precise algebraic connection.

History and connection

Boolean algebra, named after George Boole, gave algebraic form to logical operations. Set operations became a natural model of Boolean algebra. This connection is now everywhere: digital circuits, database queries, type systems, probability events, sigma-algebras, and topology all use the same underlying pattern.

12.8 Proof lab: a two-containment proof

Proof lab

Prove that for all sets A, B, and C,

$$A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C).$$

Before writing the proof, translate membership in each side:

$$x \in A \setminus (B \cap C)$$

means

$$x \in A \quad \text{and} \quad x \notin B \cap C.$$

The second part means

$$\text{not } (x \in B \text{ and } x \in C),$$

which means

$$x \notin B \quad \text{or} \quad x \notin C.$$

Now distribute:

$$x \in A \text{ and } (x \notin B \text{ or } x \notin C).$$

This is equivalent to

$$(x \in A \text{ and } x \notin B) \text{ or } (x \in A \text{ and } x \notin C).$$

That is exactly membership in

$$(A \setminus B) \cup (A \setminus C).$$

A polished proof could use the above chain directly. A slower proof can split into cases. Both are acceptable. The important habit is to translate the set notation into membership logic.

12.9 Proof lab: proving a subset of a union

Proof lab

Prove:

$$A \subseteq A \cup B.$$

Proof skeleton:
Let $x \in A$ be arbitrary. Since $x \in A$, it follows that $x \in A$ or $x \in B$. Therefore $x \in A \cup B$. Hence $A \subseteq A \cup B$.

This proof may look almost too easy, because the move from $x \in A$ to $x \in A$ or $x \in B$ is a

basic logical move:

$$P \Rightarrow (P \vee Q).$$

You are allowed to introduce a true OR statement when you already know one side is true.

This little proof appears constantly in larger arguments. For example, in topology, if U is open and V is any open set, then $U \subseteq U \cup V$. In probability, if event A occurs, then event $A \cup B$ occurs. In program analysis, if a state satisfies condition A, then it satisfies the broader condition A OR B.

12.10 Proof lab: proving equality of power sets

Theorem 29. *For all sets A and B,*

$$A \subseteq B \quad \Longleftrightarrow \quad \mathcal{P}(A) \subseteq \mathcal{P}(B).$$

Proof. First assume $A \subseteq B$. We will show $\mathcal{P}(A) \subseteq \mathcal{P}(B)$. Let $S \in \mathcal{P}(A)$ be arbitrary. By definition of power set, $S \subseteq A$. Since $A \subseteq B$, every element of S is an element of B. Thus $S \subseteq B$, so $S \in \mathcal{P}(B)$. Therefore $\mathcal{P}(A) \subseteq \mathcal{P}(B)$.

Conversely, assume $\mathcal{P}(A) \subseteq \mathcal{P}(B)$. We will show $A \subseteq B$. Since $A \subseteq A$, we have $A \in \mathcal{P}(A)$. Because $\mathcal{P}(A) \subseteq \mathcal{P}(B)$, it follows that $A \in \mathcal{P}(B)$. By definition of power set, $A \subseteq B$. □

This proof is a good test of level discipline. In the first direction, S is an element of $\mathcal{P}(A)$, so S is a subset of A. In the second direction, A itself is an element of $\mathcal{P}(A)$.

Common trap

When working with $\mathcal{P}(A)$, remember that its elements are subsets of A. If $x \in A$, that does not mean $x \in \mathcal{P}(A)$. Usually it does not.

12.11 Set proofs with indexed families

Indexed unions and intersections require more care because their definitions contain quantifiers.

Theorem 30. *Let $\{A_i : i \in I\}$ be a family of sets, and let B be a set. Then*

$$B \cap \left(\bigcup_{i \in I} A_i \right) = \bigcup_{i \in I} (B \cap A_i).$$

Proof. Let x be arbitrary. Then

$$x \in B \cap \left(\bigcup_{i \in I} A_i \right)$$

if and only if

$$x \in B \quad \text{and} \quad x \in \bigcup_{i \in I} A_i.$$

This is equivalent to

$$x \in B \quad \text{and} \quad \exists i \in I \text{ such that } x \in A_i.$$

Since $x \in B$ does not depend on i, this is equivalent to

$$\exists i \in I \text{ such that } x \in B \text{ and } x \in A_i.$$

That is equivalent to

$$\exists i \in I \text{ such that } x \in B \cap A_i.$$

Therefore

$$x \in \bigcup_{i \in I} (B \cap A_i).$$

Since this holds for arbitrary x, the sets are equal. □

There is a subtle assumption hidden in many informal presentations of indexed intersections: what if the index set I is empty? Different contexts define empty intersections carefully, usually as the whole universe under discussion. For early proof work, most indexed families you meet will have nonempty index sets, but later mathematics cannot ignore the edge case.

Common trap

When proving a statement about $\bigcap_{i \in I} A_i$, ask whether I is known to be nonempty. Empty index sets can change the statement.

12.12 A proof using contradiction

Some set proofs are cleaner by contradiction.

Theorem 31. *If $A \subseteq B$ and $B \cap C = \varnothing$, then $A \cap C = \varnothing$.*

Proof. Suppose, for contradiction, that $A \cap C \neq \varnothing$. Then there exists $x \in A \cap C$. Hence $x \in A$ and $x \in C$.

Since $A \subseteq B$ and $x \in A$, we have $x \in B$. Therefore $x \in B$ and $x \in C$, so $x \in B \cap C$.

This contradicts $B \cap C = \varnothing$. Therefore $A \cap C = \varnothing$. □

A direct proof is also possible:

Proof. Let $x \in A \cap C$. Then $x \in A$ and $x \in C$. Since $A \subseteq B$, $x \in B$. Thus $x \in B \cap C$, impossible because $B \cap C = \varnothing$. Hence no such x exists, and $A \cap C = \varnothing$. □

The two proofs are nearly the same. The difference is framing. Contradiction begins by assuming the target fails. Direct proof may show that membership in the target set is impossible.

12.13 Common traps in set proofs

Common trap

Trap 1: Proving only one containment when equality is required.
If you prove $A \subseteq B$, you have not proved $A = B$. You also need $B \subseteq A$, unless you use a valid biconditional chain.

Common trap

Trap 2: Using a diagram as a proof.
Venn diagrams are useful for intuition and counterexample hunting, but they are not usually accepted as complete proofs in rigorous writing. Convert the diagram into element-chasing logic.

Common trap

Trap 3: Confusing subset and membership.
The statement $x \in A$ and the statement $x \subseteq A$ are usually about different kinds of objects. The first says x is an element. The second says x is a set whose elements lie in A.

Common trap

Trap 4: Forgetting the universe when using complement.
The complement of A in $\mathbb{R}$ differs from the complement of A in $\mathbb{Z}$. State the universe or make sure it is already fixed.

12.14 Exercises for set proofs

Exercise 7. Prove that for all sets A and B,

$$A \cap B \subseteq A \cup B.$$

Exercise 8. Prove that for all sets A, B, and C,

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C).$$

Exercise 9. Prove that for all sets A and B,

$$A \setminus B = A \cap B^c$$

assuming a fixed universe.

Exercise 10. Disprove the claim that for all sets A, B, and C,

$$A \cap (B \setminus C) = (A \cap B) \setminus (A \cap C)$$

or prove it if it is true.

Exercise 11. Let $\{A_i : i \in I\}$ be a family of sets. Prove that if $j \in I$, then

$$\bigcap_{i \in I} A_i \subseteq A_j.$$

Walking question

When you see a set identity, do not first ask whether you remember it. Ask what it says about an arbitrary element x.

13 Functions

13.1 Functions as disciplined input-output rules

A function is often introduced as a formula, such as

$$f(x) = x^2 + 1.$$

That is useful, but it is incomplete. A function is not just a formula. A function has a domain, a codomain, and a rule assigning to each element of the domain exactly one element of the codomain.

Formal

A function $f : A \to B$ consists of:

- a domain A,
- a codomain B,
- an assignment that sends each $a \in A$ to exactly one element $f(a) \in B$.

The notation

$$f : A \to B$$

means f is a function from A to B. If $a \in A$, then $f(a)$ is the value of f at a.

The phrase "exactly one" is doing real work. A function must be defined for every element of its domain, and it must not assign two different outputs to the same input.

For example,

$$f : \mathbb{R} \to \mathbb{R}, \qquad f(x) = x^2$$

is a function. Every real number has exactly one square.

The rule

$$x \mapsto \pm\sqrt{x}$$

is not a function from $[0, \infty)$ to $\mathbb{R}$, because most positive inputs have two outputs. We can make functions by choosing one branch:

$$g : [0, \infty) \to \mathbb{R}, \qquad g(x) = \sqrt{x},$$

or

$$h : [0, \infty) \to \mathbb{R}, \qquad h(x) = -\sqrt{x}.$$

Common trap

A formula by itself is not the whole function. The same formula with a different domain or codomain can define a different function.

13.2 Domain, codomain, and range

The domain is the set of allowed inputs. The codomain is the set where outputs are declared to live. The range, also called the image of the function, is the set of outputs actually hit.

For

$$f : \mathbb{R} \to \mathbb{R}, \qquad f(x) = x^2,$$

the domain is $\mathbb{R}$, the codomain is $\mathbb{R}$, and the range is $[0, \infty)$.

For

$$g : \mathbb{R} \to [0, \infty), \qquad g(x) = x^2,$$

the domain is $\mathbb{R}$, the codomain is $[0, \infty)$, and the range is $[0, \infty)$.

The formulas are the same, but the functions are not identical because the codomains differ.

Intuition

The codomain is the promised output type. The range is what actually appears during execution.

Software analogies are helpful but imperfect. A function signature like

```
square:  Real -> Real
```

declares an input type and output type. The implementation might never return a negative number, but the declared output type still allows one. Similarly, a mathematical codomain may be larger than the range.

This distinction becomes crucial for surjectivity. The function $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = x^2$ is not surjective. The function $g : \mathbb{R} \to [0, \infty)$ given by $g(x) = x^2$ is surjective. Same formula, different codomain, different property.

13.3 Functions as relations

A relation from A to B is a subset of $A \times B$. A function is a relation satisfying two conditions:

- every $a \in A$ appears as the first coordinate of at least one ordered pair,
- no $a \in A$ appears as the first coordinate of more than one ordered pair.

In other words, $f \subseteq A \times B$ is a function from A to B if for every $a \in A$, there exists exactly one $b \in B$ such that $(a, b) \in f$.

The graph of a function

$$f : A \to B$$

is

$$\Gamma_f = \{(a, f(a)) : a \in A\} \subseteq A \times B.$$

For example, the graph of

$$f : \mathbb{R} \to \mathbb{R}, \qquad f(x) = x^2$$

is

$$\Gamma_f = \{(x, y) \in \mathbb{R}^2 : y = x^2\}.$$

This graph viewpoint is more than a picture. It lets us define functions in very general settings. In set theory, a function can literally be represented as its graph.

Focus

The vertical line test is the geometric version of the "exactly one output for each input" rule.

The relation

$$R = \{(x, y) \in \mathbb{R}^2 : y^2 = x\}$$

is not a function $\mathbb{R} \to \mathbb{R}$, because $x = 1$ relates to both $y = 1$ and $y = -1$. But it can be restricted or reinterpreted to become a function, such as

$$f : [0, \infty) \to [0, \infty), \qquad f(x) = \sqrt{x}.$$

13.4 Equality of functions

Two functions are equal when they have the same domain, the same codomain, and the same values at every input.

Formal

If $f : A \to B$ and $g : C \to D$, then $f = g$ means:

$$A = C, \qquad B = D, \qquad \forall x \in A,\ f(x) = g(x).$$

Some contexts identify functions only by their graphs, in which case codomain may be treated differently. In most proof-writing contexts for functions $A \to B$, the codomain is part of the data.

Example:

$$f : \mathbb{R} \to \mathbb{R}, \qquad f(x) = x^2,$$

and

$$g : \mathbb{R} \to [0, \infty), \qquad g(x) = x^2$$

are not equal as functions with specified codomains, even though they assign the same formula to each real input.

The functions

$$p : \mathbb{R} \to \mathbb{R}, \qquad p(x) = (x + 1)^2$$

and

$$q : \mathbb{R} \to \mathbb{R}, \qquad q(x) = x^2 + 2x + 1$$

are equal, because their domains and codomains are the same and

$$(x + 1)^2 = x^2 + 2x + 1$$

for every $x \in \mathbb{R}$.

Common trap

Do not prove two functions are equal by checking a few inputs. You must prove equality for every input in the domain.

13.5 Composition

Composition is the operation of feeding the output of one function into another. If

$$f : A \to B$$

and

$$g : B \to C,$$

then the composite

$$g \circ f : A \to C$$

is defined by

$$(g \circ f)(a) = g(f(a)).$$

The order is important. The notation $g \circ f$ means first f, then g.

Example:

$$f : \mathbb{R} \to \mathbb{R}, \quad f(x) = x^2,$$
$$g : \mathbb{R} \to \mathbb{R}, \quad g(x) = x + 1.$$

Then

$$(g \circ f)(x) = g(f(x)) = g(x^2) = x^2 + 1,$$

while

$$(f \circ g)(x) = f(g(x)) = f(x+1) = (x+1)^2.$$

Usually

$$g \circ f \neq f \circ g.$$

Composition is associative:

$$h \circ (g \circ f) = (h \circ g) \circ f$$

whenever the domains and codomains match.

Proof. Let $f : A \to B$, $g : B \to C$, and $h : C \to D$. For any $a \in A$,

$$\big(h \circ (g \circ f)\big)(a) = h((g \circ f)(a)) = h(g(f(a))).$$

Also,

$$\big((h \circ g) \circ f\big)(a) = (h \circ g)(f(a)) = h(g(f(a))).$$

The two functions have the same domain, codomain, and value at every $a \in A$, so they are equal. □

Composition is the main operation in many parts of mathematics. In linear algebra, matrix multiplication represents composition of linear maps. In abstract algebra, homomorphisms compose. In topology, continuous functions compose. In category theory, composition becomes one of the basic axioms.

13.6 Identity functions

For every set A, the identity function on A is

$$\mathrm{id}_A : A \to A, \qquad \mathrm{id}_A(a) = a.$$

It does nothing, but it does nothing in a very structured way.

If $f : A \to B$, then

$$f \circ \mathrm{id}_A = f$$

and

$$\mathrm{id}_B \circ f = f.$$

The subscripts matter. The function id_A has domain and codomain A. The function id_B has domain and codomain B.

Proof. For any $a \in A$,

$$(f \circ \mathrm{id}_A)(a) = f(\mathrm{id}_A(a)) = f(a).$$

Thus $f \circ \mathrm{id}_A = f$.

Also, for any $a \in A$,

$$(\mathrm{id}_B \circ f)(a) = \mathrm{id}_B(f(a)) = f(a).$$

Thus $\mathrm{id}_B \circ f = f$. □

Identity functions are the function-world version of neutral elements. Zero is neutral for addition, one is neutral for multiplication, and identity maps are neutral for composition.

13.7 Restrictions and extensions

If $f : A \to B$ and $S \subseteq A$, the restriction of f to S is the function

$$f|_S : S \to B$$

defined by

$$f|_S(s) = f(s)$$

for all $s \in S$.

Example:

$$f : \mathbb{R} \to \mathbb{R}, \qquad f(x) = x^2.$$

The restriction

$$f|_{[0,\infty)} : [0, \infty) \to \mathbb{R}$$

has the same formula, but only accepts nonnegative inputs.

Restrictions can change important properties. The function $f : \mathbb{R} \to \mathbb{R}$ with $f(x) = x^2$ is not injective, because $f(1) = f(-1)$. But its restriction to $[0, \infty)$ is injective.

An extension goes the other direction. If $g : S \to B$ and $S \subseteq A$, an extension of g to A is a function $f : A \to B$ such that

$$f|_S = g.$$

Extensions are central in analysis and topology. A continuous function defined on a dense subset may or may not extend continuously to a larger space. A linear map defined on a basis extends uniquely to a vector space. A local solution to a differential equation may extend only up to a finite time.

Intuition

Restriction narrows the allowed inputs. Extension fills in values on new inputs while preserving the old behavior.

13.8 Piecewise-defined functions

A function may be defined by different rules on different parts of its domain:

$$f : \mathbb{R} \to \mathbb{R}, \qquad f(x) = \begin{cases} x^2, & x \geq 0, \\ -x, & x < 0. \end{cases}$$

This is a valid function because every real number falls into exactly one of the cases.

Piecewise definitions require coverage and consistency:

- every input in the domain must be covered by at least one case,
- if an input is covered by multiple cases, the rules must give the same output.

Consider:

$$g : \mathbb{R} \to \mathbb{R}, \qquad g(x) = \begin{cases} 0, & x \leq 0, \\ 1, & x \geq 0. \end{cases}$$

This is not a well-defined function as written, because $x = 0$ falls into both cases and receives two different values.

But

$$h : \mathbb{R} \to \mathbb{R}, \qquad h(x) = \begin{cases} 0, & x < 0, \\ 1, & x \geq 0 \end{cases}$$

is well-defined.

Common trap

For piecewise functions, always check the boundary points. Most mistakes happen where cases overlap or leave a gap.

13.9 Well-definedness

To say a function is well-defined means the proposed rule really assigns exactly one valid output to each allowed input.

For ordinary formulas, this can be easy. The function

$$f : \mathbb{R} \to \mathbb{R}, \qquad f(x) = x^2 + 1$$

is well-defined because every real x gives exactly one real number $x^2 + 1$.

But well-definedness becomes subtle when the input has multiple representations.

Example: define

$$F : \mathbb{Q} \to \mathbb{Z}$$

by saying if $q = a/b$ in lowest terms, set

$$F(q) = a + b.$$

This is well-defined because every rational number has a unique lowest-terms representation up to multiplying both numerator and denominator by -1. If we require $b > 0$, then the representation is unique.

But if we define

$$G : \mathbb{Q} \to \mathbb{Z}, \qquad G(a/b) = a + b$$

using any representation a/b, then it is not well-defined:

$$\frac{1}{2} = \frac{2}{4},$$

but

$$1 + 2 = 3$$

while

$$2 + 4 = 6.$$

The same rational number would receive different outputs.

Well-definedness is especially important in quotient structures. If elements are equivalence classes, a formula using representatives must not depend on which representative is chosen.

History and connection

The demand for well-definedness is one of the quiet engines of modern algebra. Quotient groups, quotient rings, vector spaces modulo subspaces, and homology groups all rely on defining operations on equivalence classes without depending on arbitrary representatives.

13.10 Worked proof: a function is well-defined

Let $\mathbb{Z}_3$ denote the set of congruence classes modulo 3:

$$\mathbb{Z}_3 = \{[0], [1], [2]\}.$$

Define

$$F : \mathbb{Z}_3 \to \mathbb{Z}_3, \qquad F([n]) = [2n].$$

We need to check this is well-defined. The input $[n]$ may have many representatives. For example,

$$[1] = [4] = [-2].$$

The formula must give the same output no matter which representative is used.

Proof. Suppose $[n] = [m]$ in $\mathbb{Z}_3$. Then $n \equiv m \pmod 3$, so 3 divides $n - m$. Therefore 3 divides $2n - 2m = 2(n - m)$, hence

$$2n \equiv 2m \pmod 3.$$

Thus $[2n] = [2m]$. Therefore $F([n]) = [2n]$ is independent of the representative, so F is well-defined. □

Notice the proof did not merely compute three examples. It proved that any two representatives of the same class lead to the same output.

13.11 Functions in analysis, topology, and algebra

In analysis, functions are the main objects of study. Continuity, differentiability, integrability, convergence, and boundedness are properties of functions or sequences of functions. A sequence is a function

$$a : \mathbb{N} \to \mathbb{R},$$

usually written a_n instead of $a(n)$.

In topology, continuous functions are the structure-preserving maps. A topological space is a set with a notion of open subsets. A function

$$f : X \to Y$$

between topological spaces is continuous if inverse images of open sets are open. This definition may look different from the epsilon-delta definition in calculus, but it is built from the same function and set language.

In algebra, homomorphisms are functions that preserve operations. A group homomorphism

$$\varphi : G \to H$$

satisfies

$$\varphi(ab) = \varphi(a)\varphi(b).$$

A linear map

$$T : V \to W$$

satisfies

$$T(v + w) = T(v) + T(w)$$

and

$$T(cv) = cT(v).$$

Thus, functions are not merely computational rules. They are how mathematics compares structures.

Focus

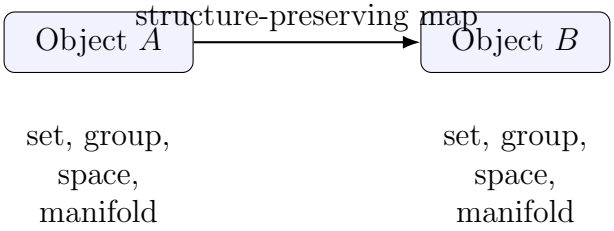
In advanced mathematics, objects matter, but maps between objects often matter more.

13.12 Structure-Preserving Maps: A First Category-Theoretic Hint

Here is a sentence worth carrying for years:

> Modern mathematics studies objects by studying the maps that preserve their structure.

This is the first tiny hint of category theory. You do not need category theory yet, but the perspective is useful immediately.



Different fields choose different structures and different maps:

Field	Object	Structure-preserving map
Set theory	sets	functions
Linear algebra	vector spaces	linear maps
Group theory	groups	homomorphisms
Topology	topological spaces	continuous maps
Differential geometry	smooth manifolds	smooth maps
Algebraic geometry	varieties/schemes	regular morphisms

This perspective explains why functions deserve so much attention before Velleman. Functions are not a side topic. They are the roads between mathematical worlds.

13.13 Exercises for functions

Exercise 12. For each rule, decide whether it defines a function with the stated domain and codomain.

$$f : \mathbb{R} \to \mathbb{R}, \quad f(x) = \frac{1}{x},$$

$$g : \mathbb{R} \setminus \{0\} \to \mathbb{R}, \quad g(x) = \frac{1}{x},$$

$$h : [0, \infty) \to \mathbb{R}, \quad h(x) = \pm\sqrt{x},$$

$$k : \mathbb{R} \to [0, \infty), \quad k(x) = x^2.$$

Exercise 13. Let $f : \mathbb{R} \to \mathbb{R}$ be given by $f(x) = 2x + 3$, and let $g : \mathbb{R} \to \mathbb{R}$ be given by $g(x) = x^2$. Compute $f \circ g$ and $g \circ f$.

Exercise 14. Prove that if $f : A \to B$, then $f \circ \mathrm{id}_A = f$ and $\mathrm{id}_B \circ f = f$.

Exercise 15. Let $f : \mathbb{R} \to \mathbb{R}$ be $f(x) = |x|$. Find a restriction of f that is injective. Find a codomain restriction that makes the resulting function surjective.

Exercise 16. Define $F : \mathbb{Z}_5 \to \mathbb{Z}_5$ by $F([n]) = [n + 2]$. Prove that F is well-defined.

Walking question

When you see a function formula, ask three questions before doing algebra: what is the domain, what is the codomain, and what values are actually hit?

14 Injective, Surjective, and Bijective Functions

14.1 Why these three words matter

Injective, surjective, and bijective describe how a function matches inputs to outputs.

Informally:

- injective means different inputs never collide,
- surjective means every target output is hit,
- bijective means both: every target output is hit exactly once.

These ideas are the beginning of mathematical comparison. If two finite sets have the same number of elements, there is a bijection between them. Cantor extended this idea to infinite sets: two sets have the same cardinality if there is a bijection between them.

History and connection

Cantor's theory of cardinality used bijections to compare infinite sets. This led to surprising results: $\mathbb{N}$ and $\mathbb{Z}$ have the same cardinality, $\mathbb{N}$ and $\mathbb{Q}$ have the same cardinality, but $\mathbb{N}$ and $\mathbb{R}$ do not. The language of injective, surjective, and bijective maps is the language behind those comparisons.

These properties are also everywhere in algebra and topology. An isomorphism is usually a bijective structure-preserving map whose inverse also preserves structure. A homeomorphism is a bijective continuous map with continuous inverse. A change of coordinates is useful when it is invertible. A database key acts like an injective identifier. A parser is unambiguous when different source structures do not collapse to the same parse result.

14.2 Injective functions

Formal

A function $f : A \to B$ is injective, or one-to-one, if

$$\forall a_1, a_2 \in A,\ f(a_1) = f(a_2) \Rightarrow a_1 = a_2.$$

This says collisions do not happen. If two inputs produce the same output, then they were not really two different inputs.

Equivalent contrapositive:

$$a_1 \neq a_2 \Rightarrow f(a_1) \neq f(a_2).$$

Example:

$$f : \mathbb{R} \to \mathbb{R}, \qquad f(x) = 2x + 3$$

is injective. If

$$f(x_1) = f(x_2),$$

then

$$2x_1 + 3 = 2x_2 + 3,$$

so

$$x_1 = x_2.$$

Example:

$$g : \mathbb{R} \to \mathbb{R}, \qquad g(x) = x^2$$

is not injective, because

$$g(1) = 1 = g(-1)$$

but

$$1 \neq -1.$$

The proof pattern for injectivity is:

> Assume $f(a_1) = f(a_2)$. Use algebra or structure to prove $a_1 = a_2$.

To disprove injectivity:

> Find $a_1 \neq a_2$ such that $f(a_1) = f(a_2)$.

Common trap

Do not prove injectivity by showing that one output has one input. You must show that every output has at most one input, or equivalently that no two different inputs collide.

14.3 Surjective functions

Formal

A function $f : A \to B$ is surjective, or onto, if

$$\forall b \in B,\ \exists a \in A \text{ such that } f(a) = b.$$

Surjectivity is about the codomain. Every element of the codomain must be hit.

Example:

$$f : \mathbb{R} \to \mathbb{R}, \qquad f(x) = 2x + 3$$

is surjective. Let $y \in \mathbb{R}$ be arbitrary. We need $x \in \mathbb{R}$ such that

$$2x + 3 = y.$$

Solving gives

$$x = \frac{y - 3}{2}.$$

This is real, and

$$f\left(\frac{y-3}{2}\right) = 2\left(\frac{y-3}{2}\right) + 3 = y.$$

So f is surjective.

Example:

$$g : \mathbb{R} \to \mathbb{R}, \qquad g(x) = x^2$$

is not surjective, because $-1 \in \mathbb{R}$ but no real x satisfies

$$x^2 = -1.$$

But

$$h : \mathbb{R} \to [0, \infty), \qquad h(x) = x^2$$

is surjective. Given $y \in [0, \infty)$, choose $x = \sqrt{y}$. Then $h(x) = y$.

Focus

Surjectivity can change when the codomain changes, even if the formula does not.

The proof pattern for surjectivity is:

> Let $b \in B$ be arbitrary. Construct $a \in A$ such that $f(a) = b$.

14.4 Bijective functions

Formal

A function $f : A \to B$ is bijective if it is both injective and surjective.

Equivalently, for every $b \in B$, there exists exactly one $a \in A$ such that

$$f(a) = b.$$

Example:

$$f : \mathbb{R} \to \mathbb{R}, \qquad f(x) = 2x + 3$$

is bijective. We already proved it is injective and surjective.

Example:

$$g : \mathbb{R} \to [0, \infty), \qquad g(x) = x^2$$

is surjective but not injective.

Example:

$$h : [0, \infty) \to [0, \infty), \qquad h(x) = x^2$$

is bijective. It is injective on $[0, \infty)$ because if $x_1^2 = x_2^2$ and $x_1, x_2 \geq 0$, then $x_1 = x_2$. It is surjective because every $y \geq 0$ has $x = \sqrt{y} \geq 0$ with $x^2 = y$.

Bijective functions are exactly the functions that have inverse functions in the ordinary two-sided sense.

14.5 Inverse functions

If $f : A \to B$ is bijective, then its inverse is the function

$$f^{-1} : B \to A$$

defined by

$$f^{-1}(b) = \text{the unique } a \in A \text{ such that } f(a) = b.$$

The word unique depends on injectivity. The existence depends on surjectivity.

If f is not surjective, some $b \in B$ has no preimage, so $f^{-1}(b)$ cannot be defined for every $b \in B$.

If f is not injective, some $b \in B$ has multiple preimages, so $f^{-1}(b)$ is ambiguous.

For a bijection,

$$f^{-1} \circ f = \mathrm{id}_A$$

and

$$f \circ f^{-1} = \mathrm{id}_B \,.$$

Proof. Let $a \in A$. Since $f(a)$ is the output corresponding to a, the unique input mapped to $f(a)$ is a. Thus

$$f^{-1}(f(a)) = a.$$

Therefore $f^{-1} \circ f = \mathrm{id}_A$.

Let $b \in B$. By definition, $f^{-1}(b)$ is the unique $a \in A$ with $f(a) = b$. Therefore

$$f(f^{-1}(b)) = b.$$

Thus $f \circ f^{-1} = \mathrm{id}_B$. □

Common trap

Do not confuse inverse functions with inverse images. The notation f^{-1} is used for both, but the meanings differ. An inverse function exists only for bijections. Inverse images of sets exist for every function.

14.6 Left inverses and right inverses

There is a useful relationship between injectivity, surjectivity, and one-sided inverses.

A function $g : B \to A$ is a left inverse of $f : A \to B$ if

$$g \circ f = \mathrm{id}_A .$$

This means g undoes f after f is applied.

A function $h : B \to A$ is a right inverse of $f : A \to B$ if

$$f \circ h = \mathrm{id}_B .$$

This means h chooses a preimage for each element of B.

Theorem 32. *If $f : A \to B$ has a left inverse, then f is injective.*

Proof. Suppose $g : B \to A$ satisfies $g \circ f = \mathrm{id}_A$. Let $a_1, a_2 \in A$ and assume $f(a_1) = f(a_2)$. Applying g to both sides gives

$$g(f(a_1)) = g(f(a_2)).$$

Since $g \circ f = \mathrm{id}_A$,

$$a_1 = a_2.$$

Thus f is injective. □

Theorem 33. *If $f : A \to B$ has a right inverse, then f is surjective.*

Proof. Suppose $h : B \to A$ satisfies $f \circ h = \mathrm{id}_B$. Let $b \in B$ be arbitrary. Choose $a = h(b) \in A$. Then

$$f(a) = f(h(b)) = b.$$

Therefore every $b \in B$ is hit, so f is surjective. □

The converses require mild set-theoretic care in full generality, especially for arbitrary surjections and choice functions. In most elementary contexts, an injective function can be given a left inverse after choosing default values, and a surjective function has right inverses by choosing one preimage for each target.

14.7 Composition and injectivity

Theorem 34. *If $f : A \to B$ and $g : B \to C$ are injective, then $g \circ f : A \to C$ is injective.*

Proof. Let $a_1, a_2 \in A$ and assume

$$(g \circ f)(a_1) = (g \circ f)(a_2).$$

Then

$$g(f(a_1)) = g(f(a_2)).$$

Since g is injective,

$$f(a_1) = f(a_2).$$

Since f is injective,

$$a_1 = a_2.$$

Therefore $g \circ f$ is injective. □

The proof shows how information travels backward through an injective composition. Equality at the end forces equality one step earlier, then one step earlier again.

Theorem 35. *If $g \circ f$ is injective, then f is injective.*

Proof. Let $a_1, a_2 \in A$ and assume $f(a_1) = f(a_2)$. Applying g gives

$$g(f(a_1)) = g(f(a_2)),$$

so

$$(g \circ f)(a_1) = (g \circ f)(a_2).$$

Since $g \circ f$ is injective, $a_1 = a_2$. Therefore f is injective. □

Notice that this theorem says nothing about g being injective. The composite may be injective even if g fails to be injective outside the range of f.

14.8 Composition and surjectivity

Theorem 36. *If $f : A \to B$ and $g : B \to C$ are surjective, then $g \circ f : A \to C$ is surjective.*

Proof. Let $c \in C$ be arbitrary. Since g is surjective, there exists $b \in B$ such that

$$g(b) = c.$$

Since f is surjective, there exists $a \in A$ such that

$$f(a) = b.$$

Then

$$(g \circ f)(a) = g(f(a)) = g(b) = c.$$

Therefore $g \circ f$ is surjective. □

Theorem 37. *If $g \circ f$ is surjective, then g is surjective.*

Proof. Let $c \in C$ be arbitrary. Since $g \circ f$ is surjective, there exists $a \in A$ such that

$$(g \circ f)(a) = c.$$

Let $b = f(a) \in B$. Then

$$g(b) = g(f(a)) = c.$$

Thus every $c \in C$ is hit by g, so g is surjective. □

Again, this says nothing about f being surjective. The composite might hit all of C even if f misses part of B, as long as the missed part is not needed by g to hit C.

14.9 Finite sets and counting

For finite sets, injectivity and surjectivity are connected by counting.

Theorem 38. *Let A and B be finite sets with $|A| = |B|$. For a function $f : A \to B$, the following are equivalent:*

$$f \text{ is injective},$$

$$f \text{ is surjective},$$

$$f \text{ is bijective}.$$

The intuition is simple. If A and B have the same finite number of elements, then a collision among inputs forces some output to be missed. Conversely, if some output is missed, then two inputs must collide.

This principle is sometimes called the pigeonhole principle in disguise.

Proof. Assume A and B are finite with $|A| = |B| = n$.

If f is injective, then the n different elements of A produce n different outputs in B. Since B has exactly n elements, every element of B is hit. Thus f is surjective.

If f is surjective, then every element of B is hit. Since B has n elements, the range has n elements. If two different elements of A had the same output, then n inputs would produce fewer than n outputs, contradicting that the range has n elements. Thus f is injective.

Therefore injectivity and surjectivity are equivalent, and either one implies bijectivity. □

For infinite sets, this theorem is false. The function

$$f : \mathbb{N} \to \mathbb{N}, \qquad f(n) = n + 1$$

is injective but not surjective if $\mathbb{N} = \{1, 2, 3, \ldots\}$, because 1 is not hit. The function

$$g : \mathbb{N} \to \mathbb{N}, \qquad g(1) = 1, \quad g(n) = n - 1 \text{ for } n \geq 2$$

is surjective but not injective, because $g(1) = g(2) = 1$.

Focus

Finite intuition is useful, but infinite sets often break it. Cantor's work begins exactly at that breaking point.

14.10 Proof lab: proving bijectivity

Proof lab

Prove that

$$f : \mathbb{R} \setminus \{1\} \to \mathbb{R} \setminus \{2\}, \qquad f(x) = \frac{2x+1}{x-1}$$

is bijective.
For injectivity, assume

$$\frac{2x_1+1}{x_1-1} = \frac{2x_2+1}{x_2-1}.$$

Cross-multiply:

$$(2x_1+1)(x_2-1) = (2x_2+1)(x_1-1).$$

Expand and simplify. The terms $2x_1x_2$ cancel, leaving

$$-2x_1 + x_2 - 1 = -2x_2 + x_1 - 1,$$

so

$$3x_2 = 3x_1,$$

and $x_1 = x_2$.
For surjectivity, let $y \in \mathbb{R} \setminus \{2\}$. Solve

$$y = \frac{2x+1}{x-1}.$$

Then

$$y(x-1) = 2x+1,$$

so

$$x(y-2) = y+1.$$

Since $y \neq 2$, define

$$x = \frac{y+1}{y-2}.$$

Check that $x \neq 1$, because $x = 1$ would imply $y+1 = y-2$, impossible. Then compute $f(x) = y$.

This lab illustrates the standard algebraic pattern. To prove surjectivity, solve $y = f(x)$ for x in terms of y, then check that the proposed x is actually in the domain.

14.11 Common traps

Common trap

Trap 1: Ignoring the codomain.
The function $x \mapsto x^2$ is not simply "surjective" or "not surjective" until its codomain is specified.

Common trap

Trap 2: Trying to prove surjectivity by starting with an arbitrary input.
For surjectivity, start with an arbitrary element of the codomain, then find a preimage in the domain.

Common trap

Trap 3: Trying to prove injectivity by solving $f(x) = y$.
That may help, but the direct injectivity pattern is to assume $f(x_1) = f(x_2)$ and prove $x_1 = x_2$.

Common trap

Trap 4: Calling every inverse-looking notation an inverse function.
For a general function f, the notation $f^{-1}(S)$ usually means inverse image of a set S, not an inverse function value.

14.12 Exercises for injective, surjective, bijective

Exercise 17. For each function, determine whether it is injective, surjective, bijective, or none.

$$f : \mathbb{R} \to \mathbb{R}, \quad f(x) = x^3,$$
$$g : \mathbb{R} \to \mathbb{R}, \quad g(x) = e^x,$$
$$h : \mathbb{R} \to (0, \infty), \quad h(x) = e^x,$$
$$p : \mathbb{Z} \to \mathbb{Z}, \quad p(n) = n + 1.$$

Exercise 18. Prove that if $f : A \to B$ and $g : B \to C$ are bijective, then $g \circ f : A \to C$ is bijective.

Exercise 19. Give an example of functions $f : A \to B$ and $g : B \to C$ such that $g \circ f$ is injective but g is not injective.

Exercise 20. Give an example of functions $f : A \to B$ and $g : B \to C$ such that $g \circ f$ is surjective but f is not surjective.

Exercise 21. Let A be a finite set. Prove that a function $f : A \to A$ is injective if and only if it is surjective.

Walking question

Look at a login system, a file path, or a database primary key. Ask which maps should be injective, which should be surjective, and which should not be either.

15 Images and Inverse Images

15.1 From points to sets

So far, $f(a)$ has meant the output of a function at one input. Images and inverse images extend this pointwise language to sets.

If

$$f : A \to B$$

and $S \subseteq A$, the image of S under f is the set of outputs reached from inputs in S:

$$f(S) = \{f(s) : s \in S\}.$$

Equivalently,

$$f(S) = \{b \in B : \exists s \in S \text{ such that } f(s) = b\}.$$

If $T \subseteq B$, the inverse image, or preimage, of T under f is

$$f^{-1}(T) = \{a \in A : f(a) \in T\}.$$

Focus

Images move subsets of the domain forward. Inverse images move subsets of the codomain backward.

Example:

$$f : \mathbb{R} \to \mathbb{R}, \qquad f(x) = x^2.$$

If

$$S = [-2, 3],$$

then

$$f(S) = [0, 9].$$

If

$$T = [1, 4],$$

then

$$f^{-1}(T) = [-2, -1] \cup [1, 2].$$

Notice that $f^{-1}(T)$ is defined even though f is not bijective. This is inverse image, not inverse function.

15.2 Image of a set

The image of $S \subseteq A$ under $f : A \to B$ is

$$f(S) = \{b \in B : \exists s \in S,\ f(s) = b\}.$$

The existential quantifier is important. An element b belongs to $f(S)$ if at least one element of S maps to b.

Example:

$$f : \mathbb{Z} \to \mathbb{Z}, \qquad f(n) = n^2.$$

If

$$S = \{-2, -1, 0, 1, 2\},$$

then

$$f(S) = \{0, 1, 4\}.$$

Different inputs can collapse to the same output. Images do not remember multiplicity.

If

$$S = 2\mathbb{Z}$$

is the set of even integers, then

$$f(S) = \{4k^2 : k \in \mathbb{Z}\}.$$

This is the set of squares of even integers.

The image of the whole domain is the range:

$$f(A) = \text{range}(f).$$

Common trap

Do not assume $f(S) = \{f(s)\}$ for one unspecified s. The image of a set contains outputs from all elements of the set.

15.3 Inverse image of a set

The inverse image of $T \subseteq B$ under $f : A \to B$ is

$$f^{-1}(T) = \{a \in A : f(a) \in T\}.$$

This definition has no existential ambiguity. It asks which inputs land inside T.

Example:

$$f : \mathbb{R} \to \mathbb{R}, \qquad f(x) = x^2.$$

Then

$$f^{-1}(\{4\}) = \{-2, 2\},$$
$$f^{-1}((-\infty, 0)) = \varnothing,$$
$$f^{-1}([0, 1]) = [-1, 1].$$

For a constant function

$$c : A \to B, \qquad c(a) = b_0,$$

we have

$$c^{-1}(T) = \begin{cases} A, & b_0 \in T, \\ \varnothing, & b_0 \notin T. \end{cases}$$

Inverse images are one of the most important constructions in all of mathematics because they behave especially well with set operations.

Intuition

The inverse image of T is the set of all inputs that pass the test "does the output lie in T?"

In software terms, if f is a function and T is a predicate on outputs, then $f^{-1}(T)$ is the predicate on inputs obtained by composing: accept a if $T(f(a))$ is true.

15.4 Images of unions and intersections

Images preserve unions exactly:

$$f(S \cup T) = f(S) \cup f(T).$$

Proof. Let $b \in f(S \cup T)$. Then there exists $a \in S \cup T$ such that $f(a) = b$. Since $a \in S \cup T$, either $a \in S$ or $a \in T$. If $a \in S$, then $b \in f(S)$. If $a \in T$, then $b \in f(T)$. Thus $b \in f(S) \cup f(T)$.

Conversely, let $b \in f(S) \cup f(T)$. Then $b \in f(S)$ or $b \in f(T)$. If $b \in f(S)$, there exists $s \in S$ such that $f(s) = b$. Since $s \in S \cup T$, $b \in f(S \cup T)$. The case $b \in f(T)$ is similar. Therefore $f(S \cup T) = f(S) \cup f(T)$. □

Images do not generally preserve intersections exactly. We always have

$$f(S \cap T) \subseteq f(S) \cap f(T),$$

but equality can fail.

Example:

$$f : \mathbb{R} \to \mathbb{R}, \qquad f(x) = x^2,$$
$$S = (-\infty, 0], \qquad T = [0, \infty).$$

Then

$$S \cap T = \{0\},$$

so

$$f(S \cap T) = \{0\}.$$

But

$$f(S) = [0, \infty)$$

and

$$f(T) = [0, \infty),$$

so

$$f(S) \cap f(T) = [0, \infty).$$

Equality fails.

The reason is collision. A value can be in both $f(S)$ and $f(T)$ because it is hit by one input from S and a different input from T.

Theorem 39. *If $f : A \to B$ is injective, then for all $S, T \subseteq A$,*

$$f(S \cap T) = f(S) \cap f(T).$$

Proof. We already know $f(S \cap T) \subseteq f(S) \cap f(T)$ for every function.

For the reverse containment, let $b \in f(S) \cap f(T)$. Then $b \in f(S)$ and $b \in f(T)$. So there exist $s \in S$ and $t \in T$ such that

$$f(s) = b = f(t).$$

Since f is injective, $s = t$. Therefore this common element lies in both S and T, so $s \in S \cap T$. Hence

$$b = f(s) \in f(S \cap T).$$

Thus $f(S) \cap f(T) \subseteq f(S \cap T)$, and equality follows. □

15.5 Inverse images and set operations

Inverse images preserve unions, intersections, and complements exactly.

Theorem 40. *Let $f : A \to B$, and let $S, T \subseteq B$. Then*

$$f^{-1}(S \cup T) = f^{-1}(S) \cup f^{-1}(T),$$

$$f^{-1}(S \cap T) = f^{-1}(S) \cap f^{-1}(T),$$

and

$$f^{-1}(B \setminus S) = A \setminus f^{-1}(S).$$

Proof. We prove the first identity. Let $a \in A$. Then

$$a \in f^{-1}(S \cup T)$$

if and only if

$$f(a) \in S \cup T.$$

This is equivalent to

$$f(a) \in S \text{ or } f(a) \in T.$$

That is equivalent to

$$a \in f^{-1}(S) \text{ or } a \in f^{-1}(T),$$

which is equivalent to

$$a \in f^{-1}(S) \cup f^{-1}(T).$$

Thus

$$f^{-1}(S \cup T) = f^{-1}(S) \cup f^{-1}(T).$$

The intersection proof is the same with "and" replacing "or." For complements:

$$a \in f^{-1}(B \setminus S)$$

if and only if

$$f(a) \in B \setminus S,$$

which means

$$f(a) \notin S.$$

That is equivalent to

$$a \notin f^{-1}(S),$$

so

$$a \in A \setminus f^{-1}(S).$$

Therefore

$$f^{-1}(B \setminus S) = A \setminus f^{-1}(S).$$

□

This exact preservation is why topology defines continuity using inverse images. If open sets in Y can be pulled back to open sets in X, then the function respects the topological structure.

History and connection

The inverse-image definition of continuity emerged from the abstraction of analysis into topology. In calculus, continuity is often described by limits and epsilons. In topology, a function $f : X \to Y$ is continuous when $f^{-1}(U)$ is open in X for every open set U in Y. This definition works in spaces where distance may not even be present.

15.6 Direct and inverse images under composition

Images and inverse images behave naturally with composition.

Let

$$f : A \to B, \qquad g : B \to C.$$

If $S \subseteq A$, then

$$(g \circ f)(S) = g(f(S)).$$

If $T \subseteq C$, then

$$(g \circ f)^{-1}(T) = f^{-1}(g^{-1}(T)).$$

Proof. For the image formula, let $c \in (g \circ f)(S)$. Then there exists $s \in S$ such that

$$(g \circ f)(s) = c.$$

That means

$$g(f(s)) = c.$$

Since $f(s) \in f(S)$, we have $c \in g(f(S))$. The reverse containment follows by unwinding the same definitions.

For inverse images, let $a \in A$. Then

$$a \in (g \circ f)^{-1}(T)$$

if and only if

$$(g \circ f)(a) \in T.$$

This means

$$g(f(a)) \in T.$$

That is equivalent to

$$f(a) \in g^{-1}(T),$$

which is equivalent to

$$a \in f^{-1}(g^{-1}(T)).$$

□

The inverse image formula reverses the order:

$$(g \circ f)^{-1} = f^{-1} \circ g^{-1}$$

as operations on subsets. This is the same order reversal familiar from inverse functions and matrix products.

15.7 The relationship between a set and its image-preimage

For $S \subseteq A$, compare S with

$$f^{-1}(f(S)).$$

Every element of S lands in $f(S)$, so

$$S \subseteq f^{-1}(f(S)).$$

But equality can fail. If f collapses points outside S onto values hit by S, those outside points also appear in the preimage.

Example:

$$f : \mathbb{R} \to \mathbb{R}, \qquad f(x) = x^2,$$

and

$$S = \{1\}.$$

Then

$$f(S) = \{1\},$$

but

$$f^{-1}(f(S)) = f^{-1}(\{1\}) = \{-1, 1\}.$$

If f is injective, then equality holds:

$$f^{-1}(f(S)) = S.$$

Proof. We already know $S \subseteq f^{-1}(f(S))$. For the reverse containment, let $a \in f^{-1}(f(S))$. Then $f(a) \in f(S)$. So there exists $s \in S$ such that

$$f(a) = f(s).$$

Since f is injective, $a = s$. Therefore $a \in S$. Hence

$$f^{-1}(f(S)) \subseteq S.$$

□

For $T \subseteq B$, compare T with

$$f(f^{-1}(T)).$$

Every element of $f(f^{-1}(T))$ lies in T, so

$$f(f^{-1}(T)) \subseteq T.$$

But equality can fail if T contains elements not hit by f.

Example:

$$f : \mathbb{R} \to \mathbb{R}, \qquad f(x) = x^2,$$

and

$$T = \{-1, 1\}.$$

Then

$$f^{-1}(T) = \{-1, 1\},$$

so

$$f(f^{-1}(T)) = \{1\} \subsetneq T.$$

If f is surjective, then equality holds for all $T \subseteq B$:

$$f(f^{-1}(T)) = T.$$

15.8 Images, inverse images, and injectivity/surjectivity

The previous subsection gives useful characterizations.

Theorem 41. *A function $f : A \to B$ is injective if and only if for every $S \subseteq A$,*

$$f^{-1}(f(S)) = S.$$

Proof. We already proved the forward direction.

Conversely, assume $f^{-1}(f(S)) = S$ for every $S \subseteq A$. We show f is injective. Suppose $f(a_1) = f(a_2)$. Let $S = \{a_1\}$. Since $f(a_2) = f(a_1)$, we have

$$a_2 \in f^{-1}(f(S)).$$

By the assumed equality,

$$f^{-1}(f(S)) = S = \{a_1\}.$$

Thus $a_2 = a_1$. Therefore f is injective. □

Theorem 42. *A function $f : A \to B$ is surjective if and only if for every $T \subseteq B$,*

$$f(f^{-1}(T)) = T.$$

Proof. Assume f is surjective. We already know $f(f^{-1}(T)) \subseteq T$. For the reverse containment, let $t \in T$. Since f is surjective, there exists $a \in A$ such that $f(a) = t$. Since $t \in T$, we have $a \in f^{-1}(T)$. Therefore $t = f(a) \in f(f^{-1}(T))$.

Conversely, assume $f(f^{-1}(T)) = T$ for every $T \subseteq B$. Let $b \in B$. Take $T = \{b\}$. Then

$$f(f^{-1}(\{b\})) = \{b\}.$$

In particular, the left side is nonempty, so there exists $a \in f^{-1}(\{b\})$. Thus $f(a) = b$. Therefore f is surjective. □

These theorems show that injectivity and surjectivity can be read as statements about how well image and inverse image undo each other.

15.9 Images and inverse images in topology and analysis

Continuity in topology is built from inverse images:

$$f : X \to Y \text{ is continuous} \quad \Longleftrightarrow \quad f^{-1}(U) \text{ is open in } X \text{ for every open } U \subseteq Y.$$

Why inverse images rather than images? Because inverse images preserve unions and intersections exactly. Topologies are defined by closure under arbitrary unions and finite intersections. If inverse images preserve those operations, the definition behaves cleanly.

Images of open sets do not generally remain open under continuous functions. For example,

$$f : \mathbb{R} \to \mathbb{R}, \qquad f(x) = x^2$$

is continuous, but

$$f((-1, 1)) = [0, 1),$$

which is not open in $\mathbb{R}$.

In measure theory and probability, inverse images are also essential. A random variable is a function

$$X : \Omega \to \mathbb{R}.$$

An event like

$$\{\omega \in \Omega : X(\omega) \leq 3\}$$

is the inverse image

$$X^{-1}((-\infty, 3]).$$

This is how statements about numerical outcomes become events in the sample space.

In analysis, level sets are inverse images:

$$f^{-1}(\{c\}) = \{x : f(x) = c\}.$$

These appear in optimization, implicit function theory, differential geometry, and partial differential equations.

History and connection

The language of images and inverse images helped unify analysis, topology, geometry, and probability. The same notation describes the range of a function, the preimage of an open set, the event determined by a random variable, and the fiber of a map in geometry.

15.10 Software analogies

Suppose a program has a function

$$\texttt{status: User -> Status}.$$

If S is a set of users, then $\texttt{status}(S)$ is the set of statuses appearing among those users.

If T is a set of statuses, then $\texttt{status}^{-1}(T)$ is the set of users whose status lies in T.

For example, if

$$T = \{\texttt{active}, \texttt{trial}\},$$

then

$$\texttt{status}^{-1}(T)$$

is the set of users whose status is active or trial.

Database queries often use inverse-image thinking. A filter such as

```
WHERE status IN ('active', 'trial')
```

selects the preimage of a subset of possible statuses under the status field.

Hashing gives another example. A hash function

$$h : \text{Keys} \to \text{Buckets}$$

maps many possible keys to a smaller set of buckets. The inverse image of a bucket is the set of keys that land in that bucket. Collisions are exactly the reason those inverse images can have more than one element.

Intuition

An inverse image is a pullback of a condition. A condition on outputs becomes a condition on inputs.

15.11 Proof lab: inverse images preserve arbitrary unions

Proof lab

Let $f : A \to B$, and let $\{T_i : i \in I\}$ be a family of subsets of B. Prove

$$f^{-1}\left(\bigcup_{i\in I} T_i\right) = \bigcup_{i\in I} f^{-1}(T_i).$$

Start with arbitrary $a \in A$:

$$a \in f^{-1}\left(\bigcup_{i\in I} T_i\right)$$

if and only if

$$f(a) \in \bigcup_{i\in I} T_i.$$

This means

$$\exists i \in I \text{ such that } f(a) \in T_i.$$

That is equivalent to

$$\exists i \in I \text{ such that } a \in f^{-1}(T_i),$$

which means

$$a \in \bigcup_{i\in I} f^{-1}(T_i).$$

The corresponding intersection identity is

$$f^{-1}\left(\bigcap_{i\in I} T_i\right) = \bigcap_{i\in I} f^{-1}(T_i),$$

with the existential quantifier replaced by a universal quantifier.

15.12 Common traps

Common trap

Trap 1: Treating $f^{-1}(T)$ as requiring an inverse function.
Inverse images exist for every function $f : A \to B$ and every subset $T \subseteq B$.

Common trap

Trap 2: Forgetting that $f(S)$ is a subset of the codomain, while $f^{-1}(T)$ is a subset of the domain.
Always ask which side of the function the set lives on.

Common trap

Trap 3: Assuming images preserve intersections.
Only the inclusion $f(S \cap T) \subseteq f(S) \cap f(T)$ holds for every function. Equality needs extra hypotheses, such as injectivity.

Common trap

Trap 4: Assuming $f(f^{-1}(T)) = T$ for every T.
This equality holds for all $T \subseteq B$ exactly when f is surjective. In general, the left side is only the part of T that f actually hits.

15.13 Exercises for images and inverse images

Exercise 22. Let $f : \mathbb{R} \to \mathbb{R}$ be $f(x) = x^2$. Compute:

$$f([-3, 2]), \qquad f^{-1}([4, 9]), \qquad f^{-1}((-1, 1)), \qquad f(\{-2, 0, 2\}).$$

Exercise 23. Let $f : A \to B$ and let $S, T \subseteq A$. Prove

$$f(S \cup T) = f(S) \cup f(T).$$

Then give an example showing that

$$f(S \cap T) = f(S) \cap f(T)$$

can fail.

Exercise 24. Let $f : A \to B$ and let $U, V \subseteq B$. Prove

$$f^{-1}(U \setminus V) = f^{-1}(U) \setminus f^{-1}(V).$$

Exercise 25. Let $f : A \to B$ and $g : B \to C$. Prove that for every $T \subseteq C$,

$$(g \circ f)^{-1}(T) = f^{-1}(g^{-1}(T)).$$

Exercise 26. Prove that $f : A \to B$ is injective if and only if for all subsets $S, T \subseteq A$,

$$f(S \cap T) = f(S) \cap f(T).$$

Exercise 27. Let $X : \Omega \to \mathbb{R}$ be a function. Express the set

$$\{\omega \in \Omega : 1 < X(\omega) \leq 5\}$$

as an inverse image of a subset of $\mathbb{R}$.

Walking question

When a theorem talks about a subset of the codomain, try pulling it back by inverse image. When it talks about a subset of the domain, try pushing it forward by image. Notice which direction preserves more structure.

16 Relations and Equivalence

Relations are one of the first places where mathematics stops treating objects one at a time. A relation is about connection: one number divides another, one set is contained in another, one function is the inverse of another, one person is an ancestor of another, one point is reachable from another by a path.

The word sounds ordinary, but the formal idea is powerful. Functions are special kinds of relations. Orders are relations with direction. Equivalence relations are relations that let us treat different objects as the same for a chosen purpose. Graphs, congruences, quotient spaces, partitions, databases, automata, and algebraic structures all use relations.

Focus

A relation is a rule for saying when two objects are connected. The most important question is not only "is a related to b?" but also "what properties does this connection have?"

16.1 From pairs to predicates

Let A and B be sets. A relation from A to B is a subset

$$R \subseteq A \times B.$$

If $(a, b) \in R$, we write

$$aRb$$

and say that a is related to b.

This definition may feel too simple. It says a relation is just a set of ordered pairs. But that simplicity is the point. A relation records which pairs pass a certain test.

For example, define a relation R from $\mathbb{Z}$ to $\mathbb{Z}$ by

$$aRb \quad \text{if and only if} \quad a \mid b.$$

Then $(3, 12) \in R$, because $3 \mid 12$, but $(5, 12) \notin R$.

We can view the same relation in several ways:

- Predicate view: $R(a, b)$ is the statement "a divides b."
- Set view: $R = \{(a, b) \in \mathbb{Z} \times \mathbb{Z} : a \mid b\}$.
- Arrow view: draw an arrow $a \to b$ when aRb.
- Table view: make a grid and mark the entries where aRb.

Intuition

A relation is like a yes-or-no machine that takes two inputs. The output is true for some ordered pairs and false for others.

16.2 Relations on one set

Most of this chapter studies relations on a single set. A relation on A is a subset

$$R \subseteq A \times A.$$

It compares elements of A with other elements of A.

Examples:

1. On $\mathbb{R}$, define xRy if $x \leq y$.
2. On $\mathbb{Z}$, define aRb if $a - b$ is even.
3. On the power set $\mathcal{P}(X)$, define $A \, R \, B$ if $A \subseteq B$.
4. On people, define xRy if x has the same birthday month as y.
5. On points in the plane, define $P \, R \, Q$ if the distance from P to Q is less than 1.

Each example has a different personality. Some are symmetric. Some are directional. Some group elements into classes. Some allow loops, where an object is related to itself. Some do not.

16.3 The three basic tests

There are three relation properties that occur constantly.

Definition 4. A relation R on a set A is *reflexive* if

$$\forall a \in A, \; aRa.$$

Reflexive means every element is related to itself. Equality is reflexive, since $a = a$. The relation $\leq$ is reflexive, since $a \leq a$. The relation $<$ is not reflexive, since $a < a$ is never true.

Definition 5. A relation R on a set A is *symmetric* if

$$\forall a, b \in A, \; aRb \Rightarrow bRa.$$

Symmetric means the relation has no preferred direction. If a is related to b, then b is related to a. The relation "has the same birthday month as" is symmetric. The relation $\leq$ is not symmetric on $\mathbb{R}$, because $2 \leq 5$ but $5 \leq 2$ is false.

Definition 6. A relation R on a set A is *transitive* if

$$\forall a, b, c \in A,\ (aRb \text{ and } bRc) \Rightarrow aRc.$$

Transitive means the relation can be chained. If a reaches b and b reaches c, then a reaches c. Equality is transitive. Divisibility is transitive. The relation "is within one mile of" is not transitive: A can be within one mile of B, and B within one mile of C, while A is not within one mile of C.

Formal

When proving a relation property, follow the quantifiers.
To prove reflexive, let $a \in A$ and prove aRa.
To prove symmetric, let $a, b \in A$ and assume aRb; prove bRa.
To prove transitive, let $a, b, c \in A$ and assume aRb and bRc; prove aRc.

16.4 Worked example: same parity

Define a relation $\sim$ on $\mathbb{Z}$ by

$$a \sim b \quad \text{if and only if} \quad a - b \text{ is even.}$$

We prove that $\sim$ is reflexive, symmetric, and transitive.

Proof. Reflexive: Let $a \in \mathbb{Z}$. Then $a - a = 0$, and $0 = 2 \cdot 0$, so $a - a$ is even. Hence $a \sim a$.

Symmetric: Let $a, b \in \mathbb{Z}$ and suppose $a \sim b$. Then $a - b$ is even, so $a - b = 2k$ for some $k \in \mathbb{Z}$. Thus

$$b - a = -(a - b) = -2k = 2(-k).$$

Since $-k \in \mathbb{Z}$, $b - a$ is even. Hence $b \sim a$.

Transitive: Let $a, b, c \in \mathbb{Z}$ and suppose $a \sim b$ and $b \sim c$. Then $a - b$ and $b - c$ are even. So $a - b = 2m$ and $b - c = 2n$ for some $m, n \in \mathbb{Z}$. Adding gives

$$a - c = (a - b) + (b - c) = 2m + 2n = 2(m + n).$$

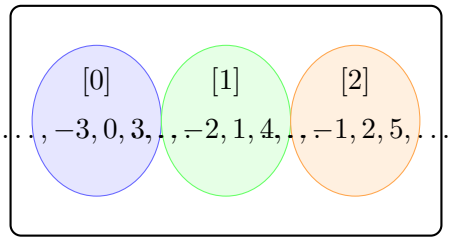
Since $m + n \in \mathbb{Z}$, $a - c$ is even. Hence $a \sim c$.

Therefore $\sim$ is reflexive, symmetric, and transitive. □

Notice the proof did not say "obvious." It unfolded the definition of even. In beginner proof writing, the habit of expanding definitions is more important than cleverness.

16.5 Visualizing Equivalence Classes

An equivalence relation cuts a set into non-overlapping regions called equivalence classes. Inside a region, objects are equivalent. Across regions, they are not.



Congruence classes modulo 3 partition $\mathbb{Z}$.

The picture suggests two crucial facts:

- every integer belongs to one of the classes,
- two classes are either identical or disjoint.

Those facts are not special to modulo 3. They follow from reflexive, symmetric, and transitive. This is one reason equivalence relations are so useful: they turn a relation into a classification system.

Common trap

False intuition: equivalence means equality. Better intuition: equivalence means equality after choosing what differences to ignore. Fractions, congruence classes, cosets, projective points, and quotient spaces all use this idea.

16.6 Common traps with relation properties

Common trap

Symmetric and reflexive are different. Symmetric says relations can be reversed when they occur. It does not say every element is related to itself. A relation may be symmetric but not reflexive.

For example, on $\mathbb{Z}$, define aRb if $a + b = 0$. This relation is symmetric: if $a + b = 0$, then $b + a = 0$. But it is not reflexive, because $a + a = 0$ is not true for every integer a.

Common trap

Transitive does not mean "eventually connected somehow." It means every two-step chain collapses to a one-step relation.

The relation "is a friend of" may be symmetric in an idealized social network, but it is not transitive. Your friend's friend need not be your friend.

16.7 Equivalence relations

Definition 7. A relation $\sim$ on a set A is an *equivalence relation* if it is reflexive, symmetric, and transitive.

An equivalence relation is a mathematically controlled version of sameness. It does not mean the objects are literally identical. It means they are the same with respect to a chosen feature.

Examples:

1. On $\mathbb{Z}$, $a \sim b$ if $a - b$ is even. This means a and b have the same parity.
2. On triangles, $T_1 \sim T_2$ if T_1 is similar to T_2. This means they have the same shape, ignoring size.
3. On strings, $s \sim t$ if s and t have the same length.
4. On people, $x \sim y$ if x and y have the same birth year.
5. On ordered pairs of integers with second coordinate nonzero, $(a, b) \sim (c, d)$ if $ad = bc$. This is the relation behind rational numbers.

Intuition

An equivalence relation says: for the question we are currently asking, these objects count as the same.

16.8 Equivalence classes

Once we have an equivalence relation, every element has a class around it.

Definition 8. Let $\sim$ be an equivalence relation on A. The *equivalence class* of $a \in A$ is

$$[a] = \{x \in A : x \sim a\}.$$

For the same-parity relation on $\mathbb{Z}$,

$$[0] = \{\ldots, -4, -2, 0, 2, 4, \ldots\}$$

and

$$[1] = \{\ldots, -3, -1, 1, 3, 5, \ldots\}.$$

There are only two equivalence classes: the even integers and the odd integers. Notice that $[0] = [2] = [-10]$ and $[1] = [3] = [-7]$.

An equivalence class is not a single object. It is a set of objects. The notation $[a]$ means "the class represented by a."

Common trap

Do not confuse a with $[a]$. The element a belongs to the original set A. The class $[a]$ is a subset of A.

16.9 The class theorem

The key theorem about equivalence classes is that they either match exactly or do not overlap at all.

Theorem 43. *Let $\sim$ be an equivalence relation on A. For any $a, b \in A$, either $[a] = [b]$ or $[a] \cap [b] = \emptyset$.*

Proof. Let $a, b \in A$. Suppose $[a] \cap [b] \neq \emptyset$. We will prove $[a] = [b]$.

Since $[a] \cap [b] \neq \emptyset$, choose $x \in [a] \cap [b]$. Then $x \in [a]$ and $x \in [b]$, so $x \sim a$ and $x \sim b$.

Because $x \sim a$, symmetry gives $a \sim x$. Since $a \sim x$ and $x \sim b$, transitivity gives $a \sim b$.

Now we prove the two inclusions.

First, let $y \in [a]$. Then $y \sim a$. Since $a \sim b$, transitivity gives $y \sim b$. Hence $y \in [b]$. Therefore $[a] \subseteq [b]$.

Second, let $y \in [b]$. Then $y \sim b$. From $a \sim b$, symmetry gives $b \sim a$. Since $y \sim b$ and $b \sim a$, transitivity gives $y \sim a$. Hence $y \in [a]$. Therefore $[b] \subseteq [a]$.

Thus $[a] = [b]$. We have shown that if the classes overlap, they are equal. Therefore two equivalence classes are either equal or disjoint. □

This theorem is one of the first examples of a very common mathematical move: instead of trying to understand every element individually, we organize the whole set into blocks.

16.10 Partitions

Definition 9. A *partition* of a set A is a collection of nonempty subsets of A such that every element of A belongs to exactly one subset in the collection.

A partition is a clean division of a set into non-overlapping blocks. The months partition the days of a year. The even and odd integers partition $\mathbb{Z}$. Letter grades partition students into groups, at least for one course.

Equivalence relations and partitions are two languages for the same idea.

Theorem 44. *Every equivalence relation on A determines a partition of A into equivalence classes.*

Proof. Let $\sim$ be an equivalence relation on A. Consider the collection of all equivalence classes

$$A/\sim = \{[a] : a \in A\}.$$

Each class is nonempty, because reflexivity gives $a \sim a$, so $a \in [a]$.

Every element of A belongs to some class: if $a \in A$, then $a \in [a]$.

Finally, by the previous theorem, two equivalence classes are either equal or disjoint. Therefore every element belongs to exactly one distinct class. Hence the equivalence classes form a partition of A. □

Conversely, every partition defines an equivalence relation: say two elements are related exactly when they belong to the same block.

Proof lab

Let $\mathcal{P}$ be a partition of A. Define $a \sim b$ if a and b lie in the same block of $\mathcal{P}$. Prove that $\sim$ is reflexive, symmetric, and transitive. The only subtle part is transitivity: if a and b are in one block and b and c are in one block, why must those two blocks be the same?

16.11 Quotient sets

The set of equivalence classes is called a quotient set.

Definition 10. If $\sim$ is an equivalence relation on A, the *quotient set* of A by $\sim$ is

$$A/\sim = \{[a] : a \in A\}.$$

The quotient set is the set whose elements are the blocks, not the original objects. This is a conceptual leap. We take a collection of objects and compress each equivalence class into a single new object.

For same parity,

$$\mathbb{Z}/\sim = \{[0], [1]\}.$$

This quotient has two elements.

For congruence modulo n, define

$$a \equiv b \pmod{n} \quad \text{if and only if} \quad n \mid (a - b).$$

Then

$$\mathbb{Z}/n\mathbb{Z} = \{[0], [1], \ldots, [n-1]\}.$$

The quotient $\mathbb{Z}/n\mathbb{Z}$ is the mathematical home of clock arithmetic.

History and connection

Congruence notation was introduced by Carl Friedrich Gauss in *Disquisitiones Arithmeticae* in 1801. Gauss gave number theory a language in which remainders could be treated structurally, not merely as leftovers after division.

16.12 Well-defined operations

Quotients become powerful when we define operations on classes. For example, in $\mathbb{Z}/n\mathbb{Z}$ we want

$$[a] + [b] = [a + b].$$

But this raises a serious question. The class $[a]$ has many representatives. If a' is another representative of $[a]$ and b' is another representative of $[b]$, do we get the same answer?

This is the issue of being well-defined.

Definition 11. An operation or function on equivalence classes is *well-defined* if its value does not depend on the choice of representatives.

Theorem 45. *Addition modulo n is well-defined: if $a \equiv a' \pmod{n}$ and $b \equiv b' \pmod{n}$, then*

$$a + b \equiv a' + b' \pmod{n}.$$

Proof. Assume $a \equiv a' \pmod{n}$ and $b \equiv b' \pmod{n}$. Then $n \mid (a - a')$ and $n \mid (b - b')$. So there are integers r, s such that

$$a - a' = nr, \qquad b - b' = ns.$$

Adding gives

$$(a + b) - (a' + b') = (a - a') + (b - b') = nr + ns = n(r + s).$$

Since $r + s \in \mathbb{Z}$, we have $n \mid ((a + b) - (a' + b'))$. Hence

$$a + b \equiv a' + b' \pmod{n}.$$

Therefore the rule $[a] + [b] = [a + b]$ is well-defined. □

Common trap

When defining something on equivalence classes, never prove only that the formula makes sense for one representative. You must prove that changing representatives does not change the class of the answer.

16.13 Rational numbers as a quotient

Fractions hide an equivalence relation. The symbols

$$\frac{1}{2}, \quad \frac{2}{4}, \quad \frac{3}{6}, \quad \frac{-5}{-10}$$

are different pairs of integers, but they represent the same rational number.

Let

$$S = \{(a, b) \in \mathbb{Z} \times \mathbb{Z} : b \neq 0\}.$$

Define a relation on S by

$$(a, b) \sim (c, d) \quad \text{if and only if} \quad ad = bc.$$

This relation says that two fractions are equal after cross multiplication.

Theorem 46. *The relation $\sim$ on S defined by $(a, b) \sim (c, d)$ if $ad = bc$ is an equivalence relation.*

Proof. Reflexive: Let $(a, b) \in S$. Then $ab = ba$, so $(a, b) \sim (a, b)$.

Symmetric: Suppose $(a, b) \sim (c, d)$. Then $ad = bc$. Since equality is symmetric, $bc = ad$, which is the same as $cb = da$. Hence $(c, d) \sim (a, b)$.

Transitive: Suppose $(a, b) \sim (c, d)$ and $(c, d) \sim (e, f)$. Then

$$ad = bc, \qquad cf = de.$$

We want to prove $af = be$. Since $d \neq 0$, multiply the desired comparison through by d. From $ad = bc$, multiply by f:

$$adf = bcf.$$

From $cf = de$, multiply by b:

$$bcf = bde.$$

Thus $adf = bde$. Since $d \neq 0$ and we are working in integers, cancellation gives $af = be$. Hence $(a, b) \sim (e, f)$.

Therefore $\sim$ is an equivalence relation. □

The rational number $\frac{a}{b}$ is really the equivalence class of the pair (a, b). In ordinary arithmetic we usually suppress the class notation because it would be exhausting to write it every time. But the quotient viewpoint explains why $\frac{1}{2} = \frac{2}{4}$ without pretending the ordered pairs $(1, 2)$ and $(2, 4)$ are identical.

16.14 Relations as data and structure

In a database, a relation is often represented as a table. In graph theory, a relation is represented by edges. In algebra, a congruence relation allows quotient structures. In topology, identifying points by an equivalence relation creates quotient spaces. In computer science, equivalence relations appear in union-find data structures, compiler optimization, type inference, minimization of automata, and equality saturation.

The same mathematical structure keeps appearing because many problems ask the same question:

> Which objects should be treated as indistinguishable for this purpose?

Walking question

Think of three situations from daily life where different objects are treated as the same: tickets for the same seat type, files with the same extension, products with the same SKU, people in the same time zone. What is the underlying equivalence relation? What are the equivalence classes?

16.15 Proof lab: building a quotient

Proof lab

Let R be the relation on $\mathbb{R}^2$ defined by

$$(x, y)R(u, v) \quad \text{if and only if} \quad x^2 + y^2 = u^2 + v^2.$$

1. Prove that R is an equivalence relation.
2. Describe the equivalence class of $(1, 0)$ geometrically.
3. Describe the quotient set $\mathbb{R}^2/R$ in plain language.
4. If you wanted to represent each class by a single number, which number would you choose?

16.16 Exercises

1. On $\mathbb{Z}$, define aRb if $a - b$ is divisible by 5. Prove that R is an equivalence relation and list the five equivalence classes.
2. On $\mathbb{R}$, define xRy if $|x - y| < 1$. Which of reflexive, symmetric, and transitive hold?

3. On $\mathcal{P}(X)$, define $A\,R\,B$ if $A \cap C = B \cap C$ for a fixed subset $C \subseteq X$. Prove that R is an equivalence relation.
4. Let $f : A \to B$ be a function. Define $a \sim a'$ if $f(a) = f(a')$. Prove that $\sim$ is an equivalence relation on A.
5. For the relation in the previous exercise, describe the equivalence classes in terms of inverse images.
6. Give an example of a relation that is reflexive and symmetric but not transitive.
7. Give an example of a relation that is symmetric and transitive but not reflexive on its whole set.
8. Explain why the equality relation on any set is an equivalence relation.
9. Explain why the universal relation $A \times A$ is an equivalence relation.
10. Let $A = \{1, 2, 3, 4\}$ and partition it as $\{\{1, 4\}, \{2\}, \{3\}\}$. Write the associated equivalence relation as a set of ordered pairs.

16.17 Kernel equivalence of a function

One of the most important sources of equivalence relations is a function. Let

$$f : A \to B.$$

Define a relation on A by

$$a \sim a' \quad \text{if and only if} \quad f(a) = f(a').$$

This says that two inputs are equivalent when the function cannot tell them apart.

Theorem 47. *For every function $f : A \to B$, the relation $a \sim a'$ if $f(a) = f(a')$ is an equivalence relation on A.*

Proof. Reflexive: Let $a \in A$. Since $f(a) = f(a)$, we have $a \sim a$.

Symmetric: Suppose $a \sim a'$. Then $f(a) = f(a')$. By symmetry of equality, $f(a') = f(a)$, so $a' \sim a$.

Transitive: Suppose $a \sim a'$ and $a' \sim a''$. Then

$$f(a) = f(a') \quad \text{and} \quad f(a') = f(a'').$$

By transitivity of equality, $f(a) = f(a'')$. Hence $a \sim a''$.

Thus $\sim$ is an equivalence relation. □

The equivalence class of a is

$$[a] = \{x \in A : f(x) = f(a)\}.$$

This is exactly the inverse image of the one-point set $\{f(a)\}$:

$$[a] = f^{-1}(\{f(a)\}).$$

These classes are sometimes called the fibers of f. The quotient $A/\sim$ is the set of fibers.

Intuition

A function partitions its domain according to output. Injective functions have tiny fibers, each with one element. Non-injective functions merge several inputs into the same fiber.

Example: Let $f : \mathbb{R} \to \mathbb{R}$ be $f(x) = x^2$. Then

$$x \sim y \quad \text{if and only if} \quad x^2 = y^2.$$

The class of 3 is $\{3, -3\}$. The class of 0 is $\{0\}$. The quotient remembers absolute value rather than sign.

This viewpoint explains one reason equivalence relations matter in algebra. A homomorphism often collapses a structure by identifying elements with the same image. The associated equivalence relation records exactly what was collapsed.

16.18 Canonical representatives

After forming a quotient, it is often convenient to choose one representative from each equivalence class. For congruence modulo n, the standard representatives are

$$0, 1, \ldots, n-1.$$

Every integer is congruent to exactly one of these.

But representatives are a convenience, not the quotient itself. In $\mathbb{Z}/5\mathbb{Z}$, the class $[2]$ can be represented by

$$2,\ 7,\ -3,\ 12,\ -8, \ldots$$

The object is the whole class. The representative is a name tag.

Common trap

When computing in a quotient, reducing to a preferred representative is allowed only after the operation on classes is known to be well-defined.

Proof lab

Let $f : \mathbb{R} \to [0, \infty)$ be $f(x) = |x|$.

1. Describe the equivalence relation induced by f.
2. Describe the equivalence class of 0.
3. Describe the equivalence class of 5.
4. Explain why the quotient $\mathbb{R}/\sim$ can be represented by $[0, \infty)$.
5. What information is lost when passing from x to $|x|$?

17 Orders

Order relations compare objects by rank, size, inclusion, refinement, information, divisibility, or dependency. The familiar order $\leq$ on numbers is only one example. Once we leave the number line, ordering becomes subtler. Some objects may be comparable, while others may not.

Focus

An order relation is a relation that lets us compare objects without necessarily forcing every pair into a single line.

17.1 Antisymmetry

Equivalence relations used reflexive, symmetric, and transitive. Orders usually replace symmetry with antisymmetry.

Definition 12. A relation R on a set A is *antisymmetric* if

$$\forall a, b \in A,\ (aRb \text{ and } bRa) \Rightarrow a = b.$$

Antisymmetry does not mean "not symmetric." It means that two different elements cannot be related in both directions.

For $\leq$ on $\mathbb{R}$, if $a \leq b$ and $b \leq a$, then $a = b$. Thus $\leq$ is antisymmetric.

For divisibility on positive integers, if $a \mid b$ and $b \mid a$, then $a = b$. Thus divisibility is antisymmetric on $\mathbb{N}^+ = \{1, 2, 3, \ldots\}$.

For subset inclusion, if $A \subseteq B$ and $B \subseteq A$, then $A = B$. Thus $\subseteq$ is antisymmetric.

Common trap

Antisymmetric does not say that aRb prevents bRa in every case. It allows both directions when $a = b$.

17.2 Partial orders

Definition 13. A relation $\preceq$ on a set P is a *partial order* if it is reflexive, antisymmetric, and transitive. A set equipped with a partial order is called a *partially ordered set*, or *poset*.

The word "partial" means not every pair must be comparable.

Examples:

1. $(\mathbb{R}, \leq)$ is a poset.
2. $(\mathcal{P}(X), \subseteq)$ is a poset.
3. Positive integers ordered by divisibility form a poset.
4. Tasks in a project ordered by "must be completed before" can form a poset after adding reflexive self-comparison.
5. Subspaces of a vector space ordered by inclusion form a poset.

17.3 Worked proof: inclusion is a partial order

Theorem 48. *For any set X, the relation $\subseteq$ is a partial order on $\mathcal{P}(X)$.*

Proof. Reflexive: Let $A \in \mathcal{P}(X)$. Every element of A is an element of A, so $A \subseteq A$.

Antisymmetric: Let $A, B \in \mathcal{P}(X)$ and suppose $A \subseteq B$ and $B \subseteq A$. To prove $A = B$, use extensionality. If $x \in A$, then $x \in B$ because $A \subseteq B$. If $x \in B$, then $x \in A$ because $B \subseteq A$. Thus A and B have exactly the same elements, so $A = B$.

Transitive: Let $A, B, C \in \mathcal{P}(X)$ and suppose $A \subseteq B$ and $B \subseteq C$. If $x \in A$, then $x \in B$, and then $x \in C$. Hence $A \subseteq C$.

Therefore $\subseteq$ is reflexive, antisymmetric, and transitive, so it is a partial order. □

17.4 Comparability

In a poset $(P, \preceq)$, two elements $a, b \in P$ are *comparable* if

$$a \preceq b \quad \text{or} \quad b \preceq a.$$

They are incomparable if neither relation holds.

In $(\mathcal{P}(\{1,2,3\}), \subseteq)$, the sets $\{1\}$ and $\{1,2\}$ are comparable because $\{1\} \subseteq \{1,2\}$. But $\{1\}$ and $\{2\}$ are incomparable because neither is a subset of the other.

In divisibility, 3 and 12 are comparable because $3 \mid 12$. But 6 and 10 are incomparable: neither divides the other.

Intuition

A partial order is not necessarily a line. It is often a network of constraints. Some elements sit above others, but many pairs may live on separate branches.

17.5 Total orders

Definition 14. A partial order $\preceq$ on P is a *total order* if every pair of elements is comparable:

$$\forall a, b \in P,\ a \preceq b \text{ or } b \preceq a.$$

The usual order $\leq$ on $\mathbb{R}$ is total. For any real numbers a and b, either $a \leq b$ or $b \leq a$.

Subset inclusion is usually not total. If X has at least two distinct elements x and y, then $\{x\}$ and $\{y\}$ are incomparable.

Common trap

Do not assume every order behaves like the number line. Many important orders are partial, not total.

17.6 Strict orders

Sometimes we use strict comparison, such as $<$ instead of $\leq$.

Definition 15. A relation $\prec$ on P is a *strict partial order* if it is irreflexive and transitive. Irreflexive means

$$\forall a \in P,\ \neg(a \prec a).$$

The relation $<$ on $\mathbb{R}$ is irreflexive and transitive. The relation $\subsetneq$ on $\mathcal{P}(X)$ is also irreflexive and transitive.

Strict and non-strict orders are closely related. From a partial order $\preceq$, define

$$a \prec b \quad \text{if and only if} \quad a \preceq b \text{ and } a \neq b.$$

This gives the associated strict order.

From many strict orders, define

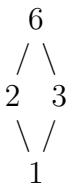
$$a \preceq b \quad \text{if and only if} \quad a \prec b \text{ or } a = b.$$

This gives the associated non-strict order.

17.7 Hasse diagrams

Finite posets can often be drawn with Hasse diagrams. A Hasse diagram places larger elements higher and draws edges only for immediate cover relations, leaving out reflexive loops and edges implied by transitivity.

For divisibility among $\{1, 2, 3, 6\}$:



The diagram says $1 \mid 2$, $1 \mid 3$, $2 \mid 6$, and $3 \mid 6$. It also implies $1 \mid 6$ by transitivity, so we do not draw a direct edge from 1 to 6 in the Hasse diagram.

17.8 Minimal, maximal, least, and greatest

Order language becomes precise because there are several kinds of extremal elements.

Definition 16. Let $(P, \preceq)$ be a poset and let $S \subseteq P$.

- An element $m \in S$ is *minimal* in S if there is no $s \in S$ with $s \prec m$.
- An element $M \in S$ is *maximal* in S if there is no $s \in S$ with $M \prec s$.
- An element $\ell \in S$ is *least* in S if $\ell \preceq s$ for every $s \in S$.
- An element $g \in S$ is *greatest* in S if $s \preceq g$ for every $s \in S$.

Least implies minimal. Greatest implies maximal. But minimal does not necessarily imply least, and maximal does not necessarily imply greatest.

Consider $S = \{\{1\}, \{2\}\}$ inside $\mathcal{P}(\{1, 2\})$ ordered by $\subseteq$. Both $\{1\}$ and $\{2\}$ are minimal. Neither is least, because neither is a subset of the other.

Common trap

"Minimal" means nothing is strictly below it. "Least" means it is below everything. In a partial order these are not the same.

17.9 Upper bounds and suprema

Definition 17. Let $(P, \preceq)$ be a poset and $S \subseteq P$. An element $u \in P$ is an *upper bound* for S if

$$\forall s \in S,\ s \preceq u.$$

A *least upper bound*, or *supremum*, is an upper bound u such that $u \preceq v$ for every upper bound v of S.

Similarly, a lower bound lies below every element of S, and a greatest lower bound is called an infimum.

In $\mathbb{R}$ with the usual order, the set $(0,1)$ has many upper bounds: 1, 2, 100. Its least upper bound is 1, even though $1 \notin (0,1)$.

In $\mathcal{P}(X)$ ordered by $\subseteq$, the supremum of a family of subsets is their union, and the infimum is their intersection.

Theorem 49. *Let $\mathcal{A} \subseteq \mathcal{P}(X)$. Under the order $\subseteq$, the least upper bound of $\mathcal{A}$ is $\bigcup_{A\in\mathcal{A}} A$.*

Proof. Let

$$U = \bigcup_{A\in\mathcal{A}} A.$$

First we show U is an upper bound. If $A \in \mathcal{A}$ and $x \in A$, then by definition of union, $x \in U$. Hence $A \subseteq U$ for every $A \in \mathcal{A}$.

Now let V be any upper bound for $\mathcal{A}$. This means $A \subseteq V$ for every $A \in \mathcal{A}$. We prove $U \subseteq V$. Let $x \in U$. Then $x \in A$ for some $A \in \mathcal{A}$. Since $A \subseteq V$, we get $x \in V$. Therefore $U \subseteq V$.

So U is an upper bound contained in every upper bound. Hence U is the least upper bound. □

17.10 Divisibility as an order

Let $\mathbb{N}^+ = \{1, 2, 3, \ldots\}$. Define

$$a \preceq b \quad \text{if and only if} \quad a \mid b.$$

This is a partial order.

Proof. Reflexive: For every $a \in \mathbb{N}^+$, $a \mid a$ because $a = 1 \cdot a$.

Antisymmetric: Suppose $a \mid b$ and $b \mid a$. Then $b = ak$ and $a = b\ell$ for positive integers k, ℓ. Substituting gives

$$a = (ak)\ell = a(k\ell).$$

Since $a > 0$, divide by a to get $1 = k\ell$. Since $k, \ell \in \mathbb{N}^+$, we must have $k = \ell = 1$. Thus $a = b$.

Transitive: Suppose $a \mid b$ and $b \mid c$. Then $b = ak$ and $c = b\ell$ for positive integers k, ℓ. Hence $c = (ak)\ell = a(k\ell)$, so $a \mid c$. □

This order is not total. For example, neither $2 \mid 3$ nor $3 \mid 2$.

In this order, greatest common divisors and least common multiples can be understood as infima and suprema. The greatest common divisor of a and b is the greatest lower bound of $\{a, b\}$ under divisibility. The least common multiple is the least upper bound.

History and connection

The Euclidean algorithm is older than modern order theory, but order language reveals what it computes structurally: the greatest element among common divisors. Later algebra generalizes this idea to rings, ideals, lattices, and modules.

17.11 Well-orders

Definition 18. A total order $\leq$ on a set W is a *well-order* if every nonempty subset of W has a least element.

The natural numbers with their usual order are well-ordered. Every nonempty subset of $\mathbb{N}$ has a least element.

The integers with their usual order are not well-ordered. The subset $\mathbb{Z}$ itself has no least element. The positive real numbers are not well-ordered under the usual order, because $(0, 1)$ has no least element.

Well-ordering is the hidden foundation under ordinary induction.

Formal

The well-ordering principle for $\mathbb{N}$ says: every nonempty subset of $\mathbb{N}$ has a least element. It is equivalent, in standard foundations of arithmetic, to mathematical induction.

17.12 Topological and algebraic echoes

Orders appear in many later fields.

- In analysis, the completeness of $\mathbb{R}$ is expressed using least upper bounds.
- In topology, open sets are ordered by inclusion.
- In algebra, ideals are ordered by inclusion, and maximal ideals are central.
- In logic, theories can be ordered by strength.
- In computer science, program states can be ordered by information content.
- In category theory, a poset can be viewed as a category with at most one morphism between any two objects.

The same theme repeats: structure is not only what objects are, but how they compare.

17.13 Proof lab: a poset of information

Proof lab

Let X be a set and let $\mathcal{P}$ be a collection of partitions of X. Define $\mathcal{A} \preceq \mathcal{B}$ if every block of $\mathcal{A}$ is contained in some block of $\mathcal{B}$.
This means $\mathcal{A}$ is finer than $\mathcal{B}$: it makes at least as many distinctions.

1. Prove that $\preceq$ is reflexive.
2. Prove that $\preceq$ is transitive.
3. Think carefully about antisymmetry. If $\mathcal{A} \preceq \mathcal{B}$ and $\mathcal{B} \preceq \mathcal{A}$, why must the partitions be the same?
4. For $X = \{1, 2, 3\}$, draw the poset of all partitions ordered by refinement.

17.14 Walking prompts

Walking question

Walk through a dependency system you know: course prerequisites, software packages, build steps, recipe steps. Which pairs are comparable? Which are not? What would a least element mean? What would a maximal element mean?

17.15 Exercises

1. Prove that $\leq$ is a total order on $\mathbb{Z}$.
2. Prove that $\subseteq$ is not a total order on $\mathcal{P}(\{1, 2\})$.
3. On $\mathbb{R}$, define $x \preceq y$ if $|x| \leq |y|$. Which partial order properties hold? Which fail?
4. On $\mathbb{Z}$, define $a \preceq b$ if $a \mid b$. Is this antisymmetric on all of $\mathbb{Z}$? Explain.
5. Give a poset with exactly three maximal elements.
6. Give a poset with a unique minimal element that is not least.
7. Find all minimal and maximal elements of $\mathcal{P}(\{1, 2, 3\}) \setminus \{\emptyset, \{1, 2, 3\}\}$ under inclusion.
8. Prove that if a least element exists in a poset, it is unique.
9. Prove that if a greatest element exists in a poset, it is unique.
10. Let $S \subseteq \mathbb{R}$ be bounded above. State what it means for u to be $\sup S$ using quantifiers.
11. In $\mathbb{N}^+$ ordered by divisibility, find the infimum and supremum of $\{12, 18\}$.
12. Explain why every finite nonempty total order is a well-order.

17.16 Chains and antichains

Partial orders become easier to see when we name two opposite patterns.

Definition 19. Let $(P, \preceq)$ be a poset. A subset $C \subseteq P$ is a *chain* if every two elements of C are comparable. A subset $A \subseteq P$ is an *antichain* if no two distinct elements of A are comparable.

In $\mathcal{P}(\{1,2,3\})$ ordered by inclusion,

$$\emptyset \subseteq \{1\} \subseteq \{1,2\} \subseteq \{1,2,3\}$$

is a chain. The set

$$\{\{1\},\{2\},\{3\}\}$$

is an antichain.

Chains are the parts of a poset that behave like total orders. Antichains are the parts where comparison fails as much as possible.

Intuition

A chain is a single line through the poset. An antichain is a horizontal layer of mutually incomparable objects.

Chains and antichains matter because they separate two kinds of complexity. A long chain means there are many strict levels of refinement or dependency. A large antichain means there are many independent alternatives that cannot be ranked by the order.

17.17 Lattices

Some posets have especially good behavior with respect to pairs.

Definition 20. A poset $(L, \preceq)$ is a *lattice* if every pair of elements $a, b \in L$ has a least upper bound and a greatest lower bound.

The least upper bound of a and b is often called their join and written

$$a \vee b.$$

The greatest lower bound is called their meet and written

$$a \wedge b.$$

In $\mathcal{P}(X)$ ordered by inclusion,

$$A \vee B = A \cup B, \qquad A \wedge B = A \cap B.$$

Union is the smallest set containing both A and B. Intersection is the largest set contained in both.

In positive integers ordered by divisibility,

$$a \vee b = \operatorname{lcm}(a, b), \qquad a \wedge b = \gcd(a, b).$$

This is a good example of abstraction paying rent: union/intersection and lcm/gcd are different operations in different settings, but they play the same order-theoretic roles.

Proof lab

Prove that $\mathcal{P}(X)$ is a lattice under $\subseteq$.

1. Show $A \cup B$ is an upper bound for $\{A, B\}$.
2. Show every upper bound of $\{A, B\}$ contains $A \cup B$.
3. Show $A \cap B$ is a lower bound for $\{A, B\}$.
4. Show every lower bound of $\{A, B\}$ is contained in $A \cap B$.

17.18 Order-preserving maps

Functions between ordered sets should sometimes respect order.

Definition 21. Let $(P, \preceq_P)$ and $(Q, \preceq_Q)$ be posets. A function $f : P \to Q$ is *order-preserving*, or *monotone*, if

$$x \preceq_P y \Rightarrow f(x) \preceq_Q f(y).$$

Examples:

1. The function $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = 3x + 1$ is order-preserving under the usual order.
2. The function $g : \mathbb{R} \to \mathbb{R}$ given by $g(x) = -x$ is order-reversing, not order-preserving.
3. If $X \subseteq Y$, then the map $\mathcal{P}(X) \to \mathcal{P}(Y)$ sending A to A is order-preserving.
4. For positive integers ordered by divisibility, the map $n \mapsto 2n$ is order-preserving.

Order-preserving maps are the structure-respecting maps of order theory. They are to posets what homomorphisms are to algebraic structures and continuous maps are to topological spaces.

Walking question

Think of a system where adding information should not reduce what you know: search filters, type constraints, permissions, prerequisites. What is the order? What would an order-preserving function between two such systems mean?

18 Induction

Mathematical induction is the proof method that turns a local step into an infinite conclusion. It is the formal version of a simple idea: if you can start, and you can always move from one case to the next, then every case is reached.

Beginners often learn induction as a template. The template matters, but induction is more than a template. It is a way of thinking about recursively generated objects, discrete time, algorithms, grammar, trees, sums, divisibility, inequalities, and definitions that refer back to earlier cases.

Focus

Induction proves infinitely many statements by proving two things: the first statement is true, and truth passes from each case to the next.

18.1 The domino picture

Imagine infinitely many dominos labeled $0, 1, 2, 3, \ldots$.

To prove all dominos fall, it is enough to prove:

1. Domino 0 falls.
2. Whenever domino n falls, domino $n+1$ falls.

The first clause starts the chain. The second clause transmits the chain.

Induction is the same idea for statements. Let $P(n)$ be a statement about n. To prove

$$\forall n \in \mathbb{N},\ P(n),$$

we prove:

1. Base case: $P(0)$ is true.
2. Inductive step: for arbitrary $n \in \mathbb{N}$, if $P(n)$ is true, then $P(n+1)$ is true.

Some books start $\mathbb{N}$ at 1 instead of 0. The method is the same. Start at the first value in the claim.

Common trap

Always match the base case to the statement. If the theorem says "for all $n \geq 5$," the base case is usually $n = 5$, not $n = 0$ or $n = 1$.

18.2 The formal principle

Formal

Principle of mathematical induction: Let $P(n)$ be a predicate on natural numbers. If

$$P(0)$$

is true and

$$\forall n \in \mathbb{N},\ P(n) \Rightarrow P(n+1)$$

is true, then

$$\forall n \in \mathbb{N},\ P(n)$$

is true.

The version starting at 1 says: if $P(1)$ is true and $P(n) \Rightarrow P(n+1)$ for all $n \geq 1$, then $P(n)$ is true for all $n \geq 1$.

The version starting at m says: if $P(m)$ is true and $P(n) \Rightarrow P(n+1)$ for all $n \geq m$, then $P(n)$ is true for all $n \geq m$.

18.3 First worked proof: sum of the first n integers

Theorem 50. *For every integer $n \geq 1$,*

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

Proof. Let $P(n)$ be the statement

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

Base case: For $n = 1$, the left side is 1, and the right side is

$$\frac{1(1+1)}{2} = 1.$$

Thus $P(1)$ is true.

Inductive step: Let $n \geq 1$ and assume $P(n)$ is true. That is, assume

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

We must prove $P(n+1)$:

$$1 + 2 + \cdots + n + (n+1) = \frac{(n+1)(n+2)}{2}.$$

Starting with the left side,

$$1 + 2 + \cdots + n + (n+1) = \frac{n(n+1)}{2} + (n+1)$$

by the induction hypothesis. Now simplify:

$$\frac{n(n+1)}{2} + (n+1) = \frac{n(n+1) + 2(n+1)}{2} = \frac{(n+1)(n+2)}{2}.$$

So $P(n+1)$ is true.

Therefore, by induction, the formula holds for every integer $n \geq 1$. □

The induction hypothesis is not a wish. It is a temporary assumption inside the inductive step. The theorem of induction tells us that if the step works for arbitrary n, then the chain is valid.

18.4 What the induction hypothesis is allowed to do

In the proof above, we used the induction hypothesis exactly once:

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

We used it to replace the sum up to n with a formula. Then we added the new term $n + 1$.

This is the pattern behind many induction proofs:

> Use the known case to handle most of the new case, then deal with the new piece.

Common trap

Do not assume $P(n+1)$ in the inductive step. You assume $P(n)$ and prove $P(n+1)$.

18.5 Induction and recursion

Induction is proof. Recursion is definition or computation. They are paired because recursive objects are built by base rules and step rules.

For example, define factorial recursively:

$$0! = 1, \qquad (n+1)! = (n+1)n!.$$

This definition tells us how to build factorial values. Induction tells us how to prove facts about all of them.

Theorem 51. *For every $n \geq 1$, $n! \geq 2^{n-1}$.*

Proof. Let $P(n)$ be the statement $n! \geq 2^{n-1}$.

Base case: For $n = 1$, we have $1! = 1$ and $2^0 = 1$, so $1! \geq 2^0$.

Inductive step: Let $n \geq 1$ and assume $n! \geq 2^{n-1}$. We need to prove

$$(n+1)! \geq 2^n.$$

Since $n + 1 \geq 2$, we have

$$(n+1)! = (n+1)n! \geq 2n!.$$

By the induction hypothesis, $n! \geq 2^{n-1}$, so

$$2n! \geq 2 \cdot 2^{n-1} = 2^n.$$

Thus $(n+1)! \geq 2^n$.

Therefore $n! \geq 2^{n-1}$ for every $n \geq 1$. □

18.6 Proving divisibility

Induction is especially useful when a statement changes predictably as n increases.

Theorem 52. *For every $n \geq 1$, the number $n^3 - n$ is divisible by 3.*

Proof. Let $P(n)$ be the statement $3 \mid (n^3 - n)$.

Base case: For $n = 1$,

$$1^3 - 1 = 0,$$

and $3 \mid 0$. So $P(1)$ is true.

Inductive step: Let $n \geq 1$ and assume $3 \mid (n^3 - n)$. We must prove

$$3 \mid ((n+1)^3 - (n+1)).$$

Compute:

$$(n+1)^3 - (n+1) = n^3 + 3n^2 + 3n + 1 - n - 1 = n^3 - n + 3n^2 + 3n.$$

So

$$(n+1)^3 - (n+1) = (n^3 - n) + 3(n^2 + n).$$

The first term is divisible by 3 by the induction hypothesis, and the second term is visibly divisible by 3. Therefore their sum is divisible by 3.

Thus $P(n+1)$ is true, and induction proves the claim. □

This theorem can also be proved without induction by factoring:

$$n^3 - n = n(n-1)(n+1),$$

the product of three consecutive integers. Among three consecutive integers, one is divisible by 3. The two proofs reveal different things. The induction proof reveals stability from n to $n+1$. The factoring proof reveals a structural divisibility reason.

Intuition

When you find a non-inductive proof of a theorem first proved by induction, do not treat one proof as wasted. Different proofs often expose different mechanisms.

18.7 Induction with inequalities

Inequality proofs by induction require care because algebraic transformations may need side conditions such as positivity.

Theorem 53. *For every integer $n \geq 4$,*

$$2^n \geq n^2.$$

Proof. Let $P(n)$ be the statement $2^n \geq n^2$.

Base case: For $n = 4$, $2^4 = 16$ and $4^2 = 16$, so $P(4)$ is true.

Inductive step: Let $n \geq 4$ and assume $2^n \geq n^2$. We must show

$$2^{n+1} \geq (n+1)^2.$$

Starting from the induction hypothesis,

$$2^{n+1} = 2 \cdot 2^n \geq 2n^2.$$

It remains to show

$$2n^2 \geq (n+1)^2.$$

This inequality is equivalent to

$$2n^2 - (n+1)^2 = n^2 - 2n - 1 \geq 0.$$

For $n \geq 4$,

$$n^2 - 2n - 1 = n(n-2) - 1 \geq 4 \cdot 2 - 1 = 7 > 0.$$

Therefore $2n^2 \geq (n+1)^2$, and hence

$$2^{n+1} \geq (n+1)^2.$$

By induction, $2^n \geq n^2$ for all $n \geq 4$. □

Common trap

In inequality induction, do not replace a difficult target by a stronger inequality unless you actually prove the stronger inequality. Keep track of the direction of every inequality.

18.8 Induction with sets

Induction is not only about formulas. It can prove statements about finite sets.

Theorem 54. *If A is a finite set with n elements, then $\mathcal{P}(A)$ has 2^n elements.*

Proof. We prove by induction on n.

Base case: If $n = 0$, then $A = \emptyset$. The only subset of $\emptyset$ is $\emptyset$ itself, so $\mathcal{P}(A)$ has $1 = 2^0$ element.

Inductive step: Assume every set with n elements has 2^n subsets. Let A be a set with $n + 1$ elements. Choose an element $a \in A$, and let $B = A \setminus \{a\}$. Then B has n elements.

Every subset of A falls into exactly one of two types:

- subsets that do not contain a,
- subsets that do contain a.

The subsets that do not contain a are exactly the subsets of B, so there are 2^n of them by the induction hypothesis.

The subsets that do contain a are obtained by taking a subset $C \subseteq B$ and forming $C \cup \{a\}$. This gives one subset of A containing a for each subset of B, so there are also 2^n of them.

Therefore

$$|\mathcal{P}(A)| = 2^n + 2^n = 2^{n+1}.$$

By induction, every finite set with n elements has 2^n subsets. □

This proof is recursive thinking: to understand subsets of a set with $n + 1$ elements, remove one element and use the subsets of the smaller set.

18.9 Induction and algorithms

Algorithm correctness proofs often use induction. A loop invariant is an induction statement in disguise. It says: after n iterations, some property holds.

Suppose an algorithm computes the sum of a list $a_1, \ldots, a_m$:

$$
\begin{aligned}
&s := 0 \\
&\text{for } i = 1 \text{ to } m : \\
&\qquad s := s + a_i
\end{aligned}
$$

The loop invariant is:

$$\text{After } k \text{ iterations, } s = a_1 + \cdots + a_k.$$

Base case: before the loop, $k = 0$ and $s = 0$, the empty sum.

Step: if after k iterations $s = a_1 + \cdots + a_k$, then after the next iteration,

$$s = (a_1 + \cdots + a_k) + a_{k+1}.$$

So the invariant holds for $k + 1$.

At the end, $k = m$, so $s = a_1 + \cdots + a_m$.

Intuition

Induction is the proof version of a loop invariant: establish the initial state, prove preservation, then conclude the final property.

18.10 False induction proofs

Here is a famous false proof:

> Claim: All horses are the same color.

The proposed induction is on the number n of horses in a group. For $n = 1$, all horses in a one-horse group have the same color. Assume every group of n horses has the same color. Given $n + 1$ horses, remove the last horse. The first n have the same color. Then remove the first horse. The last n have the same color. Since these two groups overlap, all $n + 1$ horses have the same color.

The flaw occurs when moving from $n = 1$ to $n = 2$. The two one-horse groups do not overlap. Without overlap, there is no bridge carrying color information from one horse to the other.

Common trap

In an induction proof, the inductive step must work for every allowed n. A step that works only after some later value does not connect the base case to the chain.

18.11 Proof lab: diagnose an induction

Proof lab

A student tries to prove that $n^2 + n + 1$ is even for all $n \geq 1$.
They write: "Base case: for $n = 1$, $1^2 + 1 + 1 = 3$, which is odd. Inductive step: assume $n^2 + n + 1$ is even. Then

$$(n+1)^2 + (n+1) + 1 = n^2 + 3n + 3.$$

This is close to $n^2 + n + 1$, so it is also even."

1. What is wrong with the claim?
2. What is wrong with the base case?
3. What is wrong with the phrase "close to"?
4. Find a true statement similar to this one and prove it.

18.12 Historical connections

History and connection

Inductive reasoning appears implicitly in ancient mathematics, but explicit formulations developed over time. Pascal used an argument recognizable as induction in work related to the arithmetic triangle. Modern axiomatizations of arithmetic, such as the Peano axioms, place induction at the foundation of the natural numbers.

Induction also connects to the philosophical question: what are the natural numbers? One answer is operational. The natural numbers are generated from a starting point by repeatedly applying a successor operation. Induction says that any property containing the start and closed under successor contains all natural numbers.

18.13 Writing induction clearly

A clean induction proof usually includes:

1. The predicate $P(n)$.
2. The base case.
3. The induction hypothesis.
4. The target $P(n+1)$.
5. The argument proving the target.

6. The final sentence invoking induction.

The target matters. Many confused proofs never state what must be proved in the step. Before doing algebra, write the desired statement for $n+1$.

Walking question

While walking, practice saying the skeleton aloud: "I prove $P(n)$ by induction. First $P(m)$. Now assume $P(n)$ for arbitrary $n \geq m$. I must prove $P(n+1)$." The rhythm is part of the discipline.

18.14 Exercises

1. Prove that for all $n \geq 1$,
$$1 + 3 + 5 + \cdots + (2n-1) = n^2.$$
2. Prove that for all $n \geq 1$,
$$2 + 4 + 6 + \cdots + 2n = n(n+1).$$
3. Prove that for all $n \geq 1$,
$$1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$
4. Prove that $5 \mid (6^n - 1)$ for every $n \geq 1$.
5. Prove that $3 \mid (4^n - 1)$ for every $n \geq 1$.
6. Prove that $n < 2^n$ for every $n \geq 1$.
7. Prove that $3^n \geq 2n + 1$ for every $n \geq 0$.
8. Prove that a set with n elements has $n(n-1)/2$ two-element subsets.
9. Let $a_1 = 1$ and $a_{n+1} = 2a_n + 1$. Prove that $a_n = 2^n - 1$ for all $n \geq 1$.
10. Let $F_0 = 0$, $F_1 = 1$, and $F_{n+2} = F_{n+1} + F_n$. Compute the first eight values and guess a parity pattern. Prove your pattern.

18.15 Induction as an invariant machine

A useful way to think about induction is to look for an invariant: something that remains true after each step. The base case establishes the invariant, and the inductive step proves it is preserved.

For example, suppose a process starts with one region in the plane. Each move draws a line segment that starts and ends on existing boundary and splits one existing region into two. After n moves, how many regions are there?

The answer is $n + 1$.

Proof. Let $P(n)$ be the statement: after n valid splitting moves, there are $n + 1$ regions.

Base case: Before any moves, there is $1 = 0 + 1$ region.

Inductive step: Assume after n moves there are $n + 1$ regions. The next valid move splits one existing region into two. All other regions remain unchanged. Therefore the number of regions increases by exactly 1, giving

$$(n + 1) + 1 = n + 2$$

regions after $n + 1$ moves. This is $P(n + 1)$.

By induction, after n moves there are $n + 1$ regions. □

This proof is short because the invariant is well chosen. We did not need to know the geometry of every region. We needed only the effect of one move.

Intuition

Good induction proofs often identify the part of the situation that changes predictably and ignore irrelevant detail.

18.16 Induction and definitions by stages

Many mathematical sets are defined in stages. For example, define a set S of strings over the alphabet $\{a, b\}$ as follows:

1. The empty string ϵ belongs to S.
2. If $w \in S$, then $awb \in S$.

The strings in S are

$$\epsilon,\ ab,\ aabb,\ aaabbb, \ldots$$

Every string in S has the same number of a's and b's, with all the a's first.

To prove this, induction follows the construction stages. The base object ϵ has zero a's and zero b's. If w has equal numbers of a's and b's, then awb has one more of each.

This is the beginning of structural induction. Instead of inducting on a visible number n, we induct on how an object is built.

Proof lab

Let S be the smallest set of strings such that $\epsilon \in S$ and whenever $w \in S$, the string awb belongs to S.

1. Prove every string in S has equal numbers of a's and b's.
2. Prove every nonempty string in S starts with a and ends with b.
3. Is every string with equal numbers of a's and b's in S? Give a counterexample if not.

18.17 When induction is the wrong first tool

Induction is powerful, but not every statement about natural numbers needs it. Sometimes factoring, parity, modular arithmetic, or a direct combinatorial argument is clearer.

For example, the statement

$$n^2 + n \text{ is even}$$

can be proved by induction, but the direct proof

$$n^2 + n = n(n + 1)$$

is better: one of two consecutive integers is even.

Choosing proof methods is part of mathematical taste. Induction is best when the statement has a natural "next case" structure. If the expression already factors into a known pattern, use the pattern.

Walking question

For a theorem about n, ask two questions before choosing induction: What changes from n to $n + 1$? Is there a direct structural reason that avoids stepping through the natural numbers?

19 Strong Induction

Ordinary induction lets you use the immediately previous case. Strong induction lets you use all previous cases. Despite the name, it is not logically stronger in standard arithmetic. It is a different packaging of the same principle, and it is often the natural way to prove statements where the current object breaks into smaller pieces of many possible sizes.

Focus

Use strong induction when the proof of $P(n)$ may need $P(k)$ for several earlier values $k < n$, not only $P(n-1)$.

19.1 The principle

Formal

Principle of strong induction: Let $P(n)$ be a predicate on integers $n \geq m$. If for every $n \geq m$,

$$\big(P(m) \text{ and } P(m+1) \text{ and } \cdots \text{ and } P(n-1)\big) \Rightarrow P(n),$$

then $P(n)$ holds for every $n \geq m$.

Many presentations separate a base case from the step. A common version says:

1. Prove $P(m)$.
2. For arbitrary $n > m$, assume $P(k)$ for every $m \leq k < n$, and prove $P(n)$.

Then $P(n)$ holds for all $n \geq m$.

19.2 Why this is not magic

Strong induction may feel suspicious because, in the step, we assume many cases. But for a fixed n, those are all earlier cases. The proof does not assume the conclusion $P(n)$ itself.

Think of climbing a staircase where each step can be reached from any earlier step, not only the immediately previous one. If you can prove that every step is reachable provided all lower steps are reachable, the staircase is still climbed from the bottom.

19.3 Equivalence with ordinary induction

Strong induction and ordinary induction are equivalent over the natural numbers. One way to see this is to define a new predicate

$$Q(n) : P(m), P(m+1), \ldots, P(n) \text{ are all true.}$$

If ordinary induction proves $Q(n)$ for all n, then $P(n)$ is true for all n.

Thus strong induction is not a new axiom. It is a convenient proof form.

19.4 Prime factorization existence

The classic first use of strong induction is proving that every integer greater than 1 has a prime factorization.

Definition 22. An integer $p > 1$ is *prime* if its only positive divisors are 1 and p. An integer $n > 1$ is *composite* if $n = ab$ for integers a, b with $1 < a < n$ and $1 < b < n$.

Theorem 55. *Every integer $n > 1$ is either prime or can be written as a product of primes.*

Proof. We use strong induction on n.

Let $P(n)$ be the statement: n is prime or n can be written as a product of primes.

Base case: $n = 2$. The number 2 is prime, so $P(2)$ is true.

Strong inductive step: Let $n > 2$ and assume $P(k)$ is true for every integer k with $2 \leq k < n$. We prove $P(n)$.

If n is prime, then $P(n)$ is true.

If n is not prime, then since $n > 1$, it is composite. So

$$n = ab$$

for integers a, b with $1 < a < n$ and $1 < b < n$.

By the strong induction hypothesis, $P(a)$ and $P(b)$ are true. Therefore each of a and b is either prime or a product of primes. Multiplying those prime factorizations gives a product of primes equal to n.

Thus $P(n)$ is true in all cases.

By strong induction, every integer $n > 1$ is prime or a product of primes. □

Ordinary induction is awkward here because if $n = ab$, the factors a and b are smaller than n, but they are not necessarily $n - 1$.

19.5 The role of smaller subproblems

Strong induction fits recursive decomposition:

- A number factors into smaller numbers.
- A polygon can be cut into a triangle plus a smaller polygon.
- A full binary tree can be split into smaller left and right subtrees.
- A string generated by a grammar may contain smaller generated strings.
- An algorithm on input size n may call itself on inputs of size $\lfloor n/2 \rfloor$.

When the proof needs smaller pieces whose sizes vary, strong induction is often the clean form.

19.6 Postage theorem

Theorem 56. *Every integer amount of postage $n \geq 12$ cents can be formed using only 4-cent and 5-cent stamps.*

Proof. Let $P(n)$ be the statement that n cents can be formed using 4-cent and 5-cent stamps.

We prove four base cases:

$$12 = 4 + 4 + 4,$$
$$13 = 4 + 4 + 5,$$
$$14 = 4 + 5 + 5,$$
$$15 = 5 + 5 + 5.$$

So $P(12), P(13), P(14), P(15)$ are true.

Inductive step: Let $n \geq 16$ and assume $P(k)$ is true for every $12 \leq k < n$. We prove $P(n)$.

Since $n \geq 16$, we have $n - 4 \geq 12$. Also $n - 4 < n$. By the strong induction hypothesis, $n - 4$ cents can be formed using 4-cent and 5-cent stamps. Add one more 4-cent stamp to get n cents.

Therefore $P(n)$ is true.

By strong induction, every $n \geq 12$ can be formed using 4-cent and 5-cent stamps. □

This proof shows why multiple base cases are sometimes necessary. The step moves forward by 4, so the base cases must cover all residue classes modulo 4 starting at the threshold.

Common trap

If your inductive step proves $P(n)$ from $P(n - d)$, you usually need enough base cases to bridge the gap. One base case may not start every chain.

19.7 Tiling and recursive structure

Theorem 57. *For every $n \geq 1$, a $2 \times n$ board can be tiled by dominoes in F_{n+1} ways, where $F_1 = 1$, $F_2 = 1$, and $F_{k+2} = F_{k+1} + F_k$.*

Proof. Let $T(n)$ be the number of domino tilings of a $2 \times n$ board. We prove $T(n) = F_{n+1}$.

Base cases: For $n = 1$, there is exactly one tiling, using one vertical domino. Thus $T(1) = 1 = F_2$.

For $n = 2$, there are two tilings: two vertical dominoes, or two horizontal dominoes stacked. Thus $T(2) = 2 = F_3$.

Inductive step: Let $n \geq 3$ and assume $T(k) = F_{k+1}$ for all $1 \leq k < n$. Consider a tiling of a $2 \times n$ board.

Look at the leftmost column. Either it is covered by one vertical domino, leaving a $2 \times (n-1)$ board, or it is covered by two horizontal dominoes extending into the second column, leaving a $2 \times (n-2)$ board.

These two cases are disjoint and exhaustive. Hence

$$T(n) = T(n-1) + T(n-2).$$

By the induction hypothesis,

$$T(n-1) = F_n, \qquad T(n-2) = F_{n-1}.$$

Therefore

$$T(n) = F_n + F_{n-1} = F_{n+1}.$$

By strong induction, the formula holds for every $n \geq 1$. □

The proof is not about algebra first. It is about classifying the first local choice and reducing the rest of the board to smaller boards.

19.8 Complete induction from well-ordering

Strong induction can be justified using the well-ordering principle.

Theorem 58. *Assume every nonempty subset of $\mathbb{N}$ has a least element. Then strong induction holds.*

Proof. Let $P(n)$ be a predicate for $n \geq m$. Suppose that for every $n \geq m$, if $P(k)$ holds for all $m \leq k < n$, then $P(n)$ holds.

Assume, for contradiction, that $P(n)$ fails for at least one $n \geq m$. Let

$$S = \{n \in \mathbb{N} : n \geq m \text{ and } P(n) \text{ is false}\}.$$

By assumption, S is nonempty. By well-ordering, S has a least element; call it s.

Since s is the least counterexample, every integer k with $m \leq k < s$ satisfies $P(k)$. By the induction step hypothesis, $P(s)$ must hold. This contradicts $s \in S$.

Therefore there is no counterexample, and $P(n)$ holds for all $n \geq m$. □

This proof is a model of a common strategy: assume there is a counterexample, choose the least one, and use minimality to make all smaller cases available.

19.9 Minimal counterexample method

The minimal counterexample method is strong induction in contradiction clothing.

To prove $P(n)$ for all $n \geq m$:

1. Suppose the claim is false.
2. Let s be the least counterexample.
3. Use the fact that all smaller cases are true.
4. Derive a contradiction.

This method often feels more natural in number theory.

Theorem 59. *Every integer $n \geq 2$ has a prime divisor.*

Proof. Suppose not. Then there is at least one integer $n \geq 2$ with no prime divisor. Let s be the least such integer.

If s is prime, then s is a prime divisor of itself, contradiction.

If s is composite, then $s = ab$ with $1 < a < s$ and $1 < b < s$. Since $a \geq 2$ and $a < s$, the minimality of s implies that a has a prime divisor p. Since $p \mid a$ and $a \mid s$, we have $p \mid s$. This contradicts the choice of s.

Therefore every integer $n \geq 2$ has a prime divisor. □

19.10 Recursive definitions and strong induction

Some definitions call on earlier values in more than one way. The Fibonacci sequence is the standard example:

$$F_0 = 0, \qquad F_1 = 1, \qquad F_{n+2} = F_{n+1} + F_n.$$

To prove a property of F_{n+2}, one may need both earlier cases F_{n+1} and F_n.

Theorem 60. *For every $n \geq 0$, $F_n \leq 2^n$.*

Proof. We use strong induction.

Base cases: $F_0 = 0 \leq 1 = 2^0$, and $F_1 = 1 \leq 2 = 2^1$.

Inductive step: Let $n \geq 2$ and assume $F_k \leq 2^k$ for all $0 \leq k < n$. Then

$$F_n = F_{n-1} + F_{n-2} \leq 2^{n-1} + 2^{n-2}.$$

Since

$$2^{n-1} + 2^{n-2} = 3 \cdot 2^{n-2} \leq 4 \cdot 2^{n-2} = 2^n,$$

we get $F_n \leq 2^n$.

By strong induction, $F_n \leq 2^n$ for all $n \geq 0$. □

19.11 Common traps in strong induction

Common trap

Do not say "assume true for all n" in a strong induction step. Say "assume true for all k with $m \leq k < n$." The bound matters.

Common trap

Do not use the case $P(n)$ while proving $P(n)$. Strong induction gives previous cases only, not the current case.

Common trap

Check that every smaller value you use lies inside the range where the induction hypothesis applies. If the theorem starts at $n = 12$, you cannot invoke the hypothesis at 8.

19.12 Proof lab: choose induction type

Proof lab

For each problem, decide whether ordinary induction or strong induction is more natural. Then write the induction hypothesis and the target statement.

1. Prove $1 + 2 + \cdots + n = n(n+1)/2$.
2. Prove every integer $n \geq 2$ is prime or a product of primes.
3. Prove every $2 \times n$ board has a number of domino tilings satisfying $T(n) = T(n-1) + T(n-2)$.
4. Prove $7 \mid (8^n - 1)$ for every $n \geq 1$.
5. Prove every integer $n \geq 18$ can be written as $4a+7b$ for some nonnegative integers a, b.

19.13 Historical connections

History and connection

The natural numbers are not merely a list; they are an inductive structure. Peano's axioms isolate this structure using zero, successor, and induction. Strong induction, recursion, and well-ordering are different faces of the same basic architecture.

Modern mathematics extends induction far beyond $\mathbb{N}$. Structural induction proves statements about recursively defined syntax trees. Transfinite induction proves statements over well-ordered sets larger than $\mathbb{N}$. Noetherian induction proves statements in settings where infinite descending chains are impossible.

19.14 Walking prompts

Walking question

Find something built recursively: a folder tree, a family tree, a parsed expression, a proof, a build dependency graph. What are the base objects? What are the construction rules? What would an induction proof about all such objects look like?

19.15 Exercises

1. Prove that every integer $n \geq 2$ is divisible by some prime.

2. Prove that every integer $n \geq 2$ can be written as a product of primes.
3. Prove that every integer $n \geq 8$ can be written as $3a + 5b$ for nonnegative integers a, b.
4. Prove that every integer $n \geq 18$ can be written as $4a + 7b$ for nonnegative integers a, b.
5. Let $a_0 = 1$, $a_1 = 2$, and $a_n = 3a_{n-1} + 2a_{n-2}$ for $n \geq 2$. Prove $a_n \leq 5^n$ for all $n \geq 0$.
6. Prove that a convex polygon with $n \geq 3$ sides can be triangulated into $n - 2$ triangles.
7. Prove that every full binary tree has an odd number of vertices.
8. Use the minimal counterexample method to prove that there is no smallest positive rational number.
9. Explain why the postage theorem with 4-cent and 5-cent stamps needs four base cases in the proof given above.
10. Give an example of a theorem where ordinary induction is possible but strong induction is cleaner.

19.16 Structural induction preview

Strong induction still uses natural numbers as the measuring device. Structural induction goes one step further: it proves statements about all objects generated by rules.

Consider fully parenthesized expressions built from variables using a binary operation symbol $*$:

1. Every variable is an expression.
2. If E and F are expressions, then $(E * F)$ is an expression.

Examples:

$$x, \qquad (x * y), \qquad ((x * y) * z), \qquad (x * (y * z)).$$

Suppose we want to prove that every expression has one more variable occurrence than operation symbols. Let $v(E)$ be the number of variable occurrences in E, and let $o(E)$ be the number of operation symbols in E. The claim is

$$v(E) = o(E) + 1.$$

Proof. We prove the claim by structural induction on expressions.

Base case: If E is a variable, then $v(E) = 1$ and $o(E) = 0$, so $v(E) = o(E) + 1$.

Recursive step: Suppose $E = (F * G)$, where F and G are expressions satisfying the claim. Thus

$$v(F) = o(F) + 1, \qquad v(G) = o(G) + 1.$$

For $E = (F * G)$, we have

$$v(E) = v(F) + v(G)$$

and

$$o(E) = o(F) + o(G) + 1.$$

Therefore

$$v(E) = v(F) + v(G) = (o(F) + 1) + (o(G) + 1) = o(F) + o(G) + 2.$$

But

$$o(E) + 1 = (o(F) + o(G) + 1) + 1 = o(F) + o(G) + 2.$$

Hence $v(E) = o(E) + 1$.

By structural induction, the claim holds for every expression. □

This proof has the same spirit as strong induction: assume the result for smaller pieces and prove it for the object built from those pieces. The difference is that the construction rules, not just the integer order, guide the proof.

19.17 Noetherian induction

A deeper generalization appears when a structure has no infinite descending chains. Suppose a relation $\prec$ means "is simpler than." If there is no endless sequence

$$x_0 \succ x_1 \succ x_2 \succ \cdots,$$

then one can often prove a statement for every object by assuming it for all simpler objects.

This is called Noetherian induction. It appears in algebra, rewriting systems, termination proofs, and algebraic geometry.

History and connection

Emmy Noether's work transformed abstract algebra by emphasizing structural conditions such as chain conditions on ideals. Noetherian arguments often say: a process cannot descend forever, so a minimal counterexample would have to exist, and then it can be contradicted.

You do not need Noetherian induction yet, but it is useful to know the direction of travel. Ordinary induction, strong induction, structural induction, and minimal counterexample arguments are part of one family: prove everything by showing that failure cannot first appear.

Proof lab

Let E be the set of expressions built from variables and a binary operation $*$ as above.

1. Prove that every expression has balanced parentheses.
2. Prove that every expression with at least one operation symbol starts with (and ends with).
3. Define the depth of an expression recursively and prove that every expression has finite depth.

20 Countability

Countability is where infinity becomes a precise subject. The central surprise is that infinite sets can have different sizes. Some infinities can be listed, at least in principle. Others cannot.

This chapter begins with a simple question:

> What does it mean for two sets to have the same number of elements?

For finite sets, we count. For infinite sets, counting one by one may never end, so we compare sets by functions.

Focus

Two sets have the same size when their elements can be paired perfectly: every element on each side is used exactly once.

20.1 Same size means bijection

Definition 23. Sets A and B are *equinumerous*, or have the same cardinality, if there exists a bijection $f : A \to B$.

A bijection is a perfect pairing:

- injective means no two elements of A are paired with the same element of B,
- surjective means every element of B is paired with something from A.

For finite sets, this agrees with ordinary counting. For infinite sets, it becomes surprising.

20.2 The even natural numbers

Let

$$E = \{0, 2, 4, 6, \ldots\}.$$

The function $f : \mathbb{N} \to E$ defined by

$$f(n) = 2n$$

is a bijection. Therefore $\mathbb{N}$ and the even natural numbers have the same cardinality.

This does not mean $E = \mathbb{N}$. It means the elements can be paired:

$$0 \leftrightarrow 0, \quad 1 \leftrightarrow 2, \quad 2 \leftrightarrow 4, \quad 3 \leftrightarrow 6, \ldots$$

Intuition

Infinite sets can be the same size as proper subsets of themselves. This is one reason infinity cannot be treated like a very large finite number.

20.3 Countably infinite

Definition 24. A set A is *countably infinite* if there exists a bijection $\mathbb{N} \to A$.

Equivalently, A is countably infinite if its elements can be listed as

$$a_0, a_1, a_2, a_3, \ldots$$

with every element of A appearing exactly once.

A set is *countable* if it is finite or countably infinite.

Common trap

"Countable" does not mean "small" in an everyday sense. A countably infinite set is infinite. It means listable by natural numbers.

20.4 The integers are countable

The integers extend in both directions:

$$\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots$$

At first they look bigger than $\mathbb{N}$. But they can be listed:

$$0, 1, -1, 2, -2, 3, -3, \ldots$$

One bijection $f : \mathbb{N} \to \mathbb{Z}$ is

$$f(n) = \begin{cases} n/2, & \text{if } n \text{ is even,} \\ -(n+1)/2, & \text{if } n \text{ is odd.} \end{cases}$$

Then

$$f(0) = 0, \quad f(1) = -1, \quad f(2) = 1, \quad f(3) = -2, \quad f(4) = 2.$$

This lists every integer exactly once.

Theorem 61. *The set $\mathbb{Z}$ is countably infinite.*

Proof. Define $f : \mathbb{N} \to \mathbb{Z}$ by

$$f(n) = \begin{cases} n/2, & \text{if } n \text{ is even,} \\ -(n+1)/2, & \text{if } n \text{ is odd.} \end{cases}$$

We show f is bijective.

Surjective: Let $z \in \mathbb{Z}$. If $z \geq 0$, then $z = f(2z)$. If $z < 0$, then $z = f(-2z-1)$, since $-2z-1$ is a nonnegative odd integer and

$$f(-2z-1) = -\frac{(-2z-1)+1}{2} = z.$$

Thus every integer is hit.

Injective: The even inputs map to nonnegative integers and the odd inputs map to negative integers, so an even input and an odd input cannot have the same output. Among even inputs, $f(2a) = a$, so equality of outputs gives equality of inputs. Among odd inputs, $f(2a+1) = -(a+1)$, so equality of outputs also gives equality of inputs.

Therefore f is a bijection, and $\mathbb{Z}$ is countably infinite. □

20.5 The Cartesian product $\mathbb{N} \times \mathbb{N}$

Pairs of natural numbers can be listed by diagonals:

$$(0,0);$$
$$(0,1),(1,0);$$
$$(0,2),(1,1),(2,0);$$
$$(0,3),(1,2),(2,1),(3,0);$$

and so on.

Each diagonal consists of pairs (a,b) with fixed sum $a+b=s$. Since there are finitely many pairs on each diagonal, and every pair lies on exactly one diagonal, this gives a listing of $\mathbb{N}\times\mathbb{N}$.

Theorem 62. *The set $\mathbb{N} \times \mathbb{N}$ is countably infinite.*

Proof. We describe a listing. For each $s \in \mathbb{N}$, list all pairs (a,b) with $a+b=s$:

$$(0,s),(1,s-1),\ldots,(s,0).$$

Every pair $(a,b) \in \mathbb{N} \times \mathbb{N}$ appears on the diagonal $s = a+b$. It appears only there, because the sum $a+b$ is fixed.

Thus the diagonal procedure lists every pair exactly once. Therefore $\mathbb{N} \times \mathbb{N}$ is countably infinite. □

A fully explicit formula is possible, using triangular numbers, but the listing proof is often clearer at first.

20.6 Countable unions

Theorem 63. *A countable union of countable sets is countable.*

Proof. Let

$$A = \bigcup_{n \in \mathbb{N}} A_n$$

where each A_n is countable. For each n, list the elements of A_n as

$$a_{n,0}, a_{n,1}, a_{n,2}, \ldots$$

allowing repetitions if A_n is finite.

Arrange these in an infinite table:

$$\begin{array}{cccc} a_{0,0} & a_{0,1} & a_{0,2} & \cdots \\ a_{1,0} & a_{1,1} & a_{1,2} & \cdots \\ a_{2,0} & a_{2,1} & a_{2,2} & \cdots \\ \vdots & \vdots & \vdots & \end{array}$$

Now list entries by diagonals:

$$a_{0,0};$$

$$a_{0,1}, a_{1,0};$$

$$a_{0,2}, a_{1,1}, a_{2,0};$$

and so on.

This procedure reaches every table position. Therefore every element of every A_n appears somewhere in the list. If duplicates occur, skip any element that has appeared before. The remaining list contains every element of A at least once and no element more than once.

Thus A is countable. □

Common trap

When listing a union, duplicates are not fatal. A list with repetitions can be cleaned by keeping the first occurrence of each element.

20.7 The rational numbers are countable

The rationals seem dense and endless. Between any two rational numbers there is another rational number. Yet $\mathbb{Q}$ is countable.

Each rational number can be written as a/b with $a \in \mathbb{Z}$, $b \in \mathbb{N}^+$. So $\mathbb{Q}$ is closely related to $\mathbb{Z} \times \mathbb{N}^+$. Since $\mathbb{Z}$ and $\mathbb{N}^+$ are countable, their product is countable, and the rationals are obtained from that product by identifying different fractions that represent the same number.

Theorem 64. *The set $\mathbb{Q}$ is countable.*

Proof. Consider the set

$$S = \mathbb{Z} \times \mathbb{N}^+.$$

Since $\mathbb{Z}$ and $\mathbb{N}^+$ are countable, S is countable. Define a function $f : S \to \mathbb{Q}$ by

$$f(a, b) = \frac{a}{b}.$$

This function is surjective, because every rational number can be written as a/b with $a \in \mathbb{Z}$ and $b \in \mathbb{N}^+$.

Since S is countable and f maps S onto $\mathbb{Q}$, the set $\mathbb{Q}$ is countable. □

Another proof lists fractions in a grid and skips repeats:

$$0, \quad 1, -1, \quad \frac{1}{2}, -\frac{1}{2}, 2, -2, \quad \frac{1}{3}, -\frac{1}{3}, \frac{2}{3}, -\frac{2}{3}, 3, -3, \ldots$$

There are many possible listings. The point is not the particular order. The point is that a listing exists.

Intuition

Density and countability are different ideas. The rationals are dense in the real line, but they are still countable.

20.8 Cantor's shift

Georg Cantor changed mathematics by treating infinite size as an object of rigorous study. Before Cantor, infinity often appeared as a potential process: keep going, never stop. Cantor studied completed infinite sets and compared their sizes.

The decisive idea was not just that some infinite sets are countable. It was that some are not.

History and connection

Cantor developed set theory in the late nineteenth century while studying trigonometric series and sets of exceptional points. His work was controversial, partly because it forced mathematicians to accept a hierarchy of infinite sizes rather than a single vague infinity.

20.9 The real numbers are uncountable

The most famous proof is Cantor's diagonal argument. We prove that the interval $(0, 1)$ is not countable. Since $(0, 1) \subseteq \mathbb{R}$, this implies $\mathbb{R}$ is uncountable.

Theorem 65. *The interval* $(0,1)$ *is uncountable.*

Proof. Suppose, for contradiction, that $(0,1)$ is countable. Then its elements can be listed:

$$x_1, x_2, x_3, \dots$$

Write each x_i in decimal form:

$$x_1 = 0.d_{11}d_{12}d_{13}\dots$$
$$x_2 = 0.d_{21}d_{22}d_{23}\dots$$
$$x_3 = 0.d_{31}d_{32}d_{33}\dots$$

and so on.

To avoid ambiguity from decimals such as $0.4999\ldots = 0.5000\ldots$, choose decimal expansions that do not end in an infinite string of 9s.

Now construct a new number

$$y = 0.c_1c_2c_3\dots$$

as follows. For each i, choose c_i different from d_{ii} and also not equal to 9. For instance, set

$$c_i = \begin{cases} 1, & \text{if } d_{ii} \neq 1, \\ 2, & \text{if } d_{ii} = 1. \end{cases}$$

Then $y \in (0,1)$.

But y differs from x_1 in the first decimal place, from x_2 in the second decimal place, from x_3 in the third decimal place, and in general from x_i in the ith decimal place. Therefore $y \neq x_i$ for every i.

This contradicts the claim that the list contains every element of $(0,1)$. Therefore $(0,1)$ is uncountable. □

The proof does not merely say we failed to find a list. It shows that every proposed list misses a specific number constructed from the list itself.

Common trap

The diagonal argument is not a complaint about decimal notation. Decimal notation is only a convenient coding. The core idea is: given any alleged list, build an object that differs from the ith listed object in its ith coordinate.

20.10 Binary sequences and power sets

The diagonal idea becomes cleaner for infinite binary sequences.

Let

$$\{0,1\}^{\mathbb{N}}$$

be the set of all infinite sequences of zeros and ones:

$$(a_0, a_1, a_2, \ldots).$$

Theorem 66. *The set $\{0,1\}^{\mathbb{N}}$ is uncountable.*

Proof. Suppose, for contradiction, that all binary sequences can be listed:

$$s_0, s_1, s_2, \ldots$$

where each

$$s_i = (s_i(0), s_i(1), s_i(2), \ldots).$$

Construct a new sequence t by

$$t(i) = \begin{cases} 1, & \text{if } s_i(i) = 0, \\ 0, & \text{if } s_i(i) = 1. \end{cases}$$

Then t differs from s_i at coordinate i. Hence $t \neq s_i$ for every i.

So the list did not contain every binary sequence, contradiction. Therefore $\{0,1\}^{\mathbb{N}}$ is uncountable. □

Every subset $A \subseteq \mathbb{N}$ corresponds to a binary sequence called its characteristic function:

$$\chi_A(n) = \begin{cases} 1, & \text{if } n \in A, \\ 0, & \text{if } n \notin A. \end{cases}$$

Thus $\mathcal{P}(\mathbb{N})$ has the same size as $\{0,1\}^{\mathbb{N}}$ and is uncountable.

20.11 Cantor's theorem

The power set of any set is strictly larger than the set itself.

Theorem 67 (Cantor)**.** *For every set A, there is no surjection from A onto $\mathcal{P}(A)$.*

Proof. Let $f : A \to \mathcal{P}(A)$ be any function. We show f is not surjective.

Define

$$D = \{a \in A : a \notin f(a)\}.$$

Since $D \subseteq A$, we have $D \in \mathcal{P}(A)$.

If f were surjective, there would be some $d \in A$ such that

$$f(d) = D.$$

Now ask whether $d \in D$.

By definition of D,

$$d \in D \quad \text{if and only if} \quad d \notin f(d).$$

Since $f(d) = D$, this becomes

$$d \in D \quad \text{if and only if} \quad d \notin D.$$

This is impossible.

Therefore no such d exists, so f is not surjective. Since f was arbitrary, there is no surjection from A onto $\mathcal{P}(A)$. □

This proof is the abstract diagonal argument. The set D is constructed to differ from each proposed value $f(a)$ at the element a.

History and connection

Cantor's theorem implies an endless hierarchy:

$$|A| < |\mathcal{P}(A)| < |\mathcal{P}(\mathcal{P}(A))| < \cdots .$$

There is no largest infinity. This result is one reason set theory became a foundation for modern mathematics.

20.12 Countability and algebraic numbers

A real number is algebraic if it is a root of a nonzero polynomial with integer coefficients. Examples include rational numbers, $\sqrt{2}$, and complex-looking expressions such as $\sqrt[3]{5} - 7$.

Theorem 68. *The set of algebraic real numbers is countable.*

Proof. For each degree $n \geq 0$, consider polynomials

$$a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0$$

with integer coefficients, not all zero. The set of finite coefficient lists

$$(a_0, a_1, \ldots, a_n) \in \mathbb{Z}^{n+1}$$

is countable because it is a finite product of countable sets.

The union over all degrees n is a countable union of countable sets, so the set of integer polynomials is countable.

Each nonzero polynomial of degree n has at most n real roots. Therefore the set of all real roots of all nonzero integer polynomials is a countable union of finite sets, hence countable.

Thus the algebraic real numbers are countable. □

Since $\mathbb{R}$ is uncountable, most real numbers are not algebraic. They are transcendental. This is strange at first: the numbers we can easily name are rare inside the real line.

20.13 Common countability traps

Common trap

Do not confuse "between any two elements there is another" with uncountability. The rationals have this density property, but they are countable.

Common trap

Do not think a two-dimensional grid is automatically uncountable. The grid $\mathbb{N} \times \mathbb{N}$ is countable because it can be swept by diagonals.

Common trap

Do not try to prove uncountability by saying "the list would be infinitely long." Countable lists are infinitely long. To prove uncountability, show that every proposed list misses something.

20.14 Proof lab: diagonal thinking

Proof lab

Suppose someone claims to have listed all infinite sequences of positive integers:

$$s_1, s_2, s_3, \dots$$

where

$$s_i = (s_i(1), s_i(2), s_i(3), \dots).$$

Construct a sequence t of positive integers such that $t \neq s_i$ for every i.
Questions:

1. What should $t(i)$ depend on?
2. How do you guarantee $t(i) \neq s_i(i)$?
3. Why does differing at one coordinate prove two sequences are different?
4. Does your construction need decimal notation?

20.15 Proof lab: countable constructions

Proof lab

Prove that each set is countable.

1. The set of all finite strings over the alphabet $\{0, 1\}$.
2. The set of all finite subsets of $\mathbb{N}$.
3. The set of all polynomials with rational coefficients.
4. The set of all rational points in the plane, $\mathbb{Q} \times \mathbb{Q}$.

Hint: Organize by length, maximum element, degree, or diagonal enumeration.

20.16 Walking prompts

Walking question

Imagine someone hands you an alleged complete list of all objects of a certain kind. Can you build a new object by making it disagree with the ith object at the ith test? If yes, you may have a diagonal argument.

20.17 Exercises

1. Prove that the set of odd natural numbers is countably infinite.
2. Prove that $\mathbb{N}^+$ and $\mathbb{N}$ have the same cardinality.
3. Prove that every infinite subset of $\mathbb{N}$ is countably infinite.
4. Prove that $\mathbb{Z} \times \mathbb{Z}$ is countable.
5. Prove that $\mathbb{Q} \times \mathbb{Q}$ is countable.
6. Prove that the set of all finite binary strings is countable.
7. Prove that the set of all finite sequences of natural numbers is countable.
8. Prove that the set of all finite subsets of $\mathbb{N}$ is countable.
9. Use Cantor's theorem to prove that $\mathcal{P}(\mathbb{N})$ is uncountable.
10. Prove directly by diagonalization that the set of functions $f : \mathbb{N} \to \mathbb{N}$ is uncountable.
11. Explain why the diagonal proof for $(0, 1)$ must handle decimal expansions such as $0.4999\ldots = 0.5000\ldots$.
12. Prove that if A is countable and $B \subseteq A$, then B is countable.

13. Prove that if there is a surjection from a countable set onto B, then B is countable.
14. Prove that if there is an injection from A into a countable set, then A is countable.
15. Explain why the set of transcendental real numbers is uncountable.

20.18 The continuum and the next question

Cantor's diagonal argument proves that

$$|\mathbb{N}| < |\mathbb{R}|.$$

It is natural to ask whether there is a set whose size is strictly between these two:

$$|\mathbb{N}| < |A| < |\mathbb{R}|.$$

The statement that there is no such intermediate size is called the continuum hypothesis.

The answer is one of the great surprises of modern logic. Within the usual Zermelo-Fraenkel axioms of set theory with the axiom of choice, abbreviated ZFC, the continuum hypothesis can neither be proved nor disproved if those axioms are consistent. Goedel proved in 1940 that CH cannot be disproved from ZFC, assuming ZFC is consistent. Cohen proved in 1963 that CH cannot be proved from ZFC, again assuming consistency.

History and connection

The continuum hypothesis was the first problem on Hilbert's famous 1900 list. Its later independence from ZFC showed that even precise foundational axioms can leave natural questions undecided.

For a beginning proof reader, the lesson is not to master independence results now. The lesson is that countability opens a door into foundational mathematics. Once infinite size is made precise, new questions become possible, and some of them test the limits of the axioms themselves.

20.19 Comparing sizes without explicit bijections

Sometimes constructing a bijection directly is difficult. It is often easier to build injections in both directions.

Theorem 69 (Schroeder-Bernstein)**.** *If there is an injection $f : A \to B$ and an injection $g : B \to A$, then there is a bijection between A and B.*

We will not prove the theorem here, but we will use its message: two sets have the same cardinality if each can be faithfully embedded into the other.

For example, it is clear that $(0,1) \subseteq \mathbb{R}$, so there is an injection

$$(0,1) \to \mathbb{R}.$$

There is also an injection $\mathbb{R} \to (0,1)$, such as

$$f(x) = \frac{1}{\pi}\arctan(x) + \frac{1}{2}.$$

Thus $\mathbb{R}$ and $(0,1)$ have the same cardinality.

This can feel strange: a bounded interval and the whole real line have the same number of points. But it is the same kind of phenomenon as $\mathbb{N}$ and the even natural numbers, now at an uncountable scale.

Intuition

Cardinality ignores geometry. Length, boundedness, density, and measure are different notions from size by bijection.

20.20 Measure versus cardinality

The rationals are countable, but they are dense in $\mathbb{R}$. The interval $(0,1)$ is uncountable, but it has finite length. The Cantor set is uncountable, yet it has length zero in the usual sense of measure.

These facts point toward a major theme in analysis: different theories measure different features.

- Cardinality asks whether there is a bijection.
- Order asks how elements compare.
- Topology asks about closeness and continuity.
- Measure asks about length, area, volume, and size in an analytic sense.
- Algebra asks which operations and equations are preserved.

One object can be small in one sense and large in another. The rationals are small by cardinality compared with the reals, but topologically they are everywhere in the real line. The irrationals are also everywhere. Density is not cardinality.

Common trap

Do not let one meaning of "size" silently replace another. A set can be countable and dense, uncountable and measure zero, bounded and equinumerous with an unbounded set.

20.21 A final diagonal exercise

Proof lab

Let $\mathcal{L}$ be the set of all infinite lists of natural numbers. An element of $\mathcal{L}$ has the form

$$(a_0, a_1, a_2, \ldots).$$

Prove that $\mathcal{L}$ is uncountable.
Then compare your proof with the proof that $\mathcal{P}(\mathbb{N})$ is uncountable.

1. What plays the role of the diagonal entry?
2. How do you alter the diagonal entry?
3. Why does the constructed list differ from every listed list?
4. Where is the contradiction?

20.22 Closing perspective

Relations teach us to compare and identify. Orders teach us to compare with direction. Induction teaches us to reason through the natural numbers by successor. Strong induction teaches us to reason through smaller subproblems. Countability teaches us that infinity has structure.

These are not isolated topics. Equivalence relations create quotient objects. Orders organize dependencies and bounds. Induction is powered by the well-ordering of $\mathbb{N}$. Countability depends on bijections, products, quotients, and diagonal relations between lists and objects.

Focus

The recurring move is structural: stop asking only what the objects are, and ask how they relate, how they are generated, how they are compared, and how they can or cannot be listed.

21 Preview Chapter: Real Analysis

Real analysis is the place where calculus is asked to tell the truth. In calculus, you learn to differentiate, integrate, approximate, optimize, and solve equations. In real analysis, you ask why those operations work, when they fail, and which assumptions make the familiar theorems valid.

The subject is not merely "hard calculus." It is a training ground for precise thinking about limits, functions, infinity, and approximation. The early proof concepts you have practiced already appear everywhere: quantifiers in epsilon definitions, set language in intervals and

neighborhoods, implication in continuity, contradiction in irrationality and completeness, and careful examples in every theorem.

Focus

Real analysis teaches you to separate three things that beginners often blend together: what a picture suggests, what a formula calculates, and what a proof actually guarantees.

21.1 Why analysis exists

Calculus was developed before its foundations were fully settled. Newton and Leibniz computed with infinitesimal quantities. Euler manipulated infinite series with astonishing power. Physicists and engineers obtained correct answers long before anyone had a fully rigorous account of limits.

That history is not embarrassing. It is normal. Mathematics often advances first by powerful intuition and later by careful repair. But the repair matters. Without it, statements such as

$$\lim_{x \to a} f(x) = L$$

can become gestures rather than claims.

Consider the statement:

$$\lim_{x \to 2}(3x + 1) = 7.$$

In calculus, this feels obvious. As x gets close to 2, the expression $3x + 1$ gets close to 7. Analysis asks: what does "gets close" mean? How close is close enough? Who chooses the error tolerance? What must be shown after that choice?

History and connection

The nineteenth century brought a major tightening of calculus. Cauchy, Bolzano, Weierstrass, Dedekind, Cantor, and others worked to replace geometric and infinitesimal intuition with definitions precise enough to support proofs. The epsilon language used today is part of that repair project.

21.2 The real numbers as a promise

Analysis studies functions on $\mathbb{R}$, so the nature of $\mathbb{R}$ matters. The real numbers are not just decimals. Decimals are representations. The real numbers are a number system with algebra, order, and a crucial completeness property.

Formal

The real numbers form an ordered field. This means:

- addition and multiplication satisfy the usual field laws,
- every nonzero real number has a multiplicative inverse,
- the order relation $<$ behaves compatibly with addition and multiplication,
- positive numbers multiply to positive numbers.

Those properties are already enough to prove many familiar inequalities. But they are not enough to distinguish $\mathbb{R}$ from $\mathbb{Q}$. The rational numbers are also an ordered field. What makes $\mathbb{R}$ special is completeness.

Definition 25. A number M is an *upper bound* for a set $A \subseteq \mathbb{R}$ if

$$\forall a \in A, \quad a \leq M.$$

A set is *bounded above* if it has at least one upper bound.

Definition 26. A number s is the *least upper bound*, or *supremum*, of $A \subseteq \mathbb{R}$ if:

1. s is an upper bound for A,
2. if M is any upper bound for A, then $s \leq M$.

We write $s = \sup A$.

Formal

The *least upper bound property* says: every nonempty subset of $\mathbb{R}$ that is bounded above has a supremum in $\mathbb{R}$.

This is a compact sentence with a lot inside it. It begins with a set, assumes the set is nonempty and bounded above, and concludes the existence of a special real number. The conclusion is existential. A proof using it often says: "Since A is nonempty and bounded above, let $s = \sup A$."

Intuition

The rational numbers have gaps. The set

$$A = \{q \in \mathbb{Q} : q^2 < 2\}$$

is nonempty and bounded above inside $\mathbb{Q}$, but it has no rational least upper bound. The real numbers are what you get when those gaps are filled.

Completeness is the hidden engine behind much of calculus. It explains why bounded increasing sequences converge, why continuous functions on closed intervals attain maximum values, and why Cauchy sequences of real numbers have limits.

21.3 Absolute value and inequality as proof tools

Before limits, analysis needs fluency with distance. On the real line, distance between x and y is measured by $|x - y|$.

Definition 27. For $x \in \mathbb{R}$,

$$|x| = \begin{cases} x, & x \geq 0, \\ -x, & x < 0. \end{cases}$$

Absolute value turns algebra into geometry. The inequality

$$|x - a| < r$$

means that x lies within distance r of a. Equivalently,

$$a - r < x < a + r.$$

Theorem 70. *For all real numbers x and y,*

$$|x + y| \leq |x| + |y|.$$

Proof. Since $-|x| \leq x \leq |x|$ and $-|y| \leq y \leq |y|$, adding the two inequalities gives

$$-(|x| + |y|) \leq x + y \leq |x| + |y|.$$

Therefore $|x + y| \leq |x| + |y|$. □

The triangle inequality is not just a trick. It is the basic move for controlling error. If one error is at most 0.01 and another is at most 0.03, the combined error is at most 0.04. Analysis turns that everyday idea into a proof machine.

Proof lab

Prove that

$$\big||x| - |y|\big| \leq |x - y|$$

for all $x, y \in \mathbb{R}$. Hint: write $x = y + (x - y)$ and use the triangle inequality, then reverse the roles of x and y.

21.4 Sequences and the first epsilon definition

A sequence is a function whose domain is usually $\mathbb{N}$. Instead of writing $a(n)$, we write a_n. A sequence can approach a number even if it never equals that number.

Definition 28. A sequence (a_n) of real numbers *converges* to $L \in \mathbb{R}$ if

$$\forall \varepsilon > 0,\ \exists N \in \mathbb{N},\ \forall n \in \mathbb{N}, \quad n \geq N \Rightarrow |a_n - L| < \varepsilon.$$

In that case we write $a_n \to L$ or $\lim_{n\to\infty} a_n = L$.

This definition is the first serious quantifier workout in analysis. Read it slowly:

- An error tolerance $\varepsilon > 0$ is given.
- You must find a threshold N.
- After that threshold, every term of the sequence must lie within ε of L.

The order matters. You do not get to choose ε. Your proof must work for every positive ε, even absurdly small ones. But after ε is given, you may choose N depending on ε.

Intuition

Convergence does not mean the terms stop moving. It means that every requested error band around L eventually contains the whole tail of the sequence.

21.5 Worked example: proving a sequence converges

Claim:

$$\lim_{n\to\infty} \frac{1}{n} = 0.$$

Proof. Let $\varepsilon > 0$ be arbitrary. We need to find $N \in \mathbb{N}$ such that for every $n \geq N$,

$$\left|\frac{1}{n} - 0\right| < \varepsilon.$$

Choose $N \in \mathbb{N}$ with $N > 1/\varepsilon$. This is possible by the Archimedean property. If $n \geq N$, then

$$0 < \frac{1}{n} \leq \frac{1}{N} < \varepsilon.$$

Thus $|1/n - 0| < \varepsilon$. Since $\varepsilon > 0$ was arbitrary, $1/n \to 0$. □

Notice the proof architecture. The definition told us exactly what to do. Start with arbitrary $\varepsilon > 0$. Produce N. Verify the implication for all $n \geq N$. That is not decoration; it is the structure of the proof.

Common trap

Do not write "as n gets large, $1/n$ gets small" as if that were a proof. That sentence is a good intuition. The epsilon proof is where the intuition becomes checkable.

21.6 Limit laws and what they really say

Limit laws let us combine known convergences. For example, if $a_n \to a$ and $b_n \to b$, then $a_n + b_n \to a + b$. This sounds obvious, but the proof uses the triangle inequality and the idea of splitting an error budget.

Theorem 71. *If $a_n \to a$ and $b_n \to b$, then $a_n + b_n \to a + b$.*

Proof. Let $\varepsilon > 0$ be arbitrary. Since $a_n \to a$, there exists $N_1 \in \mathbb{N}$ such that for all $n \geq N_1$,

$$|a_n - a| < \varepsilon/2.$$

Since $b_n \to b$, there exists $N_2 \in \mathbb{N}$ such that for all $n \geq N_2$,

$$|b_n - b| < \varepsilon/2.$$

Let $N = \max\{N_1, N_2\}$. If $n \geq N$, then both inequalities hold, and

$$|(a_n + b_n) - (a + b)| \leq |a_n - a| + |b_n - b| < \varepsilon/2 + \varepsilon/2 = \varepsilon.$$

Therefore $a_n + b_n \to a + b$. □

This proof is a model for many later arguments. You have two requirements, so you satisfy each with half the budget, then take a maximum threshold so both conditions hold at once.

Walking question

Where else in life do you split a tolerance budget? Think about engineering design, numerical computation, scheduling, measurement, or software latency. Analysis makes that style of reasoning exact.

21.7 Monotone convergence

Some convergence theorems do not tell you the limit explicitly. Instead, they guarantee that a limit exists.

Definition 29. A sequence (a_n) is *increasing* if $a_n \leq a_{n+1}$ for all $n \in \mathbb{N}$. It is *bounded above* if there exists $M \in \mathbb{R}$ such that $a_n \leq M$ for all $n \in \mathbb{N}$.

Theorem 72. *Every increasing sequence of real numbers that is bounded above converges.*

Proof. Let $A = \{a_n : n \in \mathbb{N}\}$. Since the sequence is bounded above, A is bounded above. Since A is nonempty, the least upper bound property gives $s = \sup A$.

We prove $a_n \to s$. Let $\varepsilon > 0$. Since s is the least upper bound of A, $s - \varepsilon$ cannot be an upper bound for A. Therefore there exists $N \in \mathbb{N}$ such that

$$a_N > s - \varepsilon.$$

Because the sequence is increasing, for every $n \geq N$,

$$s - \varepsilon < a_N \leq a_n \leq s.$$

Thus $|a_n - s| < \varepsilon$ for all $n \geq N$. Hence $a_n \to s$. □

This theorem is a major moment because completeness enters directly. The supremum exists because we are in $\mathbb{R}$. Over $\mathbb{Q}$, a bounded increasing rational sequence can approach an irrational gap.

21.8 Functions and epsilon-delta continuity

Sequences prepare you for function limits. The definition of continuity at a point says: if the input is sufficiently close to a, then the output is as close as desired to $f(a)$.

Definition 30. Let $f : D \to \mathbb{R}$ with $D \subseteq \mathbb{R}$, and let $a \in D$. The function f is *continuous at* a if

$$\forall \varepsilon > 0,\ \exists \delta > 0,\ \forall x \in D, \quad |x - a| < \delta \Rightarrow |f(x) - f(a)| < \varepsilon.$$

The quantifier pattern is similar to sequence convergence:

$$\forall \varepsilon\ \exists \delta\ \forall x.$$

The tolerance is chosen first. Then you choose a radius in the domain. Then you must control every input within that radius.

Example 1. Prove that $f(x) = 3x + 1$ is continuous at $a = 2$.

Solution. Let $\varepsilon > 0$. We need $\delta > 0$ such that $|x - 2| < \delta$ implies $|(3x + 1) - 7| < \varepsilon$. But

$$|(3x + 1) - 7| = |3x - 6| = 3|x - 2|.$$

Choose $\delta = \varepsilon/3$. If $|x - 2| < \delta$, then

$$|(3x + 1) - 7| = 3|x - 2| < 3\delta = \varepsilon.$$

Therefore f is continuous at 2.

Proof lab

Rewrite the previous worked example as a formal proof without the words "we need." Then prove that $f(x) = 5x - 4$ is continuous at an arbitrary $a \in \mathbb{R}$.

21.9 A corrected worked example in ordinary prose

The previous computation can be written in a conventional proof style as follows.

Proof. Let $\varepsilon > 0$ be arbitrary. Set $\delta = \varepsilon/3$. Suppose $x \in \mathbb{R}$ and $|x - 2| < \delta$. Then

$$|(3x + 1) - 7| = 3|x - 2| < 3\delta = \varepsilon.$$

Therefore $f(x) = 3x + 1$ is continuous at 2. □

The proof is short because the function is linear. For nonlinear functions, choosing δ may require more care.

21.10 Worked example: continuity of x^2 at a point

Let $f(x) = x^2$. We prove f is continuous at $a = 3$.

Proof. Let $\varepsilon > 0$. We want $|x^2 - 9| < \varepsilon$ when x is close enough to 3. Factor:

$$|x^2 - 9| = |x - 3||x + 3|.$$

The factor $|x - 3|$ is under our control. The factor $|x + 3|$ must be bounded near 3. If we require $|x - 3| < 1$, then $2 < x < 4$, so $5 < x + 3 < 7$ and $|x + 3| < 7$.

Choose

$$\delta = \min\{1, \varepsilon/7\}.$$

If $|x - 3| < \delta$, then $|x - 3| < 1$ and $|x - 3| < \varepsilon/7$. Hence

$$|x^2 - 9| = |x - 3||x + 3| < (\varepsilon/7) \cdot 7 = \varepsilon.$$

Therefore x^2 is continuous at 3. □

This proof introduces a common analysis pattern: first force the variable into a manageable local region, then use that region to bound a troublesome factor.

Common trap

A common mistake is to choose $\delta = \varepsilon/|x + 3|$. This is illegal because δ must be chosen before the arbitrary x is considered. Your δ may depend on ε and the fixed point a, but not on the later input x.

21.11 Open and closed sets on the line

Analysis naturally leads to topology. Even on $\mathbb{R}$, it is useful to name sets according to whether they contain room around their points or contain their boundary behavior.

Definition 31. A set $U \subseteq \mathbb{R}$ is *open* if for every $x \in U$, there exists $r > 0$ such that

$$(x - r, x + r) \subseteq U.$$

Definition 32. A set $F \subseteq \mathbb{R}$ is *closed* if whenever a sequence (x_n) in F converges to $x \in \mathbb{R}$, the limit x also belongs to F.

The interval $(0, 1)$ is open. Every point inside has some wiggle room. The interval $[0, 1]$ is closed. Any convergent sequence of points in $[0, 1]$ has a limit that remains in $[0, 1]$.

Intuition

Open sets protect local motion. Closed sets protect limits.

The set $(0, 1)$ is not closed because $x_n = 1/(n + 1)$ lies in $(0, 1)$ and converges to 0, but $0 \notin (0, 1)$. The set $[0, 1]$ is not open because the endpoint 0 has no full interval around it contained in $[0, 1]$.

21.12 Compactness as controlled infinity

Compactness is one of the central ideas of analysis and topology. It is a way of saying that a set is small enough, or finite enough in behavior, that infinite processes can be controlled.

In $\mathbb{R}$, compactness has a simple characterization.

Theorem 73 (Heine-Borel theorem in $\mathbb{R}$)**.** *A subset of $\mathbb{R}$ is compact if and only if it is closed and bounded.*

The full proof belongs in a first analysis course, but the consequences are immediate and powerful.

Theorem 74 (Extreme value theorem)**.** *If $f : [a, b] \to \mathbb{R}$ is continuous, then there exist $x_{\min}, x_{\max} \in [a, b]$ such that*

$$f(x_{\min}) \leq f(x) \leq f(x_{\max})$$

for all $x \in [a, b]$.

This theorem is used whenever optimization is meaningful. If a cost function is continuous on a closed bounded design space, then a minimum exists. It may be hard to find, but existence itself is valuable.

Common trap

The hypotheses are not cosmetic. On $(0, 1)$, the function $f(x) = x$ has no maximum. On $\mathbb{R}$, the function $f(x) = x^2$ has no maximum. On $[-1, 1]$, the function $f(x) = 1/x$ is not even defined at 0 unless the domain is changed.

21.13 Cauchy sequences and internal convergence

The definition of convergence names the limit first. A sequence converges to L if its terms eventually stay close to L. But sometimes the sequence itself shows that it is settling down before we know what the limit is.

Definition 33. A sequence (a_n) in $\mathbb{R}$ is *Cauchy* if

$$\forall \varepsilon > 0,\ \exists N \in \mathbb{N},\ \forall m, n \in \mathbb{N}, \quad m, n \geq N \Rightarrow |a_m - a_n| < \varepsilon.$$

Compare this with convergence:

$$|a_n - L| < \varepsilon.$$

Cauchy replaces closeness to a known limit with closeness among sufficiently late terms. The sequence's tail becomes internally stable.

Theorem 75. *Every convergent sequence in $\mathbb{R}$ is Cauchy.*

Proof. Suppose $a_n \to L$. Let $\varepsilon > 0$. Since $a_n \to L$, there exists $N \in \mathbb{N}$ such that if $n \geq N$, then

$$|a_n - L| < \varepsilon/2.$$

If $m, n \geq N$, then

$$|a_m - a_n| \leq |a_m - L| + |L - a_n| < \varepsilon/2 + \varepsilon/2 = \varepsilon.$$

Therefore (a_n) is Cauchy. □

The converse is where completeness enters.

Theorem 76. *Every Cauchy sequence of real numbers converges to a real number.*

This theorem is false in $\mathbb{Q}$. A rational sequence can be Cauchy while approaching $\sqrt{2}$, which is not rational. Thus Cauchy convergence is another way to feel the difference between $\mathbb{Q}$ and $\mathbb{R}$.

Intuition

A Cauchy sequence is a sequence whose tail has stopped spreading out. In a complete space, every such internally stable tail has a point it is converging to.

21.14 Pointwise versus uniform control

Continuity at a point allows the δ to depend on the point. Uniform continuity asks for one δ that works across the whole domain.

Definition 34. Let $D \subseteq \mathbb{R}$. A function $f : D \to \mathbb{R}$ is *uniformly continuous* on D if

$$\forall \varepsilon > 0,\ \exists \delta > 0,\ \forall x, y \in D, \quad |x - y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon.$$

Compare this with ordinary continuity on D:

$$\forall a \in D,\ \forall \varepsilon > 0,\ \exists \delta > 0,\ \forall x \in D, \quad |x - a| < \delta \Rightarrow |f(x) - f(a)| < \varepsilon.$$

The quantifier order changed. In uniform continuity, δ is chosen before the particular points x, y are considered. In ordinary continuity, δ may depend on the point a.

Example 2. The function $f(x) = x^2$ is continuous on $\mathbb{R}$, but not uniformly continuous on $\mathbb{R}$. Far from the origin, small changes in x can produce large changes in x^2.

Example 3. The same function $f(x) = x^2$ is uniformly continuous on $[0, 1]$. On a compact interval, continuous functions are uniformly continuous.

Common trap

Many beginners think uniform continuity means "very continuous." It does not. It means the same input tolerance works uniformly across the domain.

21.15 Series as limits of partial sums

An infinite series is not an infinite addition performed all at once. It is a limit of finite sums.

Definition 35. Given a sequence (a_n), the series

$$\sum_{n=1}^{\infty} a_n$$

converges to S if the sequence of partial sums

$$s_N = \sum_{n=1}^{N} a_n$$

converges to S.

Example 4. The geometric series satisfies

$$\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}$$

when $|r| < 1$. This means the partial sums

$$1 + r + r^2 + \cdots + r^N$$

converge to $1/(1-r)$.

Series are a source of both power and danger. Finite sums can be rearranged freely. Infinite series require conditions. Absolutely convergent series behave well under rearrangement; conditionally convergent series can behave surprisingly.

Definition 36. A series $\sum a_n$ is *absolutely convergent* if

$$\sum |a_n|$$

converges.

Focus

Whenever an infinite expression appears, ask what finite process is being limited. Analysis is often the discipline of replacing informal infinity with a controlled limiting definition.

21.16 Differentiation and integration, previewed rigorously

Derivatives and integrals are familiar from calculus, but analysis reframes them as theorems about limits.

Definition 37. The derivative of $f : \mathbb{R} \to \mathbb{R}$ at a is

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h},$$

provided this limit exists.

This definition contains a function limit. The derivative is not the symbolic rule; the symbolic rule is a theorem or computational shortcut derived from the definition.

Example 5. For $f(x) = x^2$,

$$\frac{f(a+h) - f(a)}{h} = \frac{(a+h)^2 - a^2}{h} = 2a + h.$$

As $h \to 0$, this tends to $2a$. Hence $f'(a) = 2a$.

Integration can also be made rigorous by limits, for example through Riemann sums. The integral of a bounded function on $[a, b]$ is defined by comparing upper and lower sums over partitions. Later, measure theory extends integration far beyond Riemann's setting.

History and connection

The development from Riemann integration to Lebesgue integration was not mere technical polishing. Lebesgue's theory made it possible to handle limits of functions, Fourier series, probability, and modern analysis with much better convergence theorems.

21.17 The intermediate value theorem

The intermediate value theorem says that continuous functions on intervals do not jump over values.

Theorem 77. *If $f : [a, b] \to \mathbb{R}$ is continuous and y lies between $f(a)$ and $f(b)$, then there exists $c \in [a, b]$ such that $f(c) = y$.*

This theorem is the rigorous version of a familiar graph intuition. If a continuous graph starts below the horizontal line $y = 0$ and ends above it, then it must cross the line.

Example 6. Let $f(x) = x^3 - x - 1$. Then $f(1) = -1$ and $f(2) = 5$. By the intermediate value theorem, there exists $c \in (1, 2)$ such that $c^3 - c - 1 = 0$.

This is an existence proof. It tells us a root exists without giving its exact value. Numerical methods can approximate the root; analysis explains why the search is legitimate.

21.18 Common proof habits in analysis

Analysis rewards a small set of repeated moves:

- Start with arbitrary $\varepsilon > 0$.
- Unpack the definition before trying to be clever.
- Use triangle inequalities to split errors.
- Use min to satisfy several smallness requirements.
- Use max to satisfy several eventual requirements.
- Convert local assumptions such as $|x - a| < 1$ into useful bounds.
- Watch quantifier order carefully.

Focus

In analysis, the proof often begins by letting the definition set the agenda. If you do not know what to do next, rewrite the definition of the object you are trying to prove.

21.19 Applications without losing rigor

Real analysis is not detached from applied mathematics. It supports:

- numerical approximation and error bounds,
- existence of solutions to equations,
- optimization over closed bounded domains,
- convergence of algorithms,
- Fourier analysis and signal processing,
- probability theory through measure and integration,
- differential equations through fixed point and compactness methods.

The applied use is not always immediate. A theorem about compactness may not look like engineering. But when it guarantees that a minimizing configuration exists, or that a sequence of approximations has a convergent subsequence, it becomes a structural tool.

21.20 Proof lab: the squeeze theorem

The squeeze theorem is a good beginner theorem because its proof is almost a translation of order into convergence.

Theorem 78 (Squeeze theorem for sequences)**.** *Suppose* (a_n), (b_n), *and* (c_n) *are real sequences such that*

$$a_n \le b_n \le c_n$$

for all sufficiently large n. *If* $a_n \to L$ *and* $c_n \to L$, *then* $b_n \to L$.

Proof. Let $\varepsilon > 0$. Since $a_n \to L$, there exists N_1 such that for all $n \ge N_1$,

$$L - \varepsilon < a_n < L + \varepsilon.$$

Since $c_n \to L$, there exists N_2 such that for all $n \ge N_2$,

$$L - \varepsilon < c_n < L + \varepsilon.$$

Also, there exists N_0 such that $a_n \le b_n \le c_n$ for all $n \ge N_0$. Let

$$N = \max\{N_0, N_1, N_2\}.$$

If $n \ge N$, then

$$L - \varepsilon < a_n \le b_n \le c_n < L + \varepsilon.$$

Thus $|b_n - L| < \varepsilon$. Therefore $b_n \to L$. □

Example 7. Since

$$-\frac{1}{n} \le \frac{\sin n}{n} \le \frac{1}{n}$$

and both $-1/n$ and $1/n$ converge to 0, the squeeze theorem gives

$$\frac{\sin n}{n} \to 0.$$

The proof shows several analysis habits at once. The inequalities $a_n \le b_n \le c_n$ only need to hold eventually, so the proof introduces an N_0. The two convergence hypotheses give N_1 and N_2. The maximum combines all three requirements. This is the same "satisfy several tail conditions at once" pattern seen in limit laws.

Proof lab

Use the squeeze theorem to prove that

$$\frac{\cos(n^2)}{\sqrt{n}} \to 0.$$

Then write a direct epsilon proof for the same result and compare the two arguments.

21.21 Trap lab: false converses in analysis

Analysis has many theorems whose converses are false. Beginners often remember a theorem as a vague equivalence and then use it backward.

Example 8. Every convergent sequence is bounded. The converse is false: $a_n = (-1)^n$ is bounded but does not converge.

Example 9. Every differentiable function is continuous. The converse is false: $f(x) = |x|$ is continuous at 0 but not differentiable at 0.

Example 10. Every uniformly continuous function is continuous. The converse is false on general domains: $f(x) = x^2$ is continuous on $\mathbb{R}$ but not uniformly continuous on $\mathbb{R}$.

Focus

When you learn a theorem, immediately ask whether the converse is true. If it is false, build a counterexample and keep it close.

21.22 Exercises for real analysis

Exercise 28. Prove that if $a_n \to a$, then the sequence (a_n) is bounded. Hint: use $\varepsilon = 1$ to control the tail, and handle the finite initial segment separately.

Exercise 29. Prove that $\lim_{n\to\infty}(2/n + 5) = 5$ directly from the definition.

Exercise 30. Prove that $f(x) = 2x - 1$ is continuous at an arbitrary point $a \in \mathbb{R}$.

Exercise 31. Give an example of a bounded sequence that does not converge. Explain why it does not violate the monotone convergence theorem.

Exercise 32. For each set, decide whether it is open, closed, both, or neither as a subset of $\mathbb{R}$: $(0, 1)$, $[0, 1]$, $\mathbb{R}$, $\emptyset$, $\mathbb{Q}$.

Exercise 33. Prove directly that every convergent sequence is Cauchy. Then identify exactly where the triangle inequality enters.

Exercise 34. Show that $f(x) = 3x + 1$ is uniformly continuous on $\mathbb{R}$.

Exercise 35. Use the geometric series formula to compute

$$\sum_{n=1}^{\infty} \frac{1}{2^n}.$$

Then explain what sequence is actually converging.

Walking question

Take the phrase “eventually always.” Which definitions in analysis use this idea? How is it different from “infinitely often”?

22 Preview Chapter: Abstract Algebra

Abstract algebra studies operations and the structures they create. Arithmetic, symmetry, polynomial equations, matrices, modular clocks, permutations, and geometric transformations all contain operations. Algebra asks what can be proved from the operation laws alone.

This field is often the first place where a student feels that mathematics has become abstract in a new way. In analysis, the objects are often numbers and functions on numbers. In algebra, the objects may be rotations, residue classes, matrices, permutations, polynomials, or functions. The same definition may cover all of them.

Focus

Abstract algebra is not the study of meaningless symbols. It is the study of what remains true when the same operation pattern appears in many different places.

22.1 From solving equations to studying structure

Historically, algebra grew from solving equations. Linear and quadratic equations were understood in antiquity. Cubic and quartic equations were solved in the Renaissance. Then mathematicians discovered a boundary: there is no general formula using radicals for every polynomial equation of degree five.

That fact is not merely a failure of technique. It led to a new viewpoint. The possibility of solving an equation is related to symmetries among its roots. This idea, associated with Galois, helped turn algebra from a collection of equation-solving tricks into the study of structure.

History and connection

Evariste Galois died at age twenty after developing ideas that later reshaped algebra. His work connected polynomial equations to groups of symmetries. Modern abstract algebra still carries that shift: understand the transformations that preserve structure, and the original problem becomes clearer.

22.2 Binary operations

The basic ingredient of algebra is an operation.

Definition 38. A *binary operation* on a set S is a function

$$* : S \times S \to S.$$

For $a, b \in S$, the output is usually written $a * b$.

The phrase "on a set" matters. A binary operation on S must take two elements of S and return another element of S. This is called closure.

Example 11. Addition is a binary operation on $\mathbb{Z}$, because the sum of two integers is an integer. Division is not a binary operation on $\mathbb{Z}$, because $1/2$ is not an integer and division by zero is not defined.

Common trap

Do not check operation laws before checking that the operation is actually closed on the set. Closure is often the first place an attempted example fails.

22.3 Groups

A group is a set with one operation satisfying four rules.

Definition 39. A *group* is a set G together with a binary operation $*$ such that:

1. **Closure:** for all $a, b \in G$, $a * b \in G$.
2. **Associativity:** for all $a, b, c \in G$, $(a * b) * c = a * (b * c)$.
3. **Identity:** there exists $e \in G$ such that for all $a \in G$, $e * a = a$ and $a * e = a$.
4. **Inverses:** for every $a \in G$, there exists $a^{-1} \in G$ such that $a * a^{-1} = e$ and $a^{-1} * a = e$.

This definition is a proof-writing playground. Each property has quantifiers. Each example must be checked against every condition.

Example 12. The integers $\mathbb{Z}$ form a group under addition. The identity is 0, and the inverse of n is $-n$.

Example 13. The nonzero real numbers $\mathbb{R} \setminus \{0\}$ form a group under multiplication. The identity is 1, and the inverse of x is $1/x$.

Example 14. The integers $\mathbb{Z}$ do not form a group under multiplication. The identity is 1, but most integers do not have multiplicative inverses in $\mathbb{Z}$.

Intuition

A group is a universe of reversible actions. You can combine actions, do nothing, and undo every action.

22.4 Symmetry as a group

Consider the symmetries of a square. You may rotate the square by 0°, 90°, 180°, or 270°. You may also reflect it across horizontal, vertical, and diagonal axes. If you perform one symmetry and then another, the result is still a symmetry of the square.

The set of all symmetries of a square forms a group under composition. This group is usually called D_4 or D_8, depending on convention. The important point is not the name. The important point is that group theory captures the logic of symmetry.

This example also shows that a group operation need not be commutative. If $a * b = b * a$ for all $a, b \in G$, the group is called abelian. Square symmetries are not abelian: rotating and then reflecting may differ from reflecting and then rotating.

Common trap

Associative does not mean commutative. Associativity controls parentheses: $(ab)c = a(bc)$. Commutativity controls order: $ab = ba$. These are different properties.

22.5 Modular arithmetic

Modular arithmetic is one of the best bridges from elementary number theory to abstract algebra. In clock arithmetic, 15 hours after midnight is the same position as 3 hours after midnight. We write

$$15 \equiv 3 \pmod{12}.$$

Definition 40. For integers a, b and positive integer n, we say

$$a \equiv b \pmod{n}$$

if n divides $a - b$.

The congruence classes modulo n are

$$[0], [1], \ldots, [n-1].$$

Addition modulo n turns these classes into a group. The inverse of $[a]$ is $[-a]$.

Example 15. In $\mathbb{Z}/5\mathbb{Z}$, we have

$$[3] + [4] = [7] = [2].$$

The additive inverse of $[3]$ is $[2]$, since $[3] + [2] = [5] = [0]$.

Modular multiplication is more subtle. The nonzero classes modulo a prime p form a group under multiplication. But modulo a composite number, some nonzero classes fail to have inverses.

Example 16. Modulo 6, the class $[2]$ has no multiplicative inverse. If $[2][x] = [1]$, then $2x \equiv 1 \pmod{6}$, impossible because $2x$ is even while numbers congruent to 1 modulo 6 are odd.

22.6 Subgroups

Groups often contain smaller groups.

Definition 41. Let G be a group. A subset $H \subseteq G$ is a *subgroup* if H is a group under the operation inherited from G.

Checking the full group definition every time is possible but inefficient. The following criterion is a common beginner tool.

Theorem 79 (Subgroup test)**.** *Let G be a group and let $H \subseteq G$ be nonempty. If for all $a, b \in H$, $ab^{-1} \in H$, then H is a subgroup of G.*

Proof. Since H is nonempty, choose $h \in H$. Taking $a = h$ and $b = h$, the condition gives $hh^{-1} = e \in H$. Now if $b \in H$, taking $a = e$ gives $eb^{-1} = b^{-1} \in H$. Finally, if $a, c \in H$, then $c^{-1} \in H$, so applying the condition to a and c^{-1} gives $a(c^{-1})^{-1} = ac \in H$. Thus H contains the identity, inverses, and products. Associativity is inherited from G, so H is a subgroup. □

This proof is a good example of using a compressed hypothesis several times with different substitutions. That is common in algebra.

Proof lab

Let $G = \mathbb{Z}$ under addition. Prove that $2\mathbb{Z} = \{2k : k \in \mathbb{Z}\}$ is a subgroup. Then prove that $\mathbb{N}$ is not a subgroup of $\mathbb{Z}$ under addition.

22.7 Homomorphisms

Structure-preserving maps are central in every later field. In algebra, they are called homomorphisms.

Definition 42. Let $(G, *)$ and $(H, \diamond)$ be groups. A function $\varphi : G \to H$ is a *group homomorphism* if for all $a, b \in G$,

$$\varphi(a * b) = \varphi(a) \diamond \varphi(b).$$

A homomorphism respects the operation. It does not merely send elements from one set to another; it sends products to products.

Example 17. The function $\varphi : \mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$ given by $\varphi(k) = [k]$ is a homomorphism from $(\mathbb{Z}, +)$ to $(\mathbb{Z}/n\mathbb{Z}, +)$ because

$$\varphi(a + b) = [a + b] = [a] + [b] = \varphi(a) + \varphi(b).$$

Definition 43. The *kernel* of a group homomorphism $\varphi : G \to H$ is

$$\ker \varphi = \{g \in G : \varphi(g) = e_H\}.$$

The kernel is the set of elements that the map sends to the identity. Informally, it is what the map cannot see.

Example 18. For $\varphi : \mathbb{Z} \to \mathbb{Z}/5\mathbb{Z}$ given by $\varphi(k) = [k]$, the kernel is

$$\ker \varphi = 5\mathbb{Z} = \{5m : m \in \mathbb{Z}\}.$$

Focus

Homomorphisms are as important as the objects themselves. A field becomes mature when it understands not only its objects, but also the maps that preserve the structure of those objects.

22.8 Cosets and quotient groups

Quotients are one of the first genuinely new abstractions in algebra. The idea is to treat elements as equivalent when they differ by something from a chosen subgroup.

Let G be a group and H a subgroup. A left coset of H is a set

$$gH = \{gh : h \in H\}.$$

In $\mathbb{Z}$ under addition, if $H = 3\mathbb{Z}$, then the cosets are

$$0 + 3\mathbb{Z}, \qquad 1 + 3\mathbb{Z}, \qquad 2 + 3\mathbb{Z}.$$

These are the congruence classes modulo 3.

Intuition

A quotient forgets distinctions inside a chosen subgroup. Modulo 3, the numbers $1, 4, 7, 10$ are treated as the same because their differences lie in $3\mathbb{Z}$.

Not every subgroup gives a quotient group in the noncommutative case. For that, one needs a normal subgroup. In abelian groups, every subgroup is normal, so this technical complication disappears.

Definition 44. A subgroup N of a group G is *normal* if

$$gNg^{-1} = N$$

for every $g \in G$.

If N is normal, the cosets form a group G/N. This construction appears everywhere: modular arithmetic, vector spaces modulo subspaces, quotient topologies, and coordinate rings in algebraic geometry.

22.9 Lagrange's theorem

Finite groups have arithmetic constraints.

Theorem 80 (Lagrange's theorem)**.** *If G is a finite group and H is a subgroup of G, then $|H|$ divides $|G|$.*

Proof. The left cosets of H partition G. Each coset has the same number of elements as H, because the map $H \to gH$ given by $h \mapsto gh$ is a bijection. Since G is a disjoint union of cosets, $|G|$ equals the number of cosets times $|H|$. Therefore $|H|$ divides $|G|$. □

This proof uses earlier proof concepts directly: equivalence classes, partitions, functions, bijections, and counting.

22.10 Permutations

A permutation is a rearrangement. More formally, it is a bijection from a set to itself.

Definition 45. The *symmetric group* on a set X is the group of all bijections $X \to X$ under composition. When $X = \{1, \dots, n\}$, this group is written S_n.

Example 19. In S_3, let σ send $1 \mapsto 2$, $2 \mapsto 3$, and $3 \mapsto 1$. Then applying σ three times returns every element to itself, so $\sigma^3 = e$.

Permutations are important because every finite group can be represented as a group of permutations. This is Cayley's theorem.

Theorem 81 (Cayley's theorem, preview)**.** *Every group is isomorphic to a subgroup of a symmetric group.*

The proof idea is elegant. A group G acts on itself by left multiplication. For each $g \in G$, the map

$$L_g : G \to G, \qquad L_g(x) = gx$$

is a bijection. Thus each group element gives a permutation of the underlying set of G.

Intuition

An abstract group may look like a table of symbols, but Cayley's theorem says its elements can always be understood as actions that rearrange the group itself.

22.11 Group actions

Group actions make the idea of symmetry precise.

Definition 46. Let G be a group and let X be a set. A *group action* of G on X assigns to each $g \in G$ a function $X \to X$, written $x \mapsto g \cdot x$, such that:

1. $e \cdot x = x$ for every $x \in X$,
2. $(gh) \cdot x = g \cdot (h \cdot x)$ for every $g, h \in G$ and $x \in X$.

Example 20. The rotation group of a square acts on the four vertices of the square. Each rotation permutes the vertices.

Example 21. The group $\mathbb{Z}$ acts on the real line by translations:

$$n \cdot x = x + n.$$

Definition 47. The *orbit* of $x \in X$ under a group action is

$$G \cdot x = \{g \cdot x : g \in G\}.$$

The *stabilizer* of x is

$$\mathrm{Stab}(x) = \{g \in G : g \cdot x = x\}.$$

The orbit tells where x can move. The stabilizer tells which symmetries leave x fixed. This pair of ideas is used constantly in geometry, combinatorics, number theory, and physics.

Proof lab

Let the symmetries of a square act on its four vertices. Pick one vertex. List its orbit and describe its stabilizer.

22.12 The first isomorphism theorem

The first isomorphism theorem is one of the most important organizing results in algebra. It says that a homomorphism factors through the quotient by its kernel.

Theorem 82 (First isomorphism theorem for groups)**.** *If $\varphi : G \to H$ is a group homomorphism, then*

$$G/\ker\varphi \cong \operatorname{im}\varphi.$$

The slogan is:

> quotient by what the map cannot see, and what remains is exactly what the map does see.

Example 22. Let $\varphi : \mathbb{Z} \to \mathbb{Z}/5\mathbb{Z}$ be $\varphi(n) = [n]$. The kernel is $5\mathbb{Z}$, and the image is all of $\mathbb{Z}/5\mathbb{Z}$. The theorem says

$$\mathbb{Z}/5\mathbb{Z} \cong \mathbb{Z}/5\mathbb{Z},$$

which sounds trivial here, but the mechanism is the point:

$$\mathbb{Z}/\ker\varphi = \mathbb{Z}/5\mathbb{Z}.$$

Focus

The first isomorphism theorem is a mature version of a familiar idea: two inputs become indistinguishable exactly when their difference lies in the kernel.

22.13 Rings

Groups describe one operation. Rings describe two operations, usually called addition and multiplication.

Definition 48. A *ring* is a set R with operations $+$ and $\cdot$ such that:

- $(R, +)$ is an abelian group,
- multiplication is associative,
- multiplication distributes over addition:

$$a(b+c) = ab + ac, \qquad (a+b)c = ac + bc.$$

Some texts also require a multiplicative identity 1; when needed, we say "ring with identity."

Example 23. The integers $\mathbb{Z}$, rational numbers $\mathbb{Q}$, real numbers $\mathbb{R}$, complex numbers $\mathbb{C}$, and polynomial ring $\mathbb{R}[x]$ are rings with identity.

Example 24. The set $2\mathbb{Z}$ is a ring under ordinary addition and multiplication, but it does not have a multiplicative identity inside $2\mathbb{Z}$.

Rings are the natural algebraic home for polynomial arithmetic and modular arithmetic. They are also the algebraic language behind algebraic geometry.

22.14 Ideals

Ideals are to rings what normal subgroups are to groups: they are the subsets that allow quotient structures.

Definition 49. Let R be a commutative ring. An *ideal* $I \subseteq R$ is a subset such that:

1. $0 \in I$,
2. if $a, b \in I$, then $a - b \in I$,
3. if $r \in R$ and $a \in I$, then $ra \in I$.

The third condition is the new one. An ideal absorbs multiplication by arbitrary ring elements.

Example 25. In $\mathbb{Z}$, the set $n\mathbb{Z} = \{nk : k \in \mathbb{Z}\}$ is an ideal. If a is a multiple of n and r is any integer, then ra is still a multiple of n.

Example 26. In $\mathbb{R}[x]$, the set (x^2+1) of all polynomial multiples of x^2+1 is an ideal.

Intuition

An ideal is a controlled notion of "negligible difference." In the quotient R/I, two elements become equal if their difference lies in I.

22.15 Polynomial rings and irreducibility

Polynomial rings are a main bridge from algebra to geometry.

Definition 50. Let F be a field. A nonconstant polynomial $p(x) \in F[x]$ is *irreducible* over F if it cannot be written as

$$p(x) = a(x)b(x)$$

where both $a(x)$ and $b(x)$ have positive degree.

Example 27. The polynomial $x^2 - 2$ is irreducible over $\mathbb{Q}$ but reducible over $\mathbb{R}$, since

$$x^2 - 2 = (x - \sqrt{2})(x + \sqrt{2})$$

in $\mathbb{R}[x]$.

Example 28. The polynomial $x^2 + 1$ is irreducible over $\mathbb{R}$ but reducible over $\mathbb{C}$, since

$$x^2 + 1 = (x - i)(x + i).$$

Irreducibility depends on the field. This is a recurring lesson: algebraic properties are not attached only to formulas; they are attached to formulas together with their allowed coefficients.

Common trap

Do not say a polynomial is irreducible without naming the coefficient field. The same expression can factor in one field and not in another.

22.16 Constructing fields by quotient rings

Quotient rings can create new number systems. The complex numbers can be built from real polynomials by forcing $x^2 + 1$ to become zero:

$$\mathbb{C} \cong \mathbb{R}[x]/(x^2 + 1).$$

In this quotient, the class of x behaves like i, because

$$x^2 + 1 = 0$$

inside the quotient.

Similarly, finite fields can be constructed from polynomial quotients. For example, over $\mathbb{Z}/2\mathbb{Z}$, the polynomial $x^2 + x + 1$ is irreducible. The quotient

$$(\mathbb{Z}/2\mathbb{Z})[x]/(x^2 + x + 1)$$

is a field with four elements.

Intuition

A quotient ring can be read as a world where a chosen equation is declared true. The quotient by $(x^2 + 1)$ is a world where $x^2 = -1$.

22.17 Fields

Fields are rings where division by nonzero elements is possible.

Definition 51. A *field* is a commutative ring with identity $1 \neq 0$ in which every nonzero element has a multiplicative inverse.

Example 29. $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ are fields. The integers $\mathbb{Z}$ are not a field because 2 has no multiplicative inverse in $\mathbb{Z}$.

Example 30. If p is prime, then $\mathbb{Z}/p\mathbb{Z}$ is a field. If n is composite, then $\mathbb{Z}/n\mathbb{Z}$ is not a field.

Fields are central because they are the scalars for linear algebra. Vector spaces are usually built over fields, and polynomial equations are often studied over fields.

22.18 A beginner example: units modulo n

The units modulo n are the residue classes with multiplicative inverses. They form a group under multiplication.

Definition 52. The group of units modulo n is

$$(\mathbb{Z}/n\mathbb{Z})^\times = \{[a] \in \mathbb{Z}/n\mathbb{Z} : \gcd(a, n) = 1\}.$$

Example 31. Modulo 8, the units are $[1], [3], [5], [7]$. Each is its own inverse:

$$[3]^2 = [9] = [1], \quad [5]^2 = [25] = [1], \quad [7]^2 = [49] = [1].$$

This group appears in elementary cryptography and number theory. The RSA cryptosystem, for instance, relies on arithmetic in unit groups modulo large composite numbers, though the security story requires much more detail.

22.19 Common proof habits in algebra

Algebra proofs often begin by unpacking definitions and checking structure. Common moves include:

- prove closure before using an operation freely,
- identify the identity element explicitly,
- construct inverses explicitly,
- show a map preserves operations,
- compute kernels and images,
- use equivalence classes to form quotients,
- prove two algebraic objects are the same by building an isomorphism.

Definition 53. An *isomorphism* is a bijective homomorphism. If two groups or rings are isomorphic, they have the same structure, even if their elements are named differently.

Focus

An isomorphism is a mathematical renaming that preserves all relevant structure. When algebra says two objects are "the same," it usually means isomorphic, not literally equal.

22.20 Applications of abstract algebra

Abstract algebra supports:

- symmetry in geometry, chemistry, and physics,
- cryptography and secure communication,
- error-correcting codes,
- polynomial factorization and exact computation,
- representation theory,
- algebraic topology,
- algebraic geometry,
- computer algebra systems.

The applied route is sometimes direct, as in finite fields for coding theory. Sometimes it is conceptual, as when group actions organize a complicated family of symmetric cases.

22.21 Proof lab: cyclic groups and orders

A cyclic group is generated by repeatedly applying one element.

Definition 54. Let G be a group and $g \in G$. The *subgroup generated by* g is

$$\langle g \rangle = \{g^n : n \in \mathbb{Z}\}.$$

If $G = \langle g \rangle$ for some $g \in G$, then G is called *cyclic*.

Example 32. The group $\mathbb{Z}$ under addition is cyclic, generated by 1 and also by -1. In additive notation,

$$\langle 1 \rangle = \{n \cdot 1 : n \in \mathbb{Z}\} = \mathbb{Z}.$$

Example 33. The additive group $\mathbb{Z}/6\mathbb{Z}$ is cyclic, generated by $[1]$ and by $[5]$. It is not generated by $[2]$, because repeatedly adding $[2]$ gives only $[0], [2], [4]$.

Definition 55. The *order* of an element $g \in G$ is the least positive integer n such that $g^n = e$, if such an integer exists. If no such integer exists, g has infinite order.

Theorem 83. *If G is finite and $g \in G$, then the order of g divides $|G|$.*

Proof. The set $\langle g \rangle$ is a subgroup of G. Its size is the order of g. By Lagrange's theorem, $|\langle g \rangle|$ divides $|G|$. Therefore the order of g divides $|G|$. □

This is a typical algebra proof: define a useful substructure, apply a general theorem, translate the result back into the language of the problem.

Proof lab

Find the order of each element of the additive group $\mathbb{Z}/8\mathbb{Z}$. Which elements generate the whole group?

22.22 Algebra trap lab: notation changes meaning

Algebra reuses symbols in ways that require context.

In a multiplicative group, g^3 means ggg. In an additive group, the same idea is written $3g = g + g + g$. In a ring, 0 is the additive identity and 1 is the multiplicative identity. In a quotient ring, x may mean the class of the polynomial x, not merely a variable waiting for a value.

Common trap

Before manipulating symbols, identify the structure. Are you in a group, ring, field, vector space, quotient, polynomial ring, or function space? The legal moves depend on the structure.

Example 34. In $\mathbb{Z}/6\mathbb{Z}$, the equation

$$[2][x] = [0]$$

has solutions $[0]$ and $[3]$. You cannot cancel $[2]$ because $[2]$ is not a unit modulo 6.

This is a serious difference from field algebra. In a field, $ab = 0$ implies $a = 0$ or $b = 0$. In a general ring, zero divisors may exist.

Definition 56. In a ring R, a nonzero element a is a *zero divisor* if there exists nonzero $b \in R$ such that $ab = 0$.

Example 35. In $\mathbb{Z}/6\mathbb{Z}$, $[2]$ and $[3]$ are zero divisors because

$$[2][3] = [6] = [0].$$

22.23 Exercises for abstract algebra

Exercise 36. For each set and operation, decide whether it is a group: $\mathbb{Z}$ under addition, $\mathbb{N}$ under addition, $\mathbb{R}$ under multiplication, $\mathbb{R} \setminus \{0\}$ under multiplication, and the set of even integers under addition.

Exercise 37. Prove that the identity element of a group is unique.

Exercise 38. Prove that inverses in a group are unique.

Exercise 39. Let $\varphi : \mathbb{Z} \to \mathbb{Z}$ be given by $\varphi(n) = 2n$. Is φ a group homomorphism under addition? Is it injective? Is it surjective?

Exercise 40. Find all units modulo 10 and write their multiplication table.

Exercise 41. Show that $3\mathbb{Z}$ is an ideal of $\mathbb{Z}$. Then explain why $\{0, 3, 6, 9, \ldots\}$ is not an ideal of $\mathbb{Z}$.

Exercise 42. In S_3, write the cycle (123) as a function on $\{1, 2, 3\}$. Compute its square and cube.

Exercise 43. Let $\mathbb{Z}$ act on $\mathbb{R}$ by $n \cdot x = x + n$. What is the orbit of 0? What is the stabilizer of 0?

Exercise 44. Decide whether $x^2 + 1$ is irreducible over $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$.

Walking question

When do you naturally treat different things as the same? File paths with symbolic links, numbers with the same remainder, objects with the same interface, shapes related by rotation. Which of these feel like quotient constructions?

23 Preview Chapter: Topology

Topology studies spaces through continuity, nearness, and shape, but without requiring distances or angles as primary data. It asks which properties survive continuous deformation. If geometry is concerned with measurement, topology is concerned with connection and continuity.

The subject can feel strange at first because it deliberately forgets some information. A circle and an ellipse are different geometrically but the same topologically. A coffee mug and a doughnut are often used as an informal example because each has one hole. The point is not the joke. The point is that topology classifies spaces by properties preserved under continuous change.

Focus

Topology is what remains of analysis after you remove formulas and keep the logic of open sets, continuity, compactness, and connectedness.

23.1 From metric spaces to open sets

A metric space is a set equipped with a distance function.

Definition 57. A *metric* on a set X is a function $d : X \times X \to \mathbb{R}$ such that for all $x, y, z \in X$:

1. $d(x, y) \geq 0$,
2. $d(x, y) = 0$ if and only if $x = y$,
3. $d(x, y) = d(y, x)$,
4. $d(x, z) \leq d(x, y) + d(y, z)$.

The pair (X, d) is called a *metric space*.

Example 36. The real line with $d(x, y) = |x - y|$ is a metric space. The plane $\mathbb{R}^2$ with

$$d((x_1, y_1), (x_2, y_2)) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

is also a metric space.

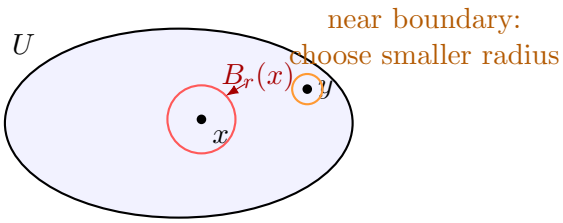
In a metric space, open balls define local neighborhoods.

Definition 58. The open ball of radius $r > 0$ around $x \in X$ is

$$B_r(x) = \{y \in X : d(x, y) < r\}.$$

Definition 59. A set $U \subseteq X$ is *open* if for every $x \in U$, there exists $r > 0$ such that $B_r(x) \subseteq U$.

This is already topological thinking. It says: each point of U has enough room to move slightly without leaving U.



The picture shows why the radius may depend on the point. A point deep inside the set can use a large ball. A point near the boundary needs a small ball. The definition does not require one radius for every point; it requires each point to have some positive radius.

Common trap

False intuition: an open set has to look round, simple, or physically open. Better intuition: an open set is one where every included point has local wiggle room inside the set. Open sets can be disconnected, jagged, infinite unions, or very strange.

23.2 Topological spaces

Topology abstracts the open sets themselves.

Definition 60. A *topology* on a set X is a collection $\mathcal{T}$ of subsets of X such that:

1. $\emptyset \in \mathcal{T}$ and $X \in \mathcal{T}$,
2. any union of sets in $\mathcal{T}$ belongs to $\mathcal{T}$,
3. any finite intersection of sets in $\mathcal{T}$ belongs to $\mathcal{T}$.

The pair $(X, \mathcal{T})$ is called a *topological space*. The sets in $\mathcal{T}$ are called open sets.

This definition may seem startling. No distances? No coordinates? Just a list of open sets? Yes. The open sets encode enough local information to define continuity, convergence in many settings, compactness, connectedness, and many notions of shape.

Intuition

A topology is a declaration of which questions count as local observations. An open set is a region you can verify membership in without landing exactly on a boundary.

23.3 Examples of topologies

Every metric space gives a topology: call a set open if every point has a small metric ball inside the set. But not every topology comes from a familiar metric.

Example 37. The *discrete topology* on a set X is $\mathcal{T} = \mathcal{P}(X)$, the set of all subsets of X. Every subset is open.

Example 38. The *indiscrete topology* on X is $\mathcal{T} = \{\emptyset, X\}$. Only the empty set and the whole space are open.

Example 39. The usual topology on $\mathbb{R}$ is generated by open intervals. A set is open if it is a union of open intervals.

These examples show that topology is not only about physical space. It is a general language for continuity.

Common trap

Open and closed are not opposites in the everyday sense. A set can be open, closed, both, or neither. In any topological space, $\emptyset$ and X are both open and closed.

Common trap

False intuition: because arbitrary unions of open sets are open, arbitrary intersections should also be open. The finite-intersection condition is the limit. In $\mathbb{R}$,

$$\bigcap_{n=1}^{\infty} (-1/n, 1/n) = \{0\},$$

and $\{0\}$ is not open. Infinite intersections can collapse all wiggle room.

23.4 Closed sets

Definition 61. In a topological space X, a set $F \subseteq X$ is *closed* if its complement $X \setminus F$ is open.

This definition is shorter than the sequence definition used in real analysis. In metric spaces the two notions agree: a set is closed if it contains limits of convergent sequences. In general topological spaces, sequences are sometimes not enough, which is one reason topology becomes more subtle.

Example 40. In $\mathbb{R}$ with the usual topology, $[0, 1]$ is closed because

$$\mathbb{R} \setminus [0, 1] = (-\infty, 0) \cup (1, \infty)$$

is open.

23.5 Subspace topology

Many spaces are studied as subsets of larger spaces. The subspace topology tells us which sets should count as open inside the subset.

Definition 62. Let X be a topological space and let $A \subseteq X$. The *subspace topology* on A declares $U \subseteq A$ open in A if there exists an open set $V \subseteq X$ such that

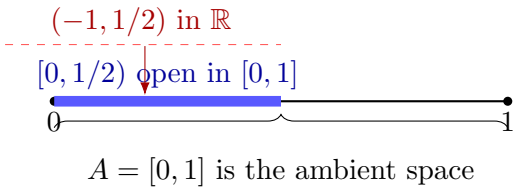
$$U = A \cap V.$$

Example 41. In $A = [0,1] \subseteq \mathbb{R}$, the set $[0,1/2)$ is open in the subspace topology on A because

$$[0,1/2) = A \cap (-1,1/2).$$

It is not open as a subset of $\mathbb{R}$.

This example is a useful warning. Openness depends on the ambient space. A set can be open in a subspace even when it is not open in the larger space.



$A = [0,1]$ is the ambient space

The subspace topology is not cheating. It says that if you live inside A, then observations are made relative to A. The point 0 has no left side inside A, so failing to contain negative numbers is not a defect.

Common trap

Never ask "is this set open?" without knowing the space in which openness is being tested. "Open in A" and "open in X" may have different answers.

23.6 Separation and Hausdorff spaces

Some topological spaces have enough open sets to distinguish points. The most important beginner separation condition is Hausdorffness.

Definition 63. A topological space X is *Hausdorff* if for every pair of distinct points $x, y \in X$, there exist open sets $U, V \subseteq X$ such that

$$x \in U, \qquad y \in V, \qquad U \cap V = \emptyset.$$

Example 42. Every metric space is Hausdorff. If $x \neq y$, let $r = d(x,y)/3$. Then $B_r(x)$ and $B_r(y)$ are disjoint open balls separating the points.

Hausdorffness is one reason compact spaces behave well. For example, compact subsets of Hausdorff spaces are closed. Also, a continuous bijection from a compact space to a Hausdorff space is automatically a homeomorphism.

Theorem 84. *If X is compact, Y is Hausdorff, and $f : X \to Y$ is a continuous bijection, then f is a homeomorphism.*

The proof is usually given after more compactness machinery, but the theorem is worth remembering because it explains why inverse continuity is sometimes automatic and sometimes not.

23.7 Sequences in topology

In metric spaces, sequences are very powerful. A set is closed exactly when it contains limits of convergent sequences. Continuity can be tested by sequences: f is continuous if $x_n \to x$ implies $f(x_n) \to f(x)$.

In general topological spaces, sequences may not detect all topological behavior. This is one reason topology introduces open sets and covers rather than relying only on sequence limits.

Definition 64. A sequence (x_n) in a topological space X *converges* to $x \in X$ if for every open set U containing x, there exists $N \in \mathbb{N}$ such that

$$n \geq N \Rightarrow x_n \in U.$$

This definition is the topological version of "eventually inside every neighborhood of the limit." In metric spaces, neighborhoods can be open balls. In general spaces, open neighborhoods are the primitive idea.

Intuition

Topological convergence says that every observation able to see the limit eventually also sees the sequence.

23.8 Continuity without epsilons

Topology gives a clean definition of continuity.

Definition 65. Let X and Y be topological spaces. A function $f : X \to Y$ is *continuous* if for every open set $V \subseteq Y$, the inverse image

$$f^{-1}(V) = \{x \in X : f(x) \in V\}$$

is open in X.

This definition may look backward at first: open sets in the target are pulled back to open sets in the domain. But this is exactly what makes it stable under composition.

Theorem 85. *If $f : X \to Y$ and $g : Y \to Z$ are continuous maps between topological spaces, then $g \circ f : X \to Z$ is continuous.*

Proof. Let $W \subseteq Z$ be open. Since g is continuous, $g^{-1}(W)$ is open in Y. Since f is continuous, $f^{-1}(g^{-1}(W))$ is open in X. But

$$f^{-1}(g^{-1}(W)) = (g \circ f)^{-1}(W).$$

Therefore $(g \circ f)^{-1}(W)$ is open in X for every open $W \subseteq Z$, so $g \circ f$ is continuous. □

This proof is pure set and function language. It is one of the rewards of learning inverse images carefully.

Focus

Topology turns continuity into a statement about inverse images. This is why early function notation matters so much.

23.9 Homeomorphism

The topological notion of sameness is homeomorphism.

Definition 66. A function $f : X \to Y$ is a *homeomorphism* if f is bijective, continuous, and its inverse $f^{-1} : Y \to X$ is continuous. If such a function exists, X and Y are *homeomorphic*.

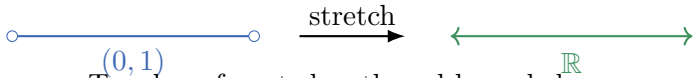
Homeomorphic spaces are the same from the viewpoint of topology. They may have different shapes in a geometric drawing, but their open-set structure is equivalent.

Example 43. The intervals $(0, 1)$ and $\mathbb{R}$ are homeomorphic. One possible map is

$$f(x) = \tan\big(\pi(x - 1/2)\big).$$

It is continuous, bijective, and has continuous inverse.

This example is important because it shows that topological sameness does not preserve length or boundedness. The bounded interval $(0, 1)$ and the unbounded line $\mathbb{R}$ are topologically the same.



Topology forgets length and boundedness here. The open interval can be stretched into the whole line without cutting or gluing.

Common trap

A continuous bijection need not be a homeomorphism unless the inverse is also continuous. On compact-to-Hausdorff spaces this issue disappears, but in general the inverse condition must be checked.

Why mathematicians care

Homeomorphism is beautiful because it separates accidental geometry from essential topology. Length, curvature, and the particular drawing may change; the continuity structure remains.

23.10 Connectedness

Connectedness formalizes the idea that a space is in one piece.

Definition 67. A topological space X is *disconnected* if there exist nonempty open sets $U, V \subseteq X$ such that

$$X = U \cup V, \qquad U \cap V = \emptyset.$$

If no such separation exists, X is *connected*.

Example 44. The interval $[0, 1]$ is connected. The set $[0, 1] \cup [2, 3]$ is disconnected in the usual topology on $\mathbb{R}$.

The proof that intervals are connected is one of the classic uses of the least upper bound property. Topology and real analysis meet here.

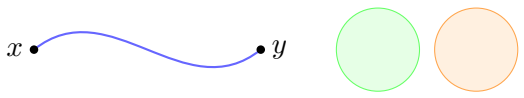
Theorem 86. *The continuous image of a connected space is connected.*

Proof. Let $f : X \to Y$ be continuous and suppose X is connected. Assume for contradiction that $f(X)$ is disconnected. Then there are nonempty open sets U and V in the subspace $f(X)$ such that

$$f(X) = U \cup V, \qquad U \cap V = \emptyset.$$

The inverse images $f^{-1}(U)$ and $f^{-1}(V)$ are open in X, nonempty, disjoint, and their union is X. This disconnects X, a contradiction. Therefore $f(X)$ is connected. □

This theorem gives the intermediate value theorem its topological heart: continuous images of intervals are intervals, and intervals contain intermediate values.



path-connected: a continuous route joins points

disconnected: separable into open pieces

Common trap

False intuition: connected means path-connected. Every path-connected space is connected, but the converse can fail. Topology has examples where a space is one piece in the open-set sense but still has no path between certain points.

Reading note

The theorem "continuous images of connected spaces are connected" is the topological skeleton of the intermediate value theorem. The calculus statement is familiar; topology shows the mechanism.

23.11 Compactness

Compactness can be introduced through open covers.

Definition 68. Let X be a topological space and $K \subseteq X$. An *open cover* of K is a collection of open sets $\{U_\alpha\}_{\alpha \in A}$ such that

$$K \subseteq \bigcup_{\alpha \in A} U_\alpha.$$

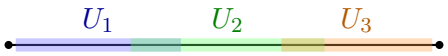
A *finite subcover* is a finite subcollection $U_{\alpha_1}, \ldots, U_{\alpha_n}$ that still covers K.

Definition 69. A set K is *compact* if every open cover of K has a finite subcover.

This definition is abstract but powerful. It says that whenever infinitely many open sets are enough to cover K, finitely many were already enough.

Intuition

Compactness is a finite-control principle. It lets you extract finite data from an apparently infinite situation.



A finite subcover controls the whole interval.

This picture is deliberately too simple. A real open cover of $[0, 1]$ may involve infinitely many open sets of wildly different sizes. Compactness says that if the cover works at all, finitely many of those open sets already do the job.

Common trap

False intuition: compact just means closed and bounded. In $\mathbb{R}^n$, Heine-Borel says compact is equivalent to closed and bounded. In general topological spaces, that slogan can fail badly. The portable definition is the open-cover definition.

Example 45. The interval $[0,1]$ is compact in $\mathbb{R}$. The interval $(0,1)$ is not compact. The open cover

$$\{(1/n, 1) : n = 2, 3, 4, \ldots\}$$

covers $(0,1)$ but no finite subcollection covers all of $(0,1)$.

Theorem 87. *The continuous image of a compact space is compact.*

Proof. Let $f : X \to Y$ be continuous and suppose X is compact. Let $\{V_\alpha\}_{\alpha \in A}$ be an open cover of $f(X)$ in Y. Then $\{f^{-1}(V_\alpha)\}_{\alpha \in A}$ is an open cover of X. Since X is compact, there exist $\alpha_1, \ldots, \alpha_n$ such that

$$X \subseteq f^{-1}(V_{\alpha_1}) \cup \cdots \cup f^{-1}(V_{\alpha_n}).$$

Applying f, we get

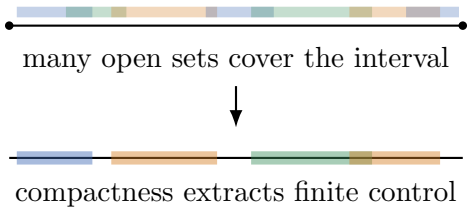
$$f(X) \subseteq V_{\alpha_1} \cup \cdots \cup V_{\alpha_n}.$$

Thus the original cover has a finite subcover. Therefore $f(X)$ is compact. □

Again, inverse images do the work. The proof pattern is central: pull a cover back to the domain, use compactness there, push the finite result forward.

Why mathematicians care

Compactness feels magical because it converts infinite information into finite control. An arbitrary cover may contain infinitely many open sets, but compactness says finitely many already suffice. This is why compactness powers existence theorems throughout analysis and geometry.



23.12 Bases

Listing every open set can be inefficient. A basis gives building blocks.

Definition 70. A *basis* for a topology on X is a collection $\mathcal{B}$ of subsets of X such that:

1. for every $x \in X$, there is $B \in \mathcal{B}$ with $x \in B$,
2. if $x \in B_1 \cap B_2$ with $B_1, B_2 \in \mathcal{B}$, then there exists $B_3 \in \mathcal{B}$ such that $x \in B_3 \subseteq B_1 \cap B_2$.

The open sets are unions of basis elements.

Example 46. The open intervals (a, b) form a basis for the usual topology on $\mathbb{R}$. Open balls form a basis for the topology of any metric space.

The basis idea previews charts in differential geometry. A manifold is built from coordinate neighborhoods that look like open subsets of $\mathbb{R}^n$.

23.13 Product spaces

If X and Y are spaces, their product $X \times Y$ should also be a space. The product topology is generated by sets of the form $U \times V$, where U is open in X and V is open in Y.

Example 47. The usual topology on $\mathbb{R}^2$ is the product topology on $\mathbb{R} \times \mathbb{R}$. Open rectangles form a basis, and open disks can be built from unions of rectangles.

Product spaces allow topology to model systems with several independent coordinates. They also appear in algebraic geometry, differential geometry, functional analysis, and probability.

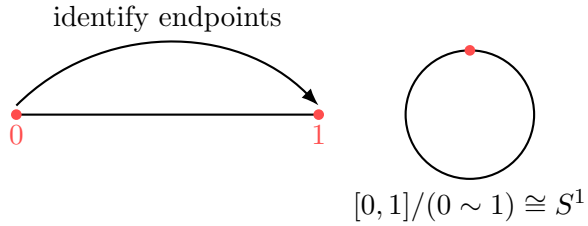
23.14 Quotient spaces

Quotients identify points according to an equivalence relation.

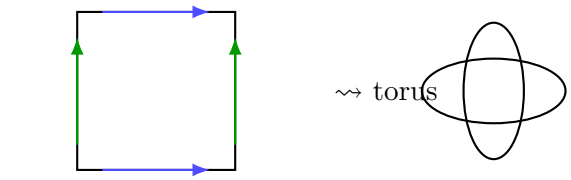
Definition 71. Let X be a topological space and let $\sim$ be an equivalence relation on X. The quotient set $X/\sim$ is the set of equivalence classes. The *quotient topology* is defined by declaring $U \subseteq X/\sim$ open if and only if its preimage under the quotient map $q : X \to X/\sim$ is open in X.

This definition mirrors the continuity definition. The quotient topology is the topology that makes the quotient map continuous while preserving as much openness as possible.

Example 48. If we take the interval $[0, 1]$ and identify $0 \sim 1$, the quotient space is a circle. This is one rigorous way to build S^1 from an interval.



Example 49. If we take a square and identify opposite edges in the same direction, the quotient space is a torus.



identify matching edges schematic

Intuition

Quotient topology is controlled gluing. Algebra quotients forget differences; topology quotients glue points while tracking which sets remain open.

Common trap

False intuition: quotient spaces are just ordinary subsets with renamed points. Better intuition: quotienting changes the space. It can create loops, cones, projective spaces, and strange identifications. The quotient topology is the rulebook that keeps continuity honest after gluing.

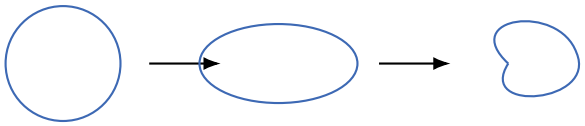
Why mathematicians care

Quotients are beautiful because they make "controlled forgetting" precise. You decide which distinctions no longer matter, then ask what structure survives that decision.

23.15 Topology Visual Gallery: Deformation, Gluing, and Pathology

Topology needs pictures, but it also teaches you not to trust pictures too quickly. The right workflow is: let the picture suggest the idea, then let the definition decide the truth.

Deformation intuition



circle ellipse deformed loop

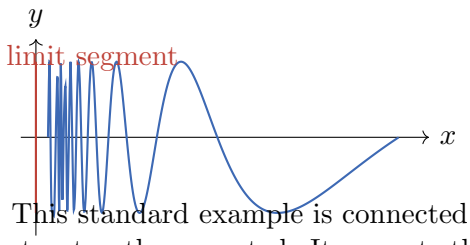
Topological sameness permits stretching and bending, but not cutting or gluing.

Pathological warning

Some spaces defeat ordinary drawing intuition. The topologist's sine curve is the closure of

$$y = \sin(1/x), \qquad 0 < x \leq 1,$$

together with a limiting vertical segment at $x = 0$.

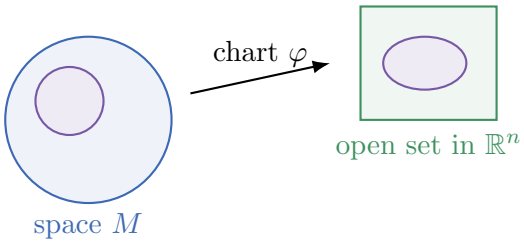


This standard example is connected but not path-connected. It corrects the naive idea that "one piece" always means "walkable between any two points."

Common trap

False intuition: every connected subset of the plane behaves like a familiar region. Pathological examples are why topology insists on definitions instead of only visual confidence.

Manifold charts as topology plus coordinates



This foreshadows differential geometry. Topology supplies open sets and local patches; calculus enters through coordinate maps.

Why mathematicians care

Topology is powerful because it keeps only enough structure to discuss continuity. By forgetting measurement, it lets deeper invariants become visible.

23.16 Topological invariants

A topological invariant is a property preserved by homeomorphism. If two spaces have different values of an invariant, they cannot be homeomorphic.

Examples of basic invariants include:

- number of connected components,
- compactness,
- connectedness,
- path-connectedness,
- whether removing a point disconnects the space,
- fundamental group, in later algebraic topology.

Example 50. The spaces $(0, 1)$ and $[0, 1]$ are not homeomorphic. One reason is compactness: $[0, 1]$ is compact, while $(0, 1)$ is not.

Example 51. The spaces $\mathbb{R}$ and $\mathbb{R}^2$ are not homeomorphic, though proving this rigorously requires more topology. A suggestive invariant is what happens when a point is removed: $\mathbb{R} \setminus \{0\}$ is disconnected, but $\mathbb{R}^2 \setminus \{(0, 0)\}$ is connected.

Invariants are proof tools. To show two spaces are the same, build a homeomorphism. To show they are different, find an invariant that separates them.

23.17 Path-connectedness

Connectedness says a space cannot be separated into two disjoint open pieces. Path-connectedness is more concrete: any two points can be joined by a continuous path.

Definition 72. A space X is *path-connected* if for every $x, y \in X$, there exists a continuous function

$$\gamma : [0, 1] \to X$$

such that $\gamma(0) = x$ and $\gamma(1) = y$.

Theorem 88. *Every path-connected space is connected.*

Proof. Suppose X is path-connected and disconnected. Then $X = U \cup V$ where U and V are nonempty disjoint open sets. Choose $u \in U$ and $v \in V$. Since X is path-connected, there is a continuous path $\gamma : [0, 1] \to X$ with $\gamma(0) = u$ and $\gamma(1) = v$.

The sets $\gamma^{-1}(U)$ and $\gamma^{-1}(V)$ form a separation of $[0, 1]$, contradicting the connectedness of the interval. Therefore X is connected. □

The converse is not always true, though it is true for many familiar spaces such as intervals and open connected subsets of $\mathbb{R}^n$ with suitable hypotheses.

23.18 A first glimpse of algebraic topology

Algebraic topology assigns algebraic objects to topological spaces. The simplest idea to preview is the fundamental group, which records loops up to continuous deformation.

Definition 73. A *loop* in a space X based at x_0 is a continuous map

$$\gamma : [0,1] \to X$$

such that $\gamma(0) = \gamma(1) = x_0$.

Loops can be concatenated: travel around one loop, then another. Up to a suitable equivalence relation called homotopy, loop classes form a group called the fundamental group. The full construction belongs later, but the intuition is useful now.

Example 52. In the plane $\mathbb{R}^2$, every loop can be continuously shrunk to a point. Around a punctured plane $\mathbb{R}^2 \setminus \{0\}$, a loop may wind around the missing point. The winding number is algebraic information extracted from topology.

Focus

Algebraic topology continues the theme of this reader: use maps and equivalence relations to convert geometric questions into algebraic ones.

23.19 Common proof habits in topology

Topology proofs often feel clean once definitions are unpacked. Common moves include:

- prove continuity by pulling back open sets,
- prove non-continuity by finding one open set whose preimage is not open,
- prove compactness by starting with an arbitrary open cover,
- prove connectedness by assuming a separation and deriving a contradiction,
- use bases to reduce "all open sets" to simpler building blocks,
- use quotient maps by checking preimages in the original space.

Common trap

Do not rely too heavily on pictures. Pictures are valuable, but topological proofs are usually about open sets, maps, covers, and equivalence relations. When a picture suggests a claim, translate it into one of those languages.

23.20 Applications of topology

Topology supports:

- qualitative analysis of dynamical systems,
- data analysis through persistent homology,
- compactness methods in analysis,
- manifolds in geometry and physics,
- fixed point theorems,
- configuration spaces in robotics,
- continuity arguments in economics and optimization,
- algebraic topology and invariants of spaces.

One reason topology is so portable is that it uses few assumptions. Whenever a problem has continuity but not necessarily a convenient formula or metric, topology may still apply.

23.21 Proof lab: compactness and closed subsets

Compactness becomes more useful when combined with ordinary set operations.

Theorem 89. *If K is compact and $F \subseteq K$ is closed, then F is compact.*

Proof. Let $\{U_\alpha\}_{\alpha \in A}$ be an open cover of F. Since F is closed in K, the complement $K \setminus F$ is open in K. Then the collection

$$\{U_\alpha\}_{\alpha \in A} \cup \{K \setminus F\}$$

is an open cover of K. Since K is compact, finitely many of these sets cover K. Removing $K \setminus F$ if it appears, the remaining finitely many U_α cover F. Therefore F is compact. □

This proof is a model for working with open covers. You start with an arbitrary cover of the smaller set, add one extra open set to cover the rest of the compact space, use compactness, and then restrict back to the original set.

Proof lab

Use the theorem to prove that any closed interval $[a, b]$ is compact once you know that a larger interval $[-M, M]$ is compact and contains $[a, b]$.

23.22 Counterexample lab: topology is context

Topology trains you to produce examples that separate similar-looking ideas.

Example 53. A compact space need not be finite. The interval $[0, 1]$ is compact and infinite.

Example 54. A connected space need not be path-connected. The topologist's sine curve is a standard example, though its full analysis is beyond this preview.

Example 55. A set can be both open and closed. In any topological space X, both $\emptyset$ and X are clopen. In a disconnected space, nontrivial clopen sets also exist.

Example 56. A topology can be too coarse to distinguish points. In the indiscrete topology on a set with at least two points, no two distinct points can be separated by disjoint open sets.

Focus

Topology is a field where examples do a lot of teaching. When two definitions feel similar, look for a space where they come apart.

23.23 Exercises for topology

Exercise 45. Let $X = \{a, b, c\}$. Decide whether

$$\mathcal{T} = \{\emptyset, X, \{a\}, \{a, b\}\}$$

is a topology on X.

Exercise 46. Prove that the union of any collection of open sets in a metric space is open.

Exercise 47. Prove that the finite intersection of open sets in a metric space is open. Where does finiteness enter?

Exercise 48. Let $f : \mathbb{R} \to \mathbb{R}$ be given by $f(x) = x^2$. Using the topological definition of continuity, explain why it is enough to know the usual epsilon-delta theorem from analysis.

Exercise 49. Give an open cover of $(0, 1)$ with no finite subcover.

Exercise 50. Explain why a constant map $f : X \to Y$ is continuous for every pair of topological spaces X and Y.

Exercise 51. In the subspace $[0, 1] \subseteq \mathbb{R}$, decide whether each set is open: $[0, 1/2)$, $(1/3, 2/3)$, $\{0\}$, and $[0, 1]$.

Exercise 52. Prove that every metric space is Hausdorff.

Exercise 53. Show that if X is path-connected and $f : X \to Y$ is continuous, then $f(X)$ is path-connected.

Walking question

What properties of an object survive stretching, bending, or reparameterizing? Which properties require measurement? Sort familiar facts into topological and geometric categories.

24 Preview Chapter: Differential Geometry

Differential geometry studies spaces that are curved but locally accessible by calculus. The central objects are manifolds: spaces that may not be linear or flat globally, but near each point resemble ordinary Euclidean space.

This field connects early proof ideas to a rich geometric setting. Definitions matter intensely. A curve is not just a drawn line; it is often a smooth map from an interval. A tangent vector is not just an arrow in a picture; it can be defined as a velocity, a derivation, or an equivalence class of curves. A manifold is not just a surface in $\mathbb{R}^3$; it is a space with compatible local coordinate systems.

Focus

Differential geometry is calculus plus topology plus linear algebra, organized so that local computations say something about global shape.

24.1 Historical motivation

Classical geometry studied lines, circles, conics, and surfaces. Calculus made it possible to study curves and surfaces through derivatives. Gauss showed that curvature could be intrinsic: a surface could have geometry detectable by beings living on it, without reference to the surrounding three-dimensional space.

Riemann then generalized geometry by allowing spaces of any dimension with a notion of length that could vary from point to point. This became essential for modern physics, especially general relativity, where gravity is modeled as curvature of spacetime.

History and connection

Gauss's *Theorema Egregium*, the "remarkable theorem," showed that Gaussian curvature is intrinsic. You can bend a sheet of paper into a cylinder without stretching it, and its intrinsic curvature remains zero. A sphere cannot be flattened without distortion because its intrinsic curvature is positive.

24.2 Curves as maps

A curve is best understood as a function, not merely as a subset.

Definition 74. A *parametrized curve* in $\mathbb{R}^n$ is a function

$$\gamma : I \to \mathbb{R}^n$$

where $I \subseteq \mathbb{R}$ is an interval. The curve is *smooth* if all component functions are differentiable as often as needed.

Example 57. The unit circle can be parametrized by

$$\gamma(t) = (\cos t, \sin t), \qquad 0 \le t < 2\pi.$$

The image is the circle, but the map also contains information about speed and direction.

The derivative of a curve is its velocity:

$$\gamma'(t) = (\gamma_1'(t), \ldots, \gamma_n'(t)) .$$

Example 58. For $\gamma(t) = (\cos t, \sin t)$,

$$\gamma'(t) = (-\sin t, \cos t).$$

At $t = 0$, the point is $(1, 0)$ and the velocity is $(0, 1)$.

Common trap

Do not confuse a curve's image with the curve as a parametrized map. The maps $\gamma(t) = (\cos t, \sin t)$ and $\eta(t) = (\cos 2t, \sin 2t)$ trace the same circle at different speeds.

24.3 Arc length

For a smooth curve $\gamma : [a, b] \to \mathbb{R}^n$, the length is

$$L(\gamma) = \int_a^b \|\gamma'(t)\| \, dt.$$

This formula is familiar from calculus, but differential geometry asks how it behaves under reparametrization.

Example 59. For the unit circle parametrization $\gamma(t) = (\cos t, \sin t)$ on $[0, 2\pi]$, we have

$$\|\gamma'(t)\| = \sqrt{(-\sin t)^2 + (\cos t)^2} = 1.$$

Therefore

$$L(\gamma) = \int_0^{2\pi} 1 \, dt = 2\pi.$$

If the same circle is traced twice as fast, the speed doubles but the time interval halves for one trip. The length remains 2π. This is an early sign of geometric invariance: a quantity should not depend on an arbitrary coordinate choice.

24.4 Surfaces and local coordinates

A surface in $\mathbb{R}^3$ can often be described by a parametrization

$$X : U \subseteq \mathbb{R}^2 \to \mathbb{R}^3.$$

For example, the sphere of radius 1 can be parametrized, away from coordinate singularities, by

$$X(\theta, \phi) = (\sin\phi\cos\theta, \sin\phi\sin\theta, \cos\phi).$$

The parameters (θ, ϕ) are coordinates. They are useful, but they are not the surface itself. At the poles, these spherical coordinates behave badly: different values of θ describe the same point. This motivates the manifold idea, where one uses many coordinate charts rather than demanding one global coordinate system.

Intuition

A chart is a local coordinate API. It lets you do Euclidean calculations near a point, but it may not cover the whole space cleanly.

24.5 Manifolds

The topological starting point is that each small region should look like an open subset of $\mathbb{R}^n$.

Definition 75. An *n-dimensional topological manifold* is a topological space M such that every point $p \in M$ has a neighborhood U homeomorphic to an open subset of $\mathbb{R}^n$. The pair (U, φ), where

$$\varphi : U \to \varphi(U) \subseteq \mathbb{R}^n$$

is a homeomorphism, is called a *chart*.

To do calculus, the charts must be compatible smoothly.

Definition 76. A *smooth manifold* is a manifold with charts whose transition maps are smooth. If (U, φ) and (V, ψ) overlap, the transition map is

$$\psi \circ \varphi^{-1} : \varphi(U \cap V) \to \psi(U \cap V).$$

This definition is a direct meeting point of topology, functions, and calculus. Charts are homeomorphisms. Transition maps are compositions. Smoothness is checked in Euclidean coordinates.

Example 60. The circle S^1 is a one-dimensional manifold. Near any point, a small arc of the circle looks like an open interval.

Example 61. The sphere S^2 is a two-dimensional manifold. No single chart covers it without distortion or singularity, but stereographic projection gives charts from the sphere minus one point to $\mathbb{R}^2$.

Common trap

An n-dimensional manifold need not live inside $\mathbb{R}^n$. The circle is one-dimensional but is usually drawn in $\mathbb{R}^2$. The sphere is two-dimensional but is usually drawn in $\mathbb{R}^3$.

24.6 Tangent spaces

At each point of a smooth manifold, there is a vector space of tangent directions.

For a surface sitting in $\mathbb{R}^3$, the tangent plane can be visualized as the best linear approximation to the surface at a point. For an abstract manifold, we need a definition that does not depend on an ambient space.

Definition 77. Let M be a smooth manifold and $p \in M$. A tangent vector at p can be viewed as an equivalence class of smooth curves $\gamma : (-\varepsilon, \varepsilon) \to M$ with $\gamma(0) = p$, where two curves are equivalent if their coordinate representations have the same derivative at 0 in some, and hence any, chart. The set of tangent vectors at p is called the *tangent space* T_pM.

This definition looks technical because it protects an intuitive idea from coordinate dependence. A tangent vector should be a direction through p, not an artifact of one chart.

Example 62. For $M = \mathbb{R}^n$, the tangent space $T_p\mathbb{R}^n$ can be identified with $\mathbb{R}^n$ itself. At every point of Euclidean space, the possible velocity vectors are ordinary vectors.

Example 63. For the unit sphere $S^2 \subseteq \mathbb{R}^3$, the tangent space at $p \in S^2$ is

$$T_pS^2 = \{v \in \mathbb{R}^3 : v \cdot p = 0\}.$$

This is the plane through the origin perpendicular to p, translated in the usual picture to touch the sphere at p.

24.7 Implicit surfaces

Many surfaces are not given first by parametrizations. They are given as level sets of functions.

Definition 78. Let $F : \mathbb{R}^3 \to \mathbb{R}$ be smooth. A set

$$S = \{(x, y, z) \in \mathbb{R}^3 : F(x, y, z) = c\}$$

is a *level surface*. If $\nabla F(p) \neq 0$ at a point $p \in S$, then near p the set S is a smooth surface.

This is an informal version of the implicit function theorem. It says that a single equation in three variables usually cuts out a two-dimensional surface, provided the gradient does not vanish.

Example 64. For the sphere,

$$F(x, y, z) = x^2 + y^2 + z^2.$$

The unit sphere is $F^{-1}(1)$. Since

$$\nabla F = (2x, 2y, 2z),$$

the gradient is nonzero at every point of the unit sphere. Therefore the sphere is a smooth surface.

The tangent plane to a level surface has a simple description:

$$T_pS = \{v \in \mathbb{R}^3 : \nabla F(p) \cdot v = 0\}.$$

This is the same calculation used earlier for the circle.

Common trap

If the gradient vanishes, the level set may still be smooth, but the simple test no longer proves it. A zero gradient is a warning that singular behavior may be present.

24.8 Differentials

A smooth map between manifolds has a derivative at each point, called its differential or pushforward.

Definition 79. Let $F : M \to N$ be a smooth map and let $p \in M$. The *differential*

$$dF_p : T_pM \to T_{F(p)}N$$

sends the velocity of a curve γ through p to the velocity of the curve $F \circ \gamma$ through $F(p)$.

In Euclidean space, this is the Jacobian matrix. On manifolds, the same idea is written in coordinate-independent language.

Example 65. Let $F : \mathbb{R}^2 \to \mathbb{R}$ be $F(x, y) = x^2 + y^2$. At (a, b), the differential is the linear map

$$dF_{(a,b)}(u, v) = 2au + 2bv.$$

This is the directional rate of change of F at (a, b) in direction (u, v).

Focus

Differential geometry repeatedly replaces nonlinear objects by linear approximations at a point. Tangent spaces are where that linearization lives.

24.9 Vector fields

A vector field assigns a tangent vector to each point.

Definition 80. A *vector field* on a smooth manifold M assigns to each $p \in M$ a vector $X_p \in T_pM$, smoothly in p.

On $\mathbb{R}^2$, a vector field can be written

$$X(x, y) = P(x, y)\frac{\partial}{\partial x} + Q(x, y)\frac{\partial}{\partial y}.$$

You have seen vector fields in differential equations and physics: velocity fields, force fields, gradient fields, and flow fields.

Example 66. The vector field

$$X(x, y) = (-y, x)$$

on $\mathbb{R}^2$ rotates around the origin. Its integral curves are circles centered at the origin.

The language of vector fields turns differential equations into geometry. A solution curve is a curve whose velocity at each point matches the vector field.

24.10 Flows

A vector field often generates a flow: a family of transformations describing where each point moves after time t.

Definition 81. Given a vector field X on a manifold M, an *integral curve* through $p \in M$ is a smooth curve $\gamma : I \to M$ such that

$$\gamma(0) = p, \qquad \gamma'(t) = X_{\gamma(t)}.$$

In Euclidean language, this is an ordinary differential equation. Differential geometry interprets it as motion on a manifold.

Example 67. For $X(x, y) = (-y, x)$ on $\mathbb{R}^2$, the curve

$$\gamma(t) = (\cos t, \sin t)$$

is an integral curve through $(1, 0)$ because

$$\gamma'(t) = (-\sin t, \cos t) = X(\gamma(t)).$$

Flows connect differential geometry with dynamical systems. The local existence and uniqueness theorem for ordinary differential equations becomes a theorem about vector fields producing local motion on manifolds.

24.11 Lie groups

Some spaces are both manifolds and groups. These are Lie groups.

Definition 82. A *Lie group* is a group G that is also a smooth manifold, such that the multiplication map

$$G \times G \to G, \qquad (g, h) \mapsto gh$$

and inverse map

$$G \to G, \qquad g \mapsto g^{-1}$$

are smooth.

Example 68. The circle

$$S^1 = \{z \in \mathbb{C} : |z| = 1\}$$

is a Lie group under complex multiplication. It is also a one-dimensional manifold.

Example 69. The group $GL(n, \mathbb{R})$ of invertible $n \times n$ real matrices is a Lie group. It is an open subset of $\mathbb{R}^{n^2}$ because invertibility is the condition $\det(A) \neq 0$.

Lie groups show how algebra and geometry combine. Their tangent spaces at the identity become Lie algebras, which encode infinitesimal symmetry.

Intuition

A Lie group is a smooth family of symmetries. Group theory describes how symmetries compose; differential geometry describes how they vary continuously.

24.12 Differential forms

Differential forms are objects that can be integrated over curves, surfaces, and higher-dimensional manifolds. A one-form eats a vector and returns a number.

Definition 83. On $\mathbb{R}^n$, a *one-form* is an expression

$$\omega = f_1\, dx_1 + \cdots + f_n\, dx_n$$

where the f_i are smooth functions. At each point, ω is a linear function on tangent vectors.

Example 70. On $\mathbb{R}^2$, let

$$\omega = y\, dx + x\, dy.$$

If $v = (u, w)$ is a tangent vector at (x, y), then

$$\omega(v) = yu + xw.$$

Forms are the natural language of integration on manifolds. They lead to Stokes' theorem, which unifies the fundamental theorem of calculus, Green's theorem, the divergence theorem, and Stokes' theorem from vector calculus.

Theorem 90 (Stokes' theorem, preview)**.** *For a suitable differential form ω on a suitable oriented manifold M,*

$$\int_M d\omega = \int_{\partial M} \omega.$$

This statement is too compressed for a first proof here, but its message is important: the integral of a derivative over a region equals the integral of the original object over the boundary.

24.13 Metrics and curvature

A Riemannian metric gives a smoothly varying inner product on tangent spaces.

Definition 84. A *Riemannian metric* on a smooth manifold M assigns to each point $p \in M$ an inner product

$$g_p : T_pM \times T_pM \to \mathbb{R}$$

that varies smoothly with p.

With a metric, one can measure lengths of tangent vectors, angles between vectors, lengths of curves, areas, volumes, gradients, and curvature.

Example 71. The usual dot product on $\mathbb{R}^n$ is a Riemannian metric. In coordinates,

$$g_p(v, w) = v \cdot w.$$

Curvature measures how a geometric space departs from being flat. For a curve, curvature measures how quickly the tangent direction changes. For a surface, Gaussian curvature captures intrinsic bending.

Example 72. A plane has Gaussian curvature 0. A sphere of radius R has constant positive Gaussian curvature $1/R^2$. A saddle surface has negative curvature near the saddle point.

Intuition

Curvature is not merely how something looks from outside. Intrinsic curvature can be detected by measurements made inside the space, such as angle sums of triangles or how circles' circumferences compare to $2\pi r$.

24.14 Geodesics

Geodesics generalize straight lines. They are locally length-minimizing paths, or equivalently paths that move straight according to the geometry of the manifold.

Example 73. On the plane, geodesics are straight lines. On the sphere, geodesics are great circles, such as the equator and lines of longitude.

This has practical consequences. Airplane routes drawn on a flat map may look curved, but on the sphere the shortest routes follow great circles.

24.15 Covariant derivatives, gently

In ordinary Euclidean space, vectors at different points can be compared directly. On a curved manifold, tangent vectors at different points live in different tangent spaces. To differentiate vector fields, one needs a rule for comparing nearby tangent spaces.

That rule is called a connection. The most important connection in Riemannian geometry is the Levi-Civita connection, determined by the metric.

Definition 85. Informally, a *covariant derivative* $\nabla_X Y$ measures the rate of change of a vector field Y in the direction of a vector field X, while respecting the manifold's geometry.

This is too advanced for full formal development here, but the reason it exists is beginner-friendly: the derivative of a vector field should again be geometric, not an artifact of coordinates or an ambient space.

Example 74. On a sphere, a tangent vector can be moved along a curve while staying tangent and changing as little as possible. This process is called parallel transport. Moving around a closed loop may return the vector rotated. That failure to return unchanged is a sign of curvature.

24.16 Orientation

Orientation is the structure needed to make consistent choices of positive direction or positive volume.

Example 75. The real line has two orientations: left-to-right and right-to-left. The plane has clockwise versus counterclockwise orientation. A sphere is orientable.

Some manifolds are not orientable. The standard beginner example is the Mobius strip, which has only one side. If you move a small oriented frame around the strip and return to the starting point, the orientation has flipped.

Orientation matters for integration. Stokes' theorem requires compatible orientations on a manifold and its boundary.

Common trap

Orientability is not the same as having a preferred orientation. A sphere is orientable, but you still must choose which orientation to use.

24.17 Common proof habits in differential geometry

Differential geometry proofs often have two layers: a coordinate calculation and an argument that the result is independent of coordinates. Common moves include:

- choose a chart near a point,
- compute using ordinary calculus in coordinates,
- check that the result does not depend on the chart,
- reduce a manifold claim locally to a Euclidean claim,
- use linear algebra on tangent spaces,
- track how objects transform under change of coordinates.

Common trap

Do not mistake coordinates for geometry. Coordinates are tools for computation. The geometric object should make sense after changing coordinates.

24.18 Applications of differential geometry

Differential geometry supports:

- general relativity and spacetime geometry,
- robotics and configuration spaces,
- computer graphics and surface processing,
- optimization on manifolds,
- mechanics and control theory,
- gauge theory and modern physics,
- shape analysis and geometric data processing.

For a software engineer, one accessible application is optimization on curved spaces. If parameters are constrained to lie on a sphere, a rotation group, or a space of positive definite matrices, ordinary Euclidean gradient descent may leave the valid space. Riemannian optimization adapts gradient methods to the manifold.

24.19 Proof lab: coordinate changes on a curve

A simple place to see coordinate dependence is a reparametrized curve. Let

$$\gamma(t) = (\cos t, \sin t)$$

and let

$$\eta(s) = \gamma(2s) = (\cos 2s, \sin 2s).$$

These curves have the same image if the parameter intervals are chosen appropriately, but their velocities differ:

$$\gamma'(t) = (-\sin t, \cos t), \qquad \eta'(s) = (-2\sin 2s, 2\cos 2s).$$

The speed of η is twice the speed of γ.

The chain rule explains this:

$$\eta'(s) = \gamma'(2s) \cdot 2.$$

Thus velocity depends on parametrization, while the geometric trace does not. Arc length remains invariant when the reparametrization is handled correctly.

Proof lab

Let $\gamma(t) = (t, t^2)$ and $\eta(s) = \gamma(s^3)$. Compute $\eta'(s)$ using the chain rule, then directly from the formula for $\eta(s)$.

24.20 Geometry trap lab: embedded versus intrinsic

A surface may be studied as a subset of Euclidean space, or intrinsically as a space in its own right. Both views are useful, but they are not the same.

Example 76. A cylinder in $\mathbb{R}^3$ looks curved from outside, but intrinsically it is flat: locally it has the same geometry as a plane. This is why a sheet of paper can be rolled into a cylinder without stretching.

Example 77. A sphere has positive intrinsic curvature. No flat map of the Earth can preserve all distances and angles globally.

Common trap

Do not assume every visual bend is intrinsic curvature. Bending without stretching can change extrinsic appearance while preserving intrinsic geometry.

This distinction matters in applications. A robot arm's configuration space may be a curved manifold because of constraints, even if it is described using many ambient coordinates. The intrinsic geometry records the legal motions, not the surrounding coordinate bookkeeping.

24.21 Beginner worked example: tangent line to a circle

Let $S^1 = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$. At a point $p = (a, b) \in S^1$, the tangent line should consist of vectors $v = (u, w)$ satisfying

$$au + bw = 0.$$

Why? Let $\gamma(t) = (x(t), y(t))$ be a curve on the circle with $\gamma(0) = p$. Since the curve stays on the circle,

$$x(t)^2 + y(t)^2 = 1$$

for all t. Differentiate at $t = 0$:

$$2x(0)x'(0) + 2y(0)y'(0) = 0.$$

Thus

$$au + bw = 0$$

where $v = \gamma'(0) = (u, w)$. The tangent vector is perpendicular to the radius.

Proof lab

Use the same method to find the tangent plane to the sphere

$$x^2 + y^2 + z^2 = 1$$

at a point (a, b, c) on the sphere.

24.22 Exercises for differential geometry

Exercise 54. Compute the velocity and speed of the curve

$$\gamma(t) = (t, t^2)$$

at time $t = 1$.

Exercise 55. Find the length of the curve $\gamma(t) = (3\cos t, 3\sin t)$ for $0 \le t \le 2\pi$.

Exercise 56. Explain why the circle S^1 is locally like $\mathbb{R}$ but not globally homeomorphic to $\mathbb{R}$.

Exercise 57. Let $F : \mathbb{R}^2 \to \mathbb{R}^2$ be $F(x, y) = (x^2 - y^2, 2xy)$. Compute the Jacobian matrix and interpret it as $dF_{(a,b)}$.

Exercise 58. For the sphere S^2, verify that $T_pS^2 = \{v : v \cdot p = 0\}$ gives a two-dimensional vector space for every $p \in S^2$.

Exercise 59. Let $F(x, y, z) = x^2 + y^2 - z$. Find the tangent plane to the level surface $F = 0$ at $(1, 1, 2)$.

Exercise 60. Show that S^1 is closed under complex multiplication and inverses. Then explain why it is reasonable to call it a group of rotations.

Exercise 61. For the vector field $X(x, y) = (1, 0)$ on $\mathbb{R}^2$, describe the integral curve through an arbitrary point (a, b).

Walking question

Think about a map of the Earth. Where do coordinates help? Where do they break down? What information belongs to the Earth itself rather than to the map?

25 Preview Chapter: Algebraic Geometry

Algebraic geometry studies shapes defined by polynomial equations. The basic idea is simple:

$$\text{polynomial equations} \quad \longleftrightarrow \quad \text{geometric sets.}$$

But this simple bridge becomes one of the deepest areas of mathematics because polynomials carry both algebraic structure and geometric meaning.

At a beginner level, algebraic geometry should not be introduced as a wall of schemes. It should begin with concrete equations: lines, circles, parabolas, intersections, and solution sets. Then the algebraic language of rings and ideals explains how equations organize geometry.

Focus

Algebraic geometry begins with a friendly question: what shape is described by these polynomial equations? The mature subject asks the reverse question too: what algebraic object records the functions on this shape?

25.1 Historical motivation

Classical analytic geometry, associated with Descartes and Fermat, connected equations with curves. A line could be written as $ax + by = c$. A circle could be written as $x^2 + y^2 = r^2$. Geometry became algebraic.

Later, mathematicians studied higher-degree curves, intersections, singularities, and solutions over different number systems. Algebraic geometry grew from this interaction. It now reaches number theory, topology, representation theory, cryptography, and mathematical physics.

History and connection

The phrase "algebraic geometry" reflects a two-way translation. Algebra gives equations. Geometry gives shapes. Major advances came from learning that neither side alone tells the whole story.

25.2 Affine space

The simplest setting is affine space.

Definition 86. Let k be a field. The affine n-space over k is

$$\mathbb{A}_k^n = k^n.$$

Its points are ordered n-tuples $(a_1, \ldots, a_n)$ with coordinates in k.

When $k = \mathbb{R}$, $\mathbb{A}^2_{\mathbb{R}}$ is the usual coordinate plane. When $k = \mathbb{C}$, it is the complex plane squared, which has real dimension four. When k is a finite field, affine space is a finite set of points.

The field matters. The equation

$$x^2 + 1 = 0$$

has no solution in $\mathbb{R}$ but has two solutions in $\mathbb{C}$. Algebraic geometry is very sensitive to the choice of ground field.

25.3 Affine varieties

Definition 87. Let k be a field and let $S \subseteq k[x_1, \ldots, x_n]$ be a set of polynomials. The *zero set* of S is

$$V(S) = \{a \in k^n : f(a) = 0 \text{ for every } f \in S\}.$$

An *affine algebraic set* is a subset of k^n of the form $V(S)$.

Some texts reserve the word *variety* for irreducible algebraic sets. In a beginner preview, it is common to use "variety" informally for a polynomial zero set, but you should know that conventions vary.

Example 78. In $\mathbb{R}^2$, the polynomial $f(x, y) = y - x^2$ defines the parabola

$$V(y - x^2) = \{(x, y) \in \mathbb{R}^2 : y = x^2\}.$$

Example 79. The polynomial $x^2 + y^2 - 1$ defines the unit circle in $\mathbb{R}^2$:

$$V(x^2 + y^2 - 1) = \{(x, y) : x^2 + y^2 = 1\}.$$

Example 80. The pair of equations

$$y - x^2 = 0, \qquad y - 1 = 0$$

defines the intersection of the parabola $y = x^2$ with the line $y = 1$. The solutions over $\mathbb{R}$ are $(-1, 1)$ and $(1, 1)$.

Intuition

One polynomial gives a hypersurface. Several polynomials give an intersection of hypersurfaces. Algebraic geometry studies those intersections using the algebra of the polynomials.

25.4 The algebra of equations

If two equations vanish on a set, then many combinations vanish too. Suppose $f(a) = 0$ and $g(a) = 0$. Then

$$(f + g)(a) = 0, \qquad (hf)(a) = 0$$

for any polynomial h.

This is the reason ideals appear.

Definition 88. Let $X \subseteq k^n$. The *ideal of polynomials vanishing on* X is

$$I(X) = \{f \in k[x_1, \dots, x_n] : f(a) = 0 \text{ for every } a \in X\}.$$

Theorem 91. *For every $X \subseteq k^n$, the set $I(X)$ is an ideal of $k[x_1, \dots, x_n]$.*

Proof. The zero polynomial vanishes on every point of X, so $0 \in I(X)$. If $f, g \in I(X)$ and $a \in X$, then

$$(f - g)(a) = f(a) - g(a) = 0 - 0 = 0,$$

so $f - g \in I(X)$. If $h \in k[x_1, \dots, x_n]$ and $f \in I(X)$, then for every $a \in X$,

$$(hf)(a) = h(a)f(a) = h(a) \cdot 0 = 0.$$

Thus $hf \in I(X)$. Therefore $I(X)$ is an ideal. □

This proof is basic but important. It shows how geometric data automatically produces algebraic structure.

25.5 From ideals back to geometry

Given an ideal $I \subseteq k[x_1, \dots, x_n]$, we can form

$$V(I) = \{a \in k^n : f(a) = 0 \text{ for every } f \in I\}.$$

Thus there are two directions:

$$X \subseteq k^n \mapsto I(X), \qquad I \subseteq k[x_1, \dots, x_n] \mapsto V(I).$$

These operations reverse inclusion:

- if $X \subseteq Y$, then $I(Y) \subseteq I(X)$,
- if $I \subseteq J$, then $V(J) \subseteq V(I)$.

The reversal is worth understanding. A bigger set of points imposes more tests on a polynomial, so fewer polynomials vanish on all of it. A bigger ideal gives more equations, so fewer points satisfy all of them.

Proof lab

Prove both inclusion-reversing statements above directly from the definitions. This is an ideal exercise for practicing "let an arbitrary element be given" proofs.

25.6 Coordinate rings

The coordinate ring records polynomial functions on an algebraic set.

Definition 89. If $X = V(I) \subseteq k^n$, the *coordinate ring* of X is

$$k[X] = k[x_1, \ldots, x_n]/I(X).$$

The quotient appears because two polynomials may define the same function on X even if they are not the same polynomial. Their difference vanishes on X.

Example 81. Let $X = V(y - x^2) \subseteq \mathbb{R}^2$, the parabola. In the coordinate ring of X, the relation $y = x^2$ holds. Thus every polynomial function on the parabola can be rewritten using x alone. Algebraically,

$$\mathbb{R}[X] \cong \mathbb{R}[x].$$

Example 82. Let $X = V(xy) \subseteq \mathbb{R}^2$. This is the union of the x-axis and the y-axis. The coordinate ring is

$$\mathbb{R}[x, y]/(xy).$$

The relation $xy = 0$ records the crossing shape algebraically.

Intuition

The coordinate ring is the algebra of polynomial measurements you can make on the shape. If two formulas always give the same value on the shape, the coordinate ring treats them as the same.

25.7 Regular functions

On an affine algebraic set, the most natural functions are polynomial functions. But because different polynomials can agree on the same set, the coordinate ring is the right home for these functions.

Definition 90. Let $X \subseteq k^n$ be an affine algebraic set. A *regular function* on X is a function obtained by restricting a polynomial $f \in k[x_1, \ldots, x_n]$ to X.

Example 83. On the parabola $X = V(y - x^2)$, the coordinate function y agrees with x^2. So as regular functions on X, y and x^2 are the same.

This is a subtle but important shift. Algebraic geometry often studies a space through the functions on it. If you know the coordinate ring, you know the polynomial observations available on the space.

Focus

In algebraic geometry, functions are not secondary. The algebra of functions is often the main way the geometry is encoded.

25.8 Maps of varieties

In algebraic geometry, the natural maps are given by polynomials.

Definition 91. A *polynomial map* $F : k^n \to k^m$ has the form

$$F(a_1, \ldots, a_n) = (f_1(a_1, \ldots, a_n), \ldots, f_m(a_1, \ldots, a_n))$$

for polynomials $f_1, \ldots, f_m \in k[x_1, \ldots, x_n]$.

Example 84. The map $F : \mathbb{R} \to \mathbb{R}^2$ given by

$$F(t) = (t, t^2)$$

parametrizes the parabola $y = x^2$.

Maps reverse algebra. If $F : X \to Y$ is a polynomial map, then a polynomial function on Y can be pulled back to a polynomial function on X by composing with F:

$$g \mapsto g \circ F.$$

This gives a homomorphism of coordinate rings in the opposite direction.

Focus

Geometry often sends points forward. Algebra often pulls functions backward. This reversal is one of the central habits of algebraic geometry.

25.9 Intersections and multiplicity

A line and a parabola may intersect in two points, one point, or no real points. For example, intersect $y = x^2$ with $y = c$. This gives

$$x^2 = c.$$

Over $\mathbb{R}$, there are two solutions if $c > 0$, one solution if $c = 0$, and no solutions if $c < 0$. Over $\mathbb{C}$, there are two solutions for $c \neq 0$ and one repeated solution for $c = 0$.

Multiplicity captures the idea of repeated intersection.

Example 85. The line $y = 0$ meets the parabola $y = x^2$ at $(0, 0)$ with multiplicity two. Algebraically, substituting $y = 0$ into $y - x^2 = 0$ gives $x^2 = 0$, a double root.

Multiplicity is one reason algebraic geometry often prefers algebraically closed fields such as $\mathbb{C}$. Over $\mathbb{C}$, polynomial equations have the expected number of roots when multiplicities are counted, at least in suitable settings.

25.10 Singularities

Some algebraic sets are smooth, while others have corners, crossings, or cusps. These special points are singularities.

Example 86. The curve

$$y^2 = x^3$$

has a cusp at $(0, 0)$.

Example 87. The curve

$$y^2 = x^2(x + 1)$$

has a node-like crossing at the origin over suitable fields.

One beginner test for singularity uses partial derivatives. For a plane curve $V(f) \subseteq k^2$, a point p is singular if

$$f(p) = 0, \qquad \frac{\partial f}{\partial x}(p) = 0, \qquad \frac{\partial f}{\partial y}(p) = 0.$$

Example 88. Let $f(x, y) = y^2 - x^3$. Then

$$f_x = -3x^2, \qquad f_y = 2y.$$

At $(0, 0)$, both partial derivatives vanish, so the cusp is singular.

Common trap

The graph may look smooth at a glance, but algebraic smoothness is a local condition checked by equations and derivatives. Always compute before trusting the drawing.

25.11 The Zariski topology

Algebraic geometry uses a topology where closed sets are polynomial zero sets.

Definition 92. The *Zariski topology* on k^n declares algebraic sets $V(I)$ to be the closed sets.

This topology is much coarser than the usual topology on $\mathbb{R}^n$ or $\mathbb{C}^n$. There are fewer open sets. For example, in $\mathbb{A}^1_k$, the proper closed sets are finite sets when k is algebraically closed.

Intuition

The Zariski topology is designed so that polynomial equations define closed conditions. It may feel strange as a topology, but it is exactly suited to algebraic geometry.

Example 89. In $\mathbb{A}^1_{\mathbb{C}}$, the set $\mathbb{C} \setminus \{0\}$ is Zariski open because its complement is $V(x)$.

The Zariski topology connects algebraic geometry back to topology, but with a different sense of nearness. Points are not near because of small distance; they are related by the polynomial conditions that vanish on them.

25.12 Groebner bases and computation

Modern algebraic geometry is not only abstract. It has computational tools for solving and simplifying polynomial systems. One of the main tools is the Groebner basis.

A full definition requires monomial orders and polynomial division in several variables, but the guiding idea is accessible: a Groebner basis is a particularly well-organized generating set for an ideal. It lets one answer questions such as:

- Does a given polynomial belong to the ideal?
- Can the system of equations be simplified?
- What are the possible values of one coordinate?
- How can variables be eliminated?

Example 90. Consider the system

$$y - x^2 = 0, \qquad y - 1 = 0.$$

Eliminating y gives $x^2 - 1 = 0$. Thus the intersection points have $x = \pm 1$ and $y = 1$.

Groebner bases generalize this kind of elimination. They are used in robotics, coding theory, symbolic computation, and automated theorem proving.

Intuition

A Groebner basis is like row-reduced echelon form for polynomial equations. The analogy is not perfect, but it is useful: both replace a system by a better organized system with the same solution set.

25.13 Finite fields and point counting

Algebraic geometry can be done over finite fields. Then a variety has finitely many points, and counting those points becomes meaningful.

Example 91. Over $\mathbb{Z}/5\mathbb{Z}$, consider the equation

$$x^2 + y^2 = 1.$$

One can count pairs $([x], [y])$ satisfying the equation by testing residues $0, 1, 2, 3, 4$. This is a finite version of studying a circle.

This may sound like a toy setting, but it is central in number theory and cryptography. Deep theorems connect point counts over finite fields to topology, complex geometry, and zeta functions.

History and connection

The Weil conjectures linked the number of solutions of polynomial equations over finite fields to cohomological invariants. Their proof required a vast expansion of algebraic geometry, including Grothendieck's scheme-theoretic methods.

25.14 Elliptic curves as a friendly advanced example

An elliptic curve is often introduced by an equation of the form

$$y^2 = x^3 + ax + b$$

with a nonzero discriminant condition ensuring smoothness.

Example 92. The curve

$$y^2 = x^3 - x + 1$$

is an affine piece of an elliptic curve, after adding a point at infinity in the projective plane.

Elliptic curves are remarkable because their points can be added. A geometric chord-and-tangent construction defines a group law. This places elliptic curves at the intersection of algebraic geometry, number theory, and abstract algebra.

Applications include elliptic curve cryptography, integer factorization methods, and the proof of Fermat's Last Theorem through modularity ideas far beyond this preview.

Common trap

The group law on an elliptic curve is not coordinatewise addition. It is a geometric operation defined by lines intersecting the curve.

25.15 Local thinking and local rings

To study a variety near a point, algebraic geometers localize. The idea is to focus on functions defined near a point and allow division by functions that do not vanish there.

Definition 93. Informally, the *local ring* of a variety X at a point p consists of regular functions near p, where two functions are identified if they agree on some neighborhood of p.

This is another local-to-global theme. A global coordinate ring sees the whole variety. A local ring sees the behavior near one point. Singularities, tangent spaces, and smoothness can be studied through local algebra.

Example 93. At a smooth point of a curve, the local ring behaves like functions in one local parameter. At a crossing point, the local algebra records that two branches pass through the same point.

25.16 Hilbert's basis theorem

Polynomial rings in finitely many variables over a field are controlled by finite data.

Theorem 92 (Hilbert basis theorem, preview)**.** *If k is a field, then every ideal in $k[x_1, \dots, x_n]$ is finitely generated.*

This means that even if an ideal is described as containing infinitely many polynomials, there are finitely many polynomials

$$f_1, \dots, f_m$$

such that every element of the ideal can be written as

$$h_1 f_1 + \cdots + h_m f_m$$

for polynomial coefficients h_i.

Geometrically, this says every algebraic set in affine n-space can be defined by finitely many equations. That is a deep finiteness principle.

History and connection

Hilbert's proof of the basis theorem was famously nonconstructive for its time. It showed existence of finite generating sets without necessarily producing them explicitly. This helped shift mathematics toward accepting structural existence proofs as legitimate and powerful.

25.17 Nullstellensatz as the bridge theorem

The central bridge between algebra and geometry is Hilbert's Nullstellensatz. The full theorem requires algebraically closed fields and more commutative algebra, but the rough message is accessible:

> Over an algebraically closed field, geometry of polynomial zero sets corresponds closely to algebra of ideals.

One version says that if k is algebraically closed, then

$$I(V(I)) = \sqrt{I},$$

where $\sqrt{I}$ is the radical of I.

Definition 94. The *radical* of an ideal I is

$$\sqrt{I} = \{f : f^m \in I \text{ for some } m \geq 1\}.$$

The radical appears because a polynomial and a power of that polynomial have the same zero set. For example, x and x^2 vanish at exactly the same points, but the ideals (x) and (x^2) are different.

Example 94. In $\mathbb{C}[x]$, let $I = (x^2)$. Then

$$V(I) = \{0\},$$

and

$$I(V(I)) = (x).$$

So $I(V(I))$ is not (x^2) but its radical (x).

25.18 Projective geometry preview

Affine geometry misses some intersection behavior. Parallel lines in the affine plane do not meet. Projective geometry adds points at infinity so that many theorems become cleaner.

Definition 95. Projective n-space over a field k, written $\mathbb{P}^n_k$, is the set of lines through the origin in k^{n+1}.

A point of projective space is written in homogeneous coordinates:

$$[x_0 : x_1 : \cdots : x_n],$$

where not all coordinates are zero and

$$[x_0 : \cdots : x_n] = [\lambda x_0 : \cdots : \lambda x_n]$$

for every nonzero $\lambda \in k$.

Example 95. In $\mathbb{P}^2$, the affine line $y = mx + b$ gains a point at infinity that records its direction. Parallel affine lines meet at the same point at infinity.

Projective space is another quotient construction: nonzero vectors are grouped by scalar multiplication. Early equivalence relation skills return again.

25.19 Applications of algebraic geometry

Algebraic geometry supports:

- number theory and Diophantine equations,
- elliptic curve cryptography,
- coding theory,

- robotics through polynomial constraint systems,
- computer vision and camera geometry,
- symbolic computation and Groebner bases,
- string theory and mathematical physics,
- algebraic statistics.

For a concrete computational example, solving polynomial systems is central in robot kinematics. A robot arm with fixed link lengths and joint constraints can lead to polynomial equations whose solution set describes possible positions.

25.20 Translation lab: one object, three languages

The parabola can be described in several mutually reinforcing ways.

Language	Description
Equation	$y - x^2 = 0$
Point set	$X = \{(a, b) \in k^2 : b = a^2\}$
Ideal	$I(X) = (y - x^2)$, over suitable fields
Coordinate ring	$k[X] = k[x, y]/(y - x^2) \cong k[x]$
Parametrization	$t \mapsto (t, t^2)$
Regular functions	Polynomial expressions in x and y, with $y = x^2$ on X

Each line emphasizes a different skill. The equation is familiar from analytic geometry. The point set trains set-builder notation. The ideal brings in abstract algebra. The coordinate ring introduces quotient thinking. The parametrization connects to functions and differential geometry.

Proof lab

Make the same translation table for the line $y = 3x + 1$. Which entries are obvious, and which require algebraic thought?

25.21 Algebraic geometry trap lab: equations are not unique

The same geometric set can be defined by many different equations.

Example 96. In $\mathbb{R}^2$, the equations

$$x = 0, \qquad 2x = 0, \qquad x^3 = 0$$

all define the y-axis as a set of real points.

But the ideals (x), $(2x)$, and (x^3) need not carry the same algebraic information in every ring or context. Over a field of characteristic not 2, (x) and $(2x)$ are the same ideal, but (x^3) is different from (x) even though its zero set is the same.

Intuition

Geometry may forget multiplicity. Algebra can remember it. This is one reason schemes were invented: they retain algebraic thickness that ordinary point sets discard.

Example 97. The equation $x = 0$ describes a line. The equation $x^2 = 0$ describes the same ordinary set of points over a field, but algebraically it suggests a doubled line. This "nilpotent" information is invisible if you only look at classical points.

25.22 Common proof habits in algebraic geometry

Beginner algebraic geometry proofs often use:

- direct evaluation of polynomials at points,
- ideal membership checks,
- inclusion reversal between sets and ideals,
- quotient rings to identify functions that agree on a variety,
- field choice awareness,
- partial derivatives for smoothness tests,
- examples over $\mathbb{R}$ and $\mathbb{C}$ to show why algebraic closure matters.

Common trap

Do not assume that the real picture tells the whole story. The equation $x^2 + y^2 = 0$ is only the origin over $\mathbb{R}$, but over $\mathbb{C}$ it has many solutions, such as $(i, 1)$ because $i^2+1 = 0$.

25.23 Beginner worked example: ideal of a line

Let $X \subseteq \mathbb{R}^2$ be the line $y = 2x$. Then $X = V(y - 2x)$. We claim that many polynomials vanish on X, for example

$$(y - 2x)(x^2 + 1), \qquad (y - 2x)(x + y), \qquad 3(y - 2x).$$

All of these belong to the ideal $(y - 2x)$.

Why should the ideal of the line be exactly $(y - 2x)$? Informally, if a polynomial vanishes whenever $y = 2x$, then after substituting $y = 2x$ the result is the zero polynomial in x. Polynomial division by $y - 2x$ then shows the original polynomial is a multiple of $y - 2x$.

This is a model of the algebra-geometry bridge:

$$\text{line } y = 2x \quad \leftrightarrow \quad \text{ideal } (y - 2x).$$

25.24 Exercises for algebraic geometry

Exercise 62. Describe the zero set in $\mathbb{R}^2$ of each polynomial:

$$x, \qquad xy, \qquad x^2 + y^2 - 1, \qquad y - x^2.$$

Exercise 63. Find the intersection points over $\mathbb{R}$ of the circle $x^2 + y^2 = 1$ and the line $y = 0$.

Exercise 64. Let $X = \{(0, 0)\} \subseteq \mathbb{R}^2$. Show that x and y belong to $I(X)$. Give three more polynomials in $I(X)$.

Exercise 65. Prove that if $I \subseteq J$ are ideals in $k[x_1, \ldots, x_n]$, then $V(J) \subseteq V(I)$.

Exercise 66. For $f(x, y) = y^2 - x^3$, compute the partial derivatives and verify that $(0, 0)$ is singular.

Exercise 67. Over $\mathbb{R}$, describe $V(x^2 + 1)$. Over $\mathbb{C}$, describe $V(x^2 + 1)$. What changed?

Exercise 68. On the parabola $X = V(y - x^2)$, show that the regular functions represented by y^2 and x^4 are equal.

Exercise 69. Eliminate y from the system

$$y - x^2 = 0, \qquad y - x - 2 = 0.$$

Find the intersection points over $\mathbb{R}$.

Exercise 70. Over $\mathbb{Z}/3\mathbb{Z}$, count the solutions to $x^2 + y^2 = 1$.

Walking question

Pick a familiar shape: a line, circle, parabola, sphere, or intersection of two surfaces. Ask which equations define it, which polynomials vanish on it, and which coordinates are genuinely independent on the shape.

26 Synthesis: How the Early Proof Ideas Reappear

The five preview chapters above may look like separate fields, but the same proof muscles return in different clothing.

26.1 Definitions are engines

In real analysis, the definition of convergence tells you to start with $\varepsilon > 0$ and produce N. In topology, the definition of continuity tells you to pull back open sets. In algebra, the definition of homomorphism tells you to compare $\varphi(ab)$ with $\varphi(a)\varphi(b)$. In differential geometry, the definition of a smooth manifold tells you to choose charts and inspect transition maps. In algebraic geometry, the definition of $I(X)$ tells you to evaluate polynomials at arbitrary points of X.

Focus

Most beginner proof blocks come from not trusting definitions enough. When you are stuck, write the relevant definition with all quantifiers visible.

26.2 Dependency Signals: This Idea Later Becomes

Long-range memory improves when you know where an idea is going. The same early tools keep reappearing under more sophisticated names.

Reading note

⇛ Compression arrow. Each row below says: this beginner idea will later compress into a more powerful structure.

Early idea	Later forms	Why it matters
Equivalence relations	quotient groups, quotient rings, quotient spaces, projective space, moduli spaces	identify objects that differ only by an irrelevant choice
Injective, surjective, bijective	embeddings, quotient maps, isomorphisms, homeomorphisms, diffeomorphisms	control whether maps lose information, hit everything, or preserve structure perfectly
Inverse images	continuity, measurable functions, pullbacks, fibers, preimages of varieties	move structure from target back to domain
Compactness	maximum/minimum theorems, finite subcovers, existence results, convergence subsequences	turn infinite problems into finite control
Open sets	continuity, manifolds, sheaves, local properties	formalize local observation and local-to-global reasoning
Induction	recursion, well-founded arguments, Noetherian induction	prove claims over objects built in stages

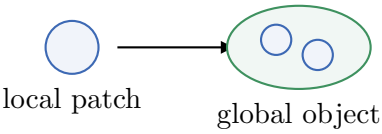
Walking question

When you learn a new definition, ask: what does this later become? A beginner definition is often the small version of a much larger machine.

26.3 Local and Global as a Recurring Motif

Local-to-global reasoning is one of the great rhythms of modern mathematics.

- In analysis, local continuity plus compactness can produce global bounds.
- In topology, local neighborhoods define global shape.
- In differential geometry, charts give local coordinates on a global manifold.
- In algebraic geometry, local rings inspect a variety near one point.



The central question: when does enough local information determine or control the whole?

Why mathematicians care

Local-to-global results are beautiful because they let small, checkable information govern large objects. Compactness, sheaves, manifolds, and many existence theorems live in this rhythm.

26.4 Mathematical Aesthetics: Why These Ideas Have Pull

Mathematics is not only a list of true statements. Some ideas keep returning because they feel unusually clarifying. That feeling is not decoration; it is a signal that an idea compresses many situations.

Idea	Why it feels powerful	What it compresses
Compactness	infinite behavior becomes finitely controllable	covers, subsequences, existence, optimization
Quotients	irrelevant distinctions are removed cleanly	congruence, symmetry, gluing, moduli
Invariants	difference can be proved without direct comparison	holes, dimension, components, algebraic signatures

Duality	one object is understood through all tests against it	linear functionals, characters, spectra, functions on spaces
Algebraic geometry	equations and shapes become two languages for one object	polynomial systems, coordinate rings, local geometry
Category viewpoint	maps reveal what structure really matters	functions, homomorphisms, continuous maps, smooth maps

Why mathematicians care

The emotional center of abstraction is recognition. You meet a new field and realize it is using an old move in a sharper language: preserve structure, identify sameness, extract invariants, pass from local data to global form.

26.5 Functions organize everything

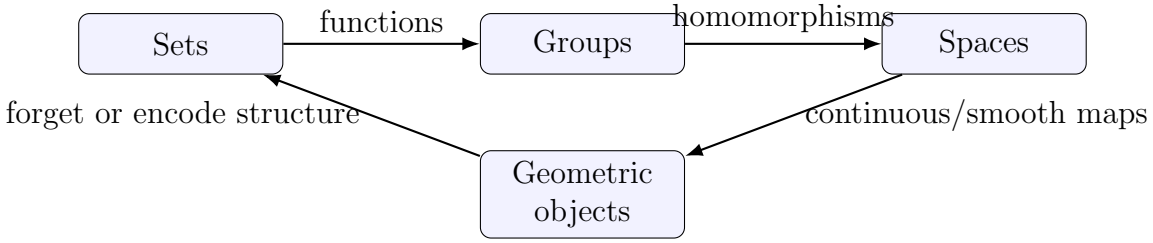
Functions appear as:

- sequences $a : \mathbb{N} \to \mathbb{R}$,
- continuous maps $f : X \to Y$,
- homomorphisms $\varphi : G \to H$,
- charts $\varphi : U \to \mathbb{R}^n$,
- smooth maps between manifolds,
- polynomial maps between varieties,
- pullbacks of functions on coordinate rings.

The early distinction among domain, codomain, image, inverse image, injective, surjective, and bijective is not beginner paperwork. It is the grammar of modern mathematics.

26.6 Structure-Preserving Maps Are the Grand Unifier

Here is the category-theoretic whisper that should be present from the beginning: mathematics studies objects together with the maps that preserve the structure we care about.



This diagram is not a formal category theory lesson. It is a memory hook. When you enter a new field, ask:

1. What are the objects?
2. What structure do they carry?
3. What maps are allowed?
4. What counts as sameness?
5. What properties survive those maps?

Those questions unify algebra, topology, geometry, logic, and much of modern mathematics.

26.7 Equivalence relations become quotients

Equivalence relations show up as:

- congruence modulo n,
- cosets of a subgroup,
- quotient groups and quotient rings,
- quotient topologies,
- projective space as nonzero vectors modulo scalar multiplication,
- tangent vectors as equivalence classes of curves.

The idea is always similar: decide which differences should be ignored, then study the structure of the resulting classes.

26.8 Local-to-global thinking

Analysis controls function behavior near points and then proves global results on compact intervals. Topology studies local neighborhoods and global shape. Differential geometry uses charts to do local calculus on manifolds. Algebraic geometry uses local behavior near points to detect smoothness and singularity.

Intuition

Local-to-global reasoning asks: if I understand every small neighborhood, when can I assemble that knowledge into a statement about the whole object?

26.9 Existence is not construction

Many theorems guarantee existence without giving an explicit object:

- supremum property,
- monotone convergence theorem,
- intermediate value theorem,
- compactness finite subcover theorem,
- Hilbert basis theorem,
- existence of quotient structures.

This is not weakness. Existence proofs tell you what is logically guaranteed. Algorithms may come later, but the proof establishes the ground.

26.10 A final set of walking questions

Walking question

When you meet a new theorem, ask:

1. What are the objects?
2. What structure do they carry?
3. What maps preserve that structure?
4. What are the examples and non-examples?
5. Which hypotheses are used?
6. What breaks if one hypothesis is removed?
7. Is the conclusion local, global, algebraic, geometric, or topological?

These questions are portable. They will not solve every problem, but they will keep you oriented in unfamiliar territory.

27 Proof Debugging: Turning Fog Into Work

A proof rarely arrives clean on the first try. Even experienced mathematicians write scratch work, notice a missing condition, rename variables, split a case, or replace a vague sentence by a definition. The difference between a beginner and a mature proof writer is not that the mature writer never gets stuck. The mature writer has a method for finding the stuck point.

This section is about that method.

Focus

When a proof feels wrong, do not ask first, "What is the trick?" Ask, "What is the first sentence whose justification I cannot defend?"

Debugging a proof is a disciplined way to convert uneasiness into work. You identify the claim, unpack the grammar, locate the first unsupported step, and repair that step with a definition, a lemma, a witness, a case split, or a counterexample.

27.1 The proof feedback workflow

Use this workflow whenever you read your own proof, review someone else's proof, or get feedback from an instructor.

1. **Restate the claim.** Write the theorem in one sentence, with all domains visible.
2. **Mark the quantifiers.** Circle every "for all," "there exists," "for some," "unique," and "there is no."
3. **Separate hypotheses from conclusion.** Put assumptions on one side and the target on the other.
4. **Type every object.** Record whether each symbol is an integer, real number, set, function, relation, subset, sequence, or something else.
5. **Open the target definition.** If the conclusion says "even," write "equals $2k$ for some $k \in \mathbb{Z}$." If it says "injective," write the exact implication to prove.
6. **Open the useful hypothesis definitions.** Do not open every definition automatically; open the ones that can produce the shape needed by the target.
7. **Choose the outer proof form.** Direct proof, contrapositive, contradiction, cases, induction, or two inclusions.
8. **Write the skeleton.** Before details, write the first and last two lines.
9. **Fill one bridge at a time.** Every line should be licensed by algebra, a definition, a hypothesis, a lemma, or a previous line.
10. **Run the failure check.** Look for undefined variables, reversed implications, hidden division by zero, lost cases, and conclusions stronger than what was proved.

Formal

A proof line is healthy when a reader can answer both questions:

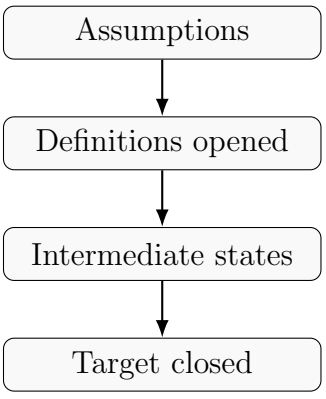
What does this line claim? Why is this line allowed?

If either answer is unclear, the proof needs repair.

27.2 Proof Anatomy Worksheet

Before debugging a proof, dissect it. This is the proof equivalent of symbolic execution.

Anatomy item	Question	Example prompt
Objects	What mathematical things are involved?	integers, sets, functions, groups
Domains	Where do the objects live?	$n \in \mathbb{Z}$, $f : A \to B$
Assumptions	What may be used?	f is injective, $A \subseteq B$
Target	What must be shown?	$g \circ f$ is injective
Definitions	Which words must be opened?	injective, subset, compact
Tools	Which previous facts apply?	contrapositive, induction, De Morgan
Strategy	What proof shape fits?	direct, cases, contradiction
Intermediate states	What bridge claims are needed?	$f(a_1) = f(a_2)$ before $a_1 = a_2$
Closure	Where is the target satisfied?	"therefore by definition..."



If a proof feels foggy, locate which arrow is unsupported.

Proof lab

Use the worksheet on this claim: if $f : A \to B$ is injective and $S, T \subseteq A$, then

$$f(S \cap T) = f(S) \cap f(T).$$

Identify where injectivity is needed. Then write a counterexample showing the claim can fail without injectivity.

27.3 The first-bad-line rule

When reviewing a proof, do not merely say that the proof is wrong. Find the first line where the argument stops being justified. Later lines may also be false, but the first bad line is the place where repair begins.

For example, consider the claim:

> If n^2 is even, then n is even.

Bad proof:

> Since n^2 is even, $n^2 = 2k$. Taking square roots gives $n = \sqrt{2k}$, so n is even.

The first two moves are not the real problem. If n^2 is even, then there is some $k \in \mathbb{Z}$ such that $n^2 = 2k$. Taking square roots is already delicate because n might be negative, but even if we replace it by $|n| = \sqrt{2k}$, the final inference is invalid. The expression $\sqrt{2k}$ need not have the form $2m$ for an integer m.

Common trap

Do not let familiar algebraic motion impersonate proof. A line can look like calculation and still fail to establish the definition you need.

A repair uses the contrapositive:

Proof. We prove the contrapositive. Suppose n is odd. Then $n = 2m + 1$ for some $m \in \mathbb{Z}$. Hence

$$n^2 = (2m + 1)^2 = 4m^2 + 4m + 1 = 2(2m^2 + 2m) + 1.$$

Since $2m^2 + 2m \in \mathbb{Z}$, n^2 is odd. Therefore, if n^2 is even, then n is even. □

27.4 A feedback vocabulary

Good feedback is specific. Here are useful phrases.

Feedback phrase	What it means	Repair question
Undefined variable	A symbol appears before introduction	Where does this object live?
Quantifier mismatch	The proof proves the wrong order of choices	Who chooses first?
Missing witness	An existence claim has no constructed object	What object actually works?

Definition not closed	The proof gestures at a property but never matches its definition	What exact form must be shown?
Converse error	The proof uses $Q \Rightarrow P$ when only $P \Rightarrow Q$ is known	Which direction is available?
Case leak	One possibility is not covered	What are all exhaustive cases?
Circular step	The proof uses what it is trying to prove	Can this line be justified without the conclusion?
Type error	The expression combines objects of incompatible kinds	What kind of object is each symbol?
Hidden assumption	The proof uses a condition not in the theorem	Is this condition given or proved?
Too strong	The proof establishes or assumes something stronger than needed	What weaker target is enough?

27.5 How to comment on a bad proof

Weak feedback:

> This is confusing.

Better feedback:

> The variable k is introduced in line 2, but its domain is not stated. If you mean $k \in \mathbb{Z}$, say so. The final line also needs to show that the expression has the form $2m$ for an integer m.

Weak feedback:

> You need more rigor.

Better feedback:

> The proof says f is "one-to-one by cancellation." To prove injectivity, start with arbitrary $a_1, a_2 \in A$ and assume $f(a_1) = f(a_2)$. Then show $a_1 = a_2$. The missing step is the cancellation calculation.

Intuition

Good proof feedback sounds like a bug report with a reproducible failure: here is the line, here is the expected definition, here is what is missing.

27.6 The repair loop

Use the following loop when revising.

1. Copy the broken line.
2. Name the reason it is broken in one phrase.
3. Write the definition or theorem that should justify it.
4. Add the missing intermediate line.
5. Check whether the new line introduced a new obligation.
6. Repeat until every obligation is discharged.

For example:

> Broken line: "Therefore $x \in A$ and $x \in B$."
> Problem: This was concluded from $x \in A \cup B$.
> Definition: $x \in A \cup B$ means $x \in A$ or $x \in B$.
> Repair: Use cases, or change the target.

27.7 Bad proof repair 1: parity

Claim: If a and b are odd integers, then ab is odd.

Bad proof:

> Odd times odd is odd because an odd number has an odd factor, so the product is odd.

Diagnosis:

- The proof repeats the claim instead of proving it.
- The phrase "has an odd factor" does not close the definition of odd.
- No variables witnessing oddness are introduced.

Repair:

Proof. Let $a, b \in \mathbb{Z}$ be odd. Then there exist $m, n \in \mathbb{Z}$ such that

$$a = 2m + 1, \qquad b = 2n + 1.$$

Thus

$$ab = (2m + 1)(2n + 1) = 4mn + 2m + 2n + 1 = 2(2mn + m + n) + 1.$$

Since $2mn + m + n \in \mathbb{Z}$, the product ab is odd. □

Commentary: The repaired proof opens the definition of odd twice and closes the definition of odd once.

27.8 Bad proof repair 2: divisibility

Claim: If $a \mid b$ and $b \mid c$, then $a \mid c$.

Bad proof:

> Since a divides b and b divides c, a is part of b and b is part of c, so a is part of c.

Diagnosis:

- "Part of" is not a divisibility definition.
- The proof gives a picture, not a witness.
- The conclusion $a \mid c$ requires an integer q such that $c = aq$.

Repair:

Proof. Suppose $a \mid b$ and $b \mid c$. Then there exist integers m, n such that

$$b = am, \qquad c = bn.$$

Substituting $b = am$ into $c = bn$ gives

$$c = (am)n = a(mn).$$

Since $mn \in \mathbb{Z}$, we have $a \mid c$. □

27.9 Bad proof repair 3: rational numbers

Claim: The sum of two rational numbers is rational.

Bad proof:

> Let $x = a/b$ and $y = c/d$. Then $x + y = (a + c)/(b + d)$, so the sum is rational.

Diagnosis:

- The addition formula is false.
- The proof does not state that $b, d \neq 0$.

- It accidentally produces an expression that can be undefined if $b + d = 0$.

Repair:

Proof. Let $x, y \in \mathbb{Q}$. Then there exist integers a, b, c, d with $b \neq 0$, $d \neq 0$, and

$$x = \frac{a}{b}, \qquad y = \frac{c}{d}.$$

Therefore

$$x + y = \frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd}.$$

Since $ad + bc \in \mathbb{Z}$ and $bd \in \mathbb{Z}$ with $bd \neq 0$, the number $x + y$ is rational. □

Common trap

In proofs about rational numbers, the nonzero denominator is not decoration. It is part of the definition.

27.10 Bad proof repair 4: set equality

Claim:

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C).$$

Bad proof:

> This is true by distribution because intersection distributes over union.

Diagnosis:

- This may be acceptable only if the distributive law has already been proved.
- If the theorem is the distributive law, the proof is circular.
- A beginner should prove it by element chasing.

Repair:

Proof. Let x be arbitrary. Then

$$x \in A \cap (B \cup C)$$

if and only if $x \in A$ and $x \in B \cup C$. This holds if and only if $x \in A$ and either $x \in B$ or $x \in C$. By distributivity of logical "and" over logical "or," this is equivalent to

$$(x \in A \text{ and } x \in B) \quad \text{or} \quad (x \in A \text{ and } x \in C).$$

That is equivalent to

$$x \in (A \cap B) \cup (A \cap C).$$

Since x was arbitrary, the sets are equal. □

27.11 Bad proof repair 5: image and preimage

Claim:

$$f^{-1}(S \cap T) = f^{-1}(S) \cap f^{-1}(T).$$

Bad proof:

> This is obvious because inverse functions preserve intersections.

Diagnosis:

- The notation $f^{-1}(S)$ does not require f to have an inverse function.
- The proof does not unpack the definition of inverse image.
- The word "obvious" is hiding all the work.

Repair:

Proof. Let x be arbitrary. Then

$$x \in f^{-1}(S \cap T)$$

if and only if $f(x) \in S \cap T$. This holds if and only if $f(x) \in S$ and $f(x) \in T$. That is equivalent to saying

$$x \in f^{-1}(S) \quad \text{and} \quad x \in f^{-1}(T).$$

Therefore $x \in f^{-1}(S) \cap f^{-1}(T)$. Since membership in the two sets is equivalent for arbitrary x, the sets are equal. □

27.12 Bad proof repair 6: injectivity

Claim: If $f : A \to B$ and $g : B \to C$ are injective, then $g \circ f : A \to C$ is injective.

Bad proof:

> Both functions are one-to-one, so no values collide. Therefore the composition is one-to-one.

Diagnosis:

- The idea is correct but not yet a proof.
- The proof does not begin with equal outputs of the composition.
- It does not use injectivity in the standard direction.

Repair:

Proof. Let $a_1, a_2 \in A$ and suppose

$$(g \circ f)(a_1) = (g \circ f)(a_2).$$

Then

$$g(f(a_1)) = g(f(a_2)).$$

Since g is injective, it follows that

$$f(a_1) = f(a_2).$$

Since f is injective, it follows that $a_1 = a_2$. Therefore $g \circ f$ is injective. □

27.13 Bad proof repair 7: surjectivity

Claim: If $f : A \to B$ and $g : B \to C$ are surjective, then $g \circ f : A \to C$ is surjective.

Bad proof:

> Let $a \in A$. Since f is onto, $f(a) = b$, and since g is onto, $g(b) = c$. Therefore every c is hit.

Diagnosis:

- To prove surjectivity onto C, start with arbitrary $c \in C$, not arbitrary $a \in A$.
- The proof uses b and c without correct quantifier order.
- Surjectivity gives preimages of chosen outputs, not arbitrary symbolic equations.

Repair:

Proof. Let $c \in C$ be arbitrary. Since g is surjective, there exists $b \in B$ such that $g(b) = c$. Since f is surjective, there exists $a \in A$ such that $f(a) = b$. Then

$$(g \circ f)(a) = g(f(a)) = g(b) = c.$$

Thus every $c \in C$ is hit by $g \circ f$, so $g \circ f$ is surjective. □

27.14 Bad proof repair 8: contrapositive

Claim: If n^2 is odd, then n is odd.

Bad proof:

> Assume n^2 is odd. Then $n^2 = 2k + 1$. So $n = \sqrt{2k+1}$, which is odd.

Diagnosis:

- Taking square roots does not produce the definition of odd.
- The expression $\sqrt{2k+1}$ need not display an integer in the form $2m + 1$.
- The contrapositive is cleaner.

Repair:

Proof. We prove the contrapositive. Suppose n is not odd. Since every integer is either even or odd, n is even. Then $n = 2m$ for some $m \in \mathbb{Z}$. Hence

$$n^2 = (2m)^2 = 4m^2 = 2(2m^2),$$

so n^2 is even. Therefore, if n^2 is odd, then n is odd. □

Commentary: A direct proof is possible, but it typically smuggles in a parity fact. Contrapositive proof uses the definition of even immediately.

27.15 Bad proof repair 9: contradiction

Claim: There are infinitely many prime numbers.

Bad proof:

> Assume there are finitely many primes $p_1, \ldots, p_n$. Then $p_1 p_2 \cdots p_n + 1$ is prime, so there is another prime. Contradiction.

Diagnosis:

- The number $p_1 p_2 \cdots p_n + 1$ need not be prime.
- What is true is that no listed prime divides it.
- The proof needs the fact that every integer greater than 1 has a prime divisor.

Repair:

Proof. Suppose, for contradiction, that there are only finitely many primes, say

$$p_1, p_2, \ldots, p_n.$$

Let

$$N = p_1 p_2 \cdots p_n + 1.$$

Since $N > 1$, it has a prime divisor q. By the assumption that our list contains all primes, $q = p_i$ for some i. Then q divides the product $p_1 p_2 \cdots p_n$, and q also divides N. Hence q divides their difference:

$$N - p_1 p_2 \cdots p_n = 1.$$

This is impossible, since no prime divides 1. Therefore there are infinitely many primes. □

27.16 Bad proof repair 10: induction

Claim:

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}$$

for all $n \in \mathbb{N}$.

Bad proof:

> It works for 1. Assume it works for n. Then
>
> $$1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$
>
> Therefore it works for $n + 1$.

Diagnosis:

- The induction step never proves the statement for $n + 1$.
- The proof repeats the induction hypothesis and stops.
- The missing move is adding $n + 1$ to both sides and simplifying to the correct target.

Repair:

Proof. For $n = 1$, the left side is 1 and the right side is $1(2)/2 = 1$.

Assume for some $k \in \mathbb{N}$ that

$$1 + 2 + \cdots + k = \frac{k(k+1)}{2}.$$

Then

$$1 + 2 + \cdots + k + (k+1) = \frac{k(k+1)}{2} + (k+1).$$

Factoring gives

$$\frac{k(k+1)}{2} + (k+1) = \frac{k(k+1) + 2(k+1)}{2} = \frac{(k+1)(k+2)}{2}.$$

This is the desired formula with $k + 1$ in place of k. Therefore the formula holds for all $n \in \mathbb{N}$. □

27.17 Bad proof repair 11: uniqueness

Claim: If an integer has an additive inverse, it is unique.

Bad proof:

> The inverse of a is $-a$, and there is only one $-a$.

Diagnosis:

- This assumes the formula for the inverse rather than proving uniqueness from the inverse property.
- A uniqueness proof should suppose there are two candidates and prove they are equal.
- The relevant law is associativity and the identity property of 0.

Repair:

Proof. Let $a \in \mathbb{Z}$. Suppose b and c are both additive inverses of a. Then

$$a + b = 0, \qquad a + c = 0.$$

Now

$$b = b + 0 = b + (a + c) = (b + a) + c = (a + b) + c = 0 + c = c.$$

Thus $b = c$, so the additive inverse is unique. □

Commentary: This proof pattern appears constantly in algebra: assume two objects have the universal property or defining property, then prove they must be equal.

27.18 Bad proof repair 12: open sets

Claim: The interval $(0, 1)$ is open in $\mathbb{R}$.

Bad proof:

> It is open because it does not contain its endpoints.

Diagnosis:

- Endpoint language works for intervals in $\mathbb{R}$, but it is not the definition of open.
- The proof must show that each point has a positive-radius neighborhood inside the interval.

- The radius must depend on the point.

Repair:

Proof. Let $x \in (0,1)$. Define

$$r = \frac{1}{2}\min\{x, 1-x\}.$$

Since $0 < x < 1$, both x and $1-x$ are positive, so $r > 0$. If $y \in \mathbb{R}$ and $|y - x| < r$, then

$$x - r < y < x + r.$$

Because $r \leq x/2$, we have $x - r \geq x/2 > 0$. Because $r \leq (1-x)/2$, we have $x + r \leq x + (1-x)/2 < 1$. Hence $0 < y < 1$, so $y \in (0,1)$. Thus every point of $(0,1)$ has a ball contained in $(0,1)$, and $(0,1)$ is open. □

27.19 Bad proof repair 13: limits

Claim:

$$\lim_{n\to\infty} \frac{4n+1}{n} = 4.$$

Bad proof:

> As n gets big, $1/n$ goes away, so the limit is 4.

Diagnosis:

- The intuition is correct, but the proof does not use the epsilon definition.
- The quantity to control is
$$\left|\frac{4n+1}{n} - 4\right|.$$
- A final proof must choose N in terms of ε.

Repair:

Proof. Let $\varepsilon > 0$. Choose $N \in \mathbb{N}$ such that $N > 1/\varepsilon$. If $n \geq N$, then

$$\left|\frac{4n+1}{n} - 4\right| = \left|4 + \frac{1}{n} - 4\right| = \frac{1}{n} \leq \frac{1}{N} < \varepsilon.$$

Therefore $\frac{4n+1}{n} \to 4$. □

27.20 Bad proof repair 14: equivalence relations

Claim: Congruence modulo m is an equivalence relation on $\mathbb{Z}$, where $m \in \mathbb{N}$ and $m \geq 2$.

Bad proof:

> This is an equivalence relation because numbers with the same remainder are equivalent.

Diagnosis:

- The sentence describes the idea but does not verify reflexive, symmetric, and transitive.
- The proof should use the definition $a \equiv b \pmod{m}$ if $m \mid (a - b)$.
- Each property needs its own argument.

Repair:

Proof. Define $a \sim b$ if $m \mid (a - b)$.

Reflexive: For any $a \in \mathbb{Z}$, we have $a - a = 0 = m \cdot 0$, so $m \mid (a - a)$. Hence $a \sim a$.

Symmetric: Suppose $a \sim b$. Then $m \mid (a - b)$, so $a - b = mk$ for some $k \in \mathbb{Z}$. Thus

$$b - a = -(a - b) = m(-k),$$

so $m \mid (b - a)$, and $b \sim a$.

Transitive: Suppose $a \sim b$ and $b \sim c$. Then $a - b = mk$ and $b - c = m\ell$ for some $k, \ell \in \mathbb{Z}$. Adding gives

$$a - c = (a - b) + (b - c) = m(k + \ell).$$

Therefore $m \mid (a - c)$, so $a \sim c$.

Thus congruence modulo m is an equivalence relation. □

27.21 Bad proof repair 15: partial orders

Claim: The subset relation $\subseteq$ is a partial order on $\mathcal{P}(X)$.

Bad proof:

> Subsets are ordered by size, so this is a partial order.

Diagnosis:

- Subset inclusion is not the same as size comparison.
- Two sets can have the same size without being equal.
- A partial order requires reflexivity, antisymmetry, and transitivity.

Repair:

Proof. We check the three properties on $\mathcal{P}(X)$.

Reflexive: For any $A \subseteq X$, every element of A is an element of A, so $A \subseteq A$.

Antisymmetric: Suppose $A \subseteq B$ and $B \subseteq A$. Then every element of A is in B, and every element of B is in A. Hence $A = B$.

Transitive: Suppose $A \subseteq B$ and $B \subseteq C$. If $x \in A$, then $x \in B$, and therefore $x \in C$. Thus $A \subseteq C$.

Therefore $\subseteq$ is a partial order on $\mathcal{P}(X)$. □

27.22 Debugging drills

For each proof below, identify the first bad line, name the gap, and write a repair.

1. Claim: If $x, y \in \mathbb{Q}$, then $xy \in \mathbb{Q}$. Bad proof: "Since rationals are fractions, their product is a fraction."
2. Claim: If $a \mid b$ and $a \mid c$, then $a \mid (b + c)$. Bad proof: "Since a goes into both, it goes into the sum."
3. Claim: If $A \subseteq B$, then $A \cap C \subseteq B \cap C$. Bad proof: "Intersections preserve inclusion."
4. Claim: If f is injective, then $f^{-1}(\{f(a)\}) = \{a\}$. Bad proof: "This follows because injective functions have inverses."
5. Claim: If n^2 is divisible by 3, then n is divisible by 3. Bad proof: "Taking square roots preserves divisibility."
6. Claim: If $a_n \to 0$ and $b_n \to 0$, then $a_n + b_n \to 0$. Bad proof: "Both are small eventually, so the sum is small."
7. Claim: If U and V are open subsets of $\mathbb{R}$, then $U \cap V$ is open. Bad proof: "The intersection of open things is open."
8. Claim: If R and S are equivalence relations on A, then $R \cap S$ is an equivalence relation. Bad proof: "Intersections preserve structure."

Proof lab

For the next week, keep a proof error log. Each entry should have four lines: the broken sentence, the gap type, the repaired sentence, and the lesson. The goal is not to collect embarrassments. The goal is to make your mistakes searchable.

28 The Velleman Readiness Plan

Velleman's *How to Prove It* is not a book of tricks. It is a book about proof architecture. You are ready to benefit from it when you can recognize the outer shape of a proof before knowing every inner move.

Focus

Readiness does not mean speed. It means you can locate the next responsible move.

The plan below is deliberately detailed. It is meant to be used, not admired. If a week takes two weeks, that is fine. The point is to build a stable proof habit before the text accelerates.

28.1 The readiness standard

Before beginning Velleman seriously, you should be able to do most of the following without looking up a template.

1. Translate a quantified English sentence into symbols.
2. Translate a symbolic quantified sentence into clear English.
3. Negate statements with one, two, and three quantifiers.
4. Identify the hypothesis and conclusion of a conditional.
5. Prove a direct implication using definitions.
6. Decide when the contrapositive is cleaner than direct proof.
7. Use contradiction without leaving the contradiction vague.
8. Prove set equality by arbitrary element or two inclusions.
9. Prove subset claims by starting with an arbitrary element.
10. Prove injectivity and surjectivity from their definitions.
11. Work with images and inverse images of sets.
12. Verify that a relation is reflexive, symmetric, transitive, antisymmetric, or total.

13. Prove a relation is an equivalence relation.
14. Describe equivalence classes and quotient sets.
15. Write ordinary induction with a clean base case and induction step.
16. Recognize when strong induction is the natural tool.
17. Produce counterexamples to false universal claims.
18. Review your own proof using the first-bad-line rule.

28.2 A six-week readiness schedule

The schedule assumes five study sessions per week. Each session should be about sixty to ninety minutes. If you have less time, preserve the order and reduce the number of exercises; do not skip the proof-revision days.

Week	Theme	Daily work	End-of-week standard
1	Mathematical sentences	Statements, predicates, domains, conditionals, negations	You can say exactly what a theorem claims
2	Quantifiers	Translation, negation, order of choice, examples and counterexamples	You no longer treat $\forall\exists$ and $\exists\forall$ as interchangeable
3	Core proof methods	Direct proof, contrapositive, contradiction, cases	You can choose a proof shape from the conclusion
4	Sets and functions	Subsets, equality, power sets, image, inverse image, injective, surjective	You can run element-chasing proofs without guessing
5	Relations and induction	Equivalence, partial order, classes, ordinary and strong induction	You can prove structural claims in organized parts
6	Portfolio and exam	Rewrite, debug, timed proofs, oral explanations	You can begin Velleman with active reading habits

28.3 Daily session template

Use this structure for most study days.

1. **Ten-minute recall.** Without notes, write definitions from yesterday.
2. **Twenty-minute reading.** Read slowly. Stop at each theorem and predict the proof shape.
3. **Thirty-minute proof writing.** Write one or two complete proofs in full sentences.

4. **Fifteen-minute debugging.** Review one proof using the feedback workflow.
5. **Five-minute log.** Record one definition, one mistake, and one question.

Walking question

At the end of a session, ask: what did I actually prove today, and which definition did I use most heavily?

28.4 Week 1: sentences and domains

Goals:

- distinguish expressions, predicates, statements, commands, and questions;
- identify the domain of every variable;
- rewrite conditionals as "if hypothesis, then conclusion";
- notice when a statement is true in one domain and false in another.

Core drills:

1. Decide whether each is a statement: $x^2 = 4$; "there exists $x \in \mathbb{R}$ such that $x^2 = 4$"; "solve $x^2 = 4$"; "n is prime."
2. For $P(x) : x^2 < 4$, determine the truth set over $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, and $\mathbb{R}$.
3. Rewrite: "A square integer is nonnegative" using a universal quantifier.
4. Rewrite: "Some integer is larger than its square" using an existential quantifier, then decide whether it is true.
5. Explain why the domain matters in the statement "Every number has a multiplicative inverse."

Portfolio task:

> Write a one-page note titled "What a Mathematical Sentence Needs." Include five examples and three nonexamples.

28.5 Week 2: quantifier discipline

Goals:

- negate quantified statements without guessing;

- understand order of dependency;
- distinguish pointwise and uniform claims;
- build counterexamples to false universal statements.

Core drills:

1. Negate: $\forall x \in \mathbb{R}, \exists y \in \mathbb{R},\ y > x$.
2. Negate: $\exists M \in \mathbb{R}, \forall x \in S,\ x \leq M$.
3. Negate: $\forall \varepsilon > 0, \exists N \in \mathbb{N}, \forall n \geq N,\ |a_n - L| < \varepsilon$.
4. Explain the difference between "every student has a favorite book" and "there is a book favored by every student."
5. Give an example where $\forall x \exists y\, R(x, y)$ is true but $\exists y \forall x\, R(x, y)$ is false.

Portfolio task:

> Create a two-column page. In the left column write ten quantified statements. In the right column write their negations in English and symbols.

Common trap

The negation of "for every x, there exists y" is not "for every x, there does not exist y." The correct negation is "there exists x such that for every y, not."

28.6 Week 3: proof methods

Goals:

- use direct proof when definitions flow forward;
- use contrapositive when the negated conclusion has strong structure;
- use contradiction when denying the conclusion creates a concrete object to defeat;
- use cases only when the cases are exhaustive.

Core drills:

1. Prove directly: If a, b are even integers, then $a + b$ is even.
2. Prove by contrapositive: If n^2 is even, then n is even.
3. Prove by contradiction: $\sqrt{2}$ is irrational.

4. Prove by cases: For every integer n, $n^2 + n$ is even.
5. For each proof above, write the proof skeleton before the final proof.

Portfolio task:

> Write four proofs of parity or divisibility claims: one direct, one contrapositive, one contradiction, one cases. Label every definition opened and closed.

28.7 Week 4: sets and functions

Goals:

- prove subset claims by arbitrary elements;
- prove set equality by membership equivalence or two inclusions;
- distinguish image from inverse image;
- prove injectivity and surjectivity using correct starting lines.

Core drills:

1. Prove $A \cap B \subseteq A$.
2. Prove $A \setminus B = A \cap B^c$.
3. Prove $f^{-1}(S \cup T) = f^{-1}(S) \cup f^{-1}(T)$.
4. Give a counterexample showing $f(S \cap T) = f(S) \cap f(T)$ can fail.
5. Prove that the composition of injective functions is injective.
6. Prove that the composition of surjective functions is surjective.

Portfolio task:

> Write a clean mini-chapter titled "Element Chasing." Include one subset proof, one equality proof, one inverse-image proof, and one counterexample involving images.

28.8 Week 5: relations and induction

Goals:

- check relation properties in separate paragraphs;
- describe equivalence classes without handwaving;

- understand quotient sets as sets of classes;
- write induction proofs with an explicit induction statement.

Core drills:

1. Prove equality modulo m is an equivalence relation on $\mathbb{Z}$.
2. Describe the equivalence classes modulo 4.
3. Prove that subset inclusion is a partial order on $\mathcal{P}(X)$.
4. Prove by induction that $1 + 3 + \cdots + (2n - 1) = n^2$.
5. Prove by induction that $2^n \geq n + 1$ for all $n \geq 1$.
6. Explain why strong induction is useful for proving every integer $n \geq 2$ has a prime factorization.

Portfolio task:

> Write one full relation proof and one full induction proof. Then write a paragraph comparing the two proof structures.

28.9 Week 6: portfolio, review, and launch

Goals:

- rewrite older proofs with cleaner definitions;
- identify recurring personal errors;
- take a timed readiness exam;
- begin Velleman with an active reading protocol.

Core work:

1. Choose five old proofs and revise them.
2. For each revision, write the first bad line from the old version.
3. Complete the final exam at the end of this file under a time limit.
4. Explain three proofs out loud without looking.
5. Start Velleman Chapter 1 by translating every displayed symbolic statement into English.

28.10 The Velleman reading protocol

When reading Velleman, do not read passively. Use the following procedure.

1. Before a section, skim headings and examples. Predict the skill being taught.
2. For each definition, write one example and one nonexample.
3. For each theorem, identify the proof shape before reading the proof.
4. After reading a proof, close the book and write the skeleton.
5. For each exercise, write the target definition before attempting the proof.
6. After solving, rewrite the proof once for clarity.
7. Keep a list of "moves" rather than isolated answers.

History and connection

Velleman's book belongs to a tradition of proof texts that teach mathematics as language, strategy, and structure rather than as a list of finished theorems. Its exercises matter because they force you to practice the transitions: English to logic, logic to proof plan, proof plan to readable argument.

28.11 Readiness self-test

Answer these without notes.

1. What is the difference between a statement and a predicate?
2. What is the negation of $P \Rightarrow Q$?
3. How do you prove a universal statement?
4. How do you prove an existential statement?
5. How do you disprove a universal statement?
6. How do you prove $A \subseteq B$?
7. How do you prove $A = B$?
8. How do you prove a function is injective?
9. How do you prove a function is surjective?
10. What are the three properties of an equivalence relation?
11. What are the three properties of a partial order?
12. What are the three required parts of an induction proof?

If you miss more than three, review before starting Velleman seriously. If you miss one or two, start but keep those topics in daily review.

29 Second Pass and Mastery Review

The first pass through proof teaches vocabulary. The second pass teaches control. You revisit the same ideas, but now you ask sharper questions:

- What is the exact logical form?
- Which definition is doing the work?
- What would a counterexample have to look like?
- Which proof method fits the conclusion?
- Where could a beginner reverse an implication?
- How will this idea return in analysis, algebra, topology, or geometry?

Focus

Mastery is not remembering a theorem. Mastery is knowing what kind of work the theorem can do.

29.1 Mastery review method

Use five passes for each major topic.

1. **Definition pass.** Write every definition from memory.
2. **Example pass.** Give three examples and two nonexamples.
3. **Proof pass.** Prove one standard theorem using the definition.
4. **Debug pass.** Repair one broken proof.
5. **Transfer pass.** Explain where the idea appears later.

This method is slower than rereading. It is also much more honest. Rereading can feel fluent even when you cannot produce anything. Mastery review forces production.

29.2 Proof Compression and Expansion

A major barrier in advanced reading is proof compression. A mature text may write:

> The result follows immediately from compactness.

That phrase can hide an entire open-cover argument. Your job is to learn how to expand compressed moves until they become checkable.

There are several compression levels:

1. **Expanded proof:** every definition is opened and every witness is named.
2. **Textbook proof:** standard definitions are used fluently, but the strategy is visible.
3. **Proof sketch:** only the main mechanism is described.
4. **Expert shorthand:** standard lemmas and idioms are invoked by name.

Formal

Expansion drill:

> Whenever a proof says "clearly," "by definition," "standard," or "immediate," replace that phrase with three explicit lines.

Compressed:

> Since $A \subseteq B$, clearly $\mathcal{P}(A) \subseteq \mathcal{P}(B)$.

Expanded:

> Let $S \in \mathcal{P}(A)$. Then $S \subseteq A$. Since $A \subseteq B$, every element of S is an element of B. Hence $S \subseteq B$, so $S \in \mathcal{P}(B)$. Therefore $\mathcal{P}(A) \subseteq \mathcal{P}(B)$.

Common trap

False intuition: if you understand each symbol, you understand the proof. Not necessarily. You also need to understand the omitted transitions between lines.

Proof lab

Take five proofs from earlier chapters. For each, write an expanded version and a compressed version. Then write one sentence naming what was compressed.

29.3 Logic review

Core definitions:

- A statement is a sentence that is true or false.
- A predicate becomes a statement when variables receive values from specified domains.
- A conditional $P \Rightarrow Q$ is false exactly when P is true and Q is false.
- The contrapositive of $P \Rightarrow Q$ is $\neg Q \Rightarrow \neg P$.
- The converse of $P \Rightarrow Q$ is $Q \Rightarrow P$.

Mastery questions:

1. Can you tell whether a sentence is a statement before trying to prove it?
2. Can you identify the domain of every variable?
3. Can you negate a conditional without saying "if not P, then not Q"?
4. Can you explain why a statement and its contrapositive are equivalent?
5. Can you produce a counterexample to the converse of a true implication?

Proof lab

Write five true implications from elementary number theory. For each one, write the converse and decide whether it is true. If false, give a counterexample.

29.4 Quantifier review

Quantifiers are not decoration. They are instructions for proof and disproof.

Statement form	To prove it	To disprove it
$\forall x \in A,\ P(x)$	Let arbitrary $x \in A$ and prove $P(x)$	Find one $x \in A$ with $\neg P(x)$
$\exists x \in A,\ P(x)$	Exhibit a particular $x \in A$ with $P(x)$	Prove every $x \in A$ fails $P(x)$
$\forall x \exists y,\ R(x,y)$	Let arbitrary x, then construct y depending on x	Find one x for which no y works
$\exists y \forall x,\ R(x,y)$	Produce one y that works for all x	Show every candidate y fails for some x

Drill:

1. Prove or disprove: $\forall x \in \mathbb{R}, \exists y \in \mathbb{R},\ x + y = 0$.
2. Prove or disprove: $\exists y \in \mathbb{R}, \forall x \in \mathbb{R},\ x + y = 0$.
3. Prove or disprove: $\forall n \in \mathbb{Z}, \exists m \in \mathbb{Z},\ n < m$.

4. Prove or disprove: $\exists m \in \mathbb{Z}, \forall n \in \mathbb{Z},\ n < m$.
5. Translate the previous two into everyday English.

29.5 Definition review

A definition is a contract. It tells you what you may use and what you must prove.

Formal

To prove an object has property P, expand the definition of P and satisfy each clause. To use the fact that an object has property P, expand the definition and extract its clauses.

Example: divisibility.

$$a \mid b \quad \text{means} \quad \exists k \in \mathbb{Z} \text{ such that } b = ak.$$

To prove $a \mid b$, produce k. To use $a \mid b$, introduce k with $b = ak$.

Example: injective.

$$f \text{ injective} \quad \text{means} \quad f(x_1) = f(x_2) \Rightarrow x_1 = x_2.$$

To prove injective, assume equal outputs. To use injective, first obtain equal outputs.

Example: open set in a metric space.

$$U \text{ open} \quad \text{means} \quad \forall x \in U, \exists r > 0,\ B_r(x) \subseteq U.$$

To prove open, start with arbitrary point and construct a radius. To use open, feed it a point and receive a radius.

29.6 Direct proof review

Direct proof is the default when the hypothesis naturally opens into the ingredients needed by the conclusion.

Claim: If $a, b, c \in \mathbb{Z}$, $a \mid b$, and $a \mid c$, then $a \mid (2b - 3c)$.

Proof. Since $a \mid b$, there exists $m \in \mathbb{Z}$ such that $b = am$. Since $a \mid c$, there exists $n \in \mathbb{Z}$ such that $c = an$. Then

$$2b - 3c = 2(am) - 3(an) = a(2m - 3n).$$

Since $2m - 3n \in \mathbb{Z}$, we have $a \mid (2b - 3c)$. □

Commentary: The proof is almost mechanical because the conclusion asks for a multiple of a, and the hypotheses already express b and c as multiples of a.

29.7 Contrapositive review

Contrapositive proof is useful when the conclusion is hard to show directly but its negation has a clean structure.

Claim: If n^2 is divisible by 3, then n is divisible by 3.

Proof. We prove the contrapositive. Suppose n is not divisible by 3. Then n has remainder 1 or 2 when divided by 3. If $n = 3q + 1$, then

$$n^2 = (3q + 1)^2 = 9q^2 + 6q + 1 = 3(3q^2 + 2q) + 1,$$

so n^2 is not divisible by 3. If $n = 3q + 2$, then

$$n^2 = (3q + 2)^2 = 9q^2 + 12q + 4 = 3(3q^2 + 4q + 1) + 1,$$

so again n^2 is not divisible by 3. Therefore, if n^2 is divisible by 3, then n is divisible by 3. □

Intuition

The contrapositive turns "a hidden factor in n^2 forces a hidden factor in n" into a visible remainder calculation.

29.8 Contradiction review

Proof by contradiction is powerful, but it should end in a named impossibility.

Weak ending:

> This is a contradiction.

Better ending:

> This contradicts the fact that p and q have no common divisor greater than 1.

Classic example:

Proof. Suppose, for contradiction, that $\sqrt{2}$ is rational. Then $\sqrt{2} = p/q$ for integers p, q with $q \neq 0$ and with the fraction in lowest terms. Squaring gives

$$2 = \frac{p^2}{q^2},$$

so

$$p^2 = 2q^2.$$

Thus p^2 is even, so p is even. Write $p = 2r$. Then

$$4r^2 = 2q^2,$$

so

$$q^2 = 2r^2.$$

Thus q^2 is even, so q is even. Hence both p and q are even, contradicting that p/q was in lowest terms. Therefore $\sqrt{2}$ is irrational. □

Commentary: The proof uses a lemma twice: if a square is even, then the original integer is even. In a polished proof, cite it. In a beginner portfolio, prove it before using it.

29.9 Cases review

Cases are useful only when they are exhaustive and each case reaches the same target.

Claim: For every integer n, $n^2 + n$ is even.

Proof. Let $n \in \mathbb{Z}$. There are two cases.

If n is even, then $n = 2k$ for some $k \in \mathbb{Z}$. Then

$$n^2 + n = n(n+1) = 2k(n+1),$$

which is even.

If n is odd, then $n = 2k + 1$ for some $k \in \mathbb{Z}$. Then

$$n + 1 = 2k + 2 = 2(k+1),$$

so

$$n^2 + n = n(n+1) = 2(n(k+1)),$$

which is even.

In both cases, $n^2 + n$ is even. □

Debugging question: where did we use that every integer is either even or odd? It appears in the phrase “There are two cases.” A fully formal proof would state the parity dichotomy as a known lemma.

29.10 Set proof review

Set proofs are logic proofs wearing membership notation.

Claim:

$$A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C).$$

Proof. Let x be arbitrary. Then $x \in A \setminus (B \cup C)$ if and only if $x \in A$ and $x \notin B \cup C$. This is equivalent to $x \in A$ and not ($x \in B$ or $x \in C$). By De Morgan's law, this is equivalent to

$$x \in A, \qquad x \notin B, \qquad x \notin C.$$

That is equivalent to $x \in A \setminus B$ and $x \in A \setminus C$, which is equivalent to

$$x \in (A \setminus B) \cap (A \setminus C).$$

Therefore the sets are equal. □

29.11 Function proof review

For functions, most beginner errors are starting with the wrong object.

Target	Correct first move	Common wrong first move
Prove f injective	Let x_1, x_2 be in the domain and assume $f(x_1) = f(x_2)$	Let y be in the codomain
Prove f surjective	Let y be arbitrary in the codomain	Let x be arbitrary in the domain
Prove $f^{-1}(S) \subseteq f^{-1}(T)$	Let $x \in f^{-1}(S)$	Assume f has an inverse
Prove $f(A) \subseteq B$	Let $y \in f(A)$ or let $x \in A$ and show $f(x) \in B$	Treat A as a number

Drill:

1. Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = 3x - 5$. Prove f is injective.
2. Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = 3x - 5$. Prove f is surjective.
3. Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = x^2$. Prove f is not injective.
4. Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = x^2$. Prove f is not surjective.
5. Let $f : A \to B$. Prove that if $S \subseteq T \subseteq B$, then $f^{-1}(S) \subseteq f^{-1}(T)$.

29.12 Relation review

Relations require property-by-property discipline. Do not compress reflexive, symmetric, and transitive into one sentence unless the relation is already routine for your reader.

Claim: On $\mathbb{R}$, define $x \sim y$ if $x - y \in \mathbb{Z}$. Then $\sim$ is an equivalence relation.

Proof. Reflexive: For any $x \in \mathbb{R}$, $x - x = 0 \in \mathbb{Z}$, so $x \sim x$.

Symmetric: Suppose $x \sim y$. Then $x - y \in \mathbb{Z}$. Since integers are closed under additive inverses,

$$y - x = -(x - y) \in \mathbb{Z}.$$

Thus $y \sim x$.

Transitive: Suppose $x \sim y$ and $y \sim z$. Then $x - y \in \mathbb{Z}$ and $y - z \in \mathbb{Z}$. Since integers are closed under addition,

$$x - z = (x - y) + (y - z) \in \mathbb{Z}.$$

Thus $x \sim z$.

Therefore $\sim$ is an equivalence relation. □

Later view: this relation identifies real numbers that differ by an integer. The quotient $\mathbb{R}/\mathbb{Z}$ can be pictured as a circle. This is a first glimpse of how equivalence relations create new spaces.

29.13 Induction review

An induction proof must make the induction statement clear. The statement is not "the formula." It is a predicate $P(n)$.

Claim:

$$1 + 3 + 5 + \cdots + (2n - 1) = n^2$$

for all $n \in \mathbb{N}$.

Proof. Let $P(n)$ be the statement

$$1 + 3 + \cdots + (2n - 1) = n^2.$$

Base case: For $n = 1$, the left side is 1 and the right side is $1^2 = 1$, so $P(1)$ is true.

Induction step: Assume $P(k)$ is true for some $k \in \mathbb{N}$, so

$$1 + 3 + \cdots + (2k - 1) = k^2.$$

Then

$$1 + 3 + \cdots + (2k - 1) + (2(k + 1) - 1) = k^2 + (2k + 1).$$

Since

$$k^2 + 2k + 1 = (k+1)^2,$$

we get

$$1 + 3 + \cdots + (2(k+1) - 1) = (k+1)^2.$$

Thus $P(k+1)$ is true. By induction, $P(n)$ is true for all $n \in \mathbb{N}$. □

29.14 Spaced repetition prompts

Use these prompts on a rotating schedule. Answer from memory first, then check notes.

Day	Prompts
1	Define statement, predicate, implication, converse, contrapositive. Give one example of each.
2	Negate three quantified statements: one $\forall$, one $\exists$, one $\forall\exists$.
3	Prove a parity claim from definitions. Then write one bad proof of the same claim and repair it.
4	Prove one subset relation and one set equality. Name each logical law used.
5	Prove one injectivity claim and one surjectivity claim. Start with the correct arbitrary object.
6	Verify one equivalence relation. Write the equivalence class of a sample element.
7	Write one induction proof. Underline the base case, hypothesis, and induction step.
14	Redo the hardest proof from week 1 without looking. Compare to the old version.
21	Explain quantifier order using your own example. Include a false reversed version.
30	Take a mini-exam: one logic question, one set proof, one function proof, one induction proof.

29.15 Mastery checklist

Mark each item as solid, shaky, or missing.

1. I can write the definition of even, odd, divides, rational, injective, surjective, equivalence relation, partial order, and open set.
2. I can prove simple parity claims without relying on slogans.
3. I can negate nested quantified statements.
4. I can choose between direct proof, contrapositive, contradiction, and cases.

5. I can prove set equalities by arbitrary elements.
6. I can avoid confusing image and inverse image.
7. I can produce counterexamples using small finite sets or simple functions.
8. I can write clean induction proofs.
9. I can find the first bad line in a flawed proof.
10. I can explain a proof aloud in ordinary mathematical English.

Focus

If a checklist item is shaky, assign it a proof task. Do not assign it more reading. Reading helps after production reveals the exact gap.

30 Portfolio Assignments

A proof portfolio is a curated record of competence. It should not contain every exercise you have ever attempted. It should contain carefully revised proofs that show you can handle the core forms of early pure mathematics.

30.1 Portfolio rules

For each portfolio proof:

1. State the theorem before proving it.
2. List definitions you expect to use.
3. Write a proof skeleton.
4. Write the full proof.
5. Add a short commentary paragraph.
6. Add one possible mistake and explain how to avoid it.

The commentary matters. It forces you to notice the architecture instead of treating the proof as a string of symbols.

30.2 Portfolio A: arithmetic and definitions

Write polished proofs of the following.

1. The sum of two even integers is even.
2. The sum of two odd integers is even.
3. The product of two odd integers is odd.
4. If $a \mid b$ and $a \mid c$, then $a \mid (b + c)$.
5. If $a \mid b$ and $b \mid c$, then $a \mid c$.
6. If $x, y \in \mathbb{Q}$, then $x - y \in \mathbb{Q}$.
7. If $x, y \in \mathbb{Q}$, then $xy \in \mathbb{Q}$.
8. If $x \in \mathbb{Q}$ and $x \neq 0$, then $1/x \in \mathbb{Q}$.

Required commentary:

> For each proof, identify the witness required by the conclusion. For example, if proving $a \mid (b + c)$, identify the integer k such that $b + c = ak$.

30.3 Portfolio B: proof methods

Write polished proofs of the following.

1. If n^2 is even, then n is even.
2. If n^2 is odd, then n is odd.
3. If n^2 is divisible by 3, then n is divisible by 3.
4. There is no largest integer.
5. There is no smallest positive real number.
6. $\sqrt{2}$ is irrational.
7. There are infinitely many primes.
8. For every integer n, $n^2 + n$ is even.

Required commentary:

> Label the proof method. If you use contradiction, state the exact contradiction. If you use contrapositive, write both the original statement and the contrapositive.

30.4 Portfolio C: sets

Write polished proofs of the following.

1. $A \cap B \subseteq A$.
2. $A \subseteq A \cup B$.
3. $A \setminus B = A \cap B^c$.
4. $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.
5. $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$.
6. If $A \subseteq B$, then $A \cap C \subseteq B \cap C$.
7. If $A \subseteq B$, then $\mathcal{P}(A) \subseteq \mathcal{P}(B)$.
8. $A \subseteq B$ if and only if $B^c \subseteq A^c$.

Required commentary:

> For each proof, translate the set statement into a logical statement about an arbitrary element x.

30.5 Portfolio D: functions

Write polished proofs or counterexamples.

1. If $f : A \to B$ and $g : B \to C$ are injective, then $g \circ f$ is injective.
2. If $f : A \to B$ and $g : B \to C$ are surjective, then $g \circ f$ is surjective.
3. If $f : A \to B$ is injective and $S, T \subseteq A$, then $f(S \cap T) = f(S) \cap f(T)$.
4. Give a counterexample showing that the previous equality can fail when f is not injective.
5. Prove $f^{-1}(S \cap T) = f^{-1}(S) \cap f^{-1}(T)$.
6. Prove $f^{-1}(S \cup T) = f^{-1}(S) \cup f^{-1}(T)$.
7. Prove that if $S \subseteq T$, then $f^{-1}(S) \subseteq f^{-1}(T)$.
8. For $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = x^2$, compute $f([-2, 1])$ and $f^{-1}([0, 4])$.

Required commentary:

> For each function proof, state whether you start with an element of the domain, codomain, image, or inverse image. Explain why.

30.6 Portfolio E: relations and induction

Write polished proofs of the following.

1. Congruence modulo m is an equivalence relation on $\mathbb{Z}$.
2. The relation on $\mathbb{R}$ defined by $x \sim y$ if $x - y \in \mathbb{Z}$ is an equivalence relation.
3. The equivalence classes of congruence modulo 4 partition $\mathbb{Z}$.
4. Subset inclusion is a partial order on $\mathcal{P}(X)$.
5. Divisibility is a partial order on $\mathbb{N}$.
6. Prove by induction that $1 + 2 + \cdots + n = n(n+1)/2$.
7. Prove by induction that $1 + 3 + \cdots + (2n-1) = n^2$.
8. Prove by induction that $2^n \geq n + 1$ for all $n \geq 1$.

Required commentary:

> For relation proofs, describe each property separately. For induction proofs, identify $P(n)$ before writing the proof.

31 Guided Solutions and Commentary

Try each problem before reading its solution. The point of a guided solution is not just to show the answer. It is to show how the proof was found, where the definitions enter, and what a reader should notice.

31.1 Solution 1: sum of two even integers

Claim: If a and b are even integers, then $a + b$ is even.

Plan: Open the definition of even for a and for b. The conclusion asks us to express $a + b$ as 2 times an integer.

Proof. Let $a, b \in \mathbb{Z}$ be even. Then there exist $m, n \in \mathbb{Z}$ such that

$$a = 2m, \qquad b = 2n.$$

Therefore

$$a + b = 2m + 2n = 2(m + n).$$

Since $m + n \in \mathbb{Z}$, the number $a + b$ is even. □

Commentary: The witness for evenness of $a + b$ is $m + n$.

31.2 Solution 2: product of odd integers

Claim: If a and b are odd integers, then ab is odd.

Plan: Odd means $2k+1$. Multiply the forms and factor the result into 2(integer) + 1.

Proof. Let $a, b \in \mathbb{Z}$ be odd. Then $a = 2m+1$ and $b = 2n+1$ for some $m, n \in \mathbb{Z}$. Hence

$$ab = (2m+1)(2n+1) = 4mn + 2m + 2n + 1 = 2(2mn + m + n) + 1.$$

Since $2mn + m + n \in \mathbb{Z}$, ab is odd. □

Commentary: Do not stop at $4mn + 2m + 2n + 1$. The final form must visibly match the definition of odd.

31.3 Solution 3: divisibility is transitive

Claim: If $a \mid b$ and $b \mid c$, then $a \mid c$.

Proof. Suppose $a \mid b$ and $b \mid c$. Then $b = am$ and $c = bn$ for some $m, n \in \mathbb{Z}$. Therefore

$$c = bn = (am)n = a(mn).$$

Since $mn \in \mathbb{Z}$, $a \mid c$. □

Commentary: Divisibility proofs are witness proofs. The witness for $a \mid c$ is mn.

31.4 Solution 4: rational subtraction

Claim: If $x, y \in \mathbb{Q}$, then $x - y \in \mathbb{Q}$.

Proof. Let $x, y \in \mathbb{Q}$. Then there exist integers a, b, c, d with $b \neq 0$ and $d \neq 0$ such that

$$x = \frac{a}{b}, \qquad y = \frac{c}{d}.$$

Thus

$$x - y = \frac{a}{b} - \frac{c}{d} = \frac{ad - bc}{bd}.$$

Since $ad - bc \in \mathbb{Z}$ and $bd \in \mathbb{Z}$ with $bd \neq 0$, the number $x - y$ is rational. □

Commentary: The denominator condition must be preserved. Since b and d are nonzero integers, bd is nonzero.

31.5 Solution 5: no largest integer

Claim: There is no largest integer.

Proof. Suppose, for contradiction, that there is a largest integer N. Then $N+1$ is an integer and $N+1 > N$. This contradicts the assumption that N is largest. Therefore there is no largest integer. □

Commentary: The contradiction is not emotional. It is the existence of an integer larger than the alleged largest integer.

31.6 Solution 6: a set containment

Claim: If $A \subseteq B$, then $A \cap C \subseteq B \cap C$.

Proof. Suppose $A \subseteq B$. Let $x \in A \cap C$. Then $x \in A$ and $x \in C$. Since $A \subseteq B$, $x \in B$. Therefore $x \in B$ and $x \in C$, so $x \in B \cap C$. Hence $A \cap C \subseteq B \cap C$. □

Commentary: The arbitrary element begins in the smaller set $A \cap C$, because the target is a subset claim.

31.7 Solution 7: power sets

Claim: If $A \subseteq B$, then $\mathcal{P}(A) \subseteq \mathcal{P}(B)$.

Proof. Suppose $A \subseteq B$. Let $S \in \mathcal{P}(A)$. Then $S \subseteq A$. Since $A \subseteq B$, it follows that $S \subseteq B$. Thus $S \in \mathcal{P}(B)$. Therefore $\mathcal{P}(A) \subseteq \mathcal{P}(B)$. □

Commentary: The elements of $\mathcal{P}(A)$ are sets. The proof respects that type.

31.8 Solution 8: inverse image of a union

Claim:

$$f^{-1}(S \cup T) = f^{-1}(S) \cup f^{-1}(T).$$

Proof. Let x be arbitrary. Then

$$x \in f^{-1}(S \cup T)$$

if and only if $f(x) \in S \cup T$. This holds if and only if $f(x) \in S$ or $f(x) \in T$. That is equivalent to $x \in f^{-1}(S)$ or $x \in f^{-1}(T)$, which is equivalent to

$$x \in f^{-1}(S) \cup f^{-1}(T).$$

Therefore the two sets are equal. □

Commentary: Inverse image proofs are usually clean because membership translates directly through f.

31.9 Solution 9: image of intersection can fail

Claim: It is not always true that

$$f(S \cap T) = f(S) \cap f(T).$$

Counterexample: Let $f : \mathbb{R} \to \mathbb{R}$ be $f(x) = x^2$, let $S = \{-1\}$, and let $T = \{1\}$. Then

$$S \cap T = \emptyset,$$

so

$$f(S \cap T) = f(\emptyset) = \emptyset.$$

But

$$f(S) = \{1\}, \qquad f(T) = \{1\},$$

so

$$f(S) \cap f(T) = \{1\}.$$

Thus $f(S \cap T) \neq f(S) \cap f(T)$.

Commentary: The failure happens because two different inputs have the same output. This is exactly the kind of collision injectivity forbids.

31.10 Solution 10: composition of injective functions

Claim: If $f : A \to B$ and $g : B \to C$ are injective, then $g \circ f : A \to C$ is injective.

Proof. Let $a_1, a_2 \in A$ and suppose $(g \circ f)(a_1) = (g \circ f)(a_2)$. Then

$$g(f(a_1)) = g(f(a_2)).$$

Since g is injective, $f(a_1) = f(a_2)$. Since f is injective, $a_1 = a_2$. Therefore $g \circ f$ is injective. □

Commentary: Injectivity is used backward through the composition: first g, then f.

31.11 Solution 11: composition of surjective functions

Claim: If $f : A \to B$ and $g : B \to C$ are surjective, then $g \circ f : A \to C$ is surjective.

Proof. Let $c \in C$ be arbitrary. Since g is surjective, there exists $b \in B$ such that $g(b) = c$. Since f is surjective, there exists $a \in A$ such that $f(a) = b$. Then

$$(g \circ f)(a) = g(f(a)) = g(b) = c.$$

Thus $g \circ f$ is surjective. □

Commentary: Surjectivity is proved by chasing a target backward from the codomain.

31.12 Solution 12: congruence modulo m

Claim: For $m \geq 2$, congruence modulo m is an equivalence relation on $\mathbb{Z}$.

Proof. For $a, b \in \mathbb{Z}$, define $a \sim b$ if $m \mid (a - b)$.

Reflexive: Since $a - a = 0 = m \cdot 0$, we have $m \mid (a - a)$, so $a \sim a$.

Symmetric: If $a \sim b$, then $a - b = mk$ for some $k \in \mathbb{Z}$. Hence $b - a = m(-k)$, so $b \sim a$.

Transitive: If $a \sim b$ and $b \sim c$, then $a - b = mk$ and $b - c = m\ell$ for some $k, \ell \in \mathbb{Z}$. Therefore

$$a - c = (a - b) + (b - c) = m(k + \ell),$$

so $a \sim c$.

Thus $\sim$ is an equivalence relation. □

Commentary: The same divisibility definition powers all three parts.

31.13 Solution 13: subset inclusion is a partial order

Claim: $\subseteq$ is a partial order on $\mathcal{P}(X)$.

Proof. Reflexive: If $A \in \mathcal{P}(X)$, then every element of A is in A, so $A \subseteq A$.

Antisymmetric: Suppose $A, B \in \mathcal{P}(X)$, $A \subseteq B$, and $B \subseteq A$. Then A and B have the same elements, so $A = B$.

Transitive: Suppose $A \subseteq B$ and $B \subseteq C$. If $x \in A$, then $x \in B$, and since $B \subseteq C$, $x \in C$. Hence $A \subseteq C$.

Therefore $\subseteq$ is a partial order on $\mathcal{P}(X)$. □

Commentary: Antisymmetry is not symmetry. Symmetry would say $A \subseteq B$ implies $B \subseteq A$, which is false.

31.14 Solution 14: induction for powers of two

Claim: For all $n \geq 1$,

$$2^n \geq n + 1.$$

Proof. Base case: If $n = 1$, then $2^1 = 2 = 1 + 1$, so the claim holds.

Induction step: Suppose $2^k \geq k + 1$ for some $k \geq 1$. Then

$$2^{k+1} = 2 \cdot 2^k \geq 2(k + 1) = 2k + 2.$$

Since $2k + 2 \geq k + 2$ for $k \geq 0$, we have

$$2^{k+1} \geq k + 2.$$

This is the desired statement for $k + 1$. Therefore $2^n \geq n + 1$ for all $n \geq 1$. □

Commentary: The target for the induction step is $2^{k+1} \geq (k + 1) + 1 = k + 2$.

31.15 Solution 15: finite intersection of open sets

Claim: If U and V are open subsets of $\mathbb{R}$, then $U \cap V$ is open.

Proof. Let $x \in U \cap V$. Then $x \in U$ and $x \in V$. Since U is open, there exists $r_1 > 0$ such that $(x - r_1, x + r_1) \subseteq U$. Since V is open, there exists $r_2 > 0$ such that $(x - r_2, x + r_2) \subseteq V$.

Let $r = \min\{r_1, r_2\}$. Then $r > 0$. If $y \in (x - r, x + r)$, then $|y - x| < r \leq r_1$, so $y \in U$, and $|y - x| < r \leq r_2$, so $y \in V$. Hence $y \in U \cap V$.

Thus every $x \in U \cap V$ has a positive-radius interval contained in $U \cap V$, so $U \cap V$ is open. □

Commentary: The smaller radius is the witness that works for both open sets.

32 Proof Repair Workshops

The previous debugging section gave individual repairs. This section gives workshops: clusters of problems that share the same kind of failure. The goal is pattern recognition. When you see a certain kind of broken proof, you should begin to know what kind of repair is likely.

Focus

Do not memorize repairs as scripts. Learn the pressure points: quantifiers, witnesses, definitions, cases, and direction of implication.

33 Reading Real Mathematics

Independent mathematical literacy begins when you can unpack compressed prose. Real books and papers often omit routine steps, change notation quickly, and expect you to infer which theorem is being used. This section trains that skill directly.

Reading note

Ⓐ Hidden quantifier. Real mathematical prose often hides "for every" and "there exists" inside phrases like "choose," "fix," "locally," "sufficiently large," and "for all small enough."

33.1 Compressed excerpt 1: set containment

Compressed text:

> Since $A \subseteq B$, it follows that $\mathcal{P}(A) \subseteq \mathcal{P}(B)$.

Expanded reading:

1. To prove $\mathcal{P}(A) \subseteq \mathcal{P}(B)$, let $S \in \mathcal{P}(A)$.
2. Then $S \subseteq A$.
3. Since $A \subseteq B$, every element of S is in B.
4. Hence $S \subseteq B$.
5. Therefore $S \in \mathcal{P}(B)$.

Hidden assumption: $\mathcal{P}(A)$ means the set of all subsets of A.

33.2 Compressed excerpt 2: continuity by preimages

Compressed text:

> The composition is continuous because preimages commute with composition.

Expanded reading:

1. Let $f : X \to Y$ and $g : Y \to Z$ be continuous.
2. To prove $g \circ f$ continuous, take an open set $W \subseteq Z$.
3. Since g is continuous, $g^{-1}(W)$ is open in Y.
4. Since f is continuous, $f^{-1}(g^{-1}(W))$ is open in X.
5. But $f^{-1}(g^{-1}(W)) = (g \circ f)^{-1}(W)$.
6. Therefore $(g \circ f)^{-1}(W)$ is open in X.

Hidden theorem: inverse images under composition satisfy

$$(g \circ f)^{-1}(W) = f^{-1}(g^{-1}(W)).$$

33.3 Compressed excerpt 3: compactness

Compressed text:

> By compactness, pass to a finite subcover.

Expanded reading:

1. There must be an open cover somewhere in the argument.
2. Compactness applies only after you verify the covered set is compact.
3. A finite subcover means finitely many sets from the original cover still cover the same compact set.
4. The proof should name those finitely many indices.

Hidden assumption: the cover consists of open sets in the correct topology.

33.4 Hidden-assumption detection

When reading a proof, ask:

1. Was a variable introduced with a domain?
2. Was a theorem applied with all hypotheses checked?
3. Was a map assumed continuous, injective, surjective, or well-defined?
4. Was a set open or closed in the correct ambient space?

5. Was division by a possibly zero quantity avoided?
6. Was a finite subcover taken only after compactness was established?

Proof lab

Notation expansion drill. Expand each phrase into three to six explicit lines:

1. "By injectivity, the result follows."
2. "The reverse inclusion is similar."
3. "Choose N sufficiently large."
4. "This is well-defined."
5. "The quotient map is continuous by definition."

33.5 Theorem Dependency Tracing

Advanced reading often fails because a proof silently leans on earlier theorems. You may understand every line locally and still miss the dependency chain.

Formal

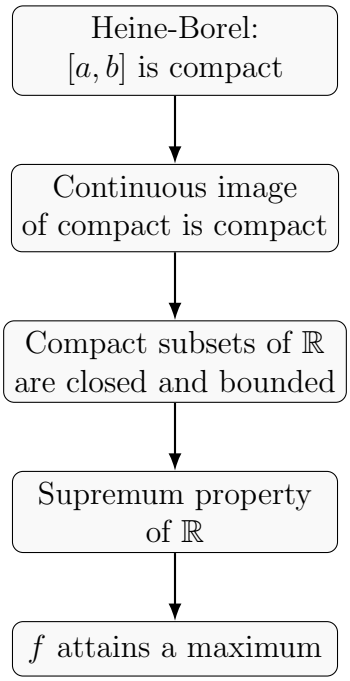
Dependency trace template:

1. Name the theorem being proved.
2. List each theorem, lemma, or definition invoked.
3. For each invocation, check its hypotheses.
4. Mark whether the dependency is local, from the current chapter, or from a previous field.
5. Ask what breaks if the dependency is unavailable.

Example:

> Every continuous function $f : [a, b] \to \mathbb{R}$ attains a maximum.

Dependency chain:



This is why compactness matters. It is not merely a definition about covers; it is a machine that feeds existence theorems.

33.6 Surviving Abstraction Jumps

Mathematical texts often move from examples to abstractions quickly:

$$\text{integers modulo } n \quad \rightsquigarrow \quad \text{quotient groups} \quad \rightsquigarrow \quad \text{quotient spaces} \quad \rightsquigarrow \quad \text{moduli spaces.}$$

When this happens, do not try to swallow the abstraction whole. Ask:

1. What was the original example?
2. What features of the example are being preserved?
3. What details are being deliberately forgotten?
4. What maps are allowed in the new setting?
5. What counts as sameness now?

Why mathematicians care

Abstraction feels powerful when it compresses many examples without erasing the reason they mattered. Bad abstraction feels like fog. Good abstraction feels like suddenly seeing the same architecture in several cities.

33.7 Paper-Style Proof Drill

Here is a deliberately compressed proof in a paper-like style.

> Let $q : X \to X/\sim$ be the quotient map. Since $U \subseteq X/\sim$ is open iff $q^{-1}(U)$ is open, q is continuous. Moreover, any map $f : X/\sim \to Y$ is continuous exactly when $f \circ q$ is continuous, provided f is compatible with the quotient structure.

Unpack it:

1. What is the definition of the quotient topology?
2. Why does that definition immediately imply q is continuous?
3. What does "compatible with the quotient structure" mean?
4. What hidden universal property is being hinted at?
5. Which direction of the continuity claim is immediate, and which requires the quotient topology?

Reading note

The phrase "universal property" is common in modern mathematics. It usually means an object is characterized by how maps into or out of it behave. This is category-theoretic thinking appearing before formal category theory.

33.8 Notation Fatigue Protocol

When notation fatigue hits, do not keep rereading the same line at full speed. Use a reset protocol.

1. Replace every symbol by its type: number, set, function, space, relation.
2. Rename the main objects in ordinary words.
3. Identify the active map, if there is one.
4. Write the target in plain English.
5. Reinsert the notation only after the sentence has meaning again.

Example:

$$q^{-1}(U) \text{ is open in } X.$$

Types:

- $q : X \to X/\sim$ is a quotient map,
- U is a subset of the quotient space,
- $q^{-1}(U)$ is the set of original points whose equivalence classes land in U.

Plain English:

> The set of original points that map into U is open upstairs.

Pause

If you understand the words but not the meaning, translate the sentence into types. Meaning often returns when every symbol has a job again.

33.9 A Gentle Recovery Chapter: One Proof, Three Readings

Here is a small proof read three ways.

Claim:

$$A \cap B \subseteq A.$$

Beginner reading:

> If x is in $A \cap B$, then x is in both A and B. So x is in A.

Formal reading:

Proof. Let $x \in A \cap B$. By definition of intersection, $x \in A$ and $x \in B$. Therefore $x \in A$. Hence $A \cap B \subseteq A$. □

Compressed reading:

> Immediate from the definition of intersection.

All three are the same mathematics at different compression levels.

Why mathematicians care

Mathematical maturity is not always knowing more facts. Sometimes it is knowing how much compression a situation can safely tolerate.

33.10 Workshop 1: quantifier inversion

Quantifier inversion happens when a proof silently changes the order of choices. This is one of the most common errors in analysis, topology, and algebra.

Claim:

$$\forall x \in \mathbb{R}, \exists y \in \mathbb{R},\ y > x.$$

Correct proof:

Proof. Let $x \in \mathbb{R}$ be arbitrary. Choose $y = x + 1$. Then $y \in \mathbb{R}$ and $y > x$. Therefore for every real x, there exists a real y with $y > x$. □

Bad proof of a different claim:

> Choose $y = 1000$. Then for every x, $y > x$.

This attempts to prove

$$\exists y \in \mathbb{R}, \forall x \in \mathbb{R},\ y > x,$$

which is false. No single real number is larger than every real number.

Repair habit: write the quantifier game.

- For $\forall x \exists y$, the opponent gives x first; you may choose y after seeing x.
- For $\exists y \forall x$, you must choose y first; it must survive every x.

Drill:

1. Prove $\forall x \in \mathbb{R}, \exists y \in \mathbb{R},\ x + y = 0$.
2. Disprove $\exists y \in \mathbb{R}, \forall x \in \mathbb{R},\ x + y = 0$.
3. Prove $\forall n \in \mathbb{Z}, \exists m \in \mathbb{Z},\ m > n$.
4. Disprove $\exists m \in \mathbb{Z}, \forall n \in \mathbb{Z},\ m > n$.

Walking question

When you hear "for every," imagine the problem handing you an input. When you hear "there exists," imagine you must produce something concrete.

33.11 Workshop 2: missing witnesses

Existence claims require witnesses. Sometimes the witness is a number. Sometimes it is a function, a set, a radius, an index, or an equivalence class.

Claim: If $a \mid b$ and $a \mid c$, then $a \mid (5b + 7c)$.

Bad proof:

> Since a divides both b and c, it divides any combination of them.

The idea is good, but the witness is missing.

Proof. Suppose $a \mid b$ and $a \mid c$. Then $b = am$ and $c = an$ for some $m, n \in \mathbb{Z}$. Hence

$$5b + 7c = 5am + 7an = a(5m + 7n).$$

Since $5m + 7n \in \mathbb{Z}$, we have $a \mid (5b + 7c)$. □

Witness inventory:

Target	Needed witness	Typical source
$a \mid b$	$k \in \mathbb{Z}$ with $b = ak$	Algebraic factoring
$x \in \mathbb{Q}$	integers p, q with $q \neq 0$ and $x = p/q$	Common denominator
f surjective	preimage a for arbitrary codomain element b	Solve $f(a) = b$
U open	radius $r > 0$ for arbitrary point $x \in U$	Distance to boundary
$a_n \to L$	index N for arbitrary $\varepsilon > 0$	Inequality scratch work
$\exists x\, P(x)$	object x satisfying P	Construction or known theorem

Drill:

1. In the proof that $x + y \in \mathbb{Q}$ for rational x, y, what is the witness numerator and denominator?
2. In the proof that $f(x) = 2x + 1$ is surjective from $\mathbb{R}$ to $\mathbb{R}$, what preimage do you choose for a given y?
3. In the proof that $(0, 3)$ is open in $\mathbb{R}$, what radius works for a point $x \in (0, 3)$?
4. In the proof that $7/n \to 0$, what N works for a given $\varepsilon > 0$?

33.12 Workshop 3: uniqueness

Uniqueness proofs have a standard shape:

1. Suppose there are two objects with the desired property.
2. Use the defining property of both.
3. Prove the two objects are equal.

Claim: If a function has an inverse, then the inverse is unique.

Bad proof:

> The inverse is unique because a function can only be undone one way.

This is intuition, not proof.

Proof. Let $f : A \to B$. Suppose $g : B \to A$ and $h : B \to A$ are both inverses of f. Then

$$g \circ f = \mathrm{id}_A, \quad f \circ g = \mathrm{id}_B,$$

and

$$h \circ f = \mathrm{id}_A, \quad f \circ h = \mathrm{id}_B .$$

We show $g = h$. Let $b \in B$. Since h is an inverse of f, we have $f(h(b)) = b$. Therefore

$$g(b) = g(f(h(b))) = (g \circ f)(h(b)) = \mathrm{id}_A(h(b)) = h(b).$$

Since this holds for every $b \in B$, $g = h$. □

Commentary: The proof does not rely on a picture of undoing. It uses the equations that define an inverse.

Drill:

1. Prove that an identity element in a group is unique.
2. Prove that inverses in a group are unique.
3. Prove that if a real number has an additive inverse, it is unique.
4. Prove that if a maximum element of a subset of $\mathbb{R}$ exists, it is unique.

33.13 Workshop 4: counterexample construction

To disprove a universal claim, one counterexample is enough. But a useful counterexample is not random; it targets the claim's weak point.

False claim:

$$f(S \cap T) = f(S) \cap f(T)$$

for every function f and all subsets S, T of the domain.

How to search:

- The left side uses points that are in both S and T.
- The right side can contain an output reached from one point in S and a different point in T.
- So make S and T disjoint, but make their images overlap.

Counterexample:

$$f : \mathbb{R} \to \mathbb{R}, \quad f(x) = x^2, \quad S = \{-1\}, \quad T = \{1\}.$$

Then $S \cap T = \emptyset$, so $f(S \cap T) = \emptyset$, but $f(S) \cap f(T) = \{1\}$.

False claim:

$$\text{If } f \circ g \text{ is injective, then } f \text{ is injective.}$$

Search:

- To make f not injective, it needs two different inputs with the same output.
- But g might never hit both of those inputs.
- Choose g with small image.

Counterexample: Let $g : \{0\} \to \{0, 1\}$ be $g(0) = 0$, and let $f : \{0, 1\} \to \{0\}$ be the constant function. Then f is not injective, but $f \circ g : \{0\} \to \{0\}$ is injective because its domain has one element.

Common trap

Counterexamples often live in tiny sets. Do not search first among complicated objects when a two-element set will expose the failure.

Drill:

1. Disprove: If $f \circ g$ is surjective, then g is surjective.

2. Disprove: If $A \cup C = B \cup C$, then $A = B$.
3. Disprove: If $A \cap C = B \cap C$, then $A = B$.
4. Disprove: Every relation that is symmetric and transitive is reflexive.
5. Disprove: Every continuous function is differentiable.

33.14 Workshop 5: hidden division by zero

Algebraic proofs often fail because they divide by a quantity that might be zero.

False proof that all numbers are equal:

Let $a = b$. Then $a^2 = ab$. Subtract b^2:

$$a^2 - b^2 = ab - b^2.$$

Factor:

$$(a - b)(a + b) = b(a - b).$$

Cancel $a - b$:

$$a + b = b.$$

Since $a = b$, this gives $2b = b$, so $b = 0$. Therefore all equal numbers are zero.

The first bad line is the cancellation of $a - b$. Since $a = b$, $a - b = 0$. Dividing or canceling by zero is invalid.

Repair habit:

- Before dividing by an expression, prove it is nonzero.
- If it may be zero, split into cases.
- If cancellation occurs in an abstract structure, check that the cancellation law is available.

Drill:

1. Identify the invalid step in the classic false proof that $1 = 2$.
2. Prove: If $ab = ac$ and $a \neq 0$ in $\mathbb{R}$, then $b = c$.
3. Explain why the condition $a \neq 0$ is necessary.
4. In a group, prove cancellation without using division.

33.15 Workshop 6: epsilon repairs

Epsilon proofs are intimidating because the proof is written forward but discovered backward.

Claim:

$$\frac{2n+3}{n} \to 2.$$

Scratch work:

$$\left|\frac{2n+3}{n} - 2\right| = \frac{3}{n}.$$

We want $\frac{3}{n} < \varepsilon$, so it is enough to choose $n > 3/\varepsilon$.

Final proof:

Proof. Let $\varepsilon > 0$. Choose $N \in \mathbb{N}$ such that $N > 3/\varepsilon$. If $n \geq N$, then

$$\left|\frac{2n+3}{n} - 2\right| = \frac{3}{n} \leq \frac{3}{N} < \varepsilon.$$

Therefore $(2n+3)/n \to 2$. □

Bad proof:

> Let $N > 3/\varepsilon$. Since n is large, $3/n < \varepsilon$.

Diagnosis:

- It does not state $n \geq N$.
- It says "large" instead of using the chosen threshold.
- It does not show the absolute-value expression.

Drill:

1. Prove $5/n \to 0$.
2. Prove $(n+1)/n \to 1$.
3. Prove $(4n-7)/n \to 4$.
4. Prove $1/(n+2) \to 0$.

34 Final Exam: From Beginner to Velleman-Ready

This exam is a diagnostic, not a performance ritual. Take it under realistic conditions: no notes for the main attempt, then a second pass with notes in a different color. The second pass is where learning becomes visible.

Focus

Scoring rule: a correct answer with unclear reasoning is not fully correct. A proof must show why the conclusion follows.

34.1 Instructions

1. Allow three hours for the first attempt.
2. Write full sentences in proof problems.
3. State definitions when you use them.
4. For any problem you cannot solve, write the proof skeleton or explain the obstruction.
5. After finishing, debug your own work before checking solutions or notes.

34.2 Part 1: logic and language

1. Define statement and predicate. Give two examples of each.
2. Explain why $x^2 = 9$ is not a statement until a domain and quantifier or value are supplied.
3. Translate into symbols: "Every integer has a larger integer."
4. Translate into English: $\exists M \in \mathbb{R}, \forall x \in S,\ x \leq M$.
5. Negate: $\forall x \in A, \exists y \in B,\ R(x, y)$.
6. Negate: $\exists N \in \mathbb{N}, \forall n \geq N,\ |a_n - L| < \varepsilon$.
7. Give an example where $\forall x \exists y\, R(x, y)$ is true but $\exists y \forall x\, R(x, y)$ is false.
8. State the contrapositive and converse of: If n^2 is even, then n is even.
9. Explain why a conditional with false hypothesis is true in classical logic.
10. Give a counterexample to the statement: If n is odd, then n is prime.

34.3 Part 2: definitions and arithmetic proofs

1. Prove that the sum of two even integers is even.
2. Prove that the sum of two odd integers is even.
3. Prove that the product of two odd integers is odd.
4. Prove that if $a \mid b$ and $a \mid c$, then $a \mid (b - c)$.
5. Prove that if $a \mid b$ and $b \mid c$, then $a \mid c$.
6. Prove that if $x, y \in \mathbb{Q}$, then $x + y \in \mathbb{Q}$.
7. Prove that if $x, y \in \mathbb{Q}$, then $xy \in \mathbb{Q}$.
8. Prove that if n^2 is even, then n is even.
9. Prove that if n^2 is divisible by 3, then n is divisible by 3.
10. Prove that $\sqrt{2}$ is irrational.

34.4 Part 3: sets

1. Prove $A \cap B \subseteq A$.
2. Prove $A \subseteq A \cup B$.
3. Prove $A \setminus B = A \cap B^c$.
4. Prove $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.
5. Prove $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$.
6. Prove that if $A \subseteq B$, then $A \cap C \subseteq B \cap C$.
7. Prove that if $A \subseteq B$, then $\mathcal{P}(A) \subseteq \mathcal{P}(B)$.
8. Prove $A \subseteq B$ if and only if $B^c \subseteq A^c$.
9. Give an example of two sets A and B such that $A \cup B = A$ but $B \neq A$.
10. Give an example showing that $A \setminus B$ need not equal $B \setminus A$.

34.5 Part 4: functions

1. State the definitions of injective and surjective.
2. Prove that $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = 3x - 5$ is injective.
3. Prove that $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = 3x - 5$ is surjective.
4. Prove that if f and g are injective, then $g \circ f$ is injective.

5. Prove that if f and g are surjective, then $g \circ f$ is surjective.
6. Prove $f^{-1}(S \cap T) = f^{-1}(S) \cap f^{-1}(T)$.
7. Prove $f^{-1}(S \cup T) = f^{-1}(S) \cup f^{-1}(T)$.
8. Give a counterexample showing $f(S \cap T) = f(S) \cap f(T)$ can fail.
9. If $f : A \to B$ is injective and $S, T \subseteq A$, prove $f(S \cap T) = f(S) \cap f(T)$.
10. For $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = x^2$, compute $f^{-1}([1, 4])$.

34.6 Part 5: relations

1. Define reflexive, symmetric, transitive, antisymmetric, and total.
2. Prove congruence modulo m is reflexive.
3. Prove congruence modulo m is symmetric.
4. Prove congruence modulo m is transitive.
5. Describe the equivalence classes modulo 5.
6. Prove that if $\sim$ is an equivalence relation, then equivalence classes are either equal or disjoint.
7. Prove subset inclusion is a partial order on $\mathcal{P}(X)$.
8. Prove divisibility is a partial order on $\mathbb{N}$.
9. Give an example of a relation that is reflexive and symmetric but not transitive.
10. Give an example of a relation that is transitive but not symmetric.

34.7 Part 6: induction

1. State the principle of mathematical induction.
2. Prove by induction that $1 + 2 + \cdots + n = n(n + 1)/2$.
3. Prove by induction that $1 + 3 + \cdots + (2n - 1) = n^2$.
4. Prove by induction that $2^n \geq n + 1$ for all $n \geq 1$.
5. Prove by induction that $3 \mid (4^n - 1)$ for all $n \geq 1$.
6. Prove by induction that $n! \geq 2^{n-1}$ for all $n \geq 1$.
7. Explain the difference between ordinary induction and strong induction.
8. Use strong induction to prove that every integer $n \geq 2$ has a prime divisor.

9. Identify the flaw: "Assume $P(n)$ is true. Therefore $P(n+1)$ is true. By induction, $P(n)$ is true for all n."
10. Write a proof skeleton for an induction proof before filling in algebra.

34.8 Part 7: proof debugging

For each bad proof, identify the first bad line and repair it.

1. Claim: If $x, y \in \mathbb{Q}$, then $x + y \in \mathbb{Q}$. Bad proof: "Write $x = a/b$ and $y = c/d$. Then $x + y = (a + c)/(b + d)$."
2. Claim: If n^2 is even, then n is even. Bad proof: "Since $n^2 = 2k$, $n = \sqrt{2k}$, so n is even."
3. Claim: If $A \subseteq B$, then $\mathcal{P}(A) \subseteq \mathcal{P}(B)$. Bad proof: "Power sets preserve size, so the result follows."
4. Claim: If f is surjective, then every element of the codomain is hit. Bad proof: "Let a be in the domain. Then $f(a)$ is in the codomain."
5. Claim: $U \cap V$ is open if U and V are open. Bad proof: "Open sets are closed under intersection, so it is open."
6. Claim: Congruence modulo m is symmetric. Bad proof: "If a and b have the same remainder, then b and a have the same remainder."
7. Claim: $f^{-1}(S \cap T) = f^{-1}(S) \cap f^{-1}(T)$. Bad proof: "Apply f^{-1} to both sides."
8. Claim: There are infinitely many primes. Bad proof: "The product of all primes plus 1 is prime."
9. Claim: If $A = B$, then $A \cup C = B \cup C$. Bad proof: "This is obvious from equality."
10. Claim: If n is even, then n^2 is even. Bad proof: "This is the converse of a true theorem."

34.9 Part 8: readiness reflection

Answer in paragraphs.

1. Which proof method feels most natural to you, and why?
2. Which proof method produces the most mistakes for you?
3. What is your most common proof error: quantifier, definition, witness, case, implication direction, or algebra?
4. Which three definitions must you review before starting Velleman?
5. Choose one proof from this exam and explain its architecture without symbols.

6. Choose one failed attempt and write a repair plan.
7. How will you structure your first two weeks with Velleman?
8. What will count as evidence that you are reading actively rather than passively?

34.10 Scoring guide

Use this guide after your first attempt.

Score band	Interpretation
85–100%	Start Velleman now. Keep a proof-debugging log and continue spaced review.
70–84%	Start Velleman slowly. Review weak topics every other day and rewrite missed exam proofs.
55–69%	Delay full-speed Velleman. Spend two weeks on the readiness plan, especially definitions and proof skeletons.
Below 55%	Return to the first-pass reader and rebuild. This is not failure; it is accurate diagnosis.

34.11 Final readiness contract

Before you move on, write and sign a short contract with yourself:

> I will not treat confusion as a verdict. I will treat it as a location. When a proof fails, I will find the first bad line, name the gap, open the relevant definition, and repair the argument.

Focus

You are ready for the next book when you can work this way without needing someone else to rescue the proof at the first sign of fog.